

SCARA Robot THE600 Robot Manual

(Compatible Controller: TS5000-MS)

Rev. 7

October 1, 2022

Important

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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Introduction

Thank you very much for your purchase of SHIBAURA MACHINE's robot system.
This manual describes the basic specifications of the industrial robot, and how to unpack and install the robot, how to connect wiring, how to install the hand, and how to maintain and inspect the robot.
Read this manual before unpacking the product.
Before beginning the work according to this manual, read through the Safety Manual so that you can understand the safety measures.

This manual has the following chapters.

Specifications
Section 1: Specifications
This section describes the basic specifications and names of respective units of the robot.

Chapter 2: Functions
This chapter describes the key points for using the functions provided with the robot.

Installation
Section 3: Unpacking and Transportation
This section describes how to unpack the robot and how to transport it to the installation site.
This section also describes precautions for temporarily storing the unpacked robot.

Section 4: Installation of the Robot
This section describes the installation environment, required space, and installation procedure for the robot.

Sections 5 and 6: Setting of the Hand
These sections describe how to install the hand on the robot arm and how to wire the hand.
Lastly, these sections describe the permissible load conditions of the hand.

Maintenance
Sections 7 and 8: Maintenance
These sections describe how to maintain respective units of the robot.

Section 9: Zero Setting and Position Detector Error
This section describes the zero setting procedure for the robot and the corrective action if the position detector error occurs.

Section 10: Replacement Parts
This section describes replacement parts of the robot.

Section 11: Disposal
This section describes disposal of the robot and parts.

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11.1 Notes on Disposal 238

Use Range

This product is the horizontally articulated type industrial robot in which the manipulator has two horizontally rotating joints and the mechanical interface on its tip moves vertically and rotates horizontally.
The product has been developed with intention of adapting it to automation of wide range of work including conveying work such as movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product with the aim to adapt it to automation of these work and incorporate it to the automation equipment.

This product is not intended to apply it to work that contacts a human (such as work related to medical care and nursing care), incorporate it to the equipment that supports human life (such as life support device and incubator), incorporate it to the equipment on which a human rides (including transport equipment such as car, aerospace equipment such as airplane, and game equipment such as roller coaster), and incorporate it to the equipment that protects human life and human body (such as safety device).
This product is not intended to incorporate it to the equipment aimed to control large quantity of heat (such as nuclear power controller and incinerator controller).
This product is prohibited from being used to apply it to work or incorporate it to the equipment that is likely to threaten human life and harm a human body directly due to a failure or malfunction. If this product is used, it will be guaranteed under customer's responsibility.

Use Method

The use conditions for the THE series robot are described. If the product is used in the state that differs from the conditions, unexpected risk may occur. Abide by the conditions for use.

1. Main Specifications of Product: See Specifications "1.4 Specifications Table" and Installation "4.1 Installation Environment."

2. Restrictions on Use: Tables 1 and 2 give "Restricted Specification Sheet" indicating restrictions on use.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer to the installation site. - Controller [1] Use a cart to transfer to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Push the robot by hand to move to a location where it is easy to perform work. [6] Mount the hand. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint axis and the origin position. [8] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power to robot, controller, and optional parts while their covers are removed - Operation in other installation state than the specified (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Entry into the hazard zone of the robot when the servo power is on - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Movable range	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: Repacking: Maintenance: Disposal: 	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m^3 shall be secured. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.

Table 3. Restricted Specifications Sheet

3. Movable range of product (restriction on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - The drive power shall be turned off whenever a person enters the robot hazard zone. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of "transfer and installation" (transportation, unpacking, transfer, installation, and repacking), "use" (teaching, operation, and maintenance), and "disposal."

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	-
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operation procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	-
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct a strict tests and inspections before delivery to ensure that the product satisfies our performance standards.
We provide warranty to cover defects of the product based on the following prerequisites.

1. Warranty period The warranty period of the product is either of the following, whichever is earlier.
   1) 18 months after delivery from our factory
   2) 12 months after installation at your site

2. Warranty scope
   1) We provide warranty for the product. Note that only the specifications and functions defined in the product specifications, catalog, and instruction manual are covered by the warranty. Any secondary or incidental damages caused by failure of the product are not covered by the warranty in any circumstances.
   2) We provide repair of a faulty product for free only if the product has been handled and used according to the instruction manual that comes with the product.
      Any repairing after the warranty period passes will be charged.

3. Disclaimer The warranty shall not apply to the following cases.
   a) Any failure or damage arising from your careless use or misuse that is not defined in the instruction manual.
   b) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables*1).
   c) Any failure regarding sensuous phenomena such as noise not affecting the functions
   d) When any modification or disassembly without our consent is confirmed
   e) Any failure or damage arising from insufficient maintenance and inspection or inappropriate repair
   f) Any failure or damage due to act of God, fires, and other force majeure
   g) Internal data such as programs and points created or modified by users
   h) When a product purchased in Japan is taken abroad

4. Precautions
   1) SHIBAURA MACHINE shall not guarantee the standard performance of the product if you use the product beyond the specifications.
   2) If the customer does not observe the “DANGER,” “WARNING,” and “CAUTION” described in this instruction manual, SHIBAURA MACHINE will not assume responsibility for any consequential accident resulting in injury or death, damage or trouble.
   3) “DANGER,” “WARNING,” “CAUTION” and other descriptions stipulated in this instruction manual are only those which can be assumed by SHIBAURA MACHINE without predicting all events in all situations. Please be aware that the range is limited.

*1): Consumable parts refer to the replacement parts for maintenance as listed in instruction Manual “10.2 List of Replacement Parts – Robot.”

Standards, laws and regulations regarding safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" defines measures for reducing risks regarding machinery.

Step 1: Fundamental safe design measures - Restrictions on force, speed, and energy

Step 2: Safety protection - Installation of protective fences, etc.
Additional protection measures - Installation of emergency stop devices

Step 3: Information for use - Warning labels, alerts, instruction manuals

Based on the above, the ISO/IEC defines the "Guide 51" that groups various standards in a hierarchal structure. Safety standards for industrial robots are standardized as separate safety standards for machinery in layer C.

- ISO 10218, ISO 10218-1

After risk assessment is carried out, provision of residual risk information to the user is standardized.

- IEC 82079-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

Safety Precautions

This manual contains important information for the safe use of the robot body and the prevention of injury to the operator and others and the prevention of damage to property.
Read and understand the meanings of the following indications and symbols first, and observe these precautions.

[Meanings of Indications]

Indication	Meaning
⚠ DANGER	Indicates the imminent danger that incorrect operation could cause death or serious injury.
⚠ WARNING	Indicates the danger that incorrect operation could cause death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1)] to operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2)] and advice about performing work.

1) Injury indicates injuries such as wounds, burns, and electric shock that do not require hospitalization or long-term outpatient treatment.
2) Damage to property indicates extensive damage related to the destruction of assets or materials.

⚠ DANGER
• In order to ensure robot safety from installation to disposal, be sure to carefully read the separate instruction manual “Safety Manual” before starting the actual work. Performing an operation different from what is described in the manual could result in harm to people.

Service Network

Please refer to our distributor's handouts.

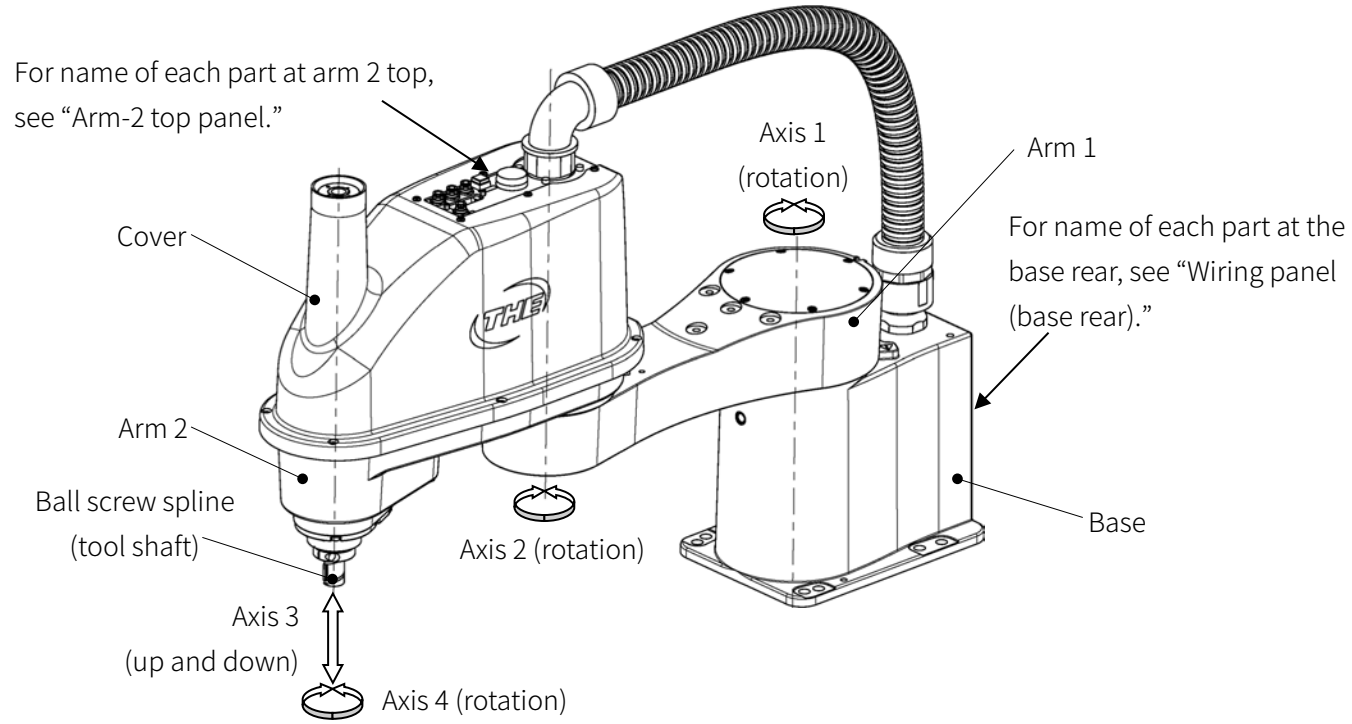
Specifications

This part describes the basic specifications, name of each part, coordinate system, and external dimensions of the THE600 robot.

1. Specifications

1.1 Name of Each Part

Fig. 1.1 shows the name of each part of the robot.



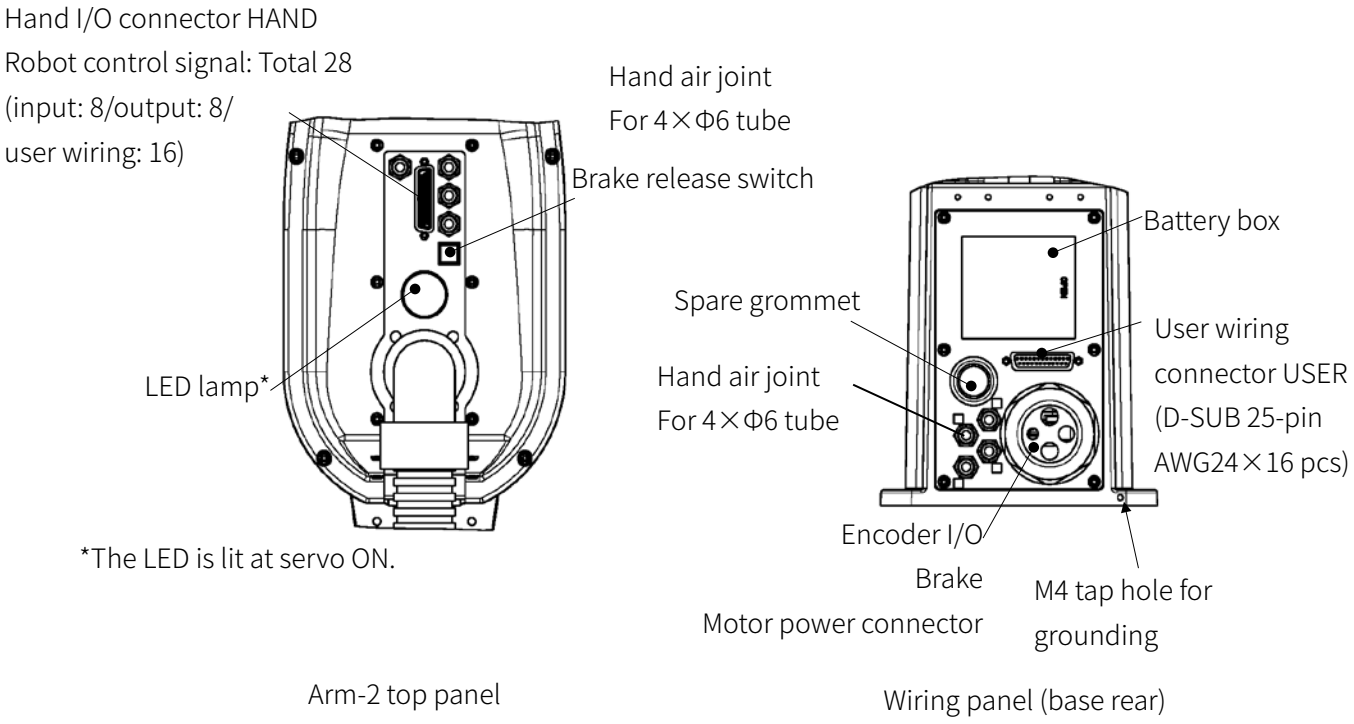
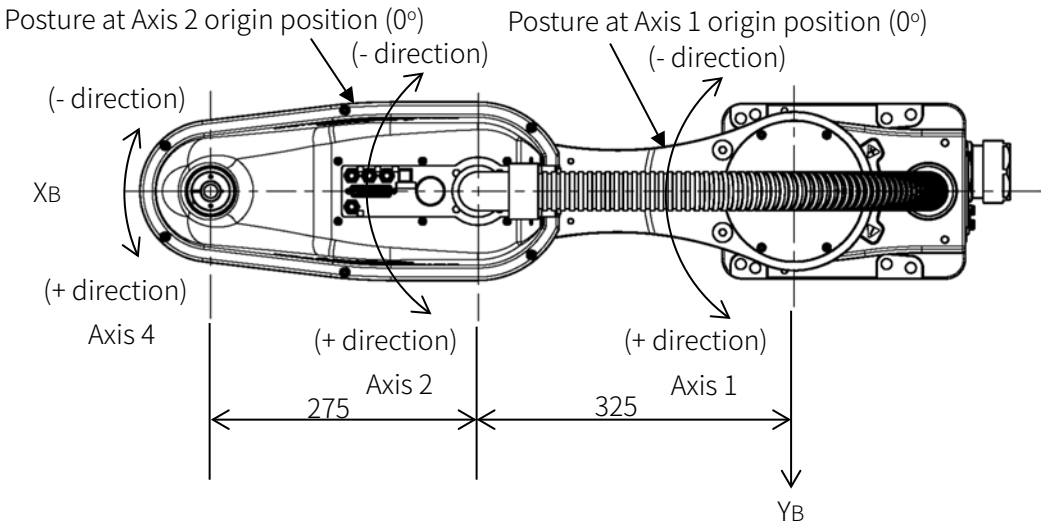
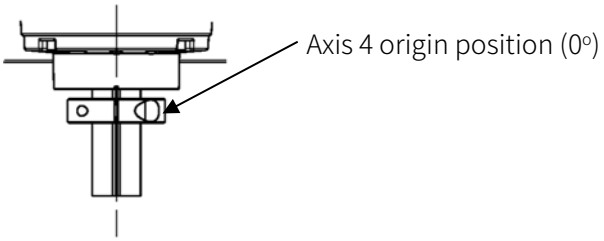


Fig. 1.1 Name of each part

1.2 Coordinate System

The robot's joint angle or stroke origin (0° or 0 mm position) is factory-calibrated according to the reference planes. Fig. 1.2 shows the base coordinate system and origin of each axis joint angle.





* The number in parentheses indicates the coordinate of each axis.

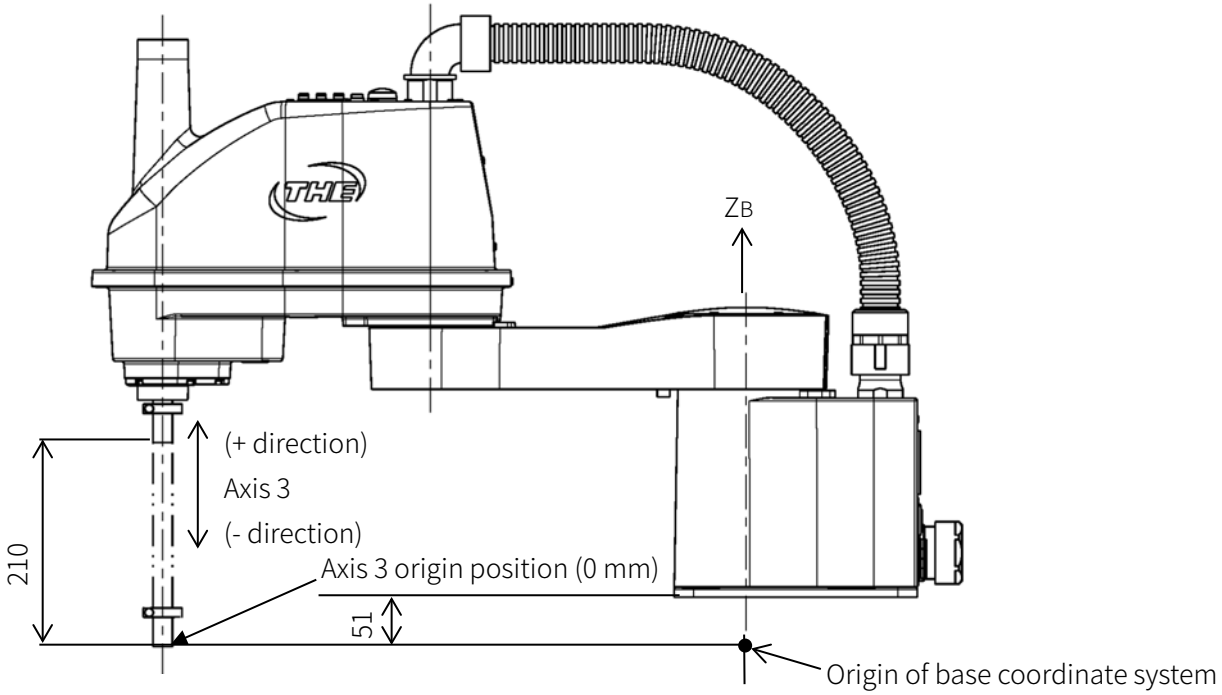


Fig. 1.2 Base coordinate system and joint angle origin

1.3 External Dimensions

Fig. 1.3 refers to the external dimensions of the THE600.

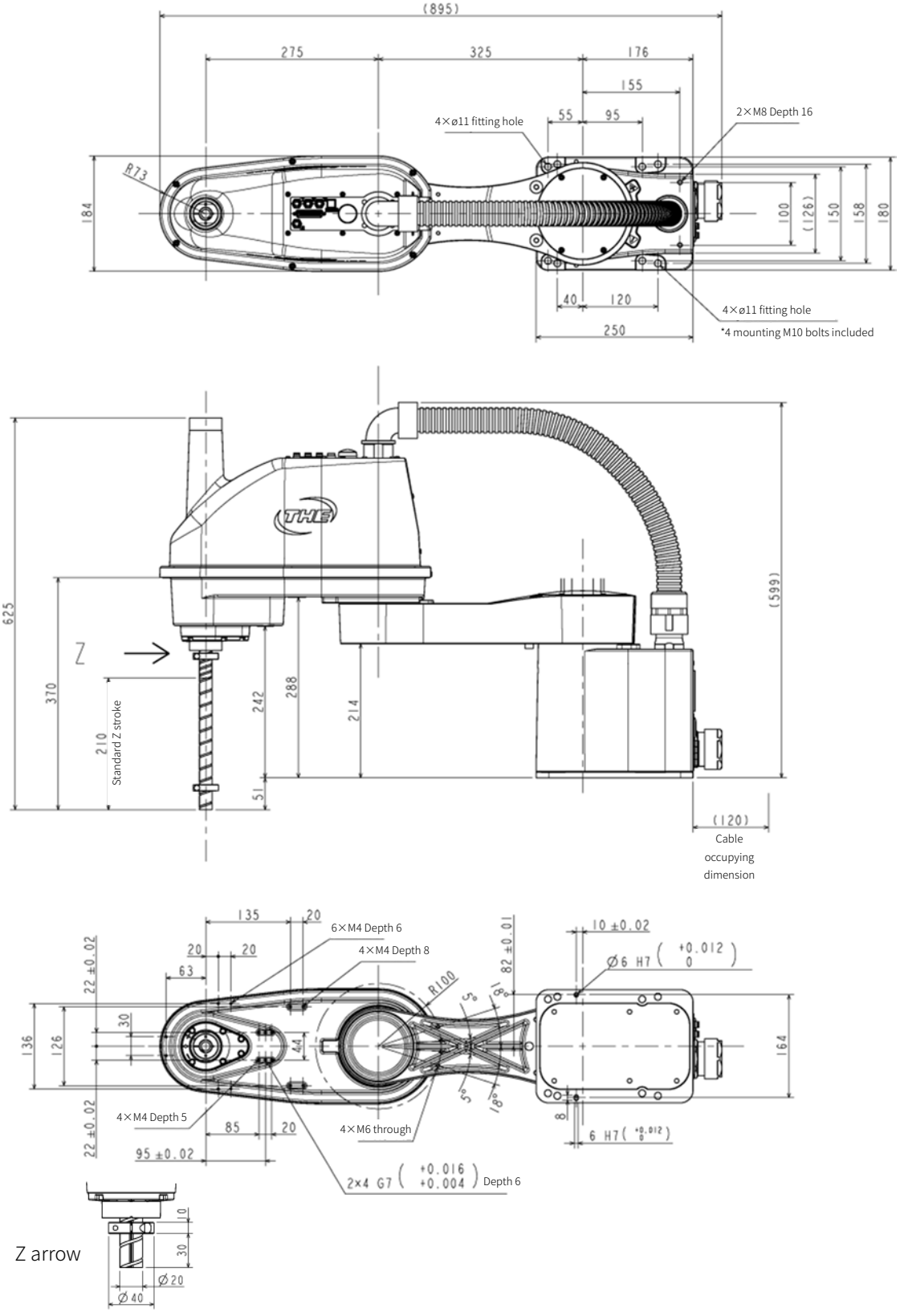


Fig. 1.3 External dimensions of the robot (THE600)

1.4 Specifications Table

Item		Specifications
Structure		Horizontally articulated type SCARA robot
Model		THE600
Applicable controller		TS5000-MS
Mass of robot body		31 kg
Number of controlled axes		Four (4)
Arm length		600 mm (275 mm + 325 mm)
Motor capacity/current limit	Axis 1	1000 W / 28.2Ao-p
	Axis 2	400 W / 14.1Ao-p
	Axis 3	400 W / 18.4.Ao-p
	Axis 4	400 W / 14.1Ao-p
Operating range	Axis 1	±140 deg
	Axis 2	±152 deg
	Axis 3	0 to 210 mm
	Axis 4	±360 deg
Maximum speed *1	Axis 1	457 deg/s
	Axis 2	672 deg/s
	Axis 3	2000 mm/s
	Axis 4	2359 deg/s
	Composite speed of axes 1 and 2	8.0 m/s
Rated payload mass*1		2 kg
Maximum payload mass*1		12 kg
Permissible load inertia*1		0.25 kg·m^2
Maximum offset of load gravity center		100 mm
Repeatability*2	Composite speed of axes 1 and 2	±0.01 mm
	Axis 3	±0.01 mm
	Axis 4	±0.005 deg
Cycle time (when payload mass is 2 kg)*3		0.31 sec
Drive system		AC servo motor
Position detection method		Absolute
Wiring for hand		Input: 8, output: 8
Paint color*4 Material color*4		Arm 2: PANTONE 293C or equivalent Base and arm 1: White AN-90 or equivalent Arm cover: White AN-90 or equivalent
Noise level*5		73 dB (A)
Power supply capacity*6		4.3 kVA

*1: The speed and acceleration are limited in accordance with the operation pattern, the load mass, and the offset value.

*2: Repeatability is the repeat accuracy in one direction at a constant ambient temperature and machine temperature. It is not absolute positioning accuracy. It may exceed the specification values depending on the operation pattern, load weight, and offset value. As for X-Y and C, they represent values at the upper limit of Z. Trajectory accuracy is not guaranteed.

*3: Continuous operation with standard cycle operation pattern and exceeding the effective load ratio cannot be performed.
During shuttle for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

*4: The color may differ according to each production lot. Be aware that there is no problem in the very nature of the product quality.

*5: The measurement conditions are as follows.

Operating condition:	Rated load, concurrent operation of axes 1 and 2, maximum speed, and maximum acceleration
Measurement position:	Height of 1.6 m from the robot installation surface at a distance of 1 m from four front, rear, right and left directions of the robot

*6: The measurement conditions are as follows.

Operating condition:	Rated load and 2 kg, concurrent operation of all axes, maximum speed, and maximum acceleration
Measuring instrument:	Power analyzer (POWER ANALYZER 3390)

NOTICE

- Never use the robot in a high-temperature environment.
  Otherwise, the performance and lifespan of the alkaline battery used for motor position detector battery are reduced.
  - If using the robot in a high-temperature environment, please consult with SHIBAURA MACHINE sales office.
- Never allow the lowered ball screw spline shaft to collide with obstacles.

Otherwise, the ball screw spline shaft will be damaged before an alarm is generated.

- When micro vibration occurs, be sure to decrease the speed and decelerate the robot.
  Otherwise, the performance and lifespan of the robot may be reduced. The robot may become faulty.
  - Use the ACCEL/DECEL command to adjust the speed to the acceleration/deceleration at which the robot does not vibrate.
  - Micro vibration may occur on the robot tip depending on the posture of the robot.
- To operate axes 1, 2, and 4, be sure to raise the ball screw spline shaft as much as possible.
  If the ball screw spline shaft is lowered, the shaft will be damaged early.
  - To operate axes 1, 2, and 4 with the ball screw spline shaft lowered, use the SPEED command and ACCEL/DECEL command to adjust the operating speed and acceleration/deceleration to prevent the ball screw spline shaft from vibrating.

2. Functions

2.1 Function for Specifying Pressing Force (N)

2.1.1 Usage Range

Figure 2.1 shows the range where the function for specifying the pressing force (N) can be used. Generally, use in the range indicated by “Usage allowed.” The range where the 3rd axis will drop is indicated by the “Drop” range. If the robot tip will be used with a light force, use it while being extremely careful about dropping. Do not use in the “Usage not allowed” range.
Table 2.1 shows the maximum margin of error when using this function. This data is intended to indicate the general trend, and so use it for reference only.

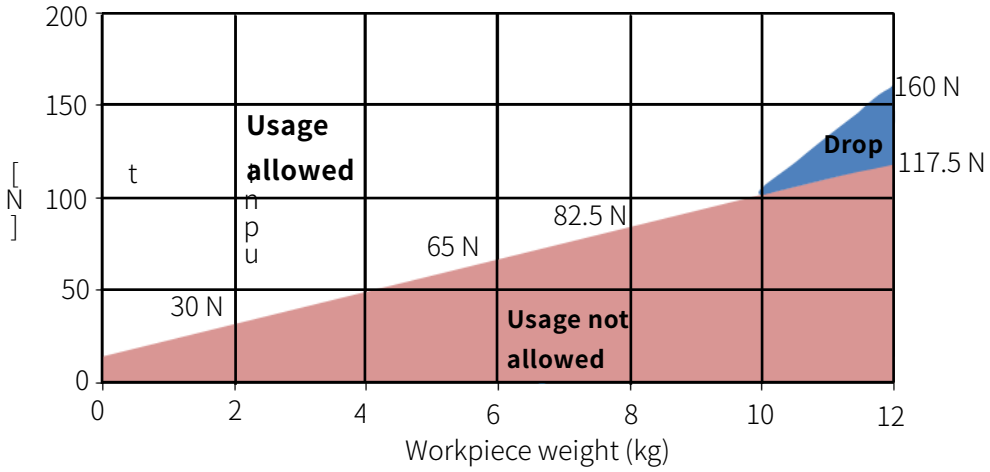


Figure 2.1 Usage range (THE600-E)

Table 2.1 Maximum margin of error in function for specifying pressing force

Model	Maximum margin of error
THE600-E	Approx. 30.0 N

2.1.2 Notes

NOTICE

- Be sure to always use the PAYLOAD command and specify a suitable setting value. If the PAYLOAD command is not set properly, a large difference could occur between the specified value and actual value.
- Use while grease is applied to the ball screw spline shaft. Usage without applying grease could lead to a large margin of error, resulting in damage to the pressed object or the ball screw spline shaft.
- If the robot was stopped for a half-day or longer, move the 3rd axis, and perform break-in operation for about 2 to 3 minutes. Usage without performing break-in operation could lead to a large margin of error, resulting in damage to the pressed object or the ball screw spline shaft.
- During the pressing operation, use with the SPEED command set to 10% or less. Although usage at conditions above 10% is possible, adjustment was performed at factory shipping so that a suitable value is obtained at 10%. If using under different conditions, a large margin of error could occur between the specified value and actual value, resulting in damage to the pressed object or the ball screw spline shaft.
- Adjustment was performed at factory shipping so that a suitable value is obtained when the override is 100%. Use at 100% for operation. However, in the operation check, operate at low speed to check that there are no problems, and then increase the override.

Installation

This part describes information you should know before unpacking, transporting, installing, and operating the robot.

3. Unpacking and Transportation

The worker should receive safety training defined by each country's laws and regulations. Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

⚠ CAUTION

- Be sure to convey the product by two or more workers.
  Working in an unnatural posture causes the packing box to drop.
  - Secure the transportation route in advance.
  - Use the cart, etc. to carry the product.
  - Wear the protective shoes.

NOTICE

- Be sure to convey the product with by two or more workers.
  Working in an unnatural posture causes the packing box to drop and the product in the box to be damaged.
  - Do not use the dropped product.

3.1 Unpacking

The robot is shipped in corrugated cardboards. Fig. 3.1 shows the packing state. Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot. After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport. For robot and controller accessories, see the accessory list packed with the controller.

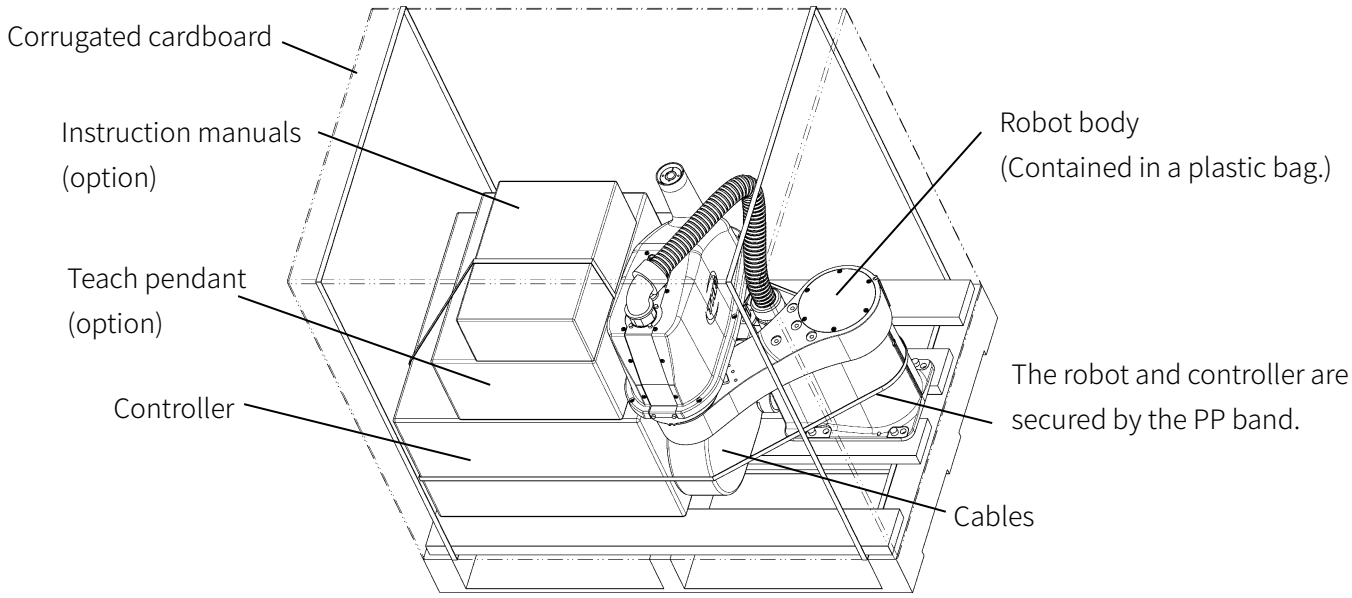


Fig. 3.1 Packaging state (THE600)

Table 3.1 Size and Mass

Packing size	630 mm (W) × 900 mm (H) × 920 mm (D)
Packing mass	About 66 kg (option not included)

⚠DANGER

- Never use the damaged robot.
  Otherwise, the faulty robot may cause unpredictable disaster.
  - Check the appearance based on the external view described in the manual.
  - Check the parts based on the accessory list attached to the controller.
  - If damage is found or parts are insufficient, please contact SHIBAURA MACHINE Sales Division.

⚠ CAUTION

- When cutting the packing rope with the cutter, never direct the blade toward a person.
  Otherwise, the person may get a cut due to the cutter.
  - Use the nippers or scissors.
  - Wear gloves.
- When putting the robot, cables, etc. in and out of the packing box, never lift them up in an unnatural posture.
  Otherwise, excessive burden may damage worker's back.
  - Secure space where work can be performed easily.
  - Put the robot, cables, etc. out of the packing box by multiple workers according to the situation.

NOTICE

- Never use the damaged robot.
  The robot is faulty.
  - Check the appearance based on the external view described in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or the parts are insufficient, please contact SHIBAURA MACHINE Sales Division.
- Be sure to dispose of the unnecessary packing materials according to the customer's in-house regulations.
  To store the packing materials, be sure to manage them to prevent them from being used for other than packing the robot.

3.2 Transportation

Move the robot very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the clamp and base set bolts completely. When transporting the product to a separate location, secure it with the band first as it was packed, and then put the product back into the corrugated cardboards.

3.3 Mass and Dimensions

Fig. 3.2 shows the mass and external dimensions of the robot.

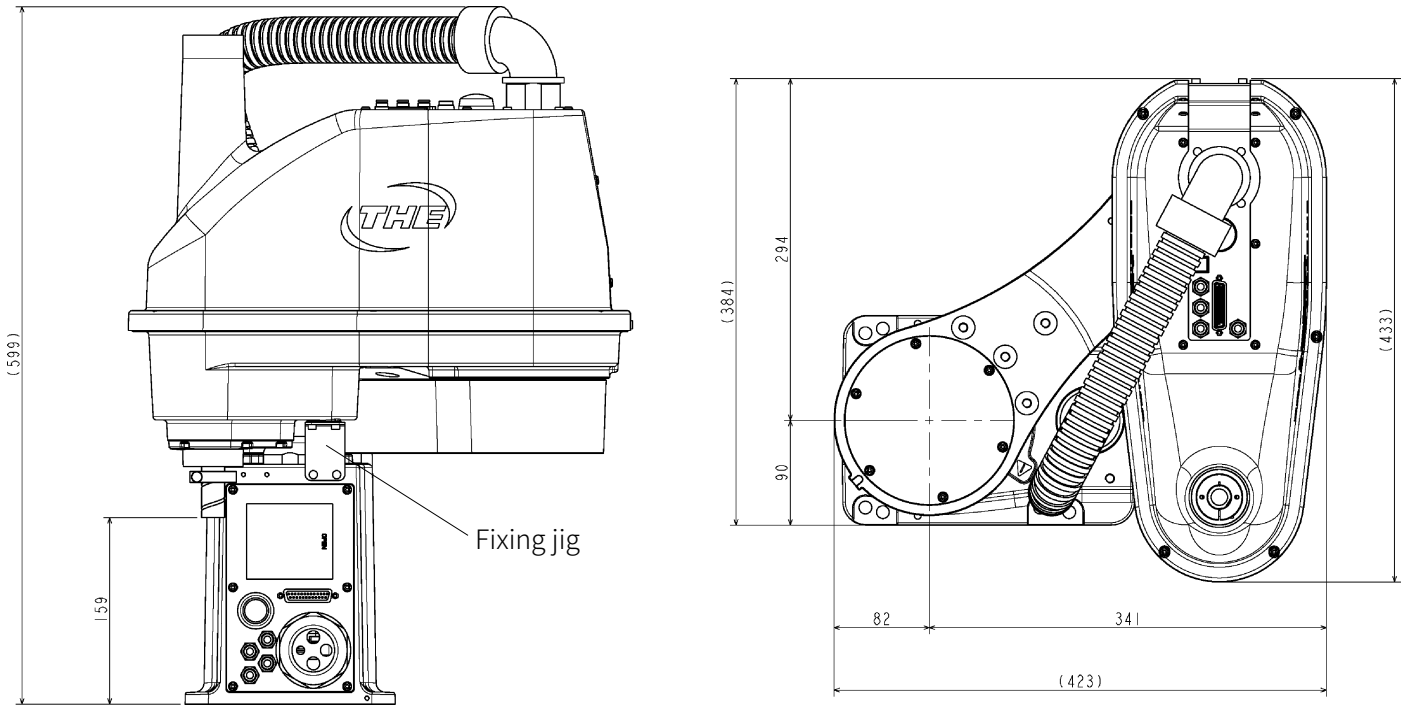


Fig. 3.2 Outer dimensions at transport (THE600)

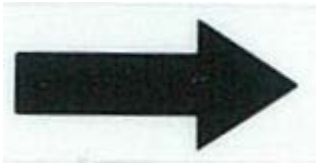
3.4 Transporting the Robot

⚠ WARNING

- When hoisting the robot, be sure to always follow the following safety guidelines.
  Otherwise, the robot could drop.
  - Use the supplied eyebolts or eyebolts with a working load of 0.79 kN or more when suspended at 45°.
  - Use a wire with a load limit that can sufficiently bear the weight of the robot.
  - Be sure to always hoist and lower the robot slowly.
  - Clear a transfer route, and make sure that no one goes under the suspended load.

The robot can be hoisted and transferred.

The following arrow marks are affixed to the base and 1st arm of the robot. Mount the two supplied M8 eyebolts into the tapped holes indicated by the arrow marks as shown below in Fig. 3.3, secure the wire to the eyebolts, and slowly hoist the unit by two-point suspension.

Label	Label description
	- Mount the supplied eye bolts into the tapped holes indicated by the arrow mark.
- When hoisting and transferring the robot by crane, be sure to mount the supplied eyebolts into the tapped holes indicated by the arrow marks. Otherwise, the robot will be unstable and could crash into a worker.	

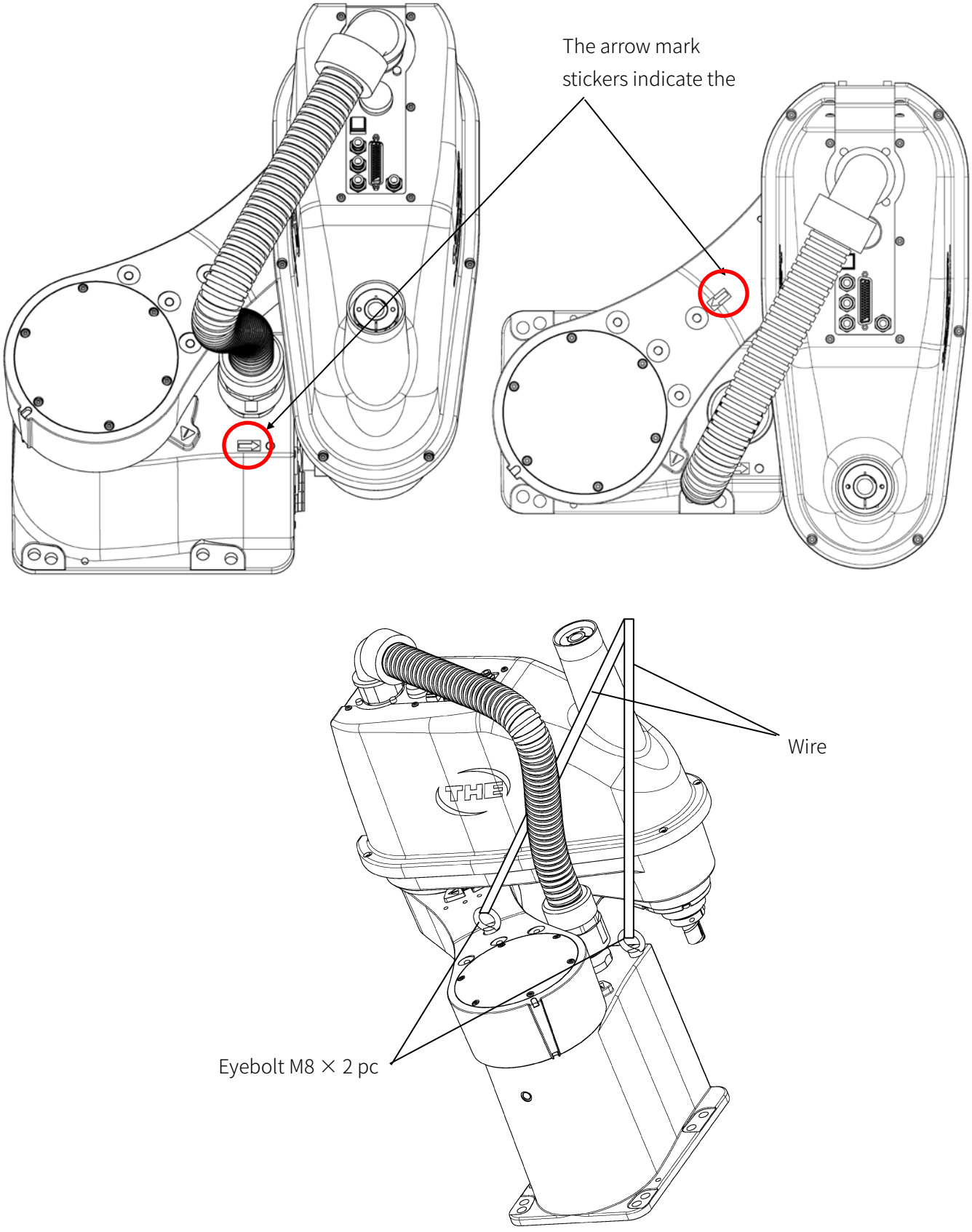
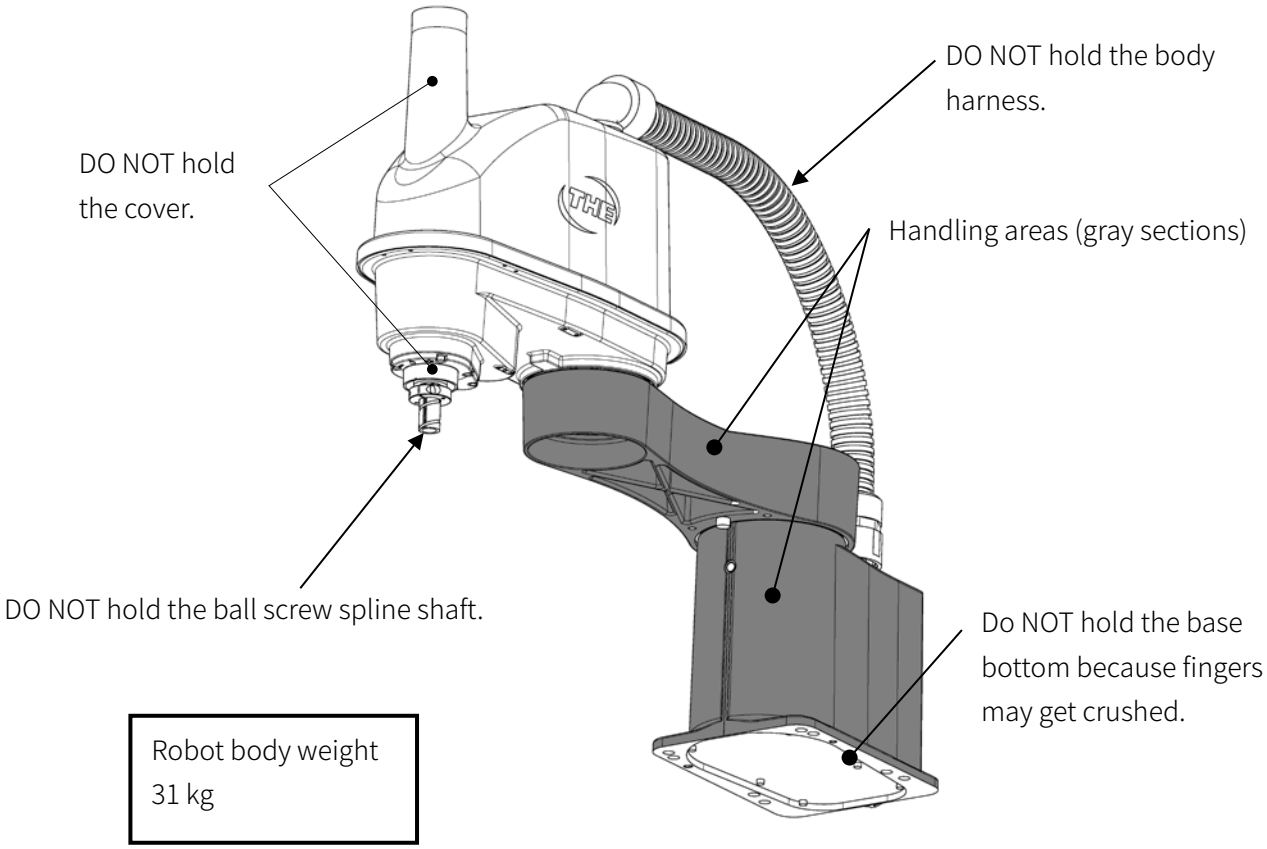


Fig. 3.3 Lifting up the robot

NOTICE

- When hoisting and transferring the robot, mount the supplied eyebolts into the tapped holes indicated by the arrow marks, and use the eyebolts to hoist the robot. Otherwise, the robot could lose its balance and fall over.
  - When hoisting with eyebolts provided by the customer, select a product with a working load of 0.79 kN or more when suspended at 45°.
- Never touch the ball screw spline shaft with your bare hands.
  Otherwise, rust could form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to hoist and lower the robot slowly.
  Otherwise, the robot could collide with the peripheral equipment, frame, or other objects, resulting in failure or damage.
- After installation is completed, be sure to remove the fixing jig and eyebolts.
  If the robot is moved while they are mounted, the robot could be damaged or fail.
  - The removed eyebolts will be used when the robot is transferred again, and so keep them in a safe place for future use.

If lifting up the robot by workers, see Fig. 3.4 to hold the robot by hands.



MEMO

This figure shows the handing areas, and does not show the conveyance posture. For the conveyance posture, see Fig. 3.2 and Fig. 3.3.

Fig. 3.4 Handling areas of the robot (THE600)

NOTICE

- Never touch the ball screw spline shaft by bare hand.
  Otherwise, rust will develop early.
  - When touching the ball screw spline shaft, wear gloves.
  - If the ball screw spline shaft is touched by bare hand, grease up the ball screw spline shaft.
- When lifting up the robot by workers, be sure to hold the gray portions shown in Fig. 3.4 by hands.
  If other portions are held, the held portions may be broken. Excessive force may be exerted on the mechanical section, resulting in a malfunction.
- After installation is completed, be sure to remove the clamp and eyebolts.
  If the robot is moved with them mounted, the robot may be broken or become faulty.

3.5 Storage

Avoid storing the robot and controller for long periods of time after unpacking them.
If this is unavoidable, however, strictly observe the following precautions for storage.

3.5.1 Storage Precautions for the Robot

⚠ CAUTION

- When storing the robot, be sure to secure the base completely.
  Otherwise, the robot will fall over.

NOTICE

- Never store the robot at a place exposed to direct sunlight and a hot and humid place.
  Otherwise, the cable duct, timing belt, and resin parts may deteriorate.
- DO NOT store the robot at a place where it is exposed to direct rain or water splashes, or at a place containing any toxic gas or liquid.
- When storing the robot, be sure to seal it in a vinyl bag and put a desiccant in the bag.
  This prevents rust from developing on and dust from adhering to the ball screw spline unit.
  - Before storing the robot, apply grease to the ball screw spline shaft.
  - For the application method, see "8.6.3 Greasing Up the Ball Screw Spline Unit and Applying Corrosion Inhibitor."
  - Before the use, apply the grease to the ball screw spline shaft.
- Before operating the robot that has been stored, be sure to perform running-in operation.
  During storage, the grease viscosity increases. If the robot is rapidly operated at a high speed, poor lubrication may cause failure.
  - Be sure to perform operation at a low speed (about 20% of override) for five (5) minutes.
- Before operating the robot that has been stored, be sure to replace the position detector battery. (See "8.3 Replacing Position Detector Battery.")
  Before periodic inspection, an alarm will occur.
  - If no-energization time is long due to storage, the time to keep data using the battery will shorten.
- When storing the robot, use the supplied fixtures to secure the arm and base, and then firmly secure the mounting sections. The robot can be unstable and fall down if it is not secured.

4. Installation of the Robot

The worker should receive safety training defined by each country's laws and regulations.
Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

4.1 Installation Environment

Table 4.1 shows the environmental conditions for the location in which the robot are to be installed.

Table 4.1 Environmental conditions for robot

Item	Specifications
Temperature	In operation: 0 to 40°C In storage: -10°C to 50°C
Humidity	20 to 80% (Non-condensing) DO NOT install the equipment where it may be subject to fluids such as water.
Altitude	1000 m or less
Vibration	In operation: 0.98 m/s^2 or less
Dust	No inductive dust should exist. Consult with SHIBAURA MACHINE first if you wish to use the robot and controller in a dusty environment.
Gas	No corrosive or combustible gas should exist.
IP (Ingress Protection) rating	Equivalent to IEC60529 IP10 (robot side), and equivalent to IP20 (for the controller)
Overvoltage category	IEC60664-1 Class III

Item	Specifications
Protection against electric shock	IEC61140 Class I
Pollution degree	IEC60664-1 Pollution Degree 3
Sunlight	The robot and controller should not be exposed to direct sunlight.
Power noise	A heavy noise source should not exist nearby.
Magnetic field	A heavy magnetic field source should not exist nearby.
Other ambient environmental requirements	No iron powder, no oil, no salt content, and no organic solvent. No water splash on the robot/controller.

⚠DANGER

- Never use the robot at a place where water exceeding the Ingress Protection ratio or liquid other than water splashes.
  Otherwise, electric leakage may result in electric shock. The electric circuit may be short-circuited to cause fire.
- Never place a flammable material near the robot.
  If the robot ignites due to failure, fire may result.
- Never use the robot in the high-temperature environment.
  Otherwise, liquid will easily leak from the alkaline battery used for the position detector battery. If liquid adheres to the skin or an eye, inflammation may result.
  - When liquid adheres to the skin, wash it away with flowing water immediately.
  - When liquid adheres to an eye, wash it away with flowing water immediately and then see a doctor.
  - Replace the parts to which liquid has adhered.

NOTICE

- Never use the robot at a place where water exceeding the Ingress Protection ratio or liquid other than water splashes.
  Otherwise, the robot may become faulty.
- When operating the robot after long-hour stop at a low temperature (10°C or less), be sure to perform operation at a low speed (about 20% of override) for five (5) minutes.
  The grease viscosity will increase in a low-temperature environment. If the robot is rapidly operated at a high speed, poor lubrication may cause failure.
  - Before the use, apply grease to the ball screw spline shaft.

4.2 Installation

Before the robot can be installed, the installation layout must be studied in consideration of the working envelope, the coordinate system, and the maintenance clearance.

⚠ DANGER

- Be sure to connect wiring after the robot is installed.
  Otherwise, a worker may stumble over the cable and fall. The worker may touch the cable that has been broken due to a dropped carried object and receive electric shock.
  - House wiring in the duct, etc.
- Be sure to connect a grounding wire.
  Otherwise, the robot that has operated unpredictably due to noise may collide with a worker. Electric leakage may result in electric shock or fire.

⚠ CAUTION

- When mounting and removing the clamp, never hold the edge of the clamp.
  Otherwise, if forcible force applies, hand or finger may be cut by the clamp.
  - Wear gloves.

NOTICE

- Be sure to connect wiring after installation is completed.
  Otherwise, a carried object may drop on the wired cable, resulting in breakage.
  - House wiring in the duct, etc.
- Be sure to connect a grounding wire.
  Otherwise, the robot may become faulty due to noise.
- The cable between the robot and controller is not a movable cable. If the robot is mounted on the traveling shaft, etc., the cable will be broken early.

4.2.1 Installing the Robot

⚠ DANGER

- Never install the robot on a frame that has low rigidity or a weak foundation.
  Otherwise, the robot may fall over together with the frame during operation.
  - See the load on the frame during robot operation (Table 4.2) to prepare the frame.

The robot is secured by using the mounting holes on the base (four (4) places).
Use attached hexagon socket head cap screws M10×25.
Table 4.2 lists the load on the frame during horizontal movement, and Fig. 4.1 shows the robot installation dimensions. The pinholes are provided on the base unit. If the robot position is to be aligned in the base coordinate system or the robot must be replaced, the robot can be positioned by the pinholes.

Table 4.2 Load on the frame during horizontal movement

Model	Load on the frame during movement (Nm)	Robot body mass (kg)
THE600	655	31

*This value is a reference value. When designing the frame, consider safety factor.

Use the frame that has the flatness and surface roughness listed in Table 4.3 for the frame on which the robot is to be installed.

Table 4.3 Flatness of robot installation part of the frame

Flatness of robot installation surface	Surface roughness of robot installation surface
0.02	Ra 6.3

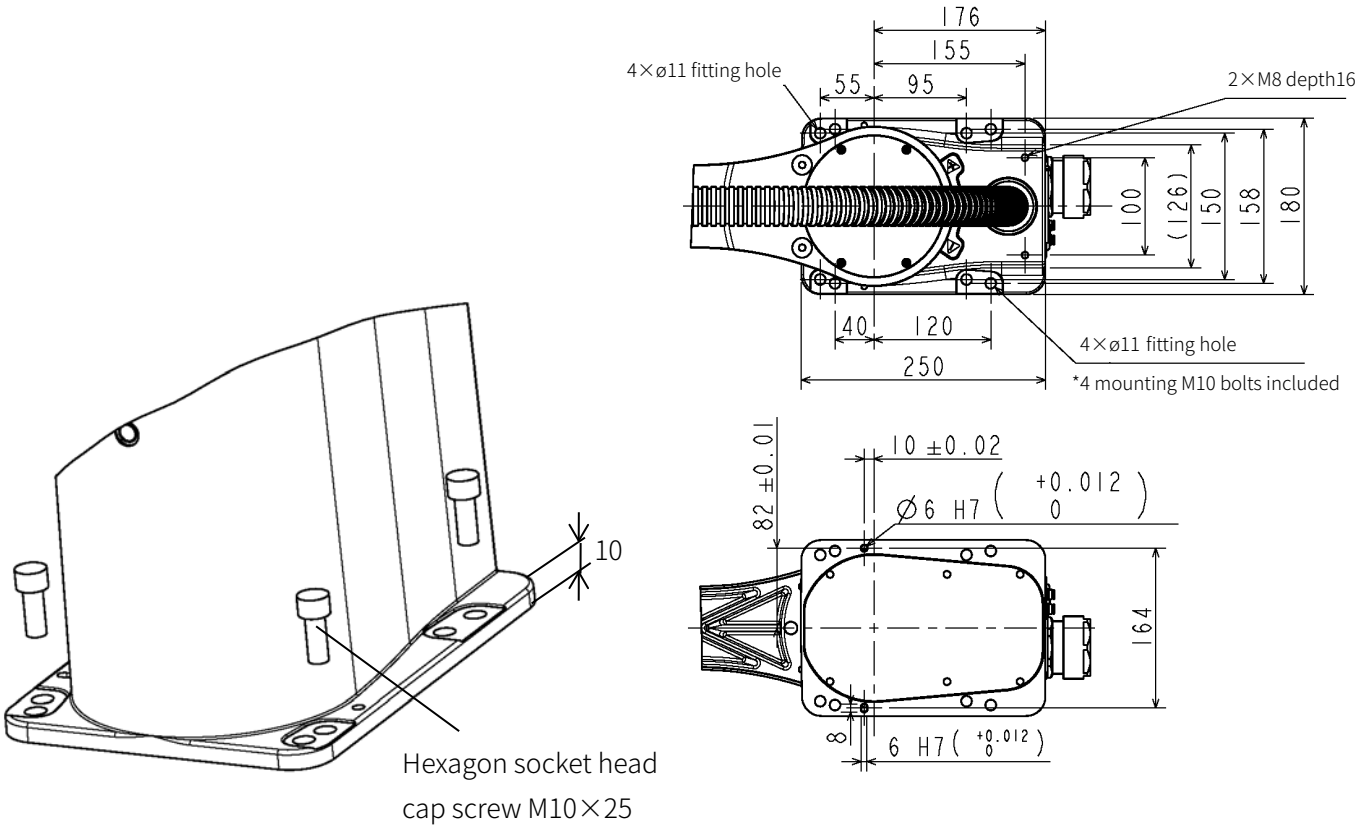


Fig. 4.1 Setting method (THE600)

The robot base has a total of eight mounting holes for installation. Use four of eight mounting holes to install the robot on the frame. The combination examples of the mounting holes are shown in Fig. 4.2.

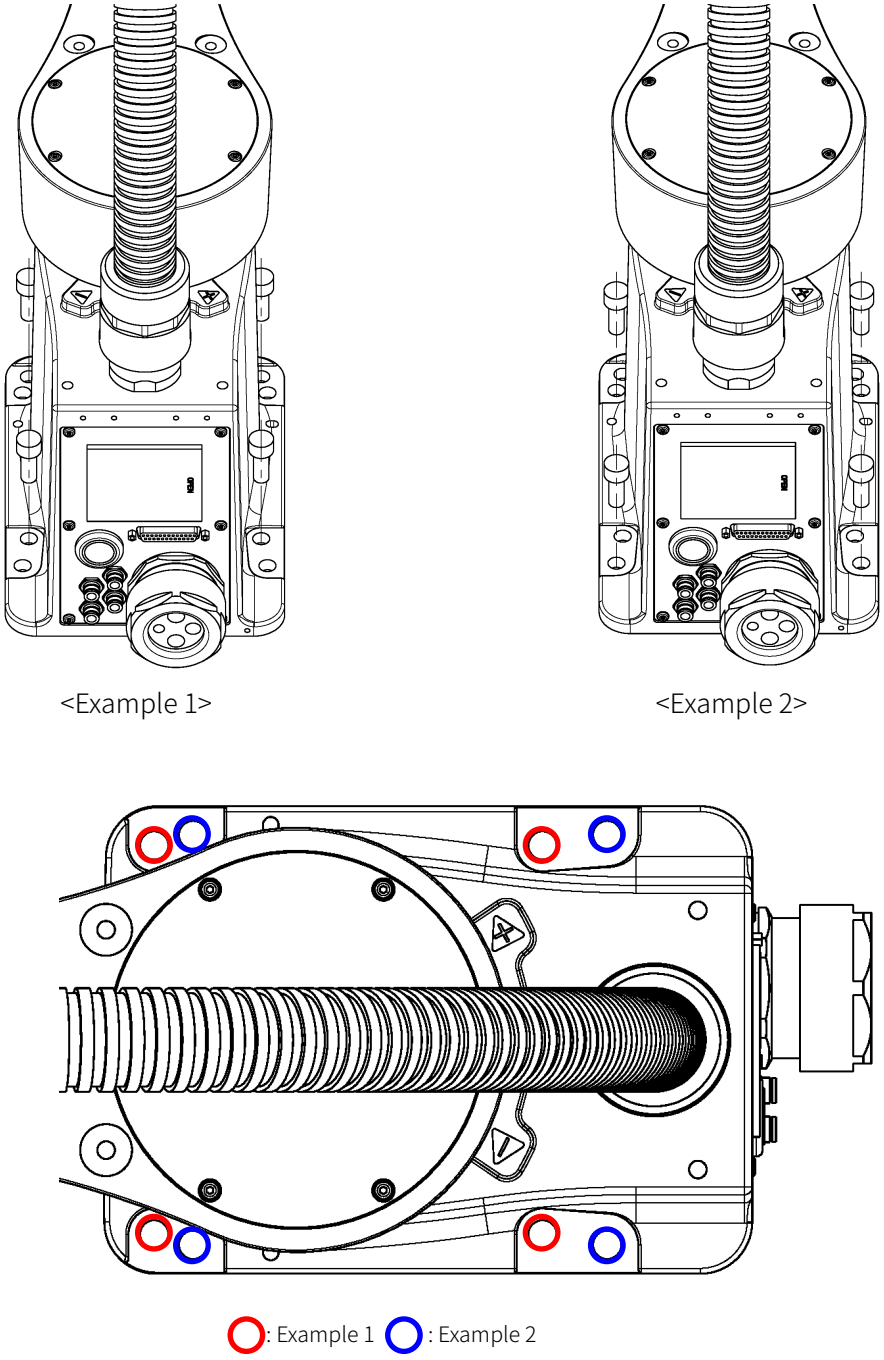


Fig. 4.2 Combination examples of mounting holes

4.2.2 Grounding the Robot

As shown in Fig. 4.3, the robot has the tap hole for grounding. When grounding the robot, use this tap hole to secure the grounding wire terminal with the bolt.

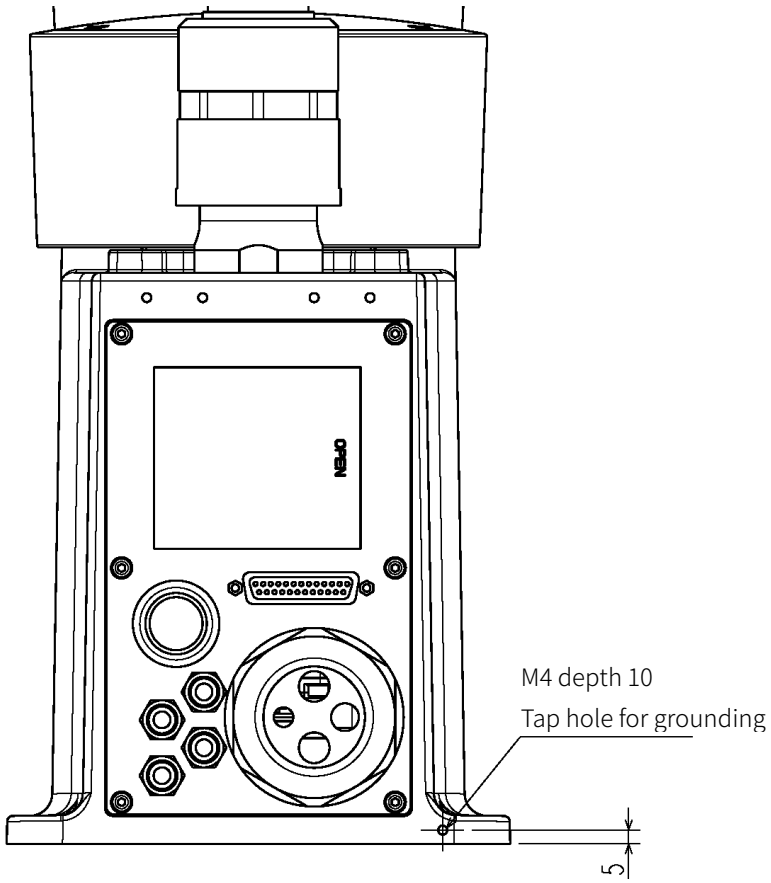


Fig. 4.3 Tap hole for grounding

NOTICE

- Never install the robot on a frame that has low rigidity or a weak foundation.
  Otherwise, the robot may vibrate during operation. It may become faulty.
  - See the load on the frame during robot operation (Table 4.2) to prepare the frame.
- Be sure to install the robot on a level place.
  Otherwise, the performance and lifespan of the robot may be reduced. The robot may become faulty.
- Be sure to secure the frame on which the robot is installed to the floor or wall.
  Otherwise, the frame will move due to the force exerted during robot operation.

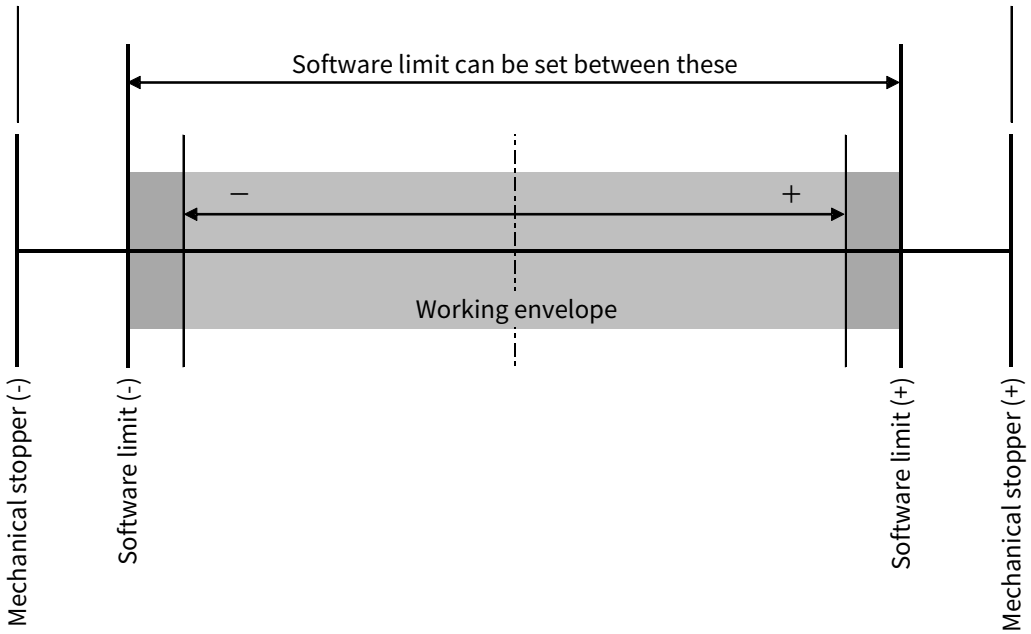
4.3 Operating Range

Fig. 4.4 shows the operating range.

Each axis can operate within the working envelope. To prevent the equipment from moving out of the working envelope by misoperation, the robot is equipped with mechanical stoppers outside the working envelope.

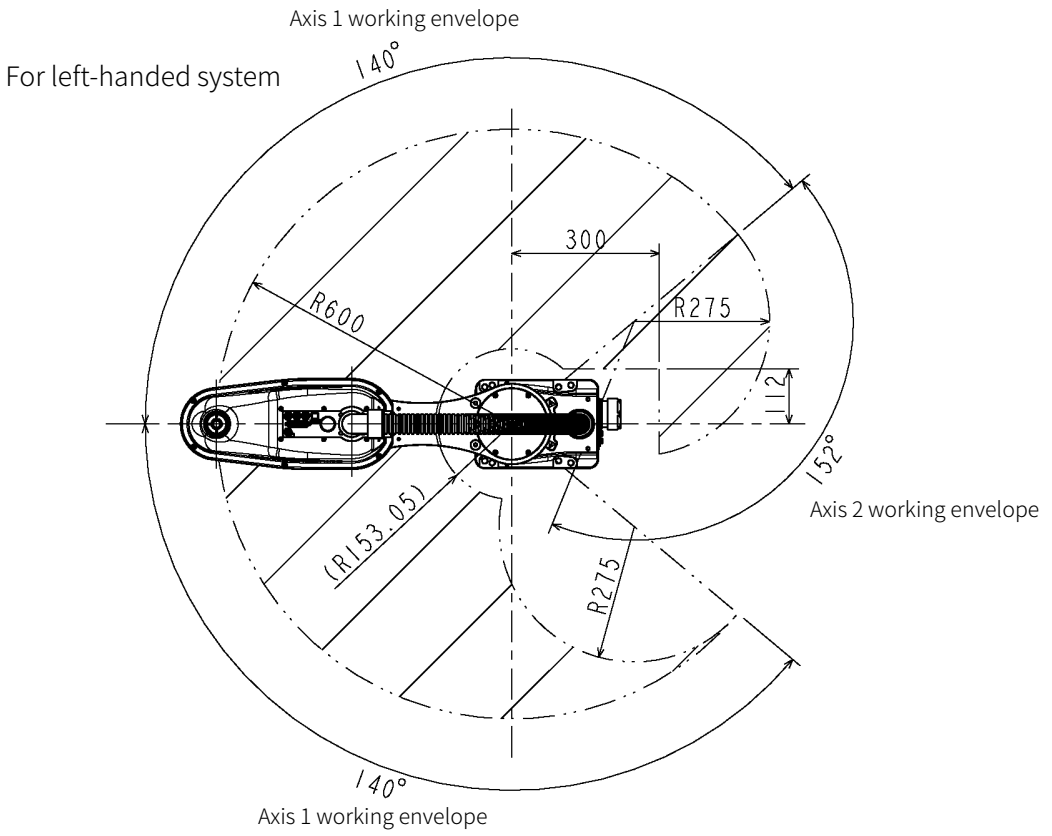
Additionally, there are software limits that the user can set up.

For details, refer to the User Parameters Instruction Manual provided separately.



MEMO

The software limit cannot be set in the range that is wider than the value set at shipment. The software limit can be set in the range that is narrower than the value set at shipment. So, to change the mechanical stopper position and narrow the working envelope, be sure to change the software limit as well.



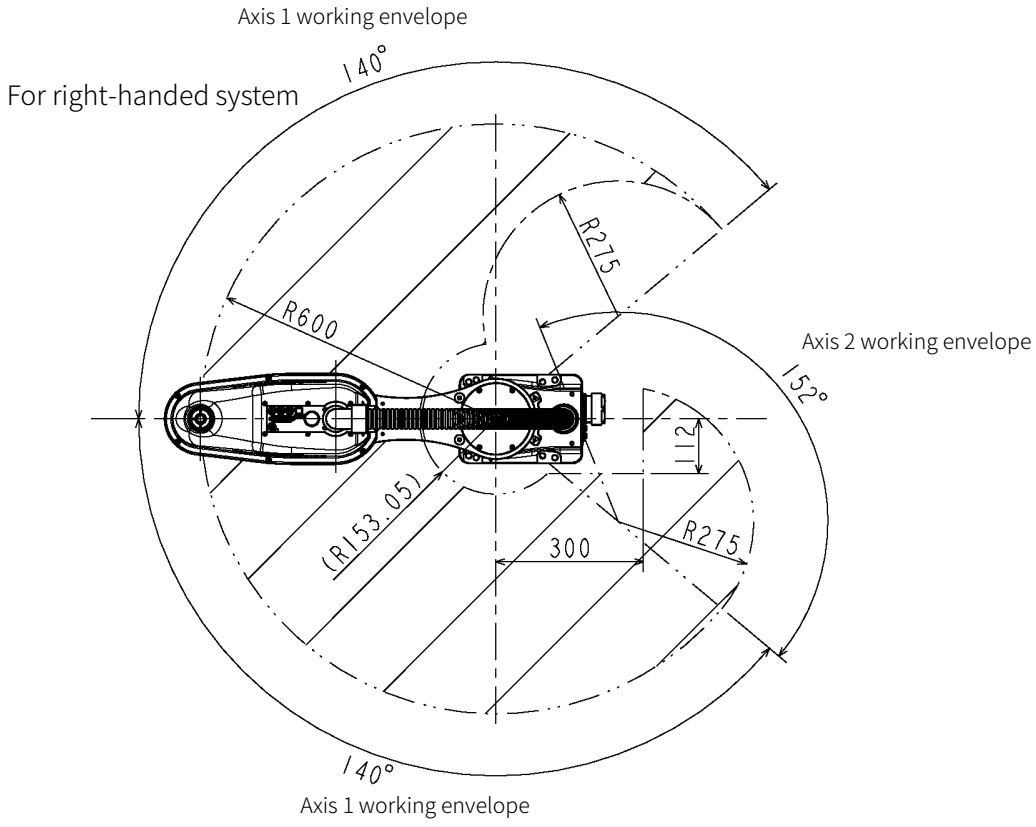


Fig. 4.4 Operating range

4.3.1 Changing axis-1 and 2 working envelope

The robot is equipped with the mechanical stoppers to restrict the working envelop of each axis mechanically. Changing the mechanical working envelope of the robot by changing this mechanical stoppers refers to “working envelope change.”
This section explains how to change axis-1 and 2 working envelope of the robot.

For how to change the axis-3 working envelope, see “4.3.2 Changing axis-3 working envelope.” Unlike other working axes, the working envelope of axis 4 is restricted by only software limit instead of mechanical stoppers. Therefore for change of axis-4 working envelope, see “4.3.3 Changing software limit.”

⚠ DANGER

- When turning on the power, never enter the dangerous range of the robot.
  Otherwise, the robot that has operated unpredictably will collide with a worker while breaking the mechanical stopper.
  - The mechanical stopper does not aim to stop the robot surely.
- Before using the robot, be sure to confirm the working envelope.
  If the working envelope is not limited as intended, the operating robot will collide with a worker.

NOTICE

- Never reuse the mechanical stopper that has collided with the robot arm and has been used to change the working envelope.
  The damaged mechanical stopper will not function normally.
- When designing, manufacturing, and mounting the mechanical stopper, be sure to observe the following items.
  - Be sure to set the changed working envelope to the working envelope indicated in the specification table and external view or narrower working envelope.
    The robot cannot be operated in the working envelope that exceeds the specifications.
  - Be sure to re-set zero position information according to "9.9③ Other Zero Setting."
    If the mechanical stopper is changed, the function described in "9.9.1.4How to Restore Data by SYSTEM1 or SYSTEM2" becomes disabled. At installation, enter zero data according to "9.9.1.2Setting HOME1 and HOME2."
  - The function is also available to restore zero using the setting tool prepared by the customer. If using "9.9.1.3How to Restore Data by HOME1 or HOME2," enter zero data according to "9.9.1.2Setting HOME1 and HOME2" at installation.
  - A robot failure caused by the mechanical stopper is excluded from application of warranty.
- Be sure to set the software limit inside the mechanical stopper.
  Otherwise, the robot will collide with the mechanical stopper during operation.

The axis-1 working envelope can be changed by changing the position of mechanical stoppers as shown in Fig. 4.5.

Table 4.4 Axis-1 working envelope before and after change

		Before change	After change
Axis-1 working envelope	＋direction	140°	110°
	－direction	140°	110°

Install hexagon socket head cap screws (M8×12) into the outside tap hole.

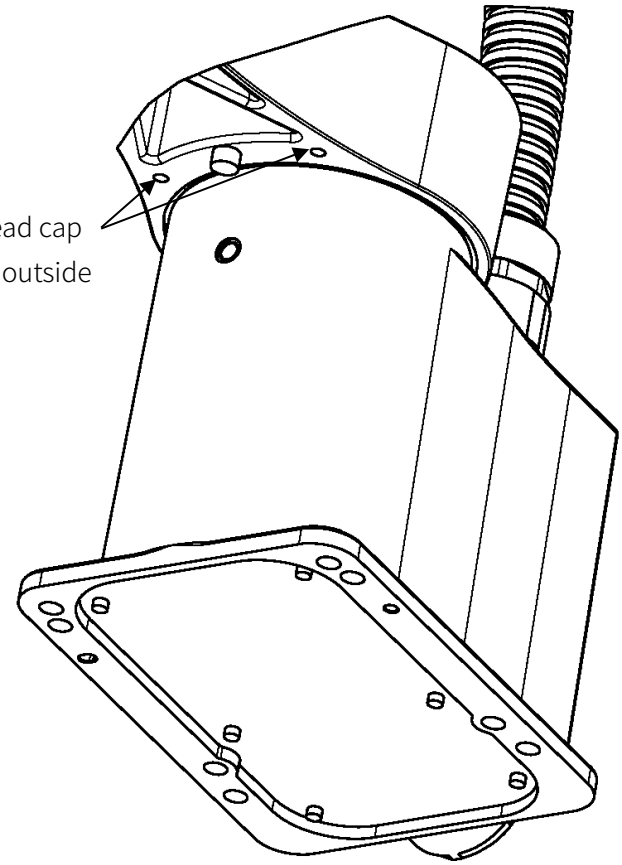


Fig. 4.5 Example of changing axis-1 working envelope

When the axis-2 working envelope is changed, it can be set to any working envelope by changing the shapes of mechanical stoppers. Fig. 4.6 shows the mounting position of the axis-2 mechanical stopper and Fig. 4.7 and Fig. 4.8 show the examples of changing the axis-2 mechanical stopper. The customer should design and manufacture the mechanical stoppers as required.

Table 4.5 Axis-2 working envelope before and after change

		Before change	After change
Axis-2 working envelope	＋direction	152°	120°
	－direction	152°	152°

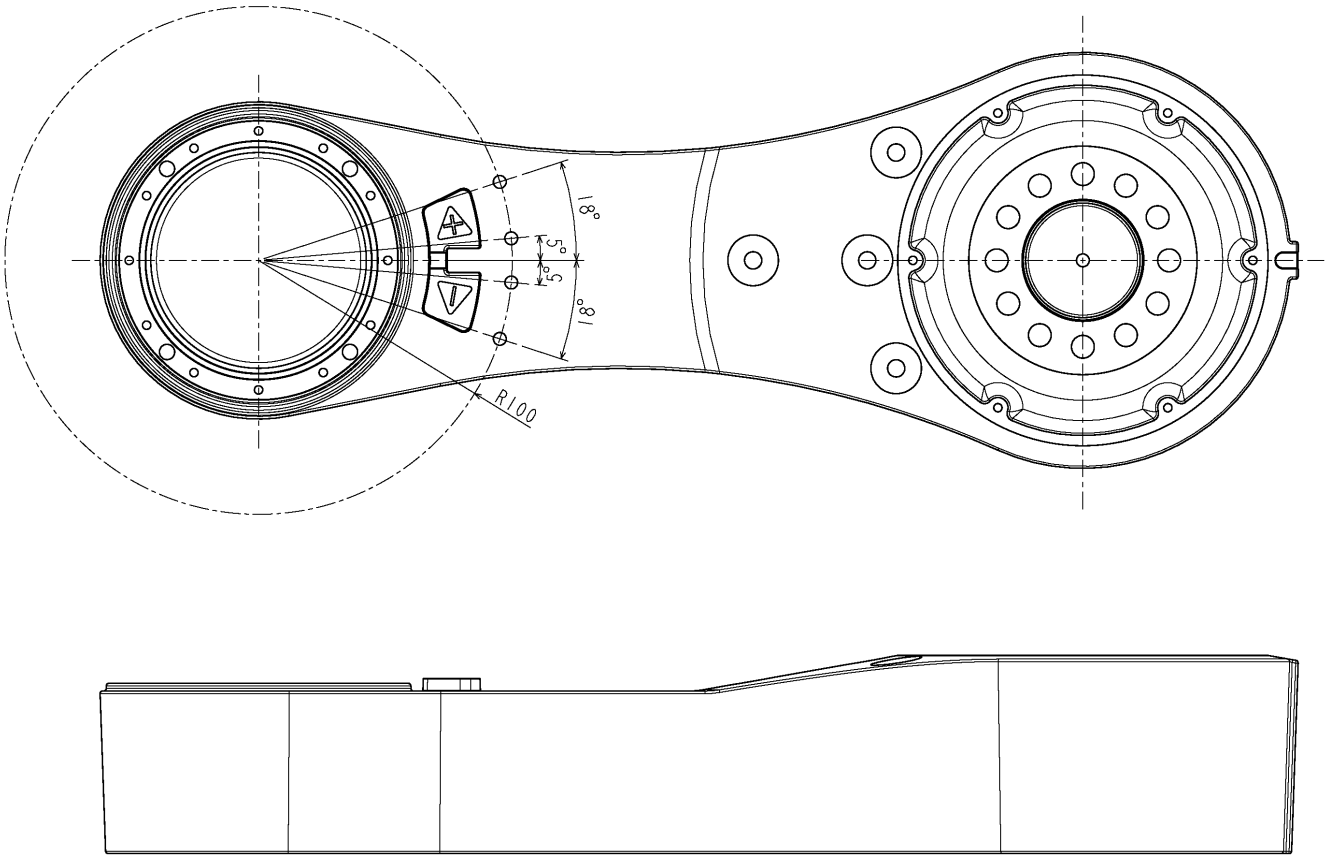


Fig. 4.6 Mounting position of axis-2 mechanical stopper

Example of changing axis-2 mechanical stopper
Set working envelope: 120°, -152°

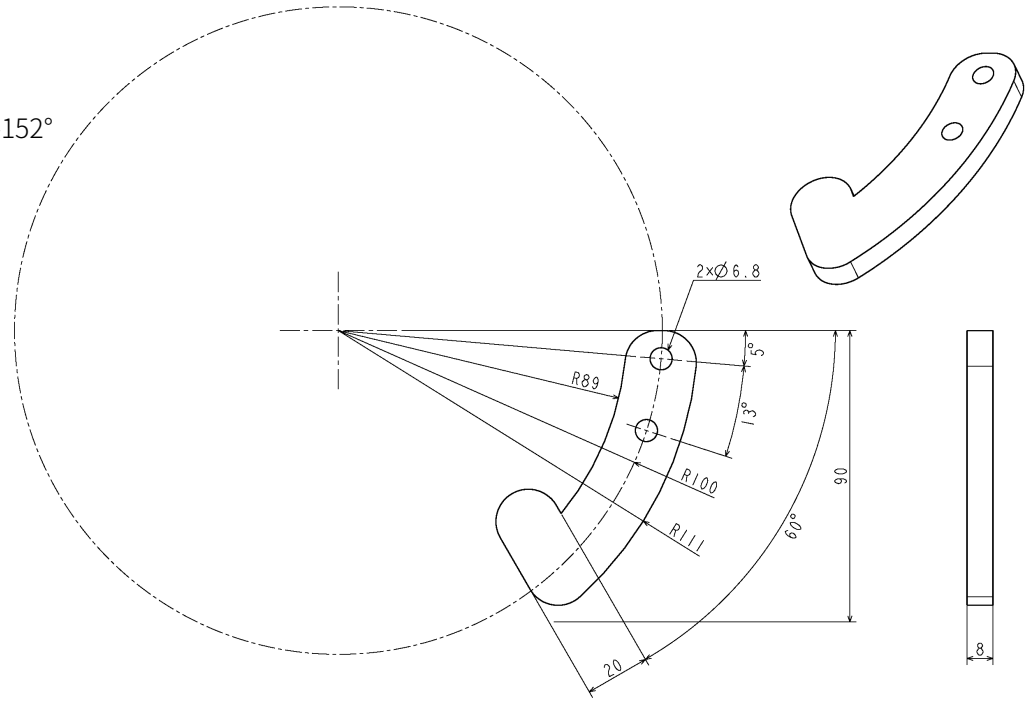


Fig. 4.7 Example of changing axis-2 mechanical stopper

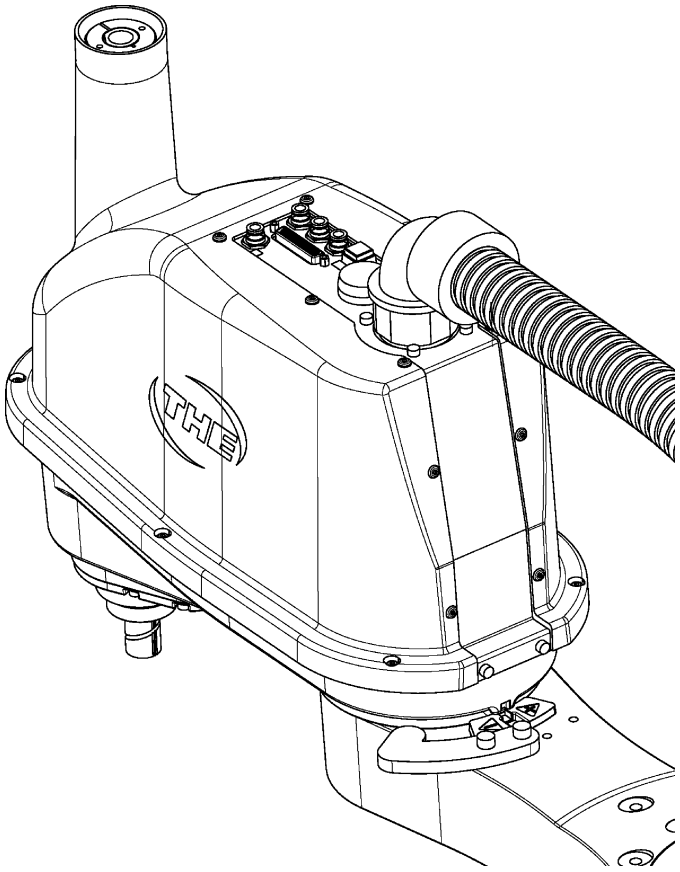


Fig. 4.8 Example of changing axis-2 working envelope

If a working envelope is changed, user parameter must be changed.
For how to change the software limit, see "4.3.3 Changing software limit."

4.3.2 Changing axis-3 working envelope

The robot is shipped while the software limits and mechanical stoppers are factory-set for Z stroke of axis 3 to travel between 0 to 210 mm. Fig. 4.9 shows the factory-set working envelope.

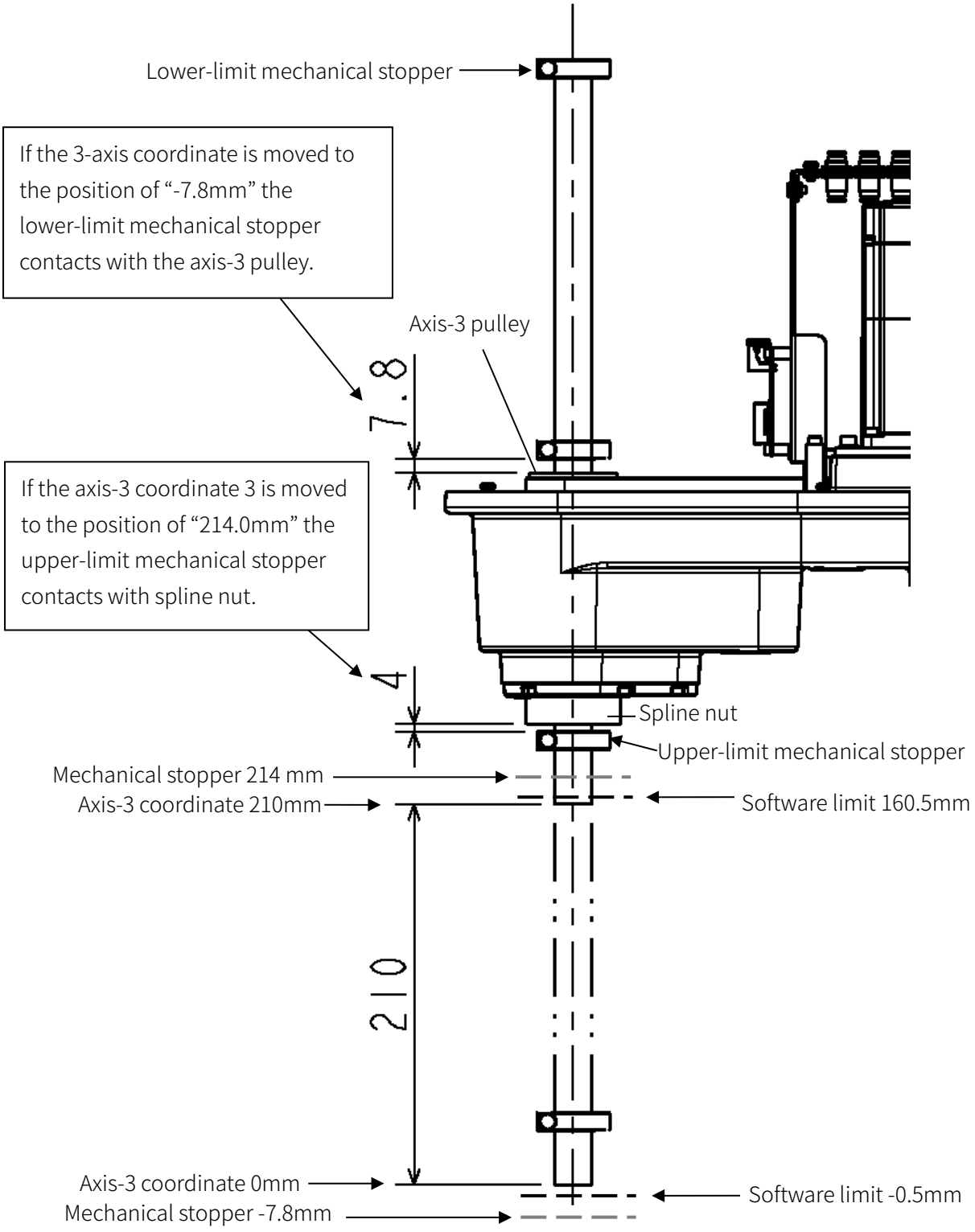


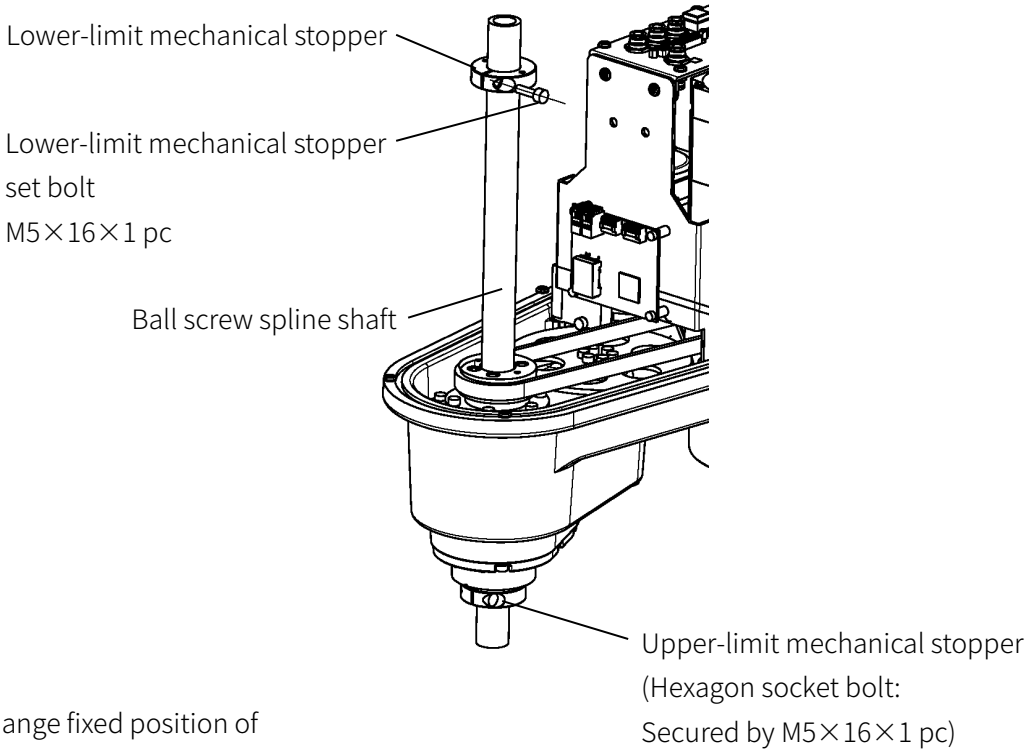
Fig. 4.9 Factory-set axis-3 working envelope

1 Remove the arm-2 cover.

For how to remove the cover, see "8.3.1 Arm-2 Cover."

2 Loosen mechanical stopper set bolts, move the mechanical stoppers to any position, and then secure them again.

When securing mechanical stoppers, be sure to apply screw locking adhesive (LOCTITE 242 medium strength) to the set bolts.



Change fixed position of mechanical stopper.
Example: If the lower-limit mechanical stopper is moved 30 mm downward

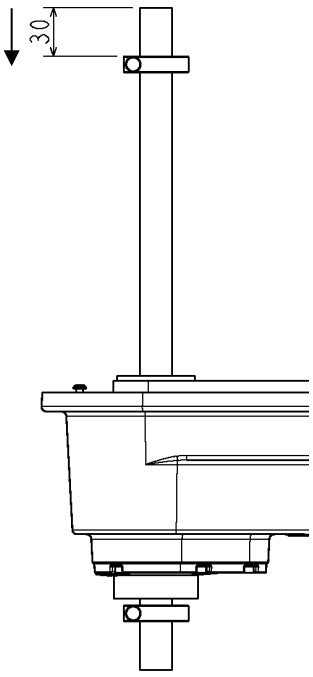


Fig. 4.10 Changing axis-3 working envelope

3 Change the software limit.

As for the change of software limits, see “4.3.3 Changing software limit” and Fig. 4.12. After the software limit is changed, move axis 3 up and down by hand while pressing the axis-3 brake release switch and make sure that the software limit is set correctly.

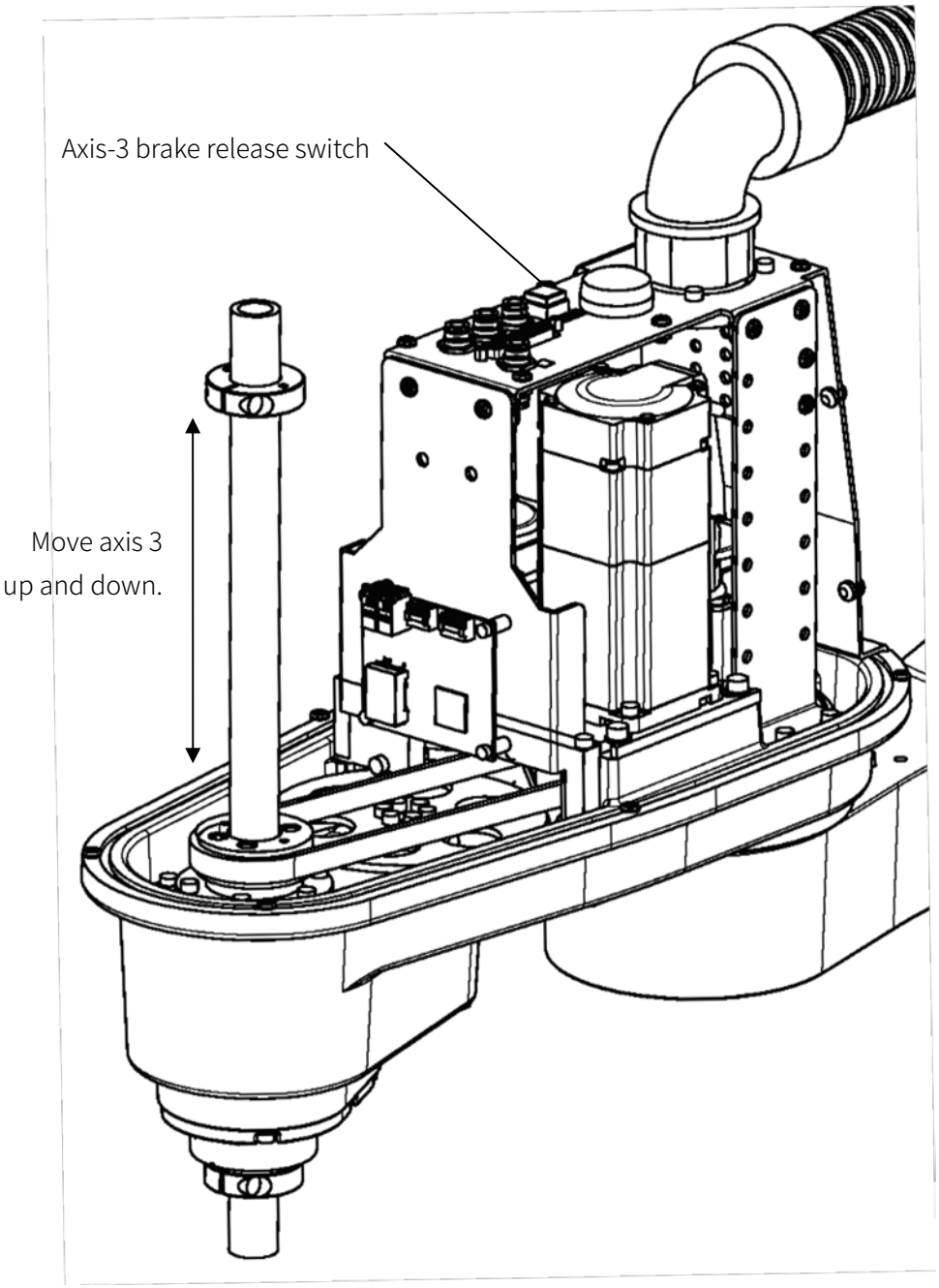


Fig. 4.11 Changing and checking the axis-3 software limit

Fig. 4.12 shows the working envelope setting if the lower-limit mechanical stopper is moved 30mm downward.

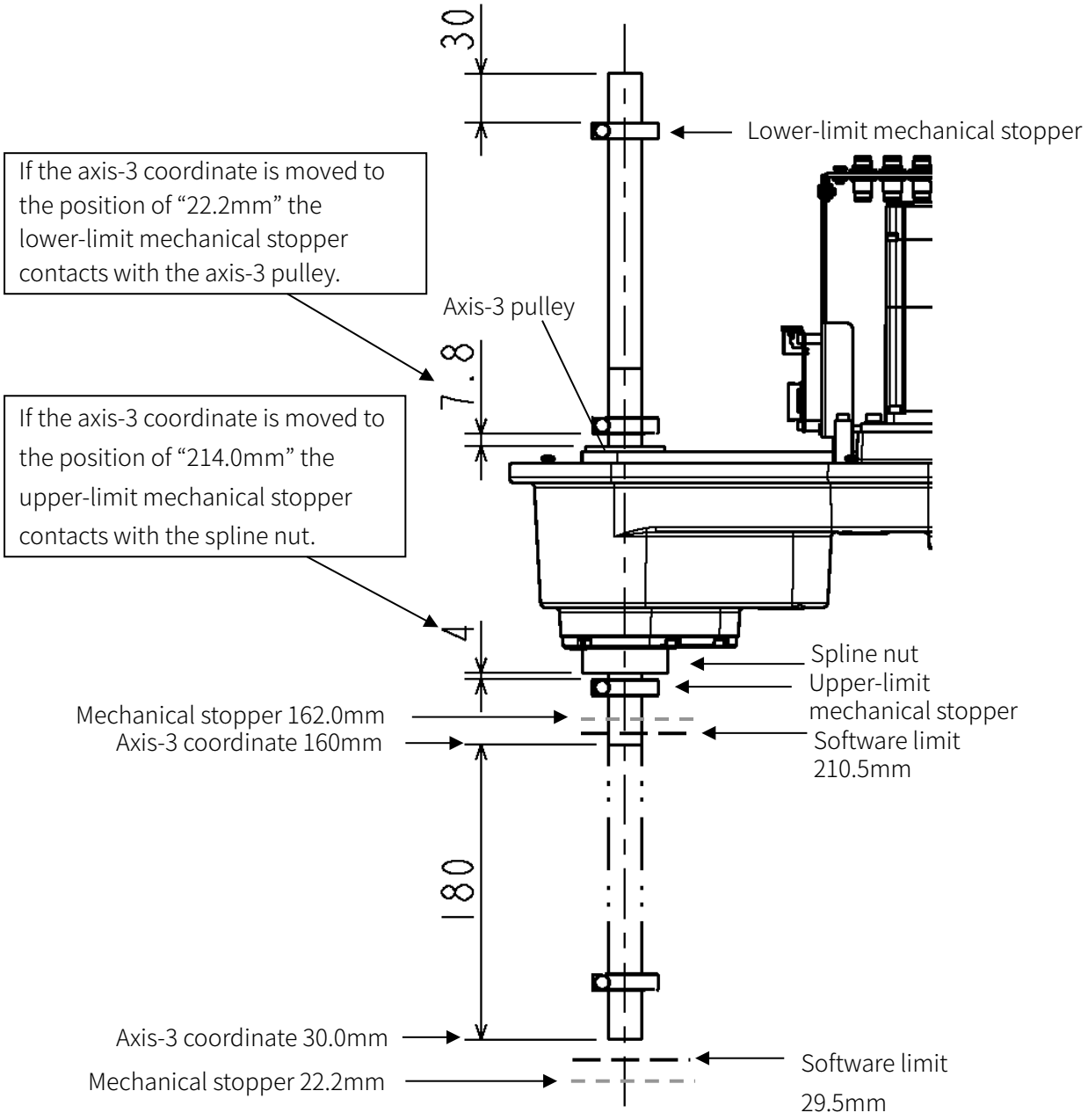


Fig. 4.12 Example of changing axis-3 working envelope

4 Mount the arm-2 cover.

For how to mount the cover, see “8.3.1 Arm-2 Cover.”

NOTICE

- To change the working envelope, be sure to use the lower-limit mechanical stopper.
  If the upper-limit mechanical stopper is used to change the working envelope, the ball screw spline shaft is always in low position. Moving Axis 1, 2, or 4 in such state can lead to premature damage to the ball screw spline.

4.3.3 Changing software limit

NOTICE

- Be sure to set the software limit inside the mechanical stopper.
  Otherwise, the robot may collide with the mechanical stopper during operation.

If the mechanical stopper is changed, be sure to change the software limit as well.
There are the following two methods for changing the software limit.

[1] Operate the teach pendant to change the software limit from the software limit setting screen.
For details, refer to the description of “7.3.1.1 Setting software limit” in “Section 7 Robot TP Setting Operation” of “STA1161 TP5000 Operation Manual” Instruction Manual provided separately.

[2] Change the setting value of “user parameter file (file name: USER.PAR).”
There are software limits that the customer can set up. For details, refer to the description of “[U14] SOFTWARE LIMIT” in “User Parameters” Instruction Manual provided separately.

When the software limit is changed using the above two types of methods, the software limit value set in the factory-set “user parameter file (file name: USER.PAR) is changed and saved.
Therefore, **before changing the software limit, be sure to back up the “user parameter file”** so that the software limit value set in the factory-set user parameter file can be identified.

If the software limit is changed using the method in [2], be sure to turn off the power and then turn it on.
Otherwise, the user parameter change is not reflected.

5. Setting Hand

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.

Be aware that failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

5.1 Mounting hand

The hand is to be mounted on the lower end of the shaft or the tool flange. Fig. 5.1 shows the dimensions of the lower end of shaft and the tool flange. If the hand is mounted on the tool flange, the tool is centered with the ϕ12H7 mating section. The hand direction is adjusted by means of the 4×4 keys and secured with four (4) M4 bolts. The tool flange is optional.

⚠DANGER

- Be sure to use the robot hand having a structure in which it does not release a workpiece, etc. even if the power of the robot or hand driving source is turned off.
  Otherwise, the workpiece, etc. will fly and drop.
  - When the power of the robot is turned off, the signal output to the robot hand is also turned off.
  - The robot hand signals (DOUT201 to DOUT208) are not reset by operation (system input signal “output signal reset” and RESET DOUT command) to reset output signals on a batch basis.
  - Perform daily inspection for the air tube connected to the hand, and perform periodic inspection for the air tube in the robot as well. If the air tube is damaged, replace it immediately.
  - Install the fence, etc. that prevents a flying object.

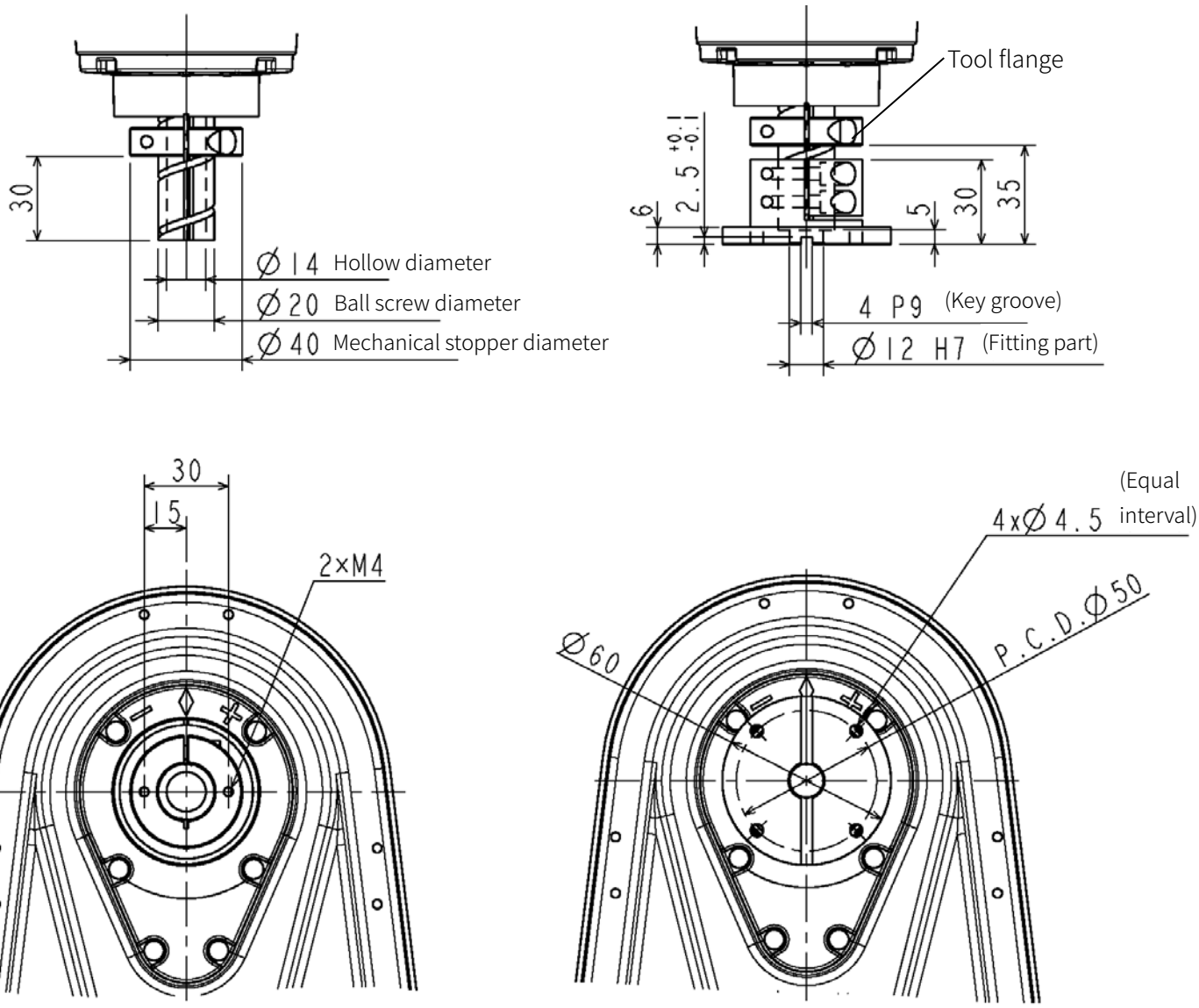


Fig. 5.1 Hand mounting dimensions

5.2 Mounting camera and air valve

The screw holes are formed in the arm 2 bottom and base as shown in Fig. 5.2. When a camera or air valve is mounted on the arm, these screw holes are used. Fig. 5.3 shows the mounting example.

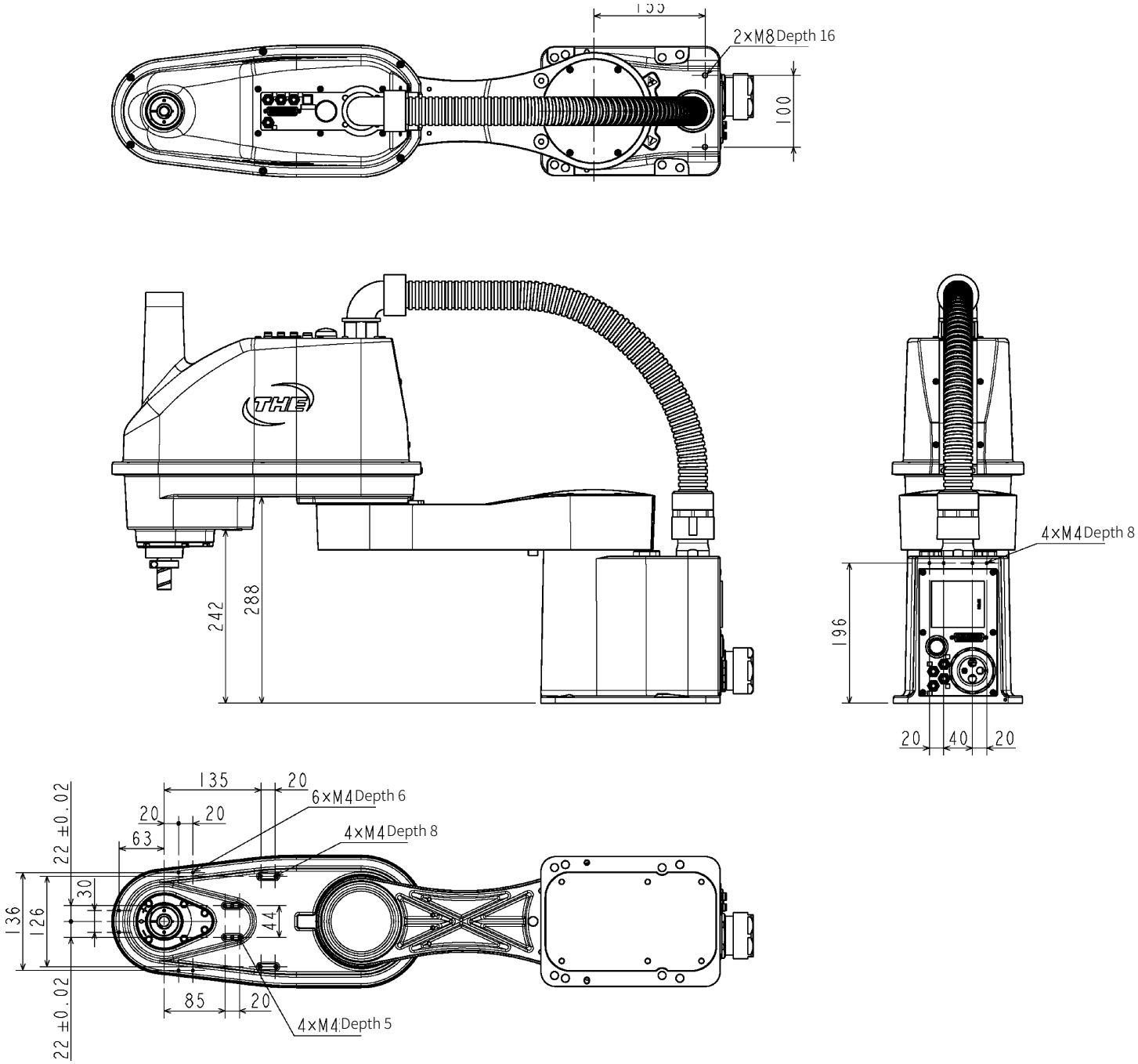
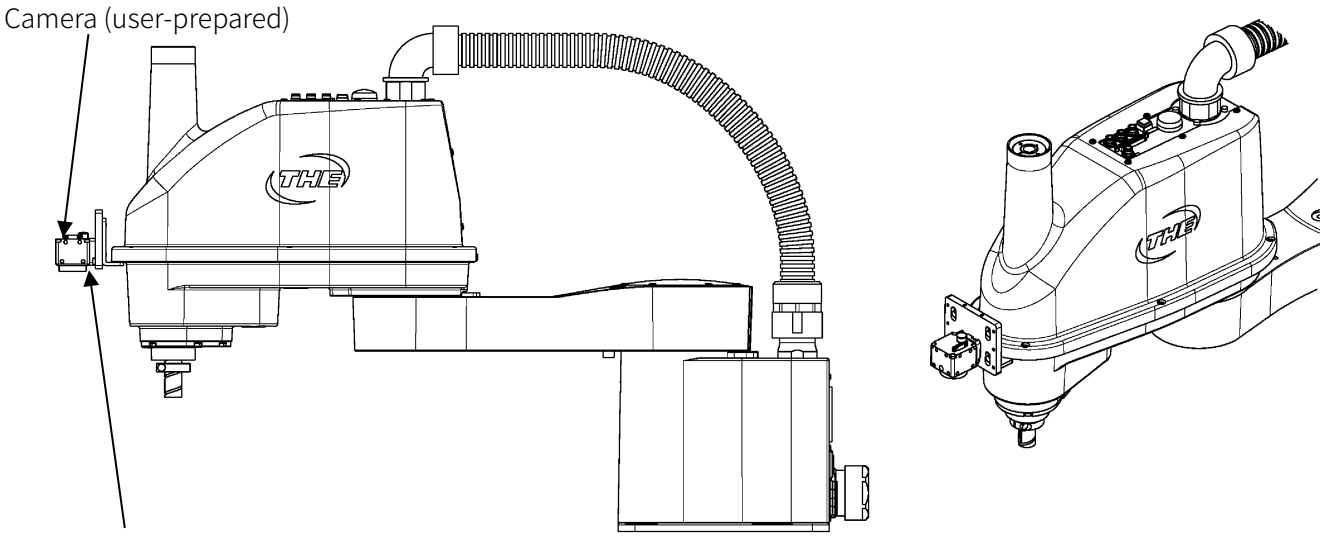


Fig. 5.2 Service tap dimensions



The camera mounting plate shown in this figure is optionally available.

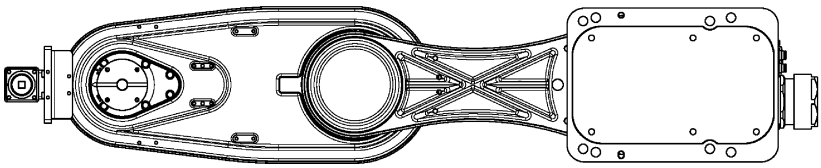


Fig. 5.3 Example of mounting the camera

NOTICE

- If the service tap is used to mount parts, be sure to check whether the parts interfere with the robot and peripheral equipment during operation. Otherwise, the robot and peripheral equipment may become faulty.

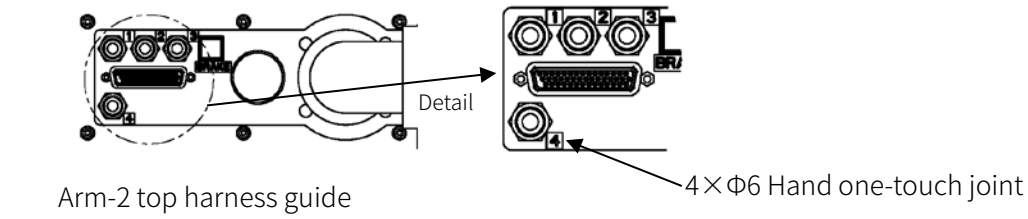
Table 5.1 lists the manufacturers of the optional camera mounting plate.

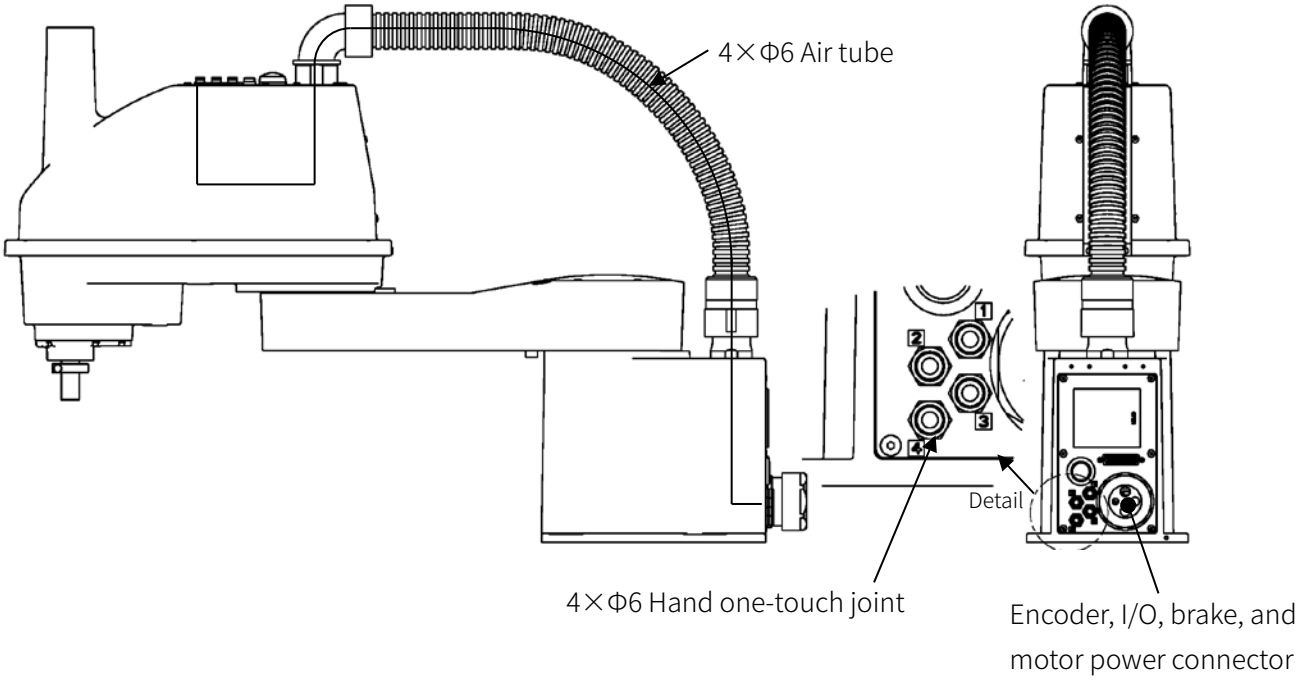
Table 5.1 Camera mounting plate (optional)

Manufacturer	Product name	Maker model Example	SHIBAURA Dwg. No.	Unit code
Keyence	Camera mounting bracket	CV-S035CH/CV-S035MH	L23754G01	H_D0E8
		CV-S200CH/CV-S200MH		
		CA-HX200C/CA-HX200M		
		CA-HX500C/CA-HX500M		
OMRON		FH-SC/FH-SM	L23754G02	H_D0E9
		FH-SC02/FH-SM02		
		FH-SC04/FH-SM04		
		FH-SM05R/FH-SC05R		
Panasonic		ANPVC2040	L23754G03	H_D0EA
		ANPVC1040		
		ANPVC5030		
		ANPVC1470		
Basler		GigE Vision air sensor Ace series	L23754G04	H_D0EB

5.3 Hand air piping

Four hand piping lines (Φ6×4) are available. The lines are connected on the base rear and the arm-2 top.





1 : Red 2 : White 3 : Blue 4 : Yellow

Fig. 5.4 Hand air piping and wiring

NOTICE

- Do not carry out piping together with the body harness.
  Otherwise, forcible force is imposed on the body harness, which can lead to breakage in the internal cable.
- The solenoid valve air tubes are not attached to the robot.
  User should prepare the air tubes.

5.4 Permissible load connections and program setting

This section describes the permissible load conditions of the robot and how to set up the program according to the load.

5.4.1 Permissible load conditions

The robot load conditions are defined by the hand mass, moment of inertia and offset value of hand gravity center from the hand shaft center as shown in Fig. 5.5.
The permissible load conditions are listed in Table 5.2.

Table 5.2 Permissible load conditions

Conditions	Permissible values
Mass	Up to 12 kg (rated 2 kg)
Moment of load inertia	Up to 0.25 kg · m^2
Offset value of load gravity center	Up to 100 mm

NOTICE

- Never use the robot under the conditions that exceed the specification values.
  Otherwise, the performance and lifespan of the robot may be reduced.
- Be sure to use the PAYLOAD command.
  Otherwise, the performance and lifespan of the robot may be reduced.
  - Even if the offset value of load gravity center is 100 mm or less, on the hand with long projection amount from the axis-4 rotation center, micro vibration at the axis center may be amplified to cause large vibration on the hand workpiece tip.
- Even within the maximum payload mass, when a heavy object is held and axis 3 is lowered, vibration may occur depending on the posture of the robot. In such case, use the SPEED command, ACCEL/DECEL command, LOCUS command, etc. to adjust the operating speed and acceleration/deceleration so that vibration does not occur.
  For the SPEED command, ACCEL/DECEL command, and LOCUS command, refer to “3 Explanation of Command Word” in “Robot Language Manual” Instruction Manual provided separately.

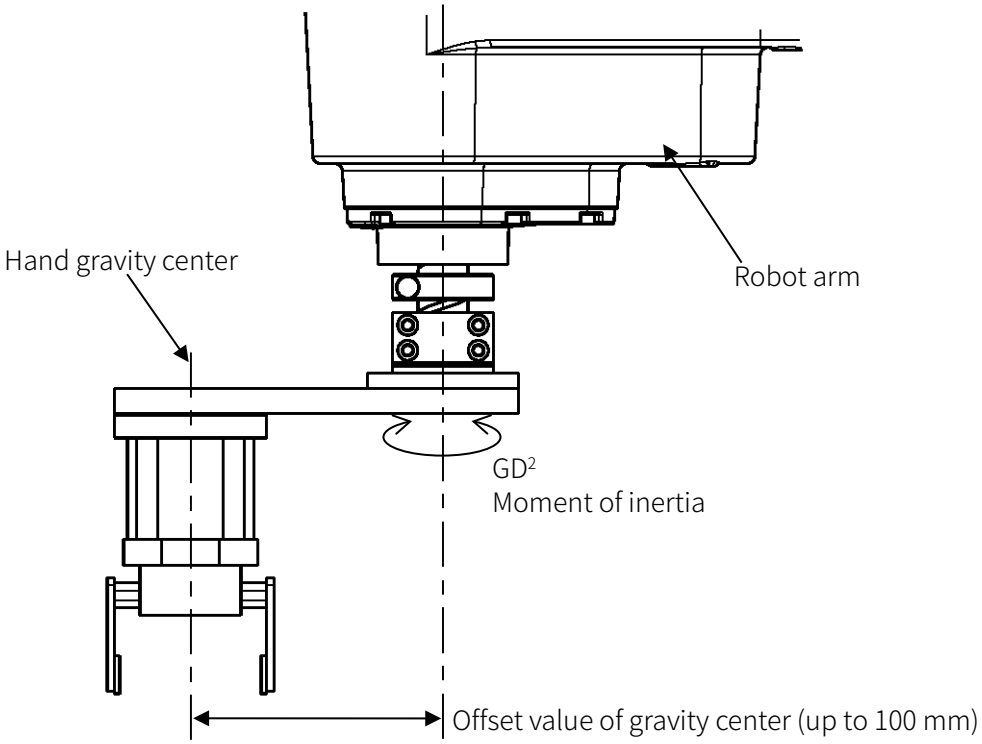


Fig. 5.5 Robot tool

5.4.2 Load conditions and program setting

This robot can automatically change the maximum speed, acceleration/deceleration and servo gain by using the PAYLOAD command in the program according to the load conditions.

Be sure to use the PAYLOAD command.

The specific method for using this function is explained below.

PAYLOAD command format

The PAYLOAD command format is written as shown below if the tool mass is M kg and the gravity center offset is L mm.

```
PAYLOAD= {M, L}
    M : Load mass    (kg)
    L  : Gravity center offset (mm)
```

The PAYLOAD command has the following functions.

The maximum speed and acceleration/deceleration of each robot axis are automatically changed according to the set load conditions.

The servo gain of each robot axis is automatically changed according to the set load conditions.

Program examples

Basic program examples using the PAYLOAD command are shown below.

For further information, refer to “3 Explanation of Command Word” in “STA1163 Robot Language Manual” Instruction Manual provided separately.

(Program example 1)

The robot is moved under the load conditions of 5 kg load mass and 100 mm gravity center offset.

```
PROGRAM SAMPLE
    SPEED= 100
    PAYLOAD={5,100}
    MOVE P1
    MOVE P2
    STOP
END
```

(Program example 2)

Assume that the hand mass becomes 5 kg and the gravity center offset becomes 50 mm when the workpiece is grasped at the hand mass of 3 kg and the gravity center offset of 30 mm.

Pick-and-place operation is executed under the above conditions.

```
PROGRAM SAMPLE
   PAYLOAD={3,30}
   ACCUR=FINE
   ENABLE NOWAIT
   RESET DOUT
   MOVE P0
   DOUT(1)
   WAIT DIN(1)
LOOP:
   MOVE P1+POINT(0,0,100)
   IF DIN(-1)THEN GOTO FIN
   MOVE P1
   WAIT MOTION>=100
   DOUT(213)
   DELAY 1
   PAYLOAD={5,50}
   MOVE P1+POINT(0,0,100)
   MOVE P2+POINT(0,0,100)
   MOVE P2
   WAIT MOTION>=100
   DOUT(-213)
   DELAY 1
   PAYLOAD={3,30}
   MOVE P2+POINT(0,0,100)
   GOTO LOOP
FIN:
   MOVE P0
   DOUT(1)
   STOP
END
```

(Program example 3)

During movement from P1 to P2 under the condition in program example 1, vibration occurs, and the ACCEL/DECEL command is used to adjust acceleration/deceleration so that vibration is reduced.

```
PROGRAM SAMPLE
   SPEED= 100
   PAYLOAD={5,100}
   MOVE P1
   MOVE P2 WITH ACCEL = 80 WITH DECEL = 80
   STOP
END
```

(Program example 4)
During linear movement from P1 to P2, vibration occurs, and the LOCUS command is used for adjustment so that vibration is reduced.

```
PROGRAM SAMPLE
   SPEED= 100
   PAYLOAD={5,100}
   MOVES P1
   MOVES P2 WITH LOCUS = 0
   STOP
END
```

Setting of PAYLOAD command

In the default state, or when the PAYLOAD command is not used, the maximum speed and acceleration/deceleration are set to 100% and the servo gain is set to the value under the minimum load.

See “5.4.3 Setting maximum speed and robot acceleration/deceleration for load conditions.”

NOTICE

- Be sure to use the PAYLOAD command.
  Otherwise, vibration or overrun may reduce the lifespan of the robot or damage the robot.
  - The moment of load inertia should be used within the tolerance listed in Table 5.2.
  - Even if there is no offset of load gravity center, when the moment of inertia is large, the robot may vibrate. In such case, calculate the virtual gravity center offset L (mm) from the moment of inertia (kg·m^2) and mass M (kg).
    $L=\sqrt{(J\times 10^6/M)}$
    Enter this calculation result.
    PAYLOAD={M,L}
  - Micro vibration may occur depending on the posture of the robot.
  - If vibration or overrun does not subside, use the SPEED command and DECEL command together.
- Even within the maximum payload mass, when a heavy object is held and axis 3 is lowered, vibration may occur depending on the posture of the robot. In such case, use the SPEED command, ACCEL/DECEL command, LOCUS command, etc. to adjust the operating speed and acceleration/deceleration so that vibration does not occur.
  For the SPEED command, ACCEL/DECEL command, and LOCUS command, refer to “3 Explanation of Command Word” in “STA1163 Robot Language Manual” Instruction Manual provided separately.
- If the robot vibrates during manual guiding, be sure to set the PAYLOAD in manual mode by referring to “2.1.7 Setting load for manual guide” in “STA1161 TP5000 Operation Manual” Instruction Manual provided separately.
  Otherwise, vibration may reduce the performance and lifespan of the robot.
  - If the load mass and gravity center offset are large, the robot may vibrate.

Weight of load mounted on the arm

If the camera or air valve is mounted on the arm, the weight of the load is converted to the equivalent weight of the shaft and is added to the load weight and set the PAYLOAD command.

Equivalent weight equation

If load is mounted on the tip from the axis-3 center: $W_M=M(A_{M2})^2/(A_2)^2$

If load is mounted between axis-1 center and axis-3 center: $W_M=M(A_{M1})^2/(A_1+A_2)^2$

W_M : Equivalent weight

M : Camera weight

A_1 : Arm-1 length

A_2 : Arm-2 length

A_{M1} : Distance from axis-1 rotation center to gravity center of camera

A_{M2} : Distance from axis-2 rotation center to gravity center of camera

Fig. 5.6 and Fig. 5.7 show the examples.

Example: If the camera of 1 kg is mounted on the arm-2 tip and the hand of 2 kg is mounted on the shaft tip

Equivalent mass W_M of the camera mounted on the arm 2 tip is

$W_M=M\ (A_{M2})^2/(A_2)^2$

$=1\times345^2/275^2=1.57$

Thus, the total load mass

$W+W_M=2+1.57=3.57$

is rounded up to become **3.6 kg**.

Set "**3.6**" in load mass of the PAYLOAD command.

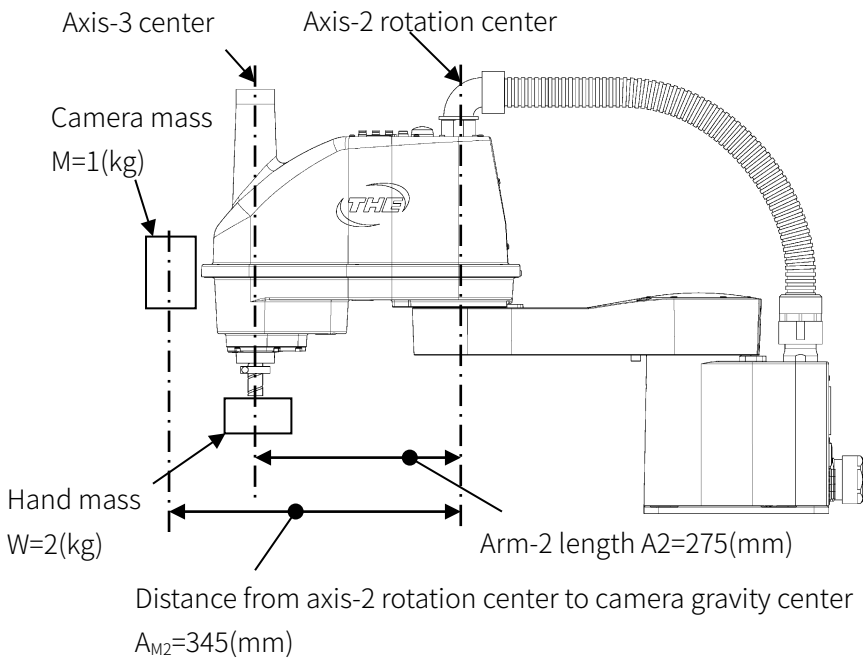


Fig. 5.6 Mounting on the tip from axis-3 center

Example) With solenoid valve of 1 kg mounted in the middle of Arm-2 and hand of 2 kg in the tip of shaft

Equivalent mass W_M of the solenoid valve mounted between axis-1 rotation center and axis-3 center is

$W_M=M\ (A_{M1})^2/(A_1+A_2)^2$

$=1\times400^2/(325+275)^2=0.44$

Thus, the total load mass

$W+W_M=2+0.44=2.44$

is rounded up to become **2.5 kg**.

Set "**2.5**" in load mass of the PAYLOAD command.

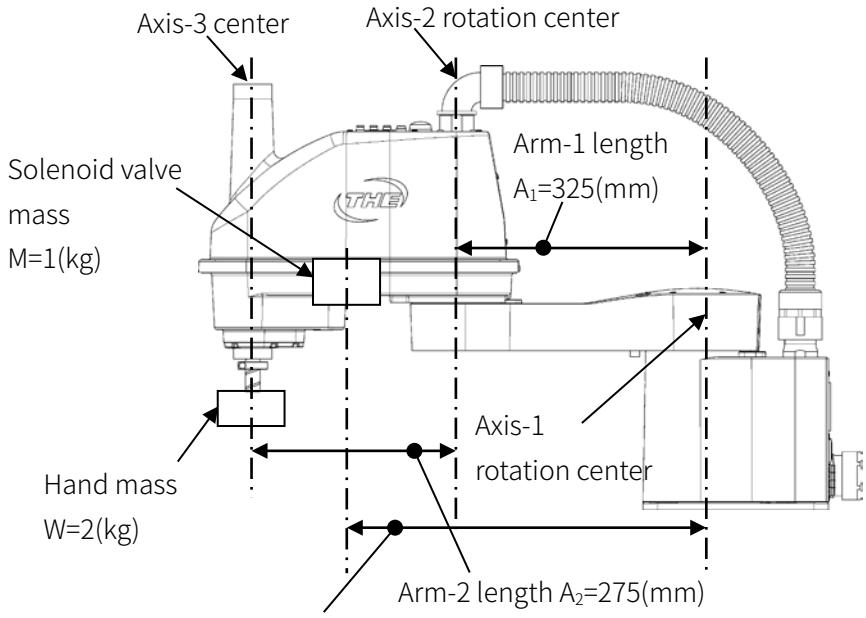
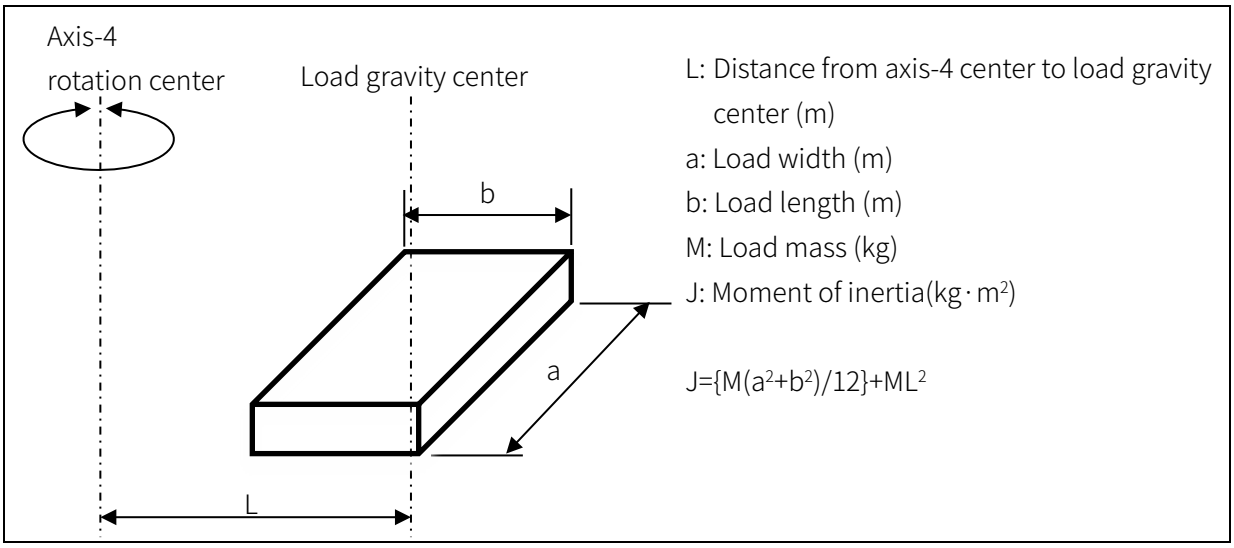


Fig. 5.7 Mounting between axis-1 rotation center and axis-3 center

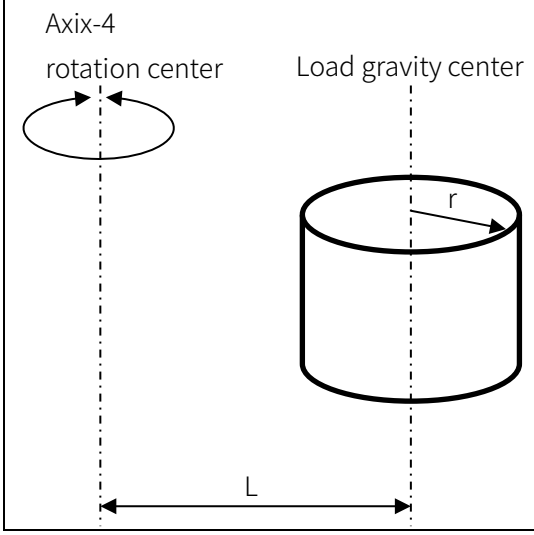
Calculating moment of inertia

The simplified model of the robot and load is used to show an example of equation of load moment of inertia below.

1) If the load shape is rectangular



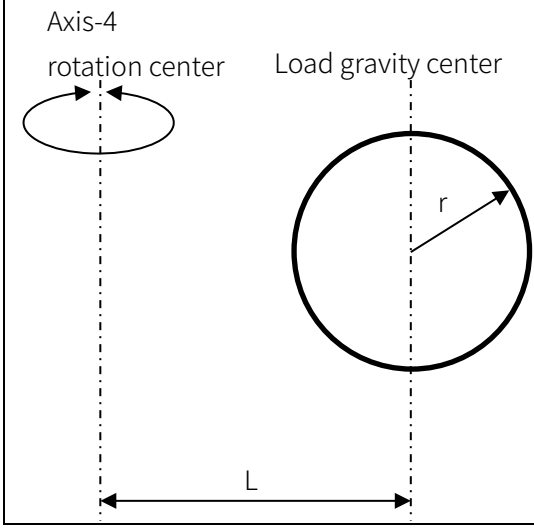
2) If the load shape is columnar



L: Distance from axis-4 center to load gravity center (m)
r: Load radius (m)
M: Load mass (kg)
J: Moment of inertia ($kg \cdot m^2$)

$J=Mr^2/2+ML^2$

3) If the load shape is spherical



L: Distance from axis-4 center to load gravity center (m)
r: Load radius (m)
M: Load mass (kg)
J: Moment of inertia ($kg \cdot m^2$)

$J=2Mr^2/5+ML^2$

Example of calculating moment of inertia

Fig. 5.8 and Fig. 5.9 show examples of calculating the moment of inertia.

The moment of inertia of the entire load can be obtained by total moment of inertia of individual parts.

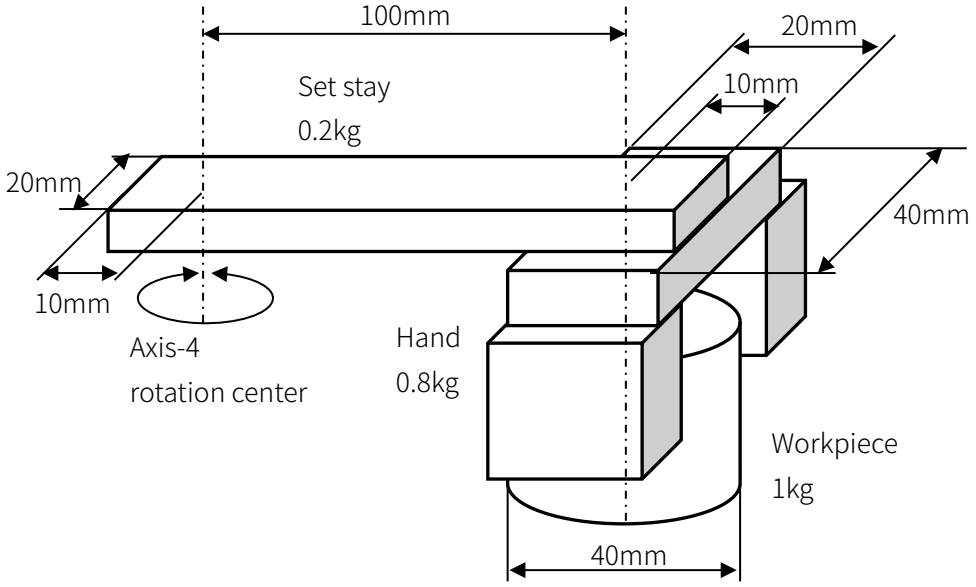
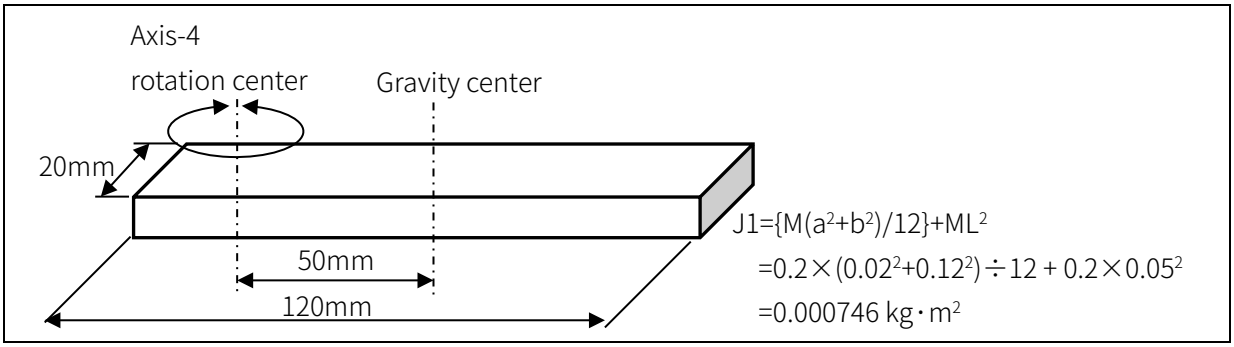
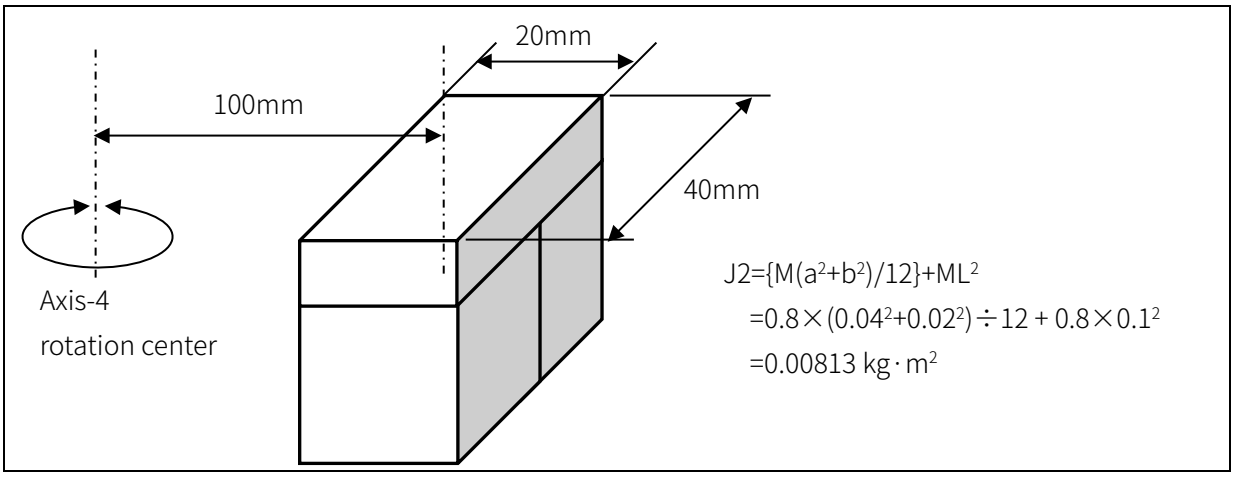


Fig. 5.8 Example of calculating the moment of inertia 1

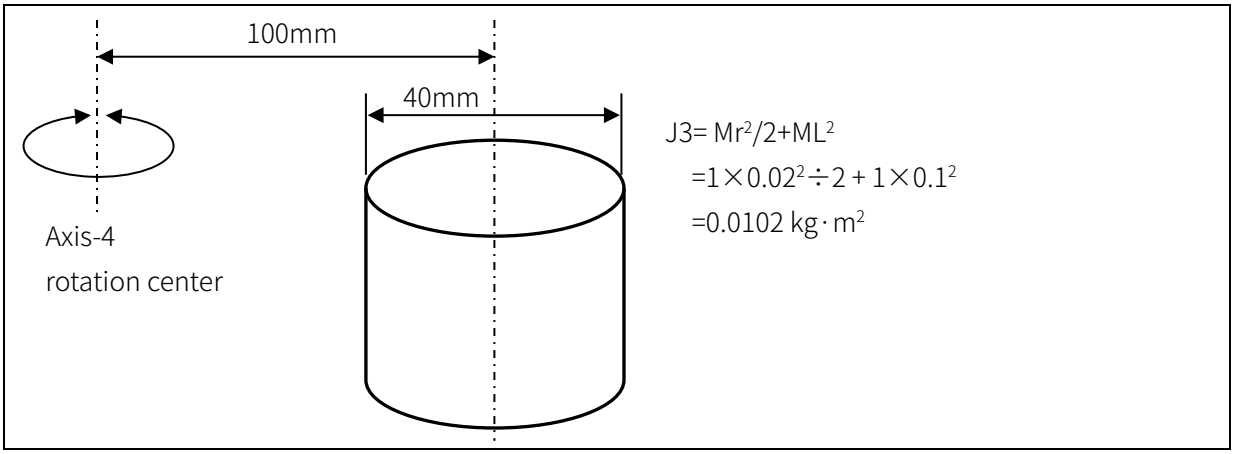
1) Moment of inertia of set stay



2) Moment of inertia of hand



3) Moment of inertia of workpiece



The entire moment of inertia is the sum of J1+J2+J3 to result in 0.019076 kg·m².

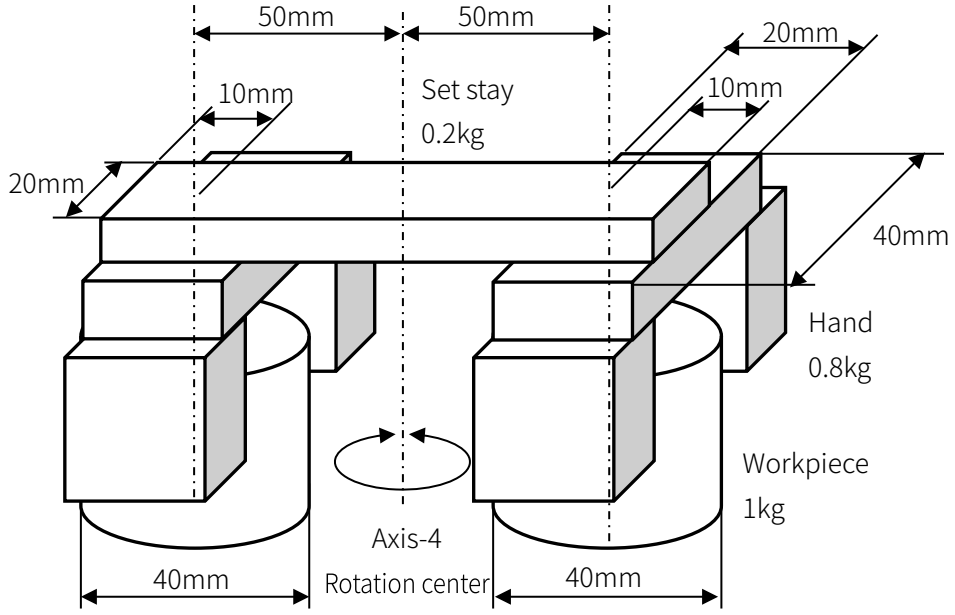
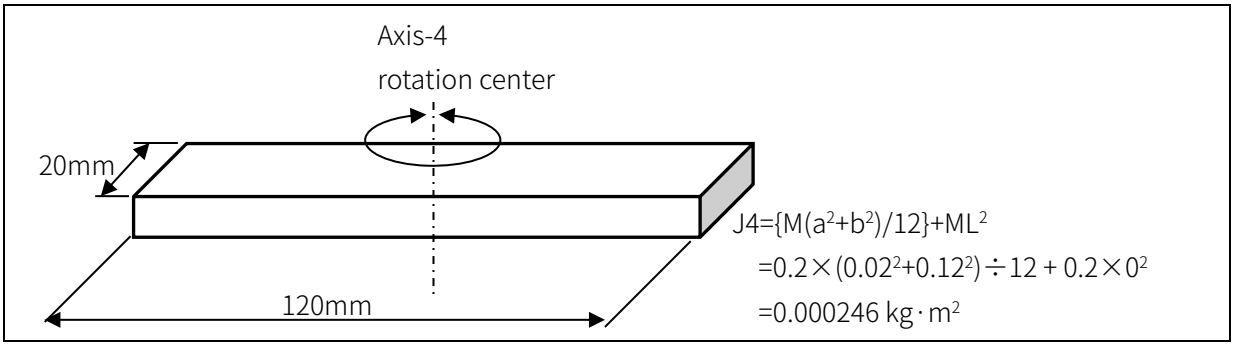
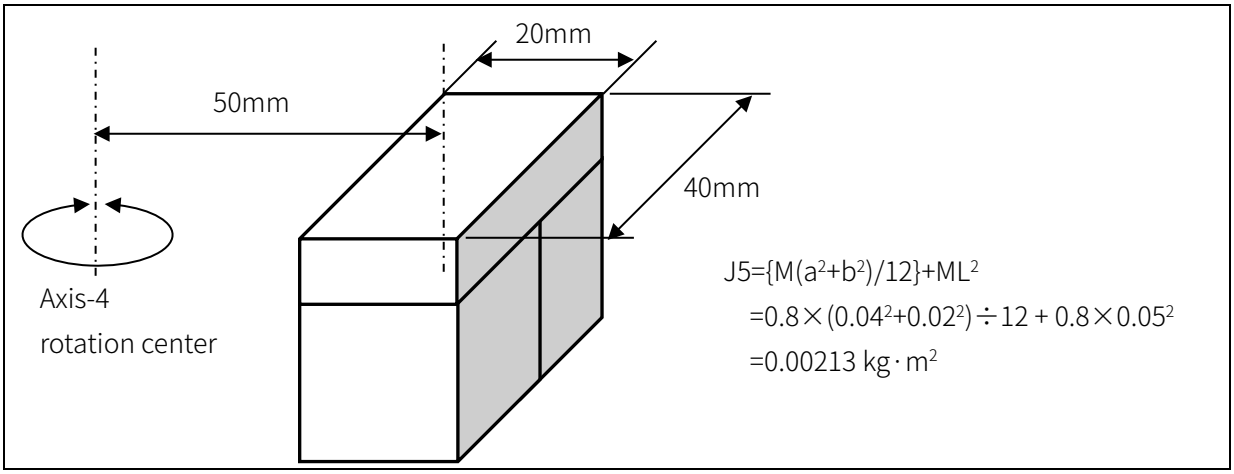


Fig. 5.9 Example of calculating the moment of inertia 2

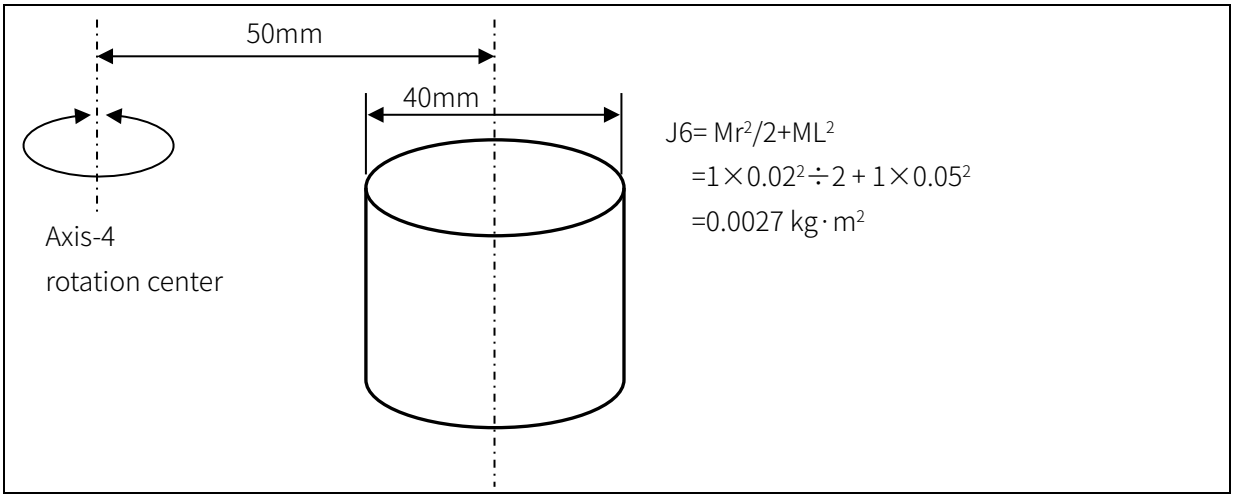
4) Moment of inertia of set stay



5) Moment of inertia of hand



6) Moment of inertia of workpiece



The entire moment of inertia is the sum of J4+J5×2+J6×2 to result in 0.009906 kg·m².

To obtain virtual gravity center offset L from the moment of inertia, see "Setting of PAYLOAD command"

In example of calculating the moment of inertia 2, the gravity center is at the axis-4 rotation center. When the virtual gravity center is obtained from the moment of inertia, the result is as follows.

$L=\sqrt{(J\times10^6/M)}$

$=\sqrt{(0.009906\times10^6/3.8)}$

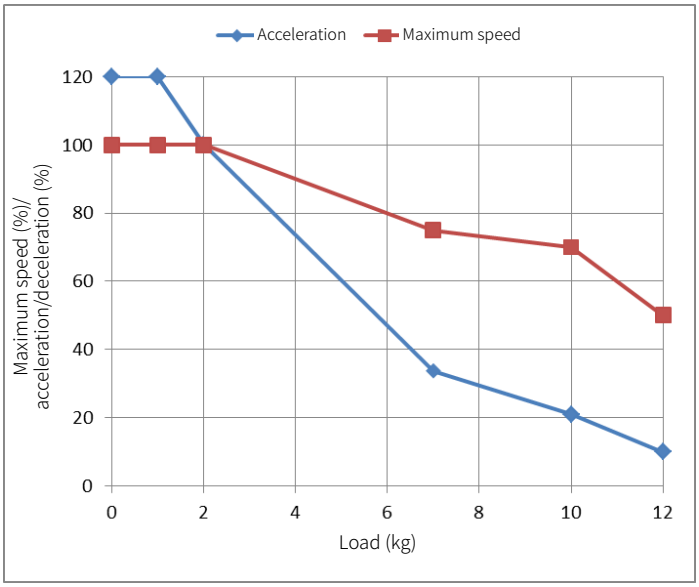
=51.057 (unit: mm)

The virtual gravity center offset is rounded up to become "51.1mm."

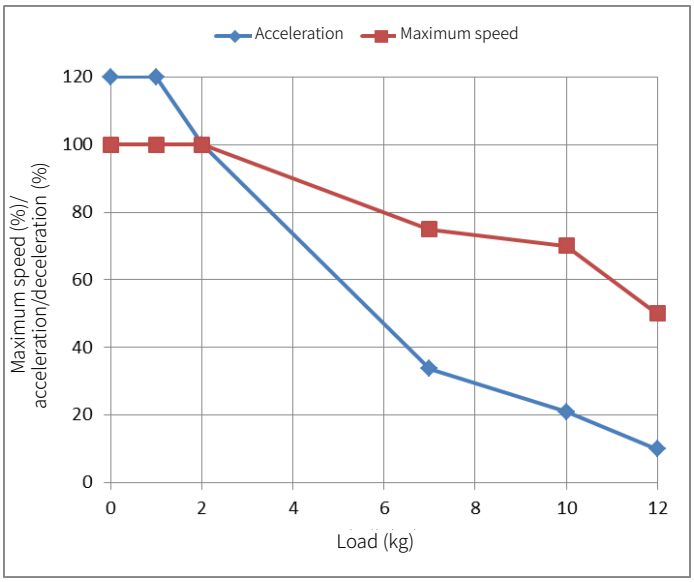
The PAYLOAD command to be set becomes PAYLOAD={3.8，51.1}.

5.4.3 Setting maximum speed and robot acceleration/deceleration for load conditions

When the PAYLOAD command is used, the maximum speed and acceleration/deceleration of the robot are automatically changed according to the load conditions. The maximum speed and acceleration/deceleration change with the load mass as shown in Fig. 5.10 and Fig. 5.11.

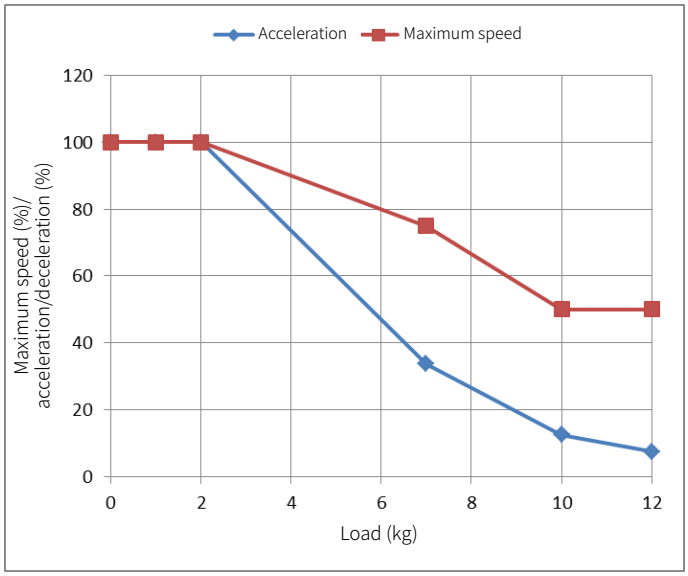


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)

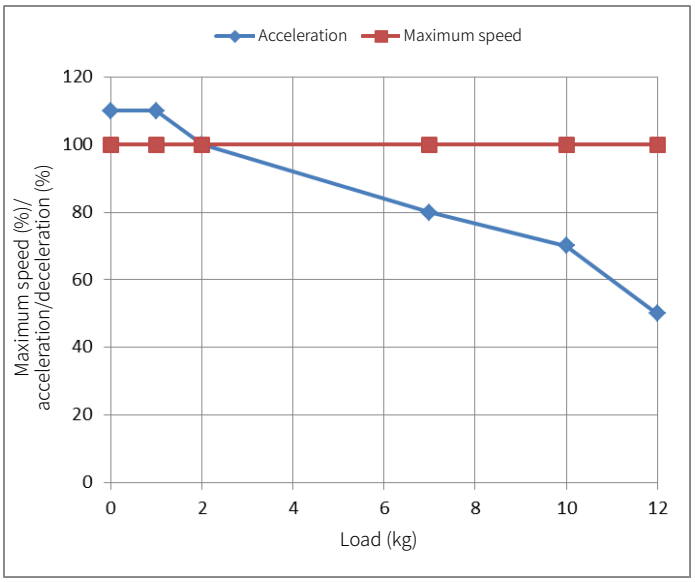


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)

Fig. 5.10 Setting of maximum speed and acceleration/deceleration in relation to load mass (Axes 1 and 2)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

Fig. 5.11 Setting of maximum speed and acceleration/deceleration in relation to load mass (Axes 3 and 4)

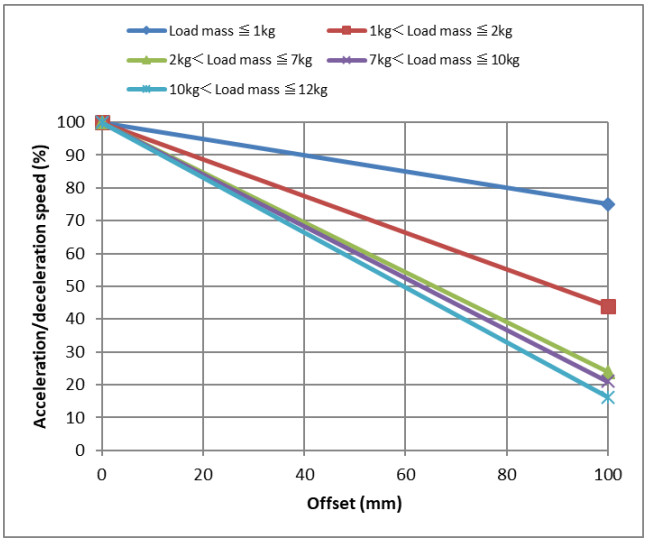
If acceleration / deceleration is to be set to 120% when load is 1 kg or less, “ACCEL = 120” and "DECEL = 120"must be written in the program.

A program example of setting acceleration to 120% is shown below.

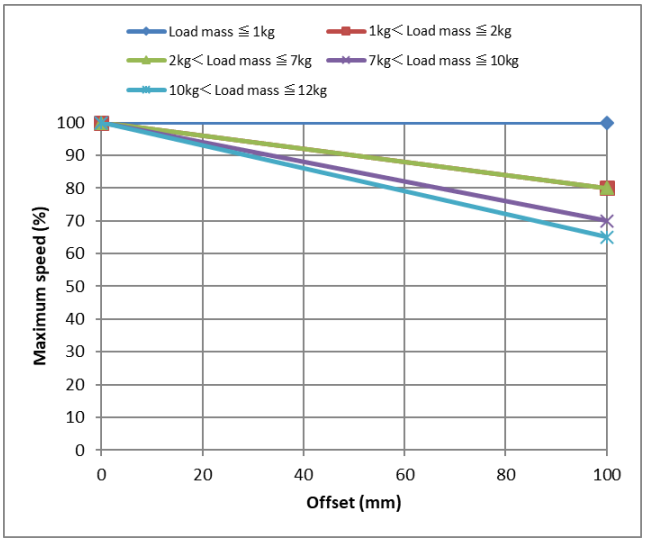
Program example of setting acceleration / deceleration to 120%

```
PROGRAM SAMPLE
  SPEED= 100
  PAYLOAD={1,0}
  ACCEL=120
  DECEL = 120
  MOVE P1
  MOVE P2
  STOP
END
```

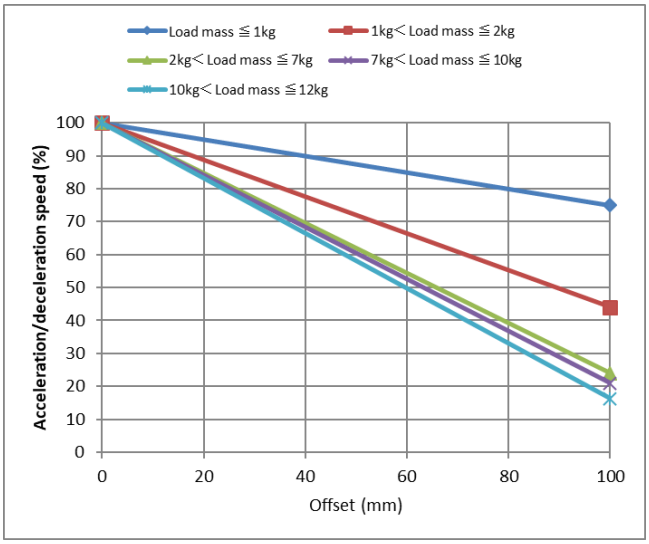
Additionally, if there is an offset of load gravity center, the acceleration/deceleration and maximum speed are changed as shown in Fig. 5.12 and Fig. 5.13.



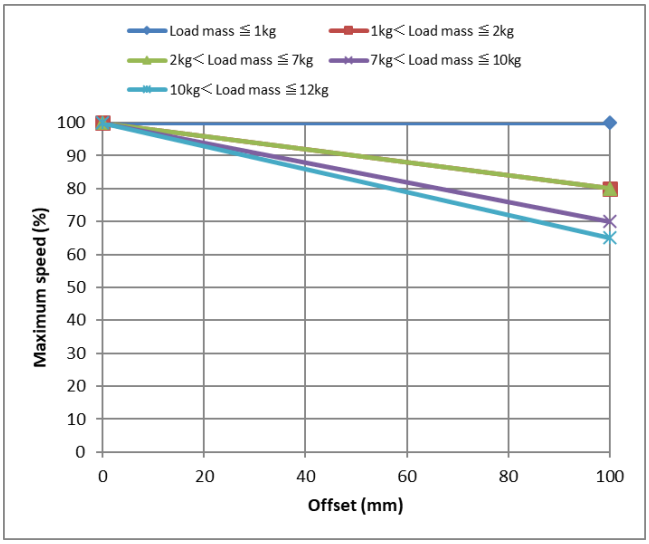
Setting of acceleration/deceleration in relation of offset (Axis 1)



Setting of maximum speed in relation of offset (Axis 1)

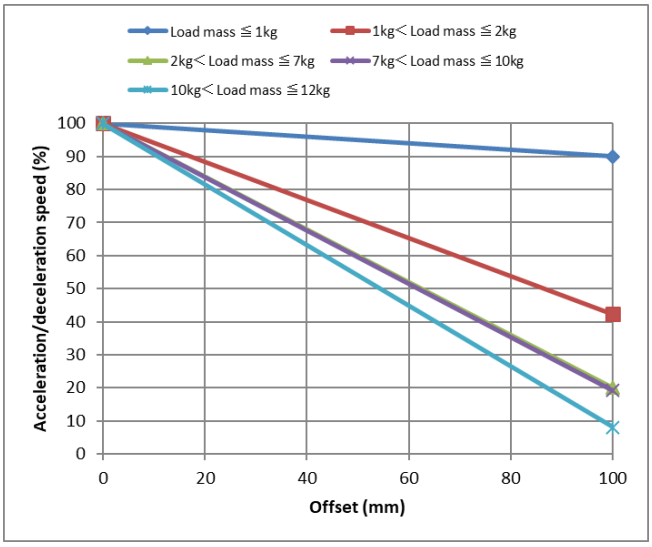


Setting of acceleration/deceleration in relation to offset (Axis 2)

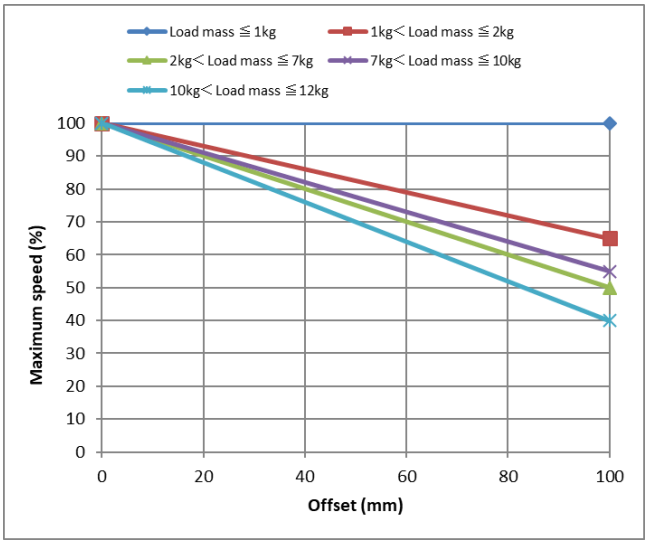


Setting of maximum speed in relation to offset (Axis 2)

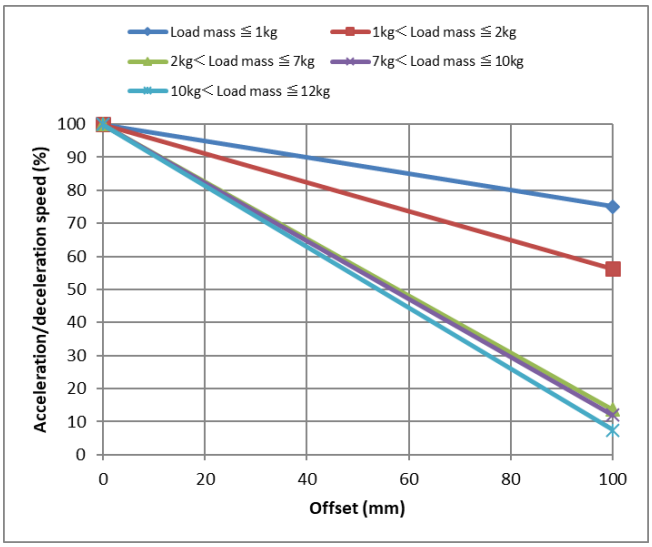
Fig. 5.12 Setting of acceleration/deceleration in relation to gravity center offset (Axes 1 and 2)



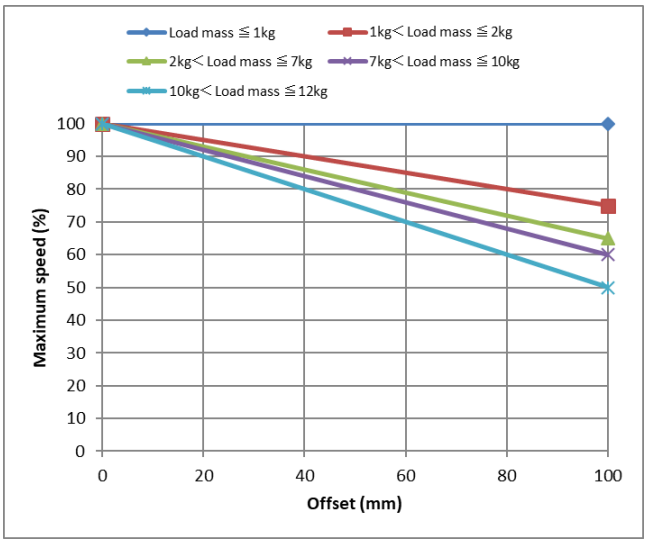
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)



Setting of acceleration/deceleration in relation to offset (Axis 4)



Setting of maximum speed in relation to offset (Axis 4)

Fig. 5.13 Setting of acceleration/deceleration in relation to gravity center offset (Axes 3 and 4)

6. User Wiring

The work should receive safety training defined by each country's laws and regulations.
Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

6.1 Hand wiring

Hand wiring is controlled from the controller or from separately placed PLC (prepared by the customer).

⚠DANGER

- When manufacturing the hand wiring cable, be sure to use the applicable wire.
  Otherwise, the cable may overheat or short circuit may occur between terminals to cause fire.
- Never connect the hand wiring cable with the power source whose voltage exceeds the specified range.
  Otherwise, the cable may overheat to cause fire.
  - Before connection, use the tester, etc. to check the voltage of the power source.

NOTICE

- Never mistake the terminal arrangement of the hand wiring.
  Otherwise, the connected parts may become faulty.
- Before the use, use the tester, etc. to check the terminal arrangement.
- When manufacturing the hand wiring cable, be sure to use the applicable wire.
  Otherwise, overheat or short circuit between terminals may occur to burn the cable.
- Be sure to use a highly flexible and twisting-resistant robot cable for the hand wiring cable.
  If the general-purpose cable is used, the wire may be broken during robot operation.
  - Secure the cable to the arm with the saddle, etc. to prevent the cable from being swung around.
- Never connect the hand wiring cable with the power source whose voltage exceeds the specified range.
  Otherwise, the cable may overheat to cause fire.
  - Before connection, use the tester, etc. to check the voltage of the power source.
- Be sure to install hand wiring and piping while checking the following status during robot operation.
  Otherwise, the robot or hand may become faulty early.
  - Check whether a load is applied to the hand connector.
  - Check whether hand wiring and piping contact the robot or peripheral equipment and are worn.

6.1.1 Hand control wiring

Eight input signals, eight output signals, and 24 VDC/PGND are provided as hand wiring and connected to the controller. Fig. 6.6 and Fig. 6.7 are the wiring diagrams. The wiring is connected to the connector on the arm-2 top.

The user should prepare the connector and connect the cable.

Connector (standard) type:	<Hood>	09670250494 (manufacturer: HARTING)
	<Connector>	09563005601 (manufacturer: HARTING)
	<Terminal>	09560008177 (manufacturer: HARTING)

Applicable cross sectional area of wire conductor AWG 24 to 26 (plain braided wire and twisted wire)

*Connecting the crimp terminal and wire requires special crimp tool.

Tool type <Crimp tool> 09990000596 (Manufacturer: HARTING)

MEMO

Hand I/O cable is also optionally available. For optional hand I/O cable, see "6.2 Hand I/O cable (option) wiring."

The signal line from the controller passes inside the robot and is wired to hand wiring connector “HAND.” Connect the manufactured cable connector to “HAND.”

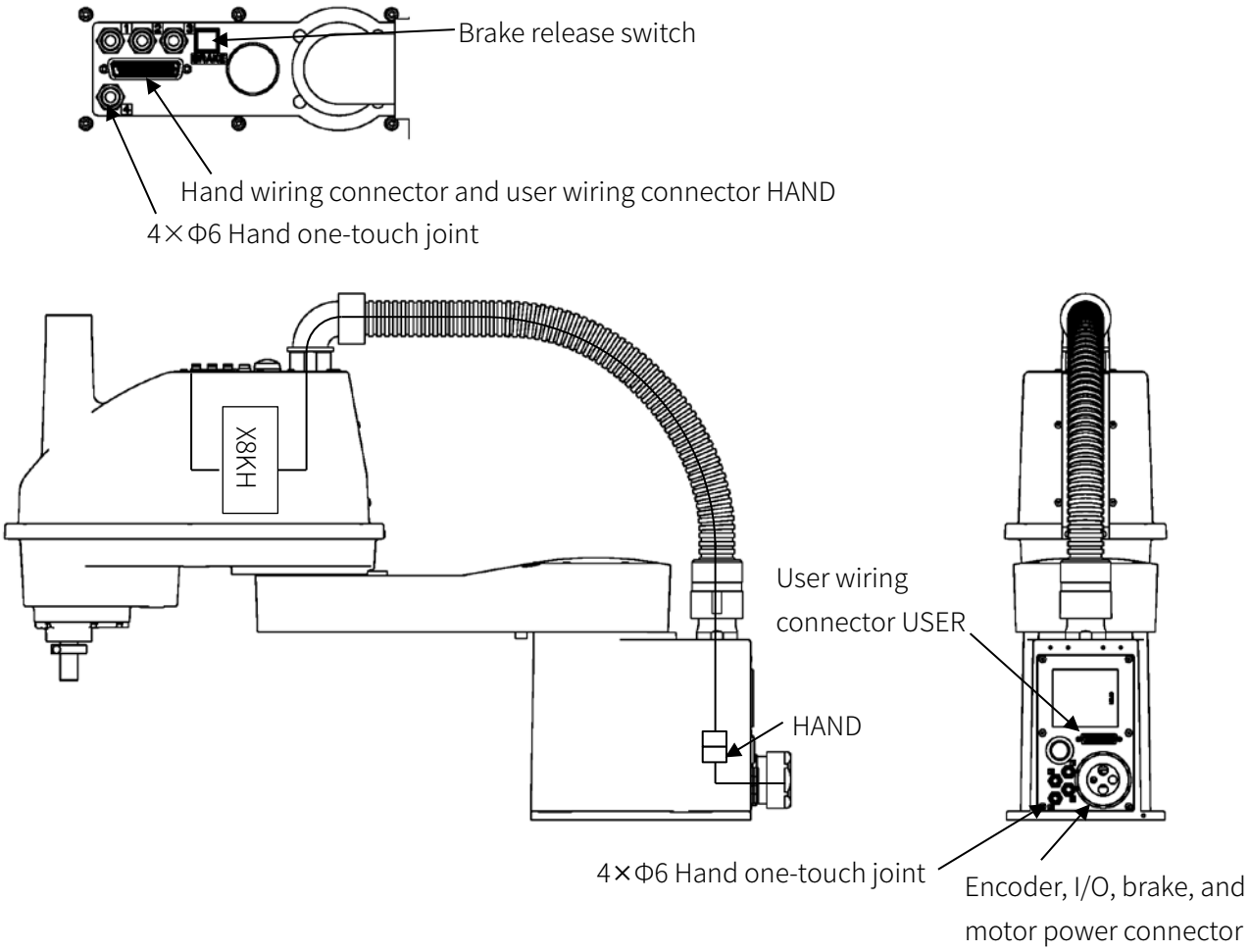


Fig. 6.1 Hand wiring

6.1.2 User wiring

Sixteen (16) signals are provided as user wiring that is controlled by electric gripper, separately placed PLC, etc.
The wiring is connected to the connector on the arm-2 top and on the base rear.
The user should prepare the connector and connect the cable.

Connector (standard) type: <Hood> XM2S-2511 (manufacturer: OMRON)
<Connector> XM3A-2521 (manufacturer: OMRON)
Applicable cross sectional area of wire conductor: AWG 24 to 26 (plain braided wire and twisted wire)
*Connect the connector and wire by soldering.

For the arm-2 side connector, see the connector in "6.1.1Hand control wiring."

The signal line from the base rear connector "USER" passes inside the robot and is wired to the hand wiring connector "HAND."
Connect the manufactured cable connector to "USER" and "HAND."

MEMO
Hand I/O cable to be connected to the connector "HAND" is also optionally available. For optional hand I/O cable, see "6.2 Hand I/O cable (option) wiring."

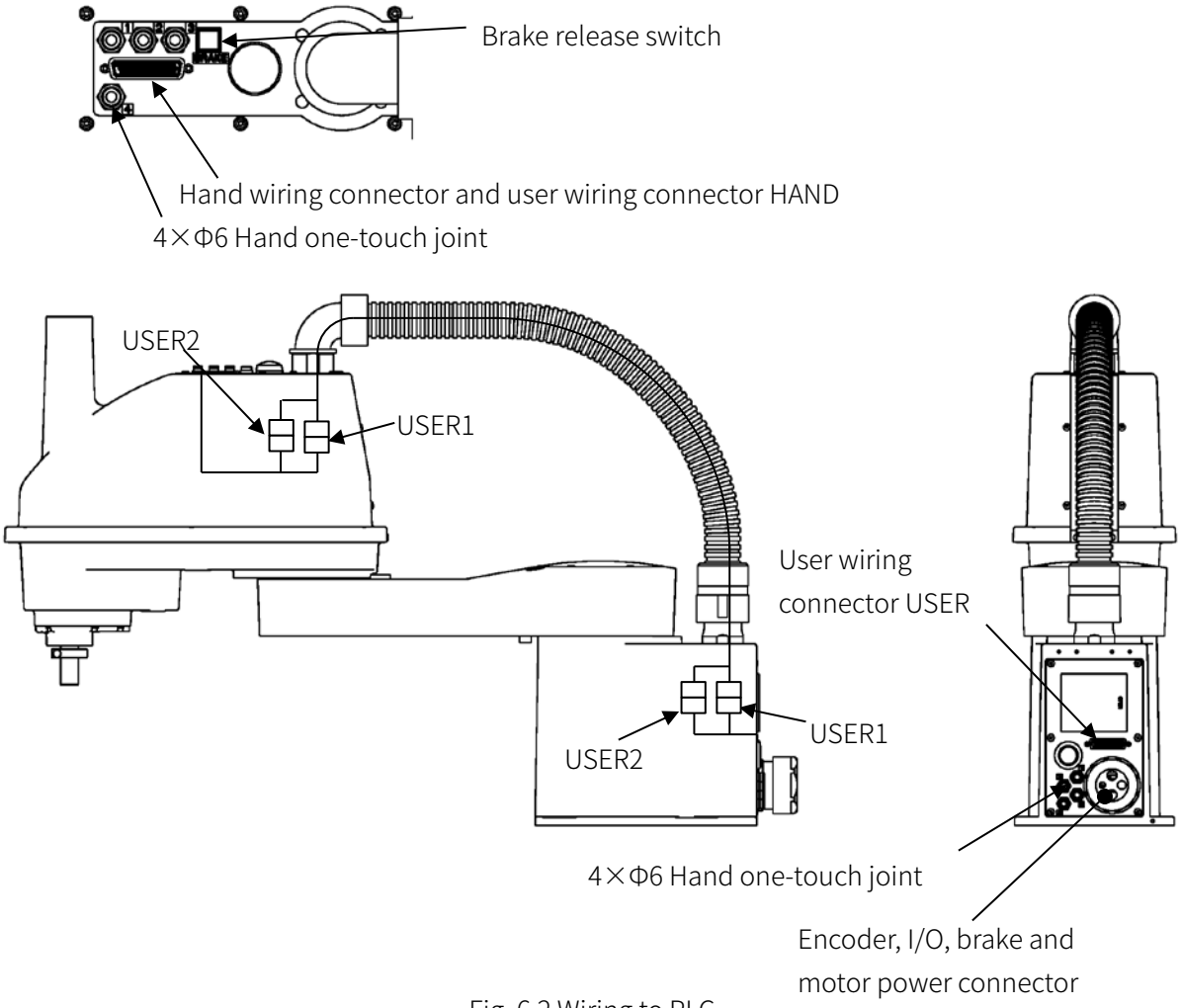


Fig. 6.2 Wiring to PLC

User wiring is as shown in Fig. 6.3.

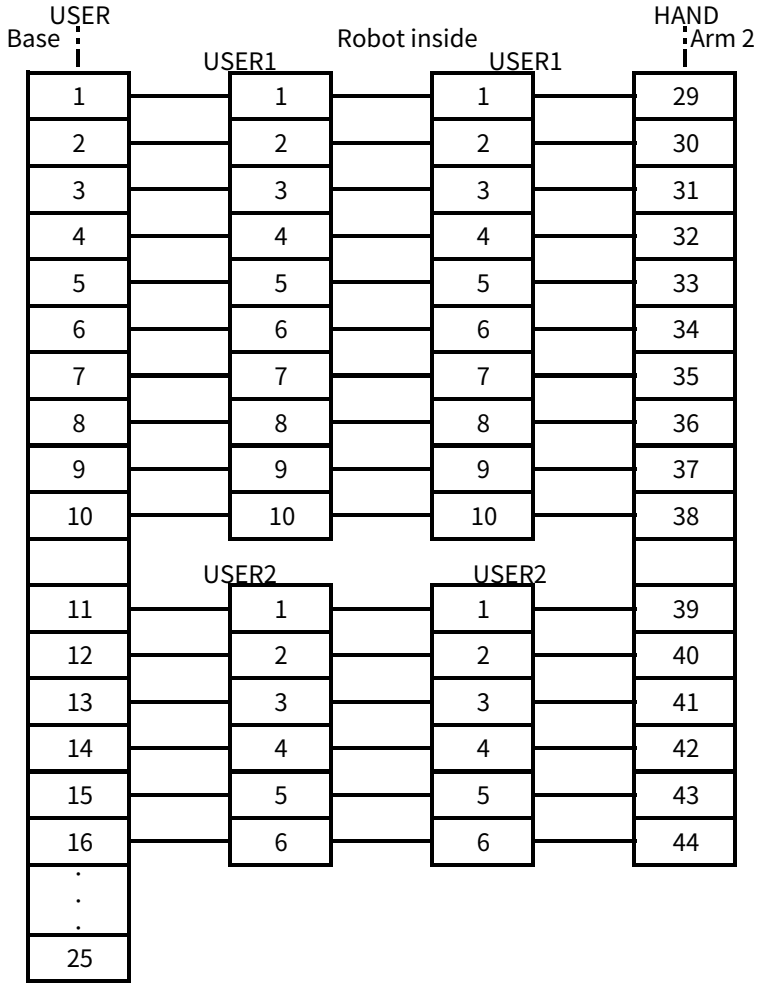


Fig. 6.3 User wiring diagram

6.1.3 Specifications of hand connector HAND

Table 6.1 Input/output signal connector HAND (TS5000-MS, Type-N)

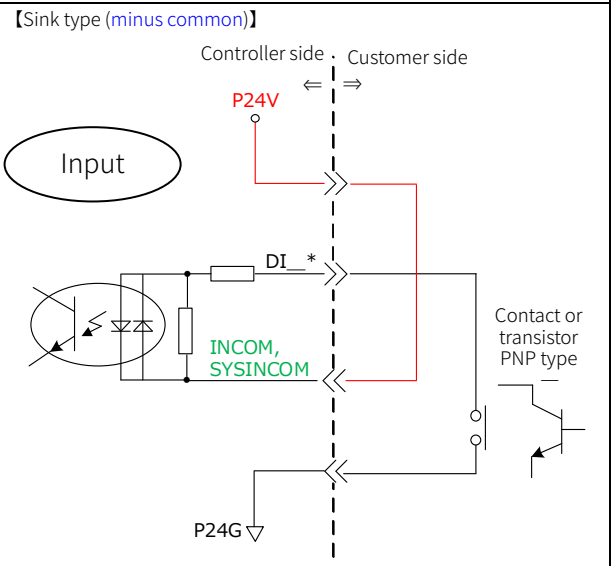
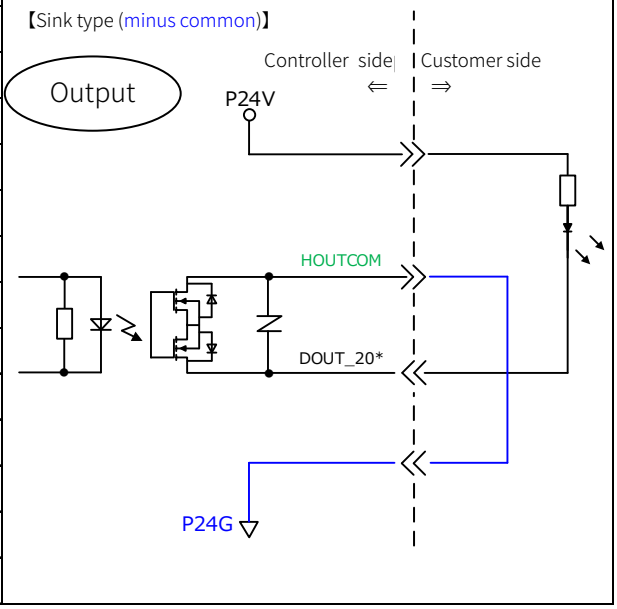
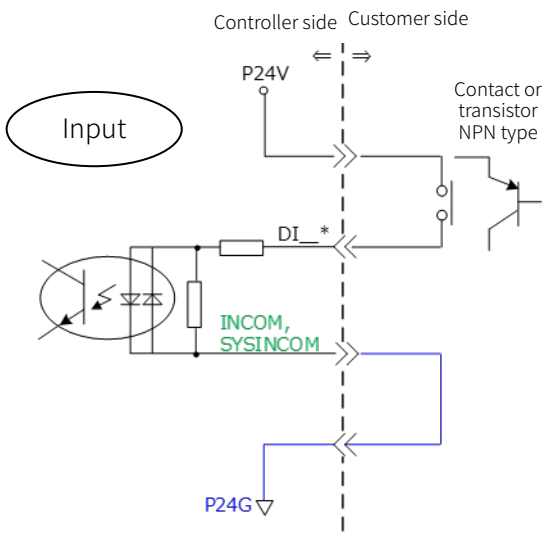
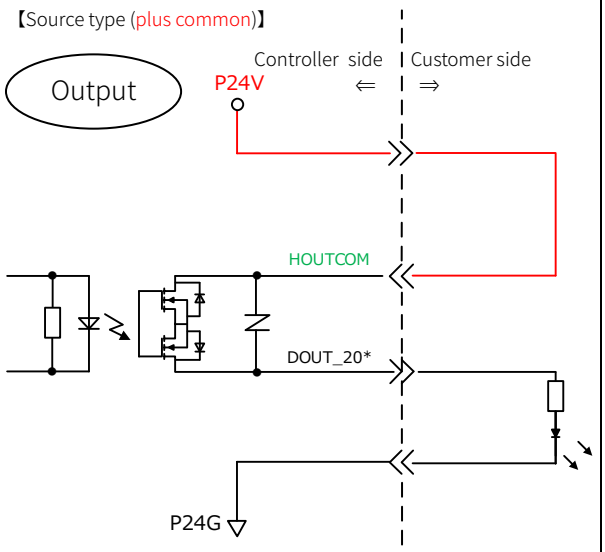
Pin	Signal name		Signal No	Input/output circuit and example of connections
1	HAND-IN0	Input signal 1	201	
2	HAND-IN1	Input signal 2	202	
3	HAND-IN2	Input signal 3	203	
4	HAND-IN3	Input signal 4	204	
5	HAND-IN4	Input signal 5	205	
6	HAND-IN5	Input signal 6	206	
7	HAND-IN6	Input signal 7	207	
8	HAND-IN7	Input signal 8	208	
9	P24V			
10				
11	HINCOM1			
12	HINCOM2			
13	P24VGND			
14				
15	HAND-OUT0	Output signal 1	201	
16	HAND -OUT1	Output signal 2	202	
17	HAND -OUT2	Output signal 3	203	
18	HAND -OUT3	Output signal 4	204	
19	HAND -OUT4	Output signal 5	205	
20	HAND -OUT5	Output signal 6	206	
21	HAND -OUT6	Output signal 7	207	
22	HAND -OUT7	Output signal 8	208	
23	P24V			
24				
25	HOUTCOM1			
26	HOUTCOM2			
27	P24VGND			
28				

Table 6.2 Input/output signal connector HAND (TS5000-MS, Type-P)

Pin	Signal name		Signal No.	Input/output circuit and example of connections
1	HAND-IN0	Input signal 1	201	【Source type (plus common)】
2	HAND-IN1	Input signal 2	202	
3	HAND-IN2	Input signal 3	203	
4	HAND-IN3	Input signal 4	204	
5	HAND-IN4	Input signal 5	205	
6	HAND-IN5	Input signal 6	206	
7	HAND-IN6	Input signal 7	207	
8	HAND-IN7	Input signal 8	208	
9	P24V			
10				
11	HINCOM1			
12	HINCOM2			
13	P24VGND			
14				
15	HAND-OUT0	Output signal 1	201	【Source type (plus common)】
16	HAND -OUT1	Output signal 2	202	
17	HAND -OUT2	Output signal 3	203	
18	HAND -OUT3	Output signal 4	204	
19	HAND -OUT4	Output signal 5	205	
20	HAND -OUT5	Output signal 6	206	
21	HAND -OUT6	Output signal 7	207	
22	HAND -OUT7	Output signal 8	208	
23	P24V			
24				
25	HOUTCOM1			
26	HOUTCOM2			
27	P24VGND			
28				





Input signal

As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating:	24 VDC, 10 mA or over (circuit current: approx. 7 mA)
Minimum contact current:	24 VDC, 1 mA
Contact impedance:	100 Ω or less

Transistor specifications:

Withhold voltage between collector and emitter:	30 V or over
Current between collector and emitter:	10 mA or over (circuit current: approx. 7 mA)
Leak current between collector and emitter:	100 μA or less

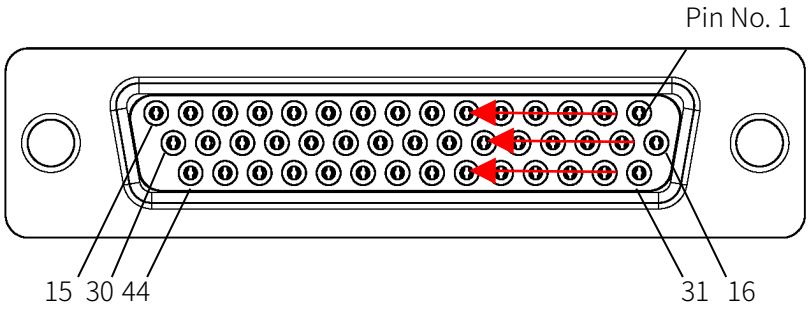
Output signal

By using the P24V of the controller, a relay, solenoid valve, etc. can be driven.

To use external power supply, connect GND of the external power supply with the GND (P24G) of the robot controller (GND common connection).

Transistor specifications:	Rated voltage:	24 VDC
	Rated current:	100 mA

- Limit P24V supplied from the robot controller to within 3 A in total.
- To use external power supply, limit the total current value to within 3 A as well
- To connect a relay, solenoid valve, etc., use a surge killer or diode to absorb the surge voltage.



[Pin arrangement with robot side HAND connector viewed]

Fig. 6.4 Hand connector pin arrangement

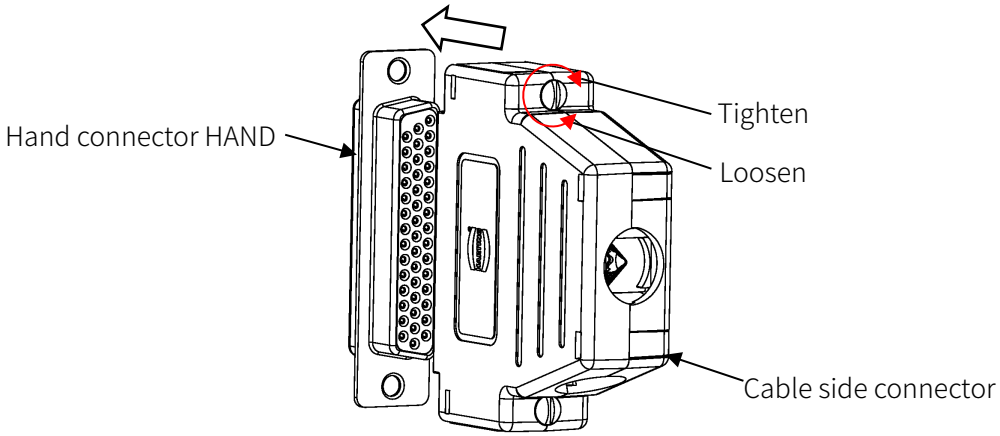


Fig. 6.5 Connecting/disconnecting hand input/output signal connector

⚠ CAUTION

- Be sure to connect the connectors securely.
  If the connectors almost come off, the robot hand may move unpredictably.
  - Since the hand wiring cable is swung around during robot operation, secure it to the arm with the clamp, etc.
  - Adjust the robot wiring and piping while checking the status during operation to prevent them from contacting the robot and peripheral equipment and applying load to the connector.

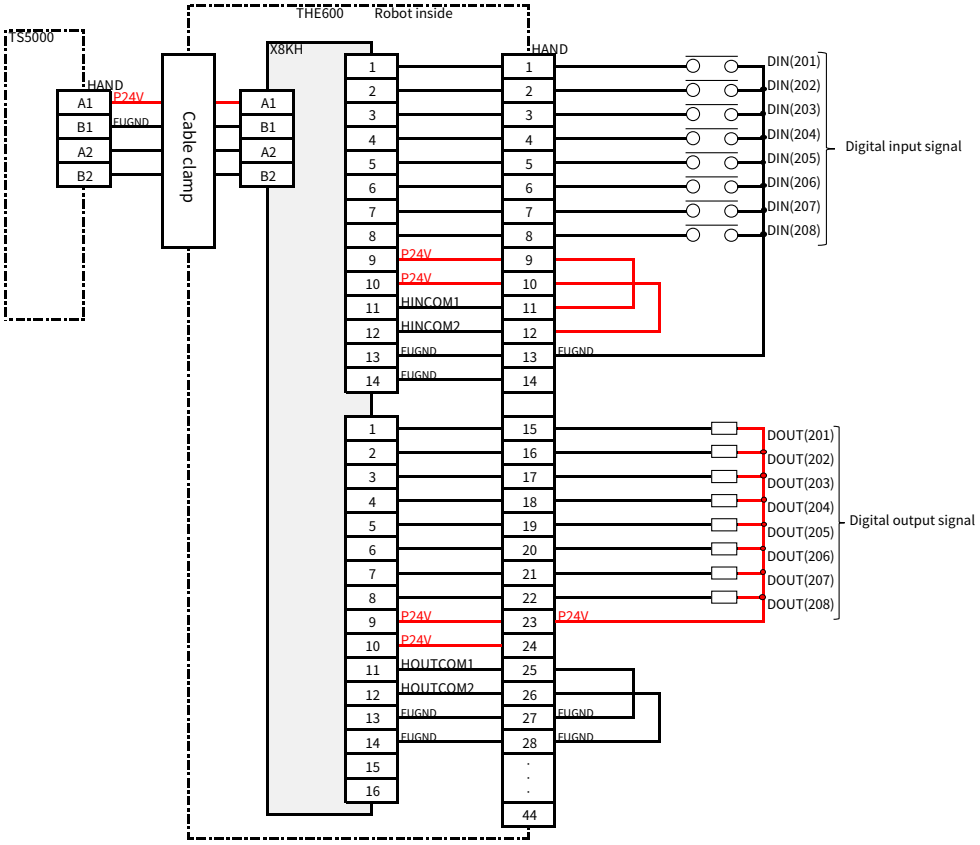


Fig. 6.6 Hand wiring diagram (common to Type-N)

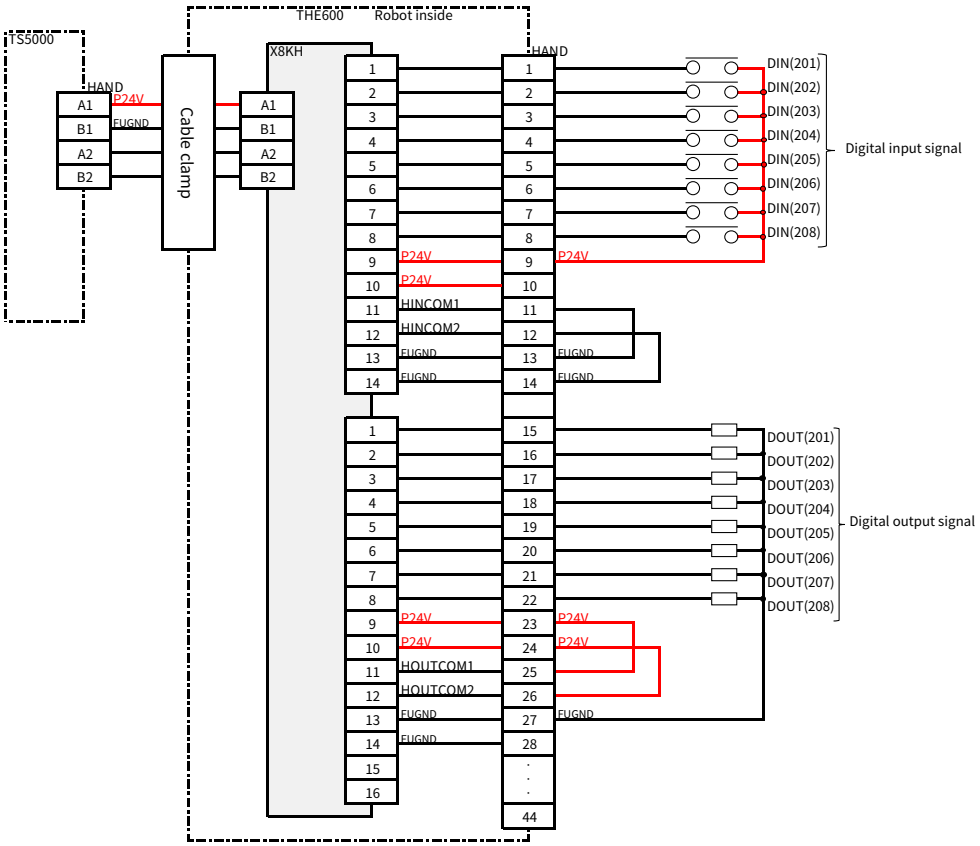
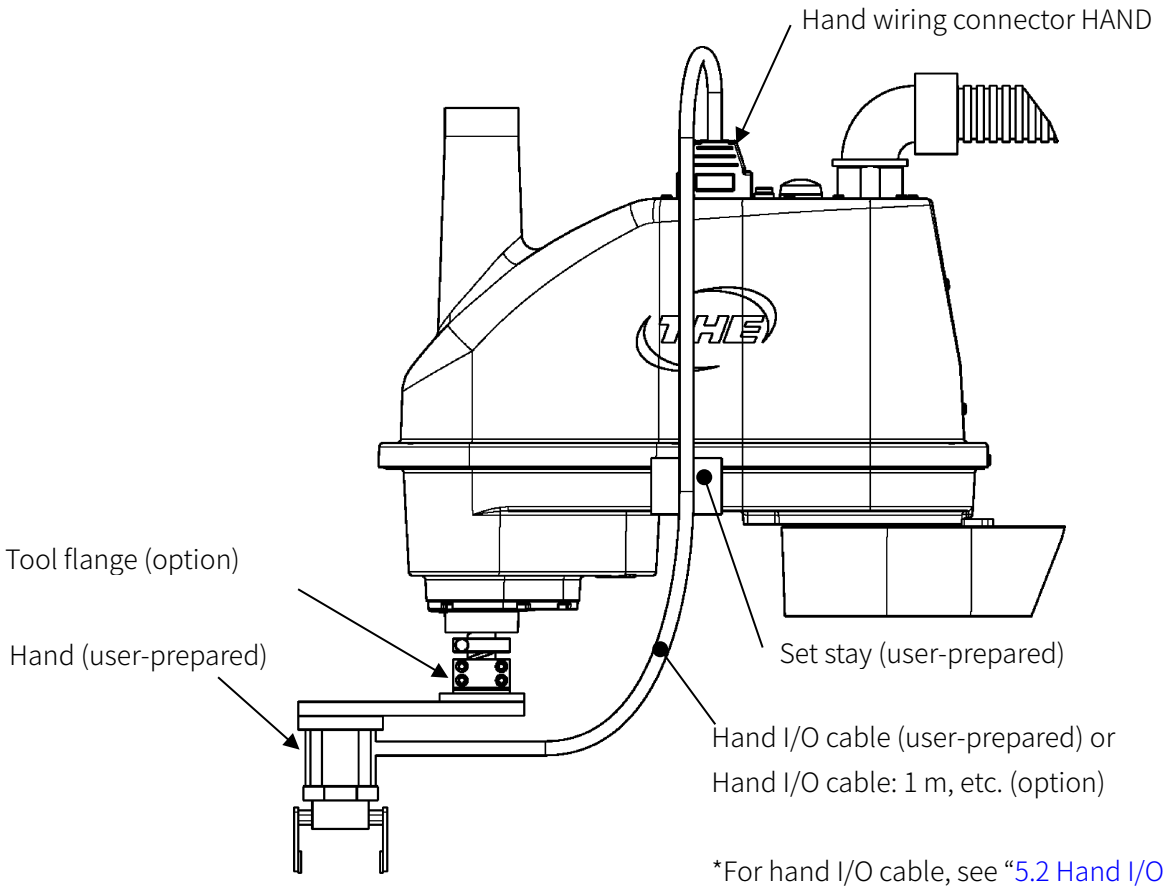


Fig. 6.7 Hand wiring diagram (common to Type-N)

6.1.4 Connection example

Fig. 6.8 shows an example of hand wiring.



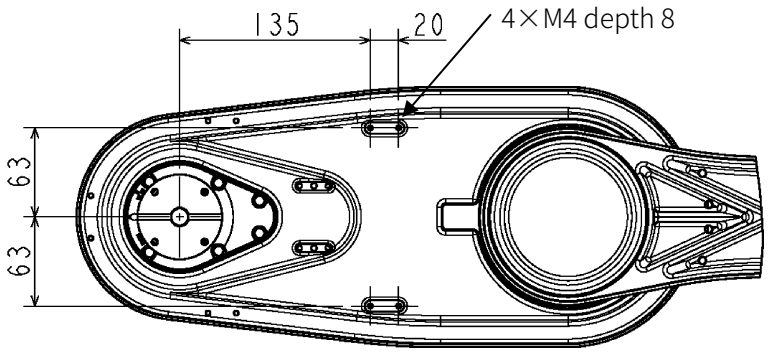


Fig. 6.8 Example of hand I/O cable wiring to arm-2 top

For the mounting dimensions of the set stay, see Fig. 5.2.

Fig. 6.9 shows a connection example of hand piping.

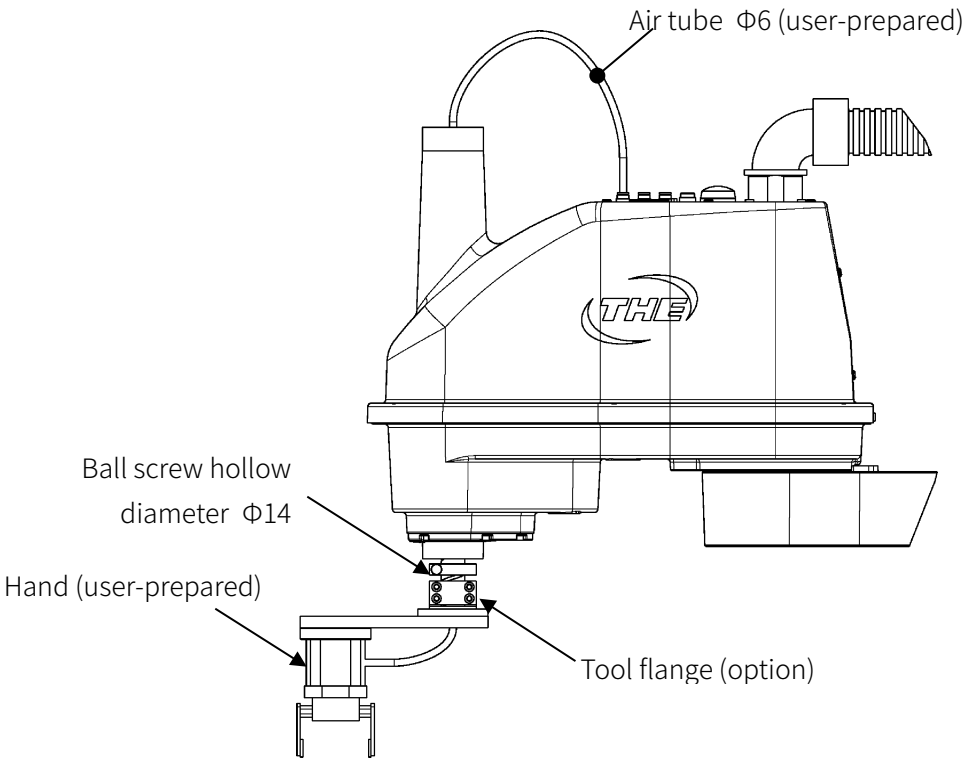


Fig. 6.9 Example of air tube piping to arm-2 top

NOTICE

- Be sure to perform hand wiring and piping while checking the following status during robot operation. Otherwise, the robot and hand will become faulty early.
  - Check whether a load is applied to the hand connector.
  - Check whether hand wiring and piping contact the robot or peripheral equipment and are worn.
  - Check whether the set stay interferes with the arm 1, etc.

6.2 Hand I/O cable (option) wiring

Fig. 6.10 and Fig. 6.11 are external view and wiring diagram of hand I/O cable (option). Table 6.3 Hand I/O cable parts used lists parts used, and Table 6.4 Hand I/O cable (option) lists SHIBAURA drawing number of hand I/O cable (option) for reference.

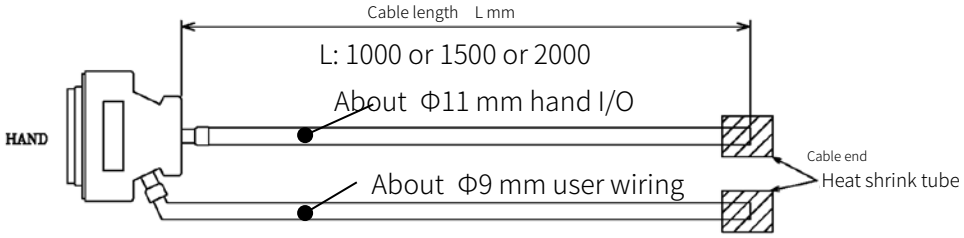


Fig. 6.10 External view drawing of hand I/O cable (option)

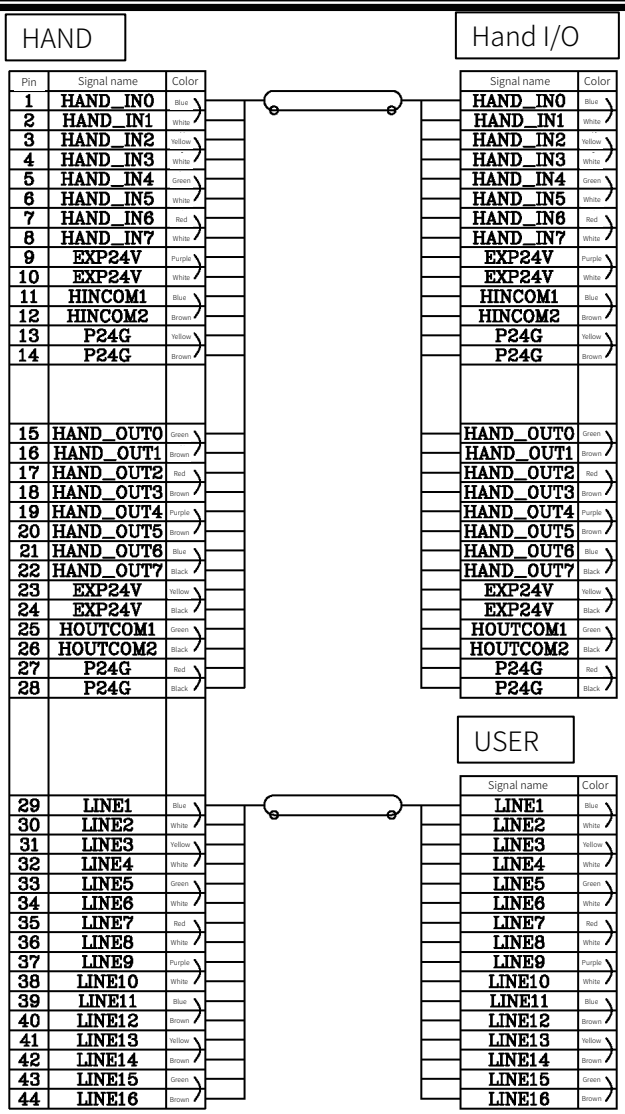


Fig. 6.11 Hand I/O cable (option) wiring diagram

Table 6.3 Hand I/O cable parts used

No.	Product name	Type	Manufacturer	Use qt.
1	Cable	RMFEV-SB(CL3) 0.2SQ-15P	DYDEN Corp.	1
2		RMFEV-SB(CL3) 0.2SQ-8P		1
3	D-Sub connector	09563005601	HARTING	1
4	D-Sub hood	09670250494		1
5	D-Sub contact	09560008177		44

Table 6.4 Hand I/O cable (option)

Product name	Type	Cable length	SHIBAURA Dwg. No.	Unit code
Hand I/O cable		1m	396N7701	Y610D0DY0
		1.5m	396N7702	Y610D0DZ0
		2m	396N7703	Y610D0E00

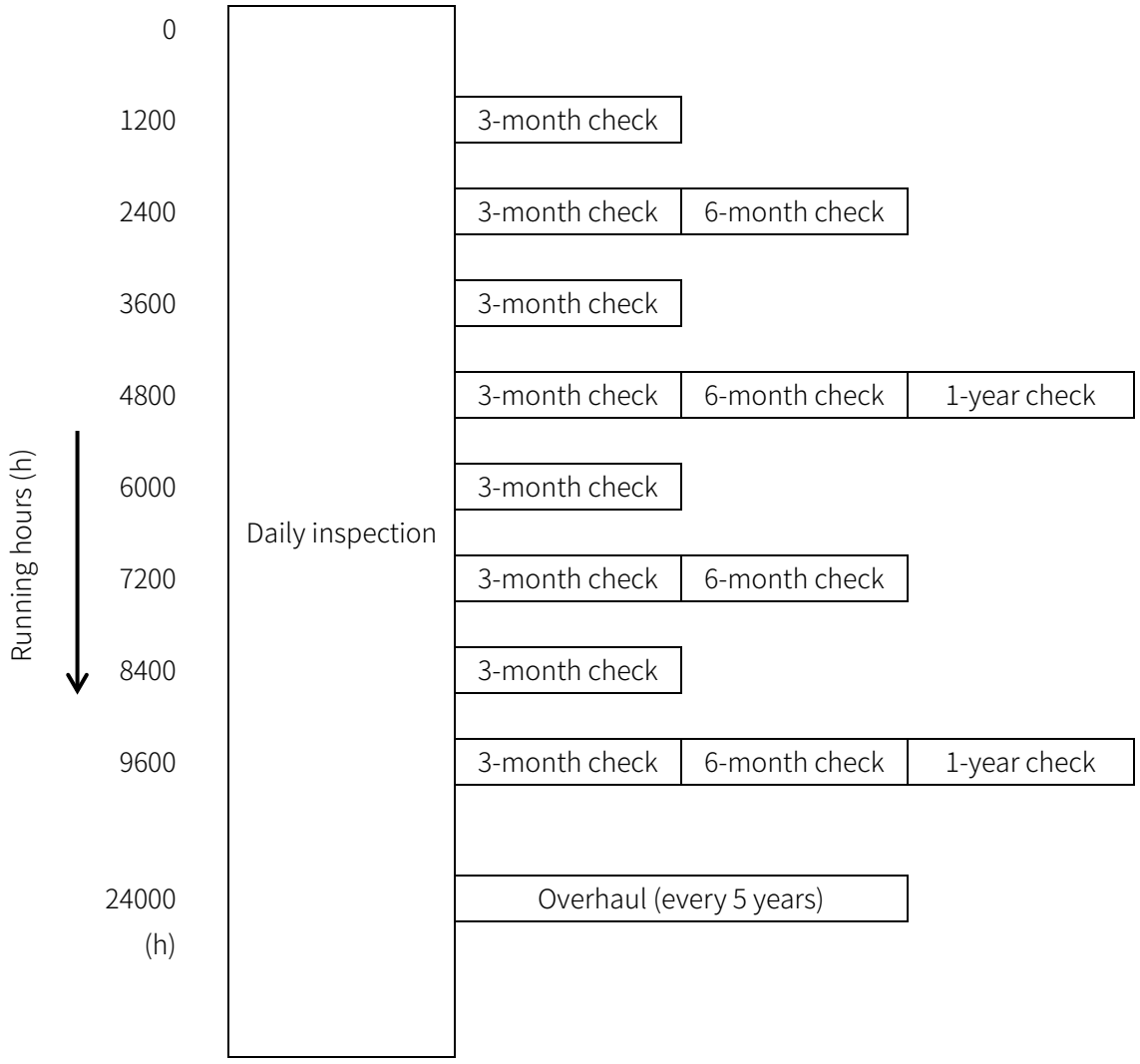
Maintenance Manual

This manual describes how to maintain respective units of the THE series robot.

7. Maintenance

7.1 Maintenance Schedule

Maintenance comes in two types: daily inspection, and regular inspection and maintenance.
For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



MEMO

When the system is operated for 16 hours per day: 1,200 hours ÷ 16 hours ÷ 25 days = 3 months
When the system is operated for 24 hours per day: 1,200 hours ÷ 24 hours ÷ 25 days = 2 months

When the system is operated for 16 hours per day, regular inspection should be conducted every three months. However, if the running hours per day are increased, regular inspection must be done at shorter intervals.

Fig. 7.1 Maintenance schedule

7.2 Items for Maintenance and Inspection

Maintenance comes in two types: daily inspection, and regular maintenance and inspection that are performed every time a predetermined period elapses.

For the executing procedures, see the relevant paragraphs listed in the table below.

7.2.1 Inspection at Power OFF

Table 7.1 Inspection at Power OFF

D: Daily inspection Q: Quarterly inspection
S: Semi-annual inspection A: Annual inspection

Description	Inspection point	D	Q	S	A	Refer to
Make sure that all bolts are completely tightened and secured. If there are any loosened or unsecured bolts, re-tighten them.	Hand set bolts	○	○	○	○	8.1.1
	Robot installation bolts	○	○	○	○	
	Motor set bolts				○	
Check the cable clamp for tightening.	Upper part of arm 2 and base				○	8.1.4
Make sure that no cut or scratch is found. Clean the contaminant, if any.	Entire robot	○	○	○	○	
	Inside of arm 2			○	○	
Check state of corrosion inhibitor and apply corrosion inhibitor.	Ball spline nut and ball screw shaft end	○	○	○	○	8.6.3
Make sure that each timing belt is not worn off or cracked. Check each timing belt for tension.	Inside of arm 2			○	○	8.5.1
Make sure that the cables and air tubing are not worn out.	Cables and air tubing of each unit		○	○	○	8.1.3
Check each axis operation. Move each axis by hand for check.	Entire robot			○	○	8.1.5

7.2.2 Inspection at Power ON

Table 7.2 Inspection at Power ON

D: Daily inspection Q: Quarterly inspection
S: Semi-annual inspection A: Annual inspection

Description	Inspection point	D	Q	S	A	Refer to
Check grease state and grease.	Ball screw shaft	○	○	○	○	8.6.3
Push each arm by hand in the servo ON condition to make sure that the arm is secured.	Each joint				○	
Make sure that no abnormal vibration or noise is caused.	Entire robot	○	○	○	○	
Make sure that the teaching points are in place.	Robot tip		○	○	○	
Replace the batteries for position detection with new ones.	Battery box				○	9.3.2

7.2.3 Overhaul

Table 7.3 Overhaul

Description	Maintenance parts	Every 5 years	Refer to
Check the consumable parts and replace as necessary.	Motors, reduction gears, belts, ball screw, body harness, LED lamp	○	8.3 to 8.8

7.3 Cautions on Maintenance and Inspection

⚠ DANGER

- When performing maintenance with the power turned ON in the dangerous range of the robot, be sure to switch the master mode key to “MANUAL” to pull out the key and carry it with the teach pendant. Otherwise, the robot that has operated unpredictably may collide with a worker.
  - The worker should receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.
  - Display the fact that work is in progress near the controller.
- When performing maintenance with the power turned ON in the dangerous range of the robot, be sure to arrange a work observer outside the dangerous range of the robot.
  Otherwise, when a worker in the dangerous range of the robot suffers damage, the worker cannot be rescued.
  - The observer should secure safety of the worker and the third party who approaches the dangerous range of the robot.
  - The observer should also receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.
- When approaching the robot being stopped, be sure to check the robot operation status.
  If the robot is under operation, the robot that has restarted may collide with the worker.
  - Check the operation status indicated on the indicator lamp of the controller and on the teach pendant.
  - The robot operation status is output by signals (servo operation preparation completion, external mode, operation preparation completion, automatic operation, failure, and emergency stop). Connect to the display equipment, etc.
- Be sure to perform maintenance and inspection periodically.
  If maintenance and inspection are neglected, the faulty robot may cause an unpredictable disaster.

⚠ WARNING

- When rotating axis 4 by hand, never insert your hand or finger between the upper or lower mechanical stopper and the robot or under the robot hand.
  Otherwise, your hand or finger may get caught in the mechanical stopper and robot hand.
  - When axis 4 rotates, the ball screw spline shaft will move up and down.
- The axis-4 motor is not equipped with the brake. If the robot hand is touched by hand when the servo power is off, axis 4 will rotate.
- When releasing the motor brake, be sure to support the ball screw spline shaft.
  Otherwise, the shaft will drop by its own weight concurrently with release of the brake.
  - When a worker supports the shaft, be sure to prevent hand or finger from getting caught.
  - When something strange is felt during work, release the brake release switch immediately. The brake will also be applied by turning off the power to the controller.
  - The brake release switch is located on the robot base rear. Do not press the switch by mistake.
  - The worker should receive safety training regarding industrial robot inspection-related work defined by each country's laws and restrictions.

⚠ CAUTION

- Be sure to start maintenance when the surface temperature becomes less than 50°C after power off.
  If hot motor or periphery of the motor is touched, burn injury may be received.

NOTICE

- Be sure to perform maintenance and inspection periodically.
  If maintenance and inspection are neglected, the performance and lifespan of the robot may be reduced and the robot may malfunction.

7.4 Maintenance Tools

We recommend using the following tools for maintaining the robot and controller.

- Hexagonal wrench key, nominal size: 2.5 mm to 8 mm
- Torque wrench, nominal size: 3 mm to 8 mm
- Spanner, nominal size: 46 mm and 58 mm
- Torque wrench extension bar
- Plastic hammer
- Scale
- Nipper
- Longnose pliers
- Pliers

Recommended provisions other than the tools:

- Alcohol (for cleaning)
- Waste cloth
- Vise (fixing tool)
- Spatula (for applying grease)
- Screw locking adhesive Recommended: Loctite (241 medium strength) (221 low strength)(425)
- Belt tension meter Recommended: GATES UNITTA ASIA
- Corrosion inhibitor, Kluber A20 or WD-40
- TSAssist program creation/instruction support software (optional)
- Grease gun
- Grease (for Axis-1 reduction gear) Multemp FZ NO.00
- Grease (for Axis-2 reduction gear) SFB No.1
- Tension tool 34 assembly (option)
- Tension tool intermediate assembly (option)
- Axis-1 and Axis-2 zero setting tool (option)
- Axis-3 and Axis-4 zero setting tool (option)

7.5 Clamping Hexagon Socket Head Cap Screws

This robot uses hexagon socket head cap screws at places requiring mechanical strength.
At the time of factory-assembly, screw locking adhesives are applied and each screw is tightened with the following clamping torque.
When these screws are retightened during replacement and inspection described in this manual, use a torque wrench and torque driver and apply Loctite (medium strength) to assure the appropriate clamping torque.
The portions that are not listed in the following table are described in the sections of replacement procedure.

Table 7.4 Clamping torque

Hexagon socket head cap screws	M3	M4	M5
Clamping torque	2.0 N·m	4.7 N·m	9.0 N·m

Hexagon socket head cap screws	M6	M8	M10	M12
Clamping torque	15 N·m	37 N·m	75 N·m	128 N·m

To mount the reduction gear, for the screws arranged on a circle, tighten them in the diagonal order, as shown below.

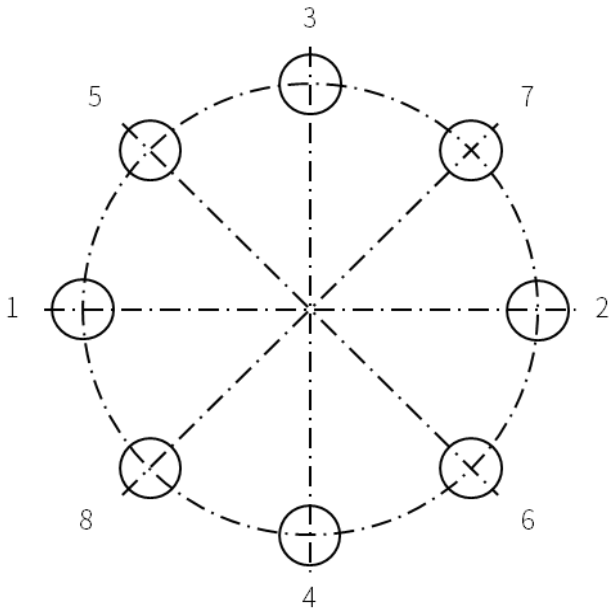


Fig. 7.2 Clamping bolts on a circle

DO NOT tighten one screw at a time. Tighten each screw in multiple steps, using a hexagonal wrench key, and secure with appropriate clamping torque by means of a torque wrench.

7.6 Maintenance Contract and Repair

7.6.1 Maintenance Contract

SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.
We recommend that the user contact SHIBAURA MACHINE Service Department and conclude an after-sale service contract with us.

7.6.2 Repair

If a fault has occurred or if repair is necessary, turn off the controller power and contact the SHIBAURA MACHINE Service Department. At this time, advise us of the details of the fault and the following information stated on the robot and controller.
(For contact information of SHIBAURA MACHINE Service Department, see Service Network.)

- Model name of the robot
- Manufacture number of the robot
- Manufacture number of the controller
- Date of manufacture of the robot
- Running hours

For the manufacture number and the date of manufacture, see the following label on the robot body base.

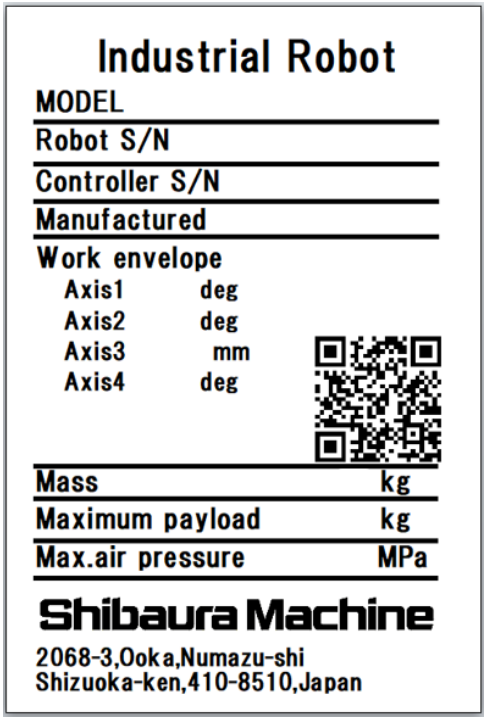


Fig. 7.3 Label

7.6.3 Modification

This robot and controller MUST NOT be modified or disassembled without a prior consent from SHIBAURA MACHINE.

⚠DANGER

- Never replace and modify the parts for other than permitted portions.
  Otherwise, the faulty robot may cause unpredictable disaster.

NOTICE

- Never replace and modify the parts for other than the permitted portions.
  Otherwise, the performance and lifespan of the robot may be reduced, and the robot may malfunction.

8. Maintenance of the Robot Body

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

8.1 Details of Inspection

8.1.1 Check of Each Bolt (or Screw) for Clamping

8.1.1.1 Check of Hand Set bolts

If the tool flange (option) is not provided

Use the hexagonal wrench key, etc. to make sure that the bolts clamping the user-used hand to the ball screw spline shaft are tightened completely. If they are loosened, tighten them completely.

If the tool flange (option) is not provided

Use the hexagonal wrench key to make sure that the four tool flange split clamp bolts (M4) of the tool flange (option) split clamped to the ball screw spline shaft are tightened completely. If they are loosened, tighten them completely.

Use the hexagonal wrench key to make sure that the four hand set bolts (M4) clamping the hand to the tool flange are tightened completely. If they are loosened, tighten them completely. (Screw locking adhesive is unnecessary.)

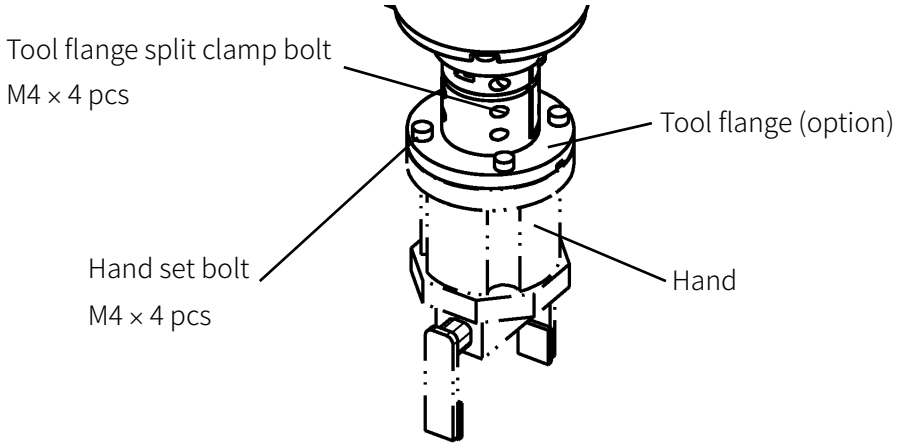


Fig. 8.1 Hand set bolts

8.1.1.2 Robot installation bolts

Use the hexagonal wrench key to make sure that the installation bolts of the robot body base are tightened completely. If they are loosened, tighten them completely. (Screw locking adhesive is unnecessary.)

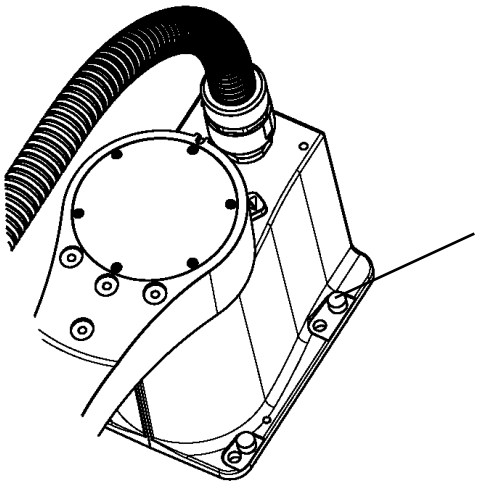


Fig. 8.2 Robot installation bolts

8.1.2 Check of Motor Set Bolts

Use the hexagonal wrench key to make sure that axis drive motor clamp bolts are tightened completely. Also, check the motor and motor plate clamp bolts. For the following bolt use positions, see "8.4 Replacement of Motors."

Table 8.1 Motor and motor plate clamp bolts

Use position	Type	No. of bolts	Reference destination
Axis-1 motor clamp bolt	M5×16	4	8.4.4 Fitting the Axis-1 Motor
Axis-2 motor clamp bolt	M4×12SUS	4	8.4.6 Fitting the Axis-2 Motor
Axis-2 motor base clamp bolt	M4×10SUS	8	8.4.6 Fitting the Axis-2 Motor
Axis-3 motor clamp bolt	M4×12SUS	4	8.4.8 Fitting the Axis-3 Motor
Axis-3 motor base clamp bolt	M5×16SUS	3	8.4.8 Fitting the Axis-3 Motor
Axis-4 motor clamp bolt	M4×12SUS	4	8.4.10 Fitting the Axis-4 Motor
Axis-4 motor base clamp bolt	M5×16SUS	3	8.5.4Replacing the Axis-4 Timing Belt B

MEMO

- *For the recommended clamping torque, see "7.5 Clamping Hexagon Socket Head Cap Screws."*
- *It is unnecessary to apply screw locking adhesive to the hexagon socket head cap screws for the axis-3 and axis-4 motor plate clamp bolts.*

8.1.3 Check of the Wearing Out of Cables and Air tube

Remove the arm-2 cover and base rear cover to make sure that the cable is not worn out, bent or cracked. Particular care should be taken in checking locations near the ends of the cable. For how to remove the cover, see "8.3 Dismounting and Mounting Each Cover."

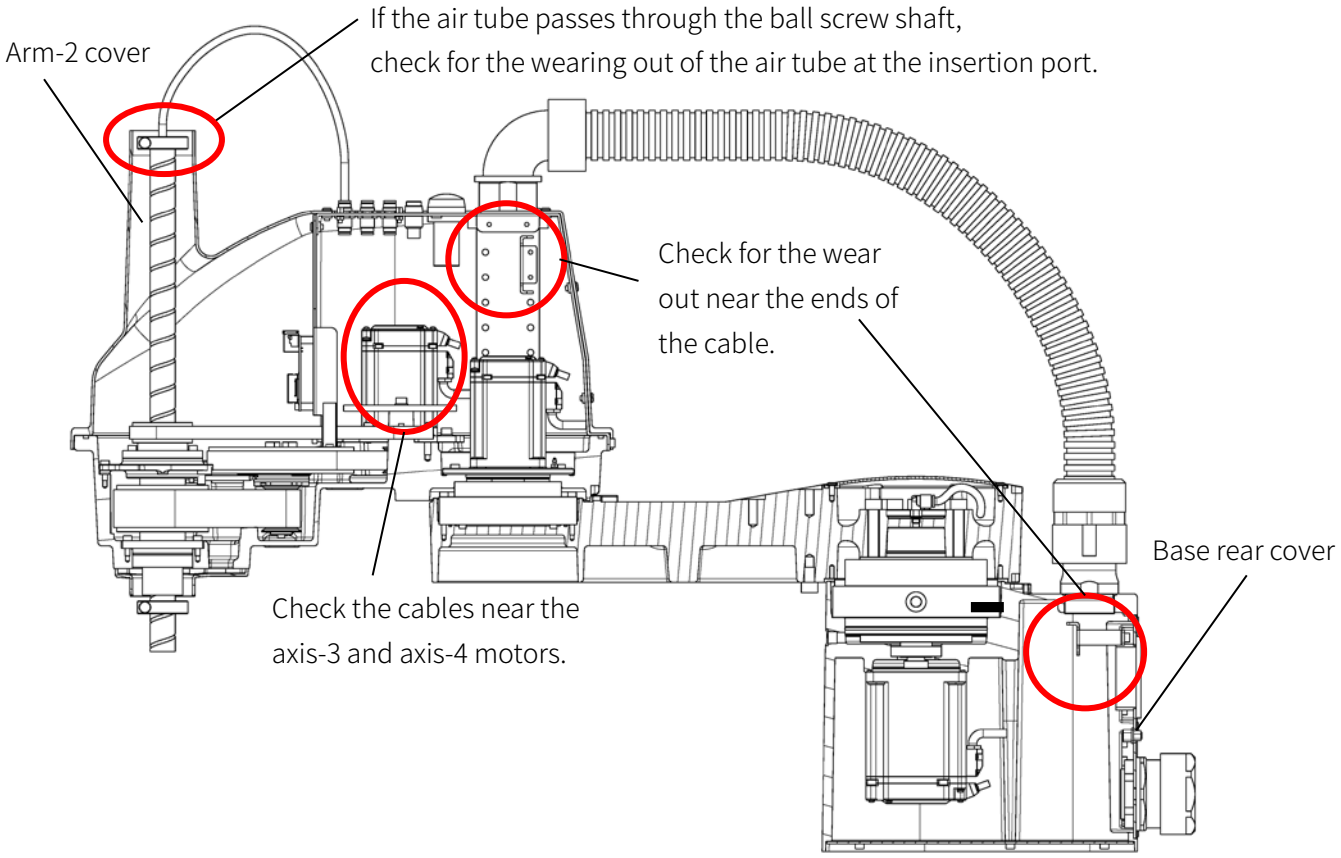


Fig. 8.3 Inspection points of cable

NOTICE

- Be sure to perform daily inspection for the air tube.
  If the air tube is damaged, replace it immediately.

8.1.4 Check of Duct Connector and Cable Clamps for Clamping

Use the wrench to make sure that the clamps securing the cable and duct and duct connector are tightened completely. There are duct connectors at two positions: Arm-2 side and base side. If the duct connectors are loosened, apply Loctite (low strength) to them and tighten them. Also, tighten the duct connectors in a position shown in Fig. 8.4. If the duct connectors are tightened with the arm bent, the cable cannot be connected accurately.

Check the duct connector for looseness. If the duct connector is loose, tighten it.
At this time, be careful to prevent the cable from being twisted.

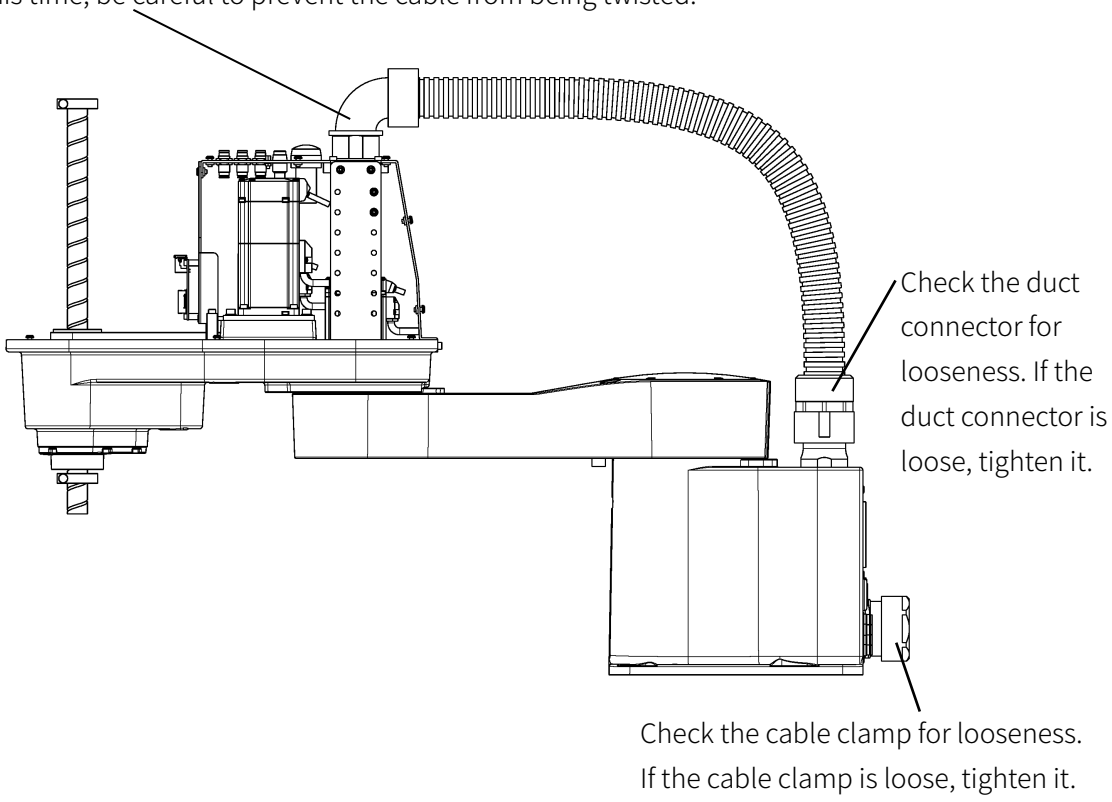


Fig. 8.4 Inspection spot of cable clamp

8.1.5 Check of Operation of Each Axis

After connecting the power plug of the controller to the power supply, keep the emergency stop switch pressed. Make sure that each axis is smoothly moved by hand. Axis 3 enters a non-braking state when the brake release switch is pressed. Note that the tool shaft will drop due to the weight of the hand.

8.2 Configuration of Mechanical Components

Fig. 8.5 shows the configuration of the mechanical components.

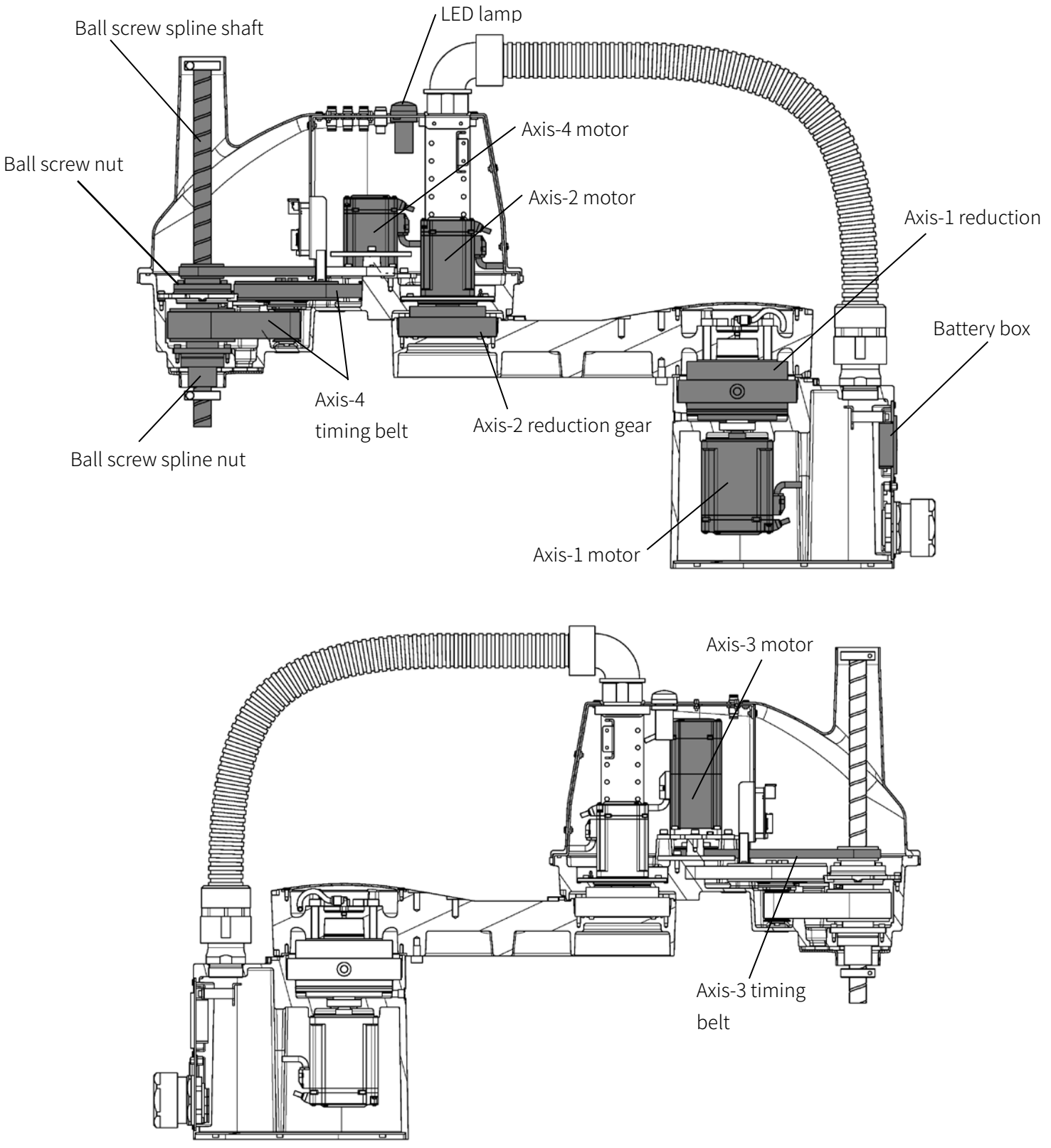


Fig. 8.5 Configuration of mechanical components

8.3 Dismounting and Mounting Each Cover

This paragraph describes the dismounting and mounting of the covers, which are common to the maintenance and replacement of each unit.

⚠DANGER

- When opening the cover, never allow entry of moisture or contaminant into the robot.
  If the power is turned on while moisture or contaminant is left, an electric shock may result.

NOTICE

- When opening the cover, never allow entry of moisture or contaminant into the robot.
  If the power is turned on while moisture or contaminant is left, the robot may malfunction.
- When mounting the covers, prevent the cables from being caught.
  Otherwise, the cables will be broken.
- Be sure to restore where to route the cables pulled out by removing the cover and where to secure with the cable straps to their original state.
  If the cables are pushed by force or where to secure changes, the cables will be worn out or broken.
  - Before pulling out the cables, take pictures for restoration

8.3.1 Arm-2 Cover

The arm-2 cover is secured to arm 2 and harness guide with 16 hexagon socket head cap button screws (M4×8).

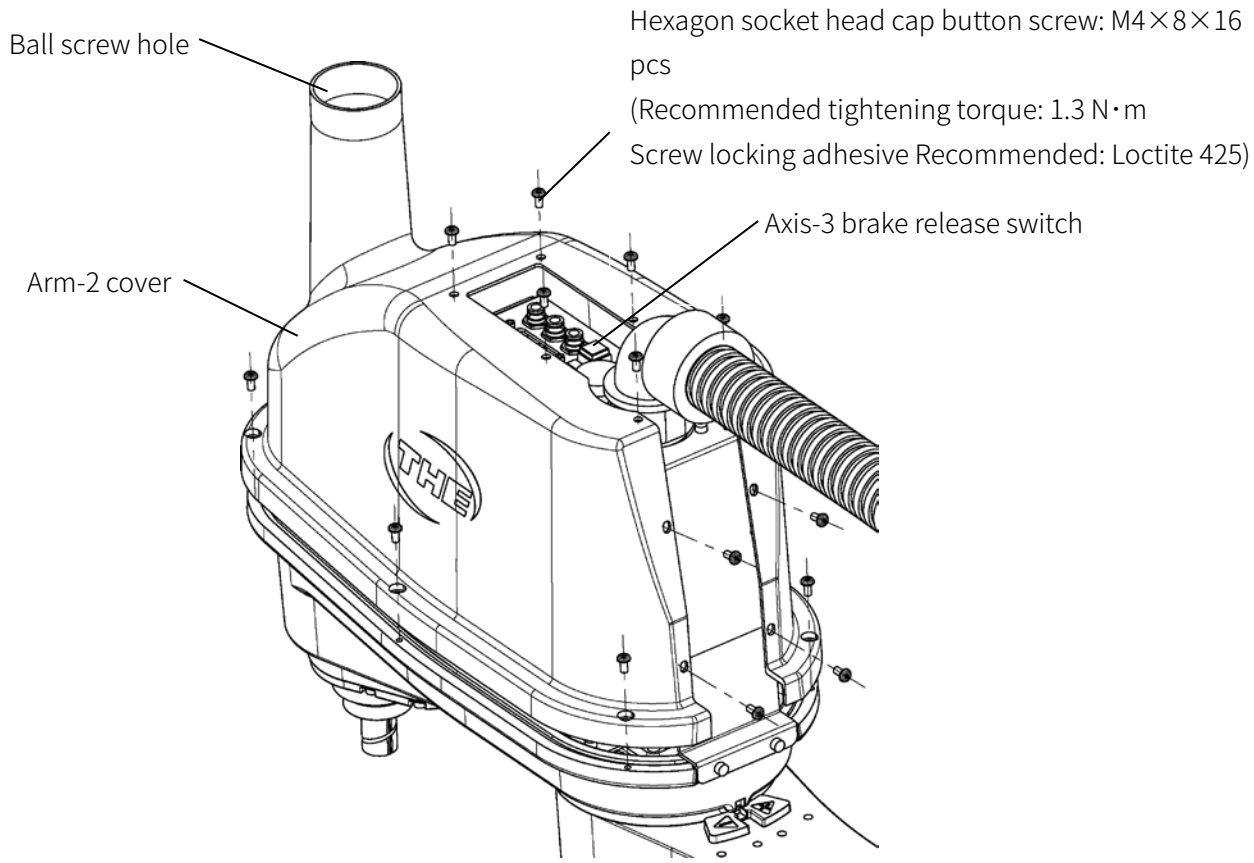


Fig. 8.6 Arm-2 covers

After mounting the cover, move the ball screw spline shaft up and down manually while pressing the brake release switch, and make sure that the hole for ball screw of the arm-2 cover does not interfere with the stopper.

8.3.2 Arm-1 Cover

The arm-1 cover is secured to arm 1 with six hexagon socket head cap button screws (M4×8). (During mounting, application of screw locking adhesive is not required.)

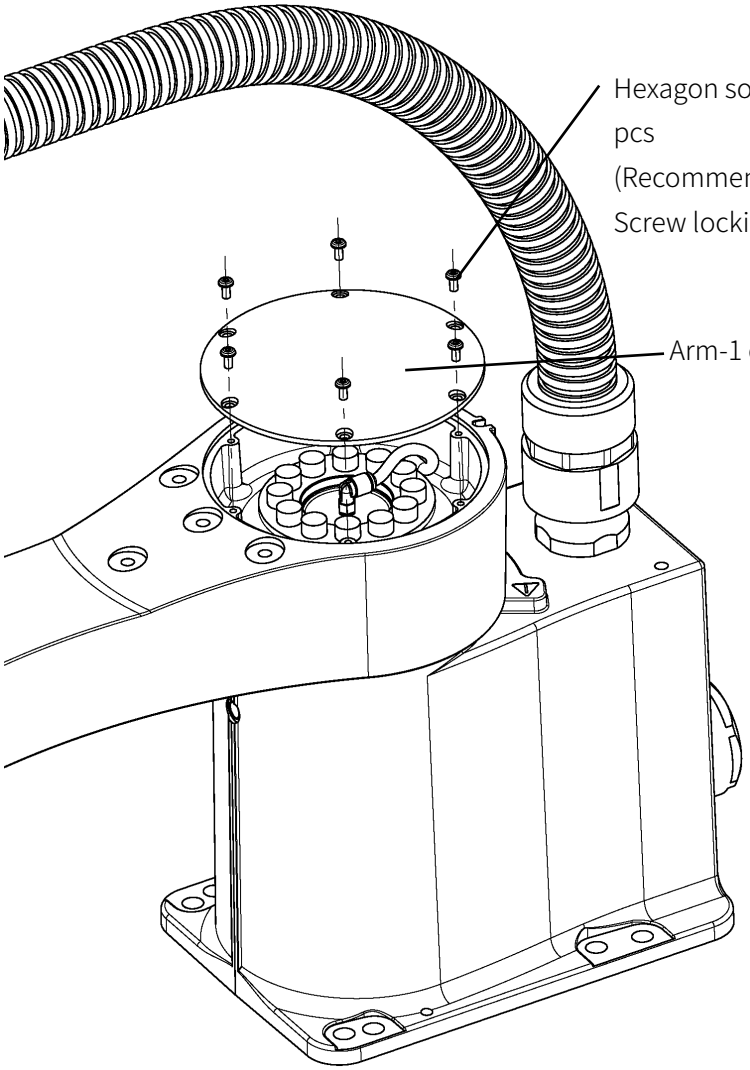
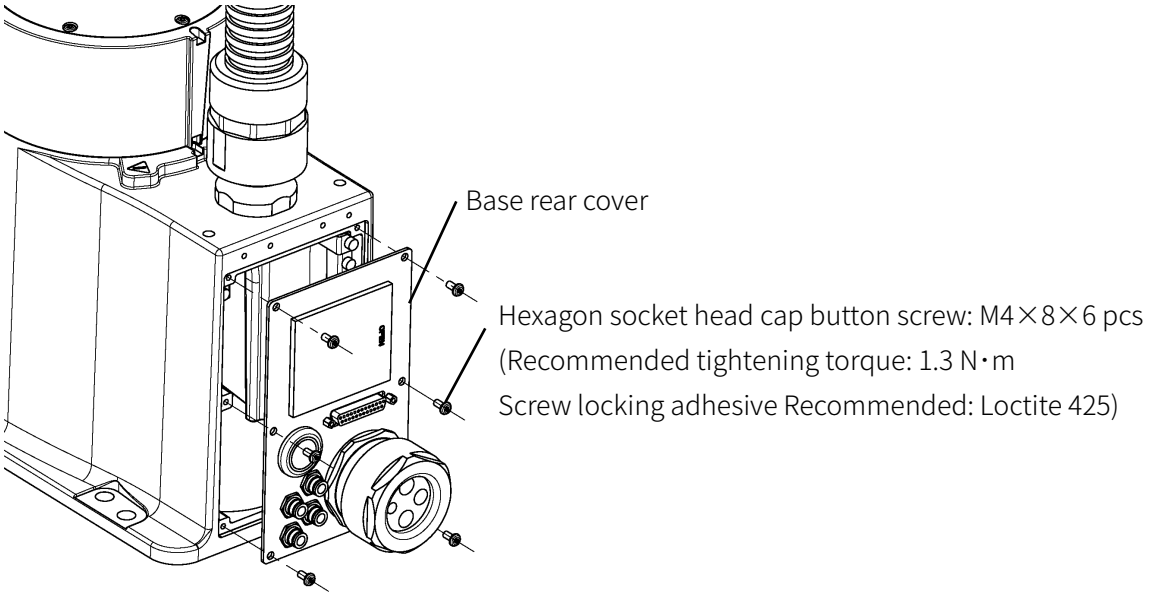


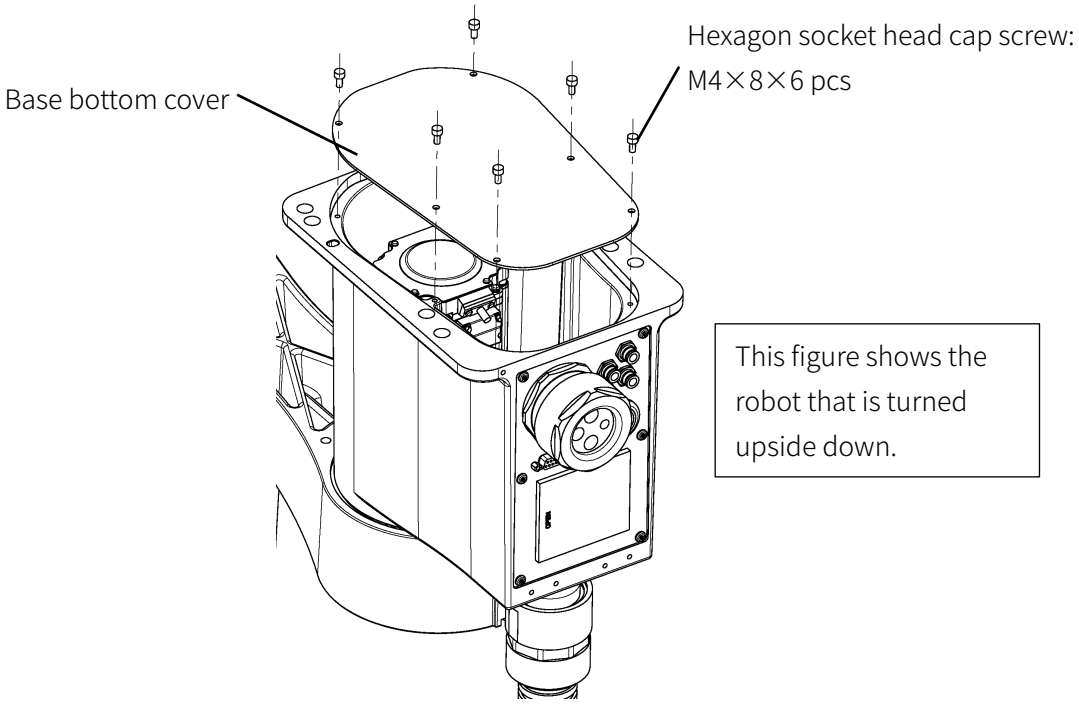
Fig. 8.7 Arm-1 cover

8.3.3 Base Cover

The base cover is provided in two types: base rear cover and base bottom cover.
The base rear cover is secured to the base with six hexagon socket head cap button screws (M4×8).
When the fixing screws are removed, the rear cover is separated from the base. However, the cover is connected to the connectors inside; therefore, do not pull it forcibly.
The base bottom cover is secured to the base with six hexagon socket head cap button screws (M4×8).



<Base rear cover>



<Base bottom cover>

Fig. 8.8 Base Cover

8.4 Replacement of Motors

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that failure and accident resulting from the work done by the customer will not be under our warranty.

⚠DANGER

- When replacing a motor, be sure to remove the power plug of the controller.
  Otherwise, the removed cable terminal may be touched to result in an electric shock. The worker may be entangled in rotation of the motor that has made unpredictable motion.

NOTICE

- Never cause heavy impact on the motor shaft.
  The motor and position detector may malfunction due to impact.
- Never disassemble the motor and position detector.
  Otherwise, they may be disabled due to difference in position or intrusion of foreign material.
- When replacing a motor, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.
- After the motor is replaced, be sure to re-set zero position.
  Replacing a motor causes deviation from mechanical zero position; therefore, the robot cannot make normal motion.
  - For how to re-set the zero position, see "9 Robot Zero Point and Position Detector Error."

8.4.1 Types of Motors

The motors used by this robot are listed in the table below. When ordering a replacement motor, check the robot model name (THE600) and manufacture number on the label (see Fig. 7.3), and then check applicable axis, SHIBAURA drawing number, and unit code according to Table 8.2 Types of motors. For the location of the manufacture number label, refer to "STA1179 Safety Manual" Instruction Manual.

Table 8.2 Types of motors

Part name	Applicable axis	SHIBAURA Dwg. No.	Unit code
AC servo motor	Axis 1	S968416	Y610D0CW0
	Axis 2/Axis 4	S968417	Y610D0CX0
	Axis 3	S968418	Y610D0CY0

8.4.2 Locations of Motors

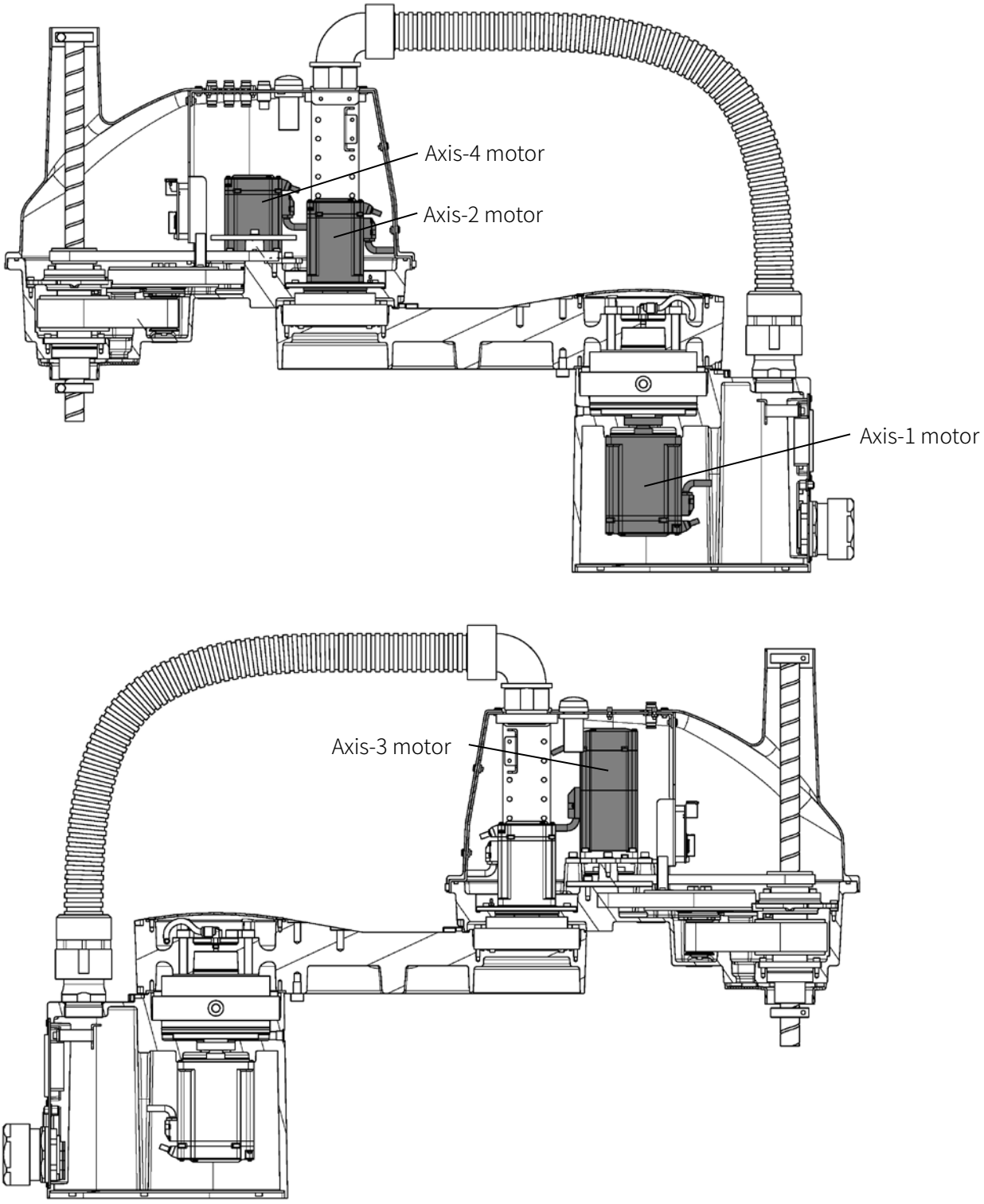


Fig. 8.9 Motor locations

8.4.3 Removing the Axis-1 Motor

1 Remove the base rear cover and base bottom cover.

(See "8.3.3 Base Cover.") The rear cover is connected to the connectors inside; therefore, do not pull it forcibly.

2 Remove the axis-1 motor.

Remove the hexagon socket head cap screws (M5×16×4 pcs) securing the axis-1 motor, and pull out the axis-1 motor assembly axially. When taking out the motor assembly, do not pull it forcibly as the motor connector is connected to it.

3 Remove the cable connectors of the axis-1 motor.

Disconnect cable connectors of the axis-1 motor, J1AS and J1AP and J1BS and J1BP.

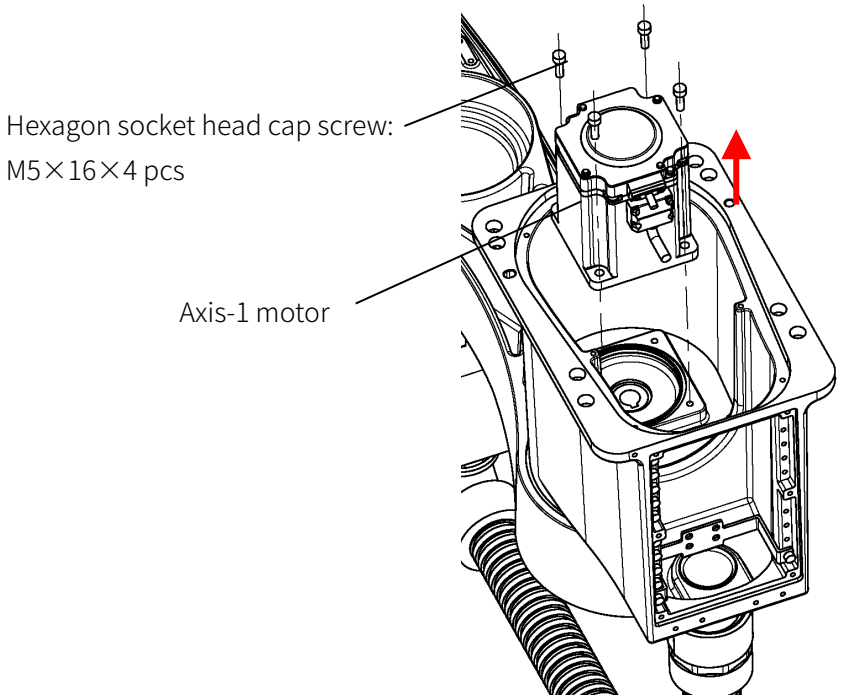


Fig. 8.10 Removing axis-1 motor assembly

8.4.4 Fitting the Axis-1 Motor

1 Insert the axis-1 motor into the base.

Fit the motor key into the key groove of the reduction gear. At this time, remove the key once, and apply fretting inhibitor to the key side face (see Fig. 8.11) (Recommended fretting grease: LGAF 3E)

2 Secure the axis-1 motor assembly with the hexagon socket head cap screws (M5×16×4 pcs).

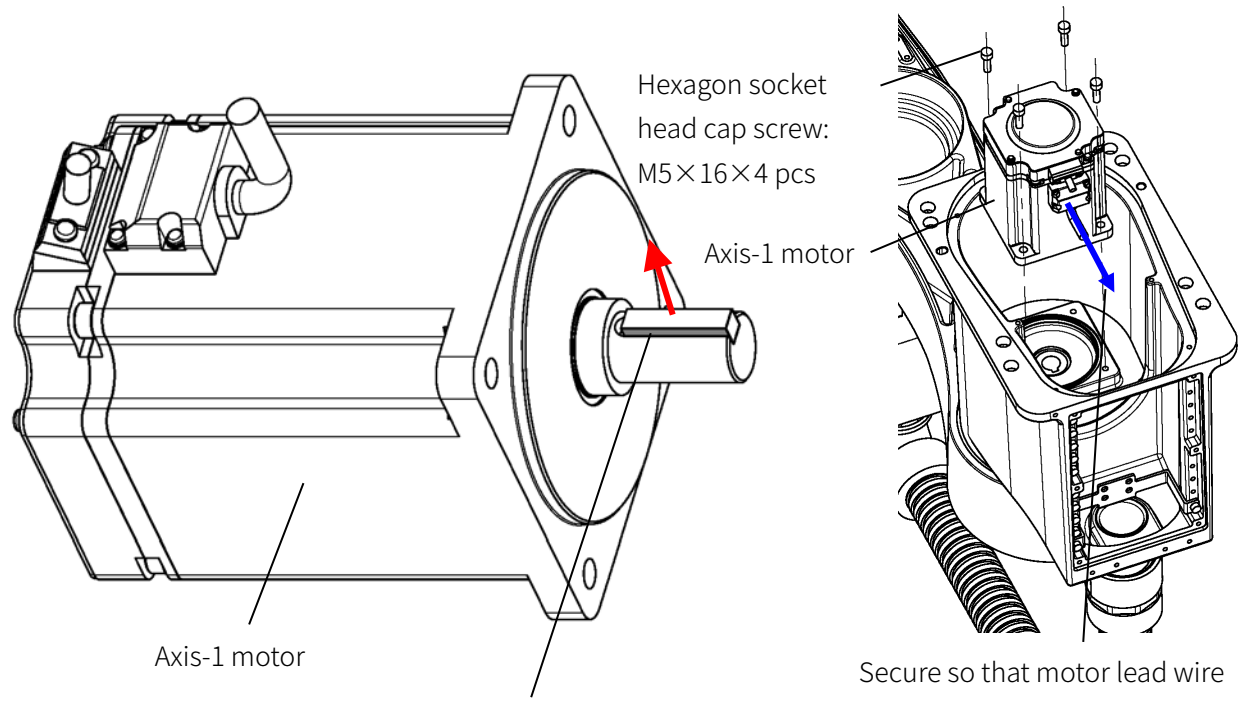


Fig. 8.11 Fitting axis-1 motor assembly

3 Fit the cable connectors of the axis-1 motor that have been removed.

Connect the cable connectors of the axis-1 motor, J1AS and J1AP and J1BS and J1BP.

4 Fit the base rear cover and base bottom cover.

(See "8.3.3 Base Cover.")

5 Move the arm 1 by hand.

Make sure that there is no abnormal sound.

6 Turn on the power and set the zero position of the axis 1.

The replacement of the axis-1 motor is completed. (Check the coordinates of axes 2, 3, and 4 and set zero position as required. See "9 Robot Zero Point and Position Detector Error.")

7 Perform operation at a low speed.

Perform running-in operation at 20% of override for about one hour to check whether there is any problem with operation of each part.

8.4.5 Removing the Axis-2 Motor

1 Remove the arm-2 cover.

See “8.3.1 Arm-2 Cover.”

2 Use a nipper or the like to cut off the cable straps of the cables.

It is recommended to take pictures beforehand so that where to route the cables and where to use the cable straps are identifiable in restoration work.

3 Remove the cable connectors of the axis-2 motor.

Disconnect the cable connectors of the axis-2 motor, J2AS and J2AP and J2BS and J2BP.

4 Remove the harness guide and support plate.

Remove the harness guide secured with the hexagon socket head cap screws (M4×8×4 pcs) and the support plate secured with the hexagon socket head cap screws (M4×8×4 pcs).

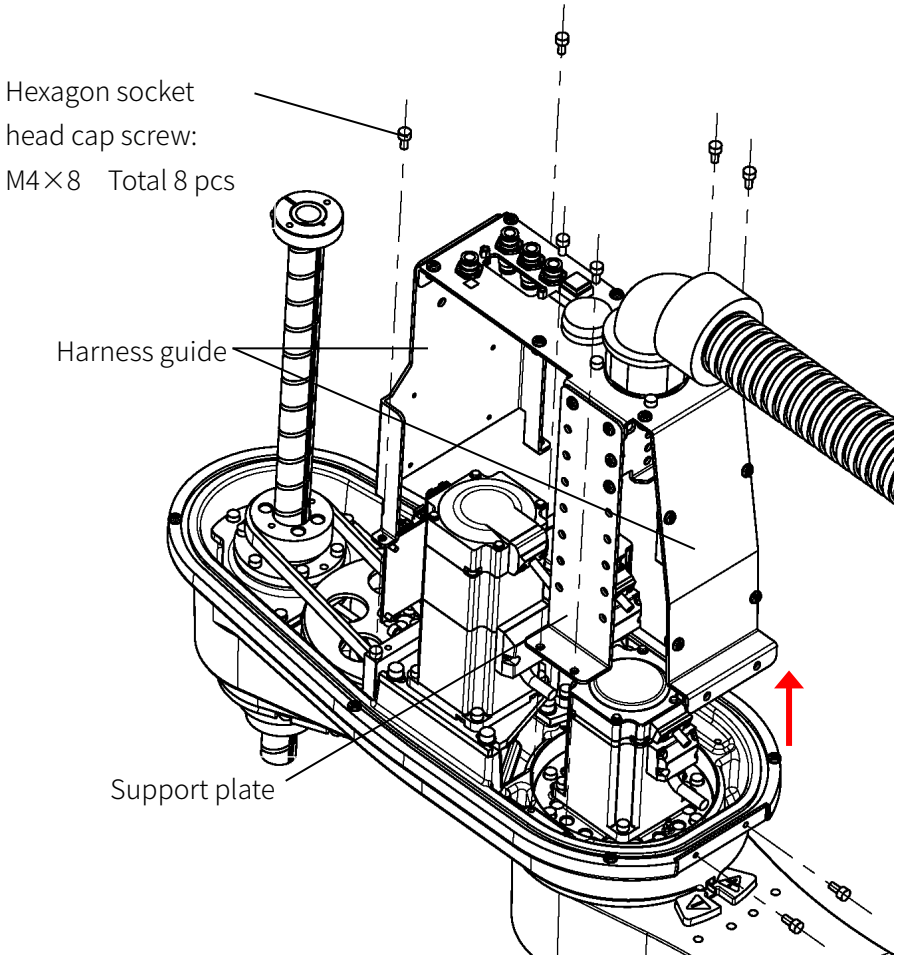


Fig. 8.12 Removing harness guide and support plate

5 Remove the axis-2 motor assembly.

Remove the hexagon socket head cap screws (M4×10×8 pcs) securing the axis-2 motor base and pull out the axis-2 motor assembly.

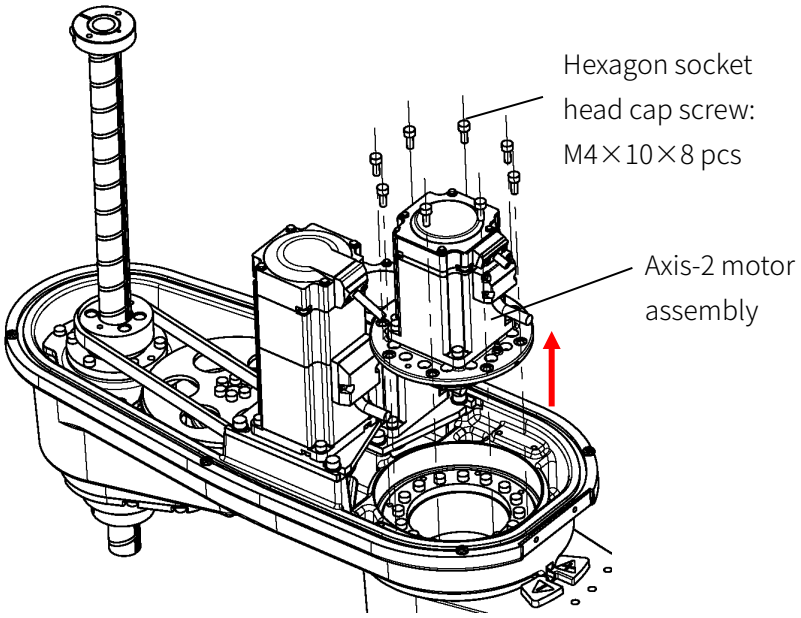


Fig. 8.13 Removing the axis-2 motor assembly

6 Remove the elliptical cam.

Secure the protrusion part of elliptical cam with a plier, etc., and then remove the hexagon socket head cap screw (M4×10×1 pc) at the end of the shaft. When securing the elliptical cam, place a cushioning material such as waste cloth, etc. between it and the plier to prevent it from being damaged. After removing the hexagon socket head cap screw, pull out the press plate and elliptical cam.

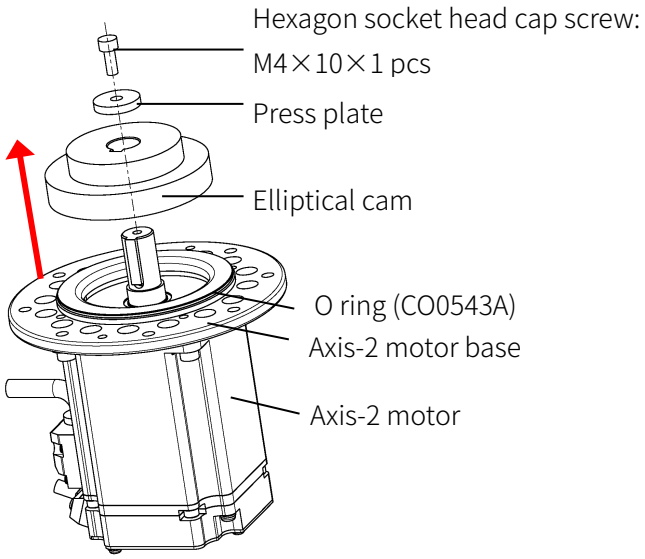


Fig. 8.14 Removing the elliptical cam

MEMO

The O ring (CO0543A) fitted on the axis-2 motor base is removed if it is seriously damaged such as crush.

7 Remove the axis-2 motor from the axis-2 motor base.

Remove the hexagon socket head cap screws (M4×12×4 pcs) securing the axis-2 motor to the axis-2 motor base, and remove the axis-2 motor from the axis-2 motor base.

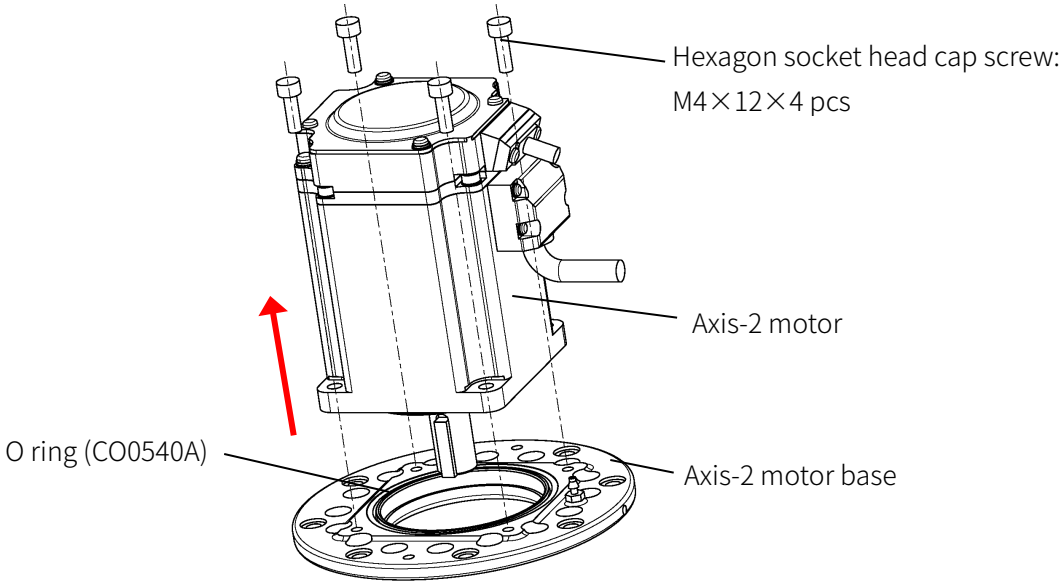


Fig. 8.15 Removing the axis-2 motor

MEMO

The O ring (CO0540A) fitted on the axis-2 motor base is removed if it is seriously damaged such crush.

8.4.6 Fitting the Axis-2 Motor

1 Fit the O ring.

If the O ring that has been fitted is damaged and removed, apply grease to a new O ring (CO0540A) and fit it into the groove on the top surface of the axis-2 motor base.

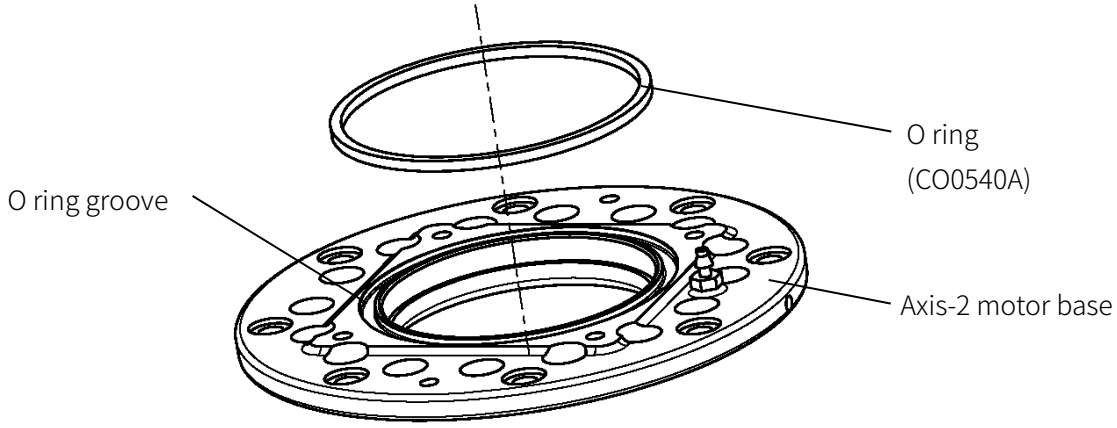


Fig. 8.16 Fitting O ring

NOTICE

- Never forget to fit the O ring, and never use the cut or damaged O ring.
  Otherwise, grease will leak from the fitting surface.

2 Fit the axis-2 motor on the axis-2 motor base.

Use the hexagon socket head cap screws (M4×12×4 pcs) to secure the axis-2 motor to the axis-2 motor base.

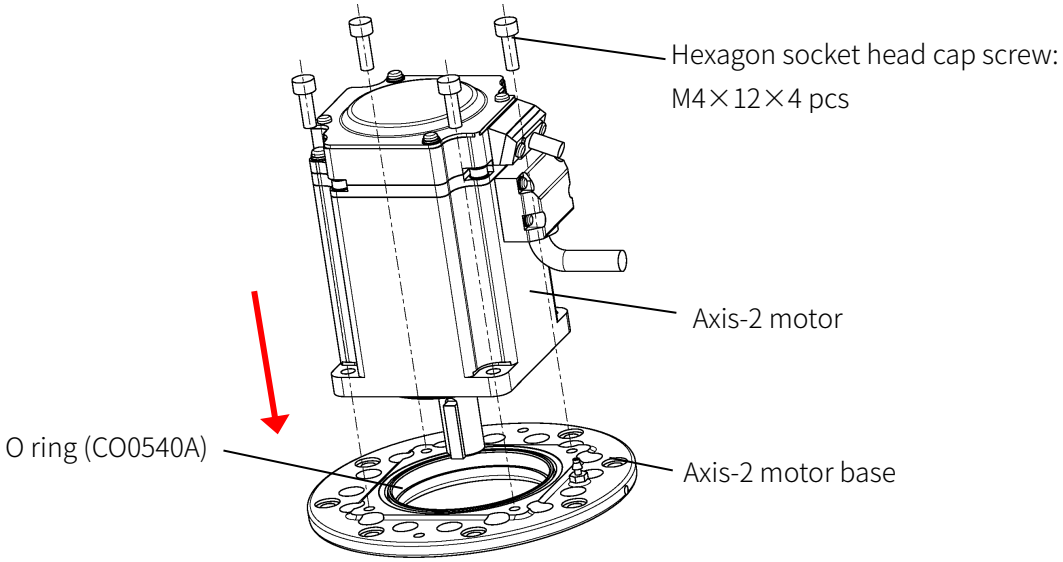


Fig. 8.17 Fitting the axis-2 motor

3 Fit the O ring.

If the O ring that has been fitted is damaged and removed, apply grease to a new O ring (CO0543A), and fit the O ring (CO0543A) to the axis-2 motor base (gray portion in Fig. 8.18)

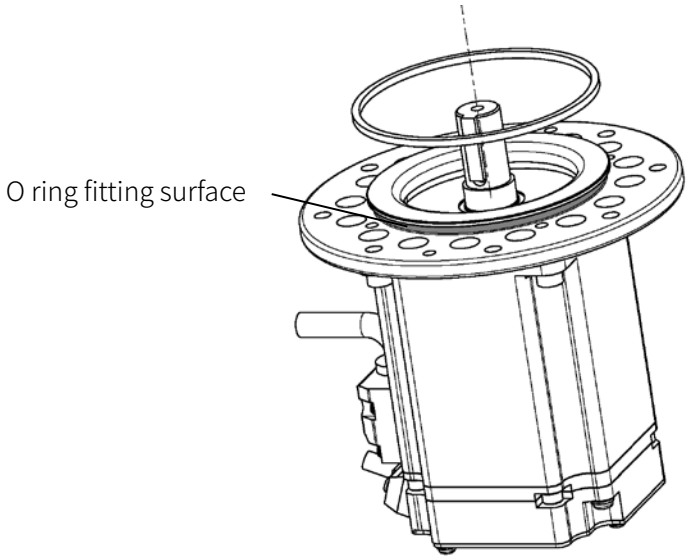


Fig. 8.18 Fitting the O ring

4 Fit the elliptical cam.

Remove the key that has been fitted to the axis-2 motor once, apply fretting inhibitor (recommended fretting grease: LGAF 3E) to the side face (see Fig. 8.19), and then fit the elliptical cam and secure it with the hexagon socket head cap screw (M4×10×1 pc) at the end of the shaft and the press plate. Secure the protrusion part of elliptical cam with a plier, etc., and then tighten the hexagon socket head cap screw (M4×10×1 pc) at the end of the shaft. When securing the elliptical cam, place a cushioning material such as waste cloth, etc. between it and the plier to prevent it from being damaged.

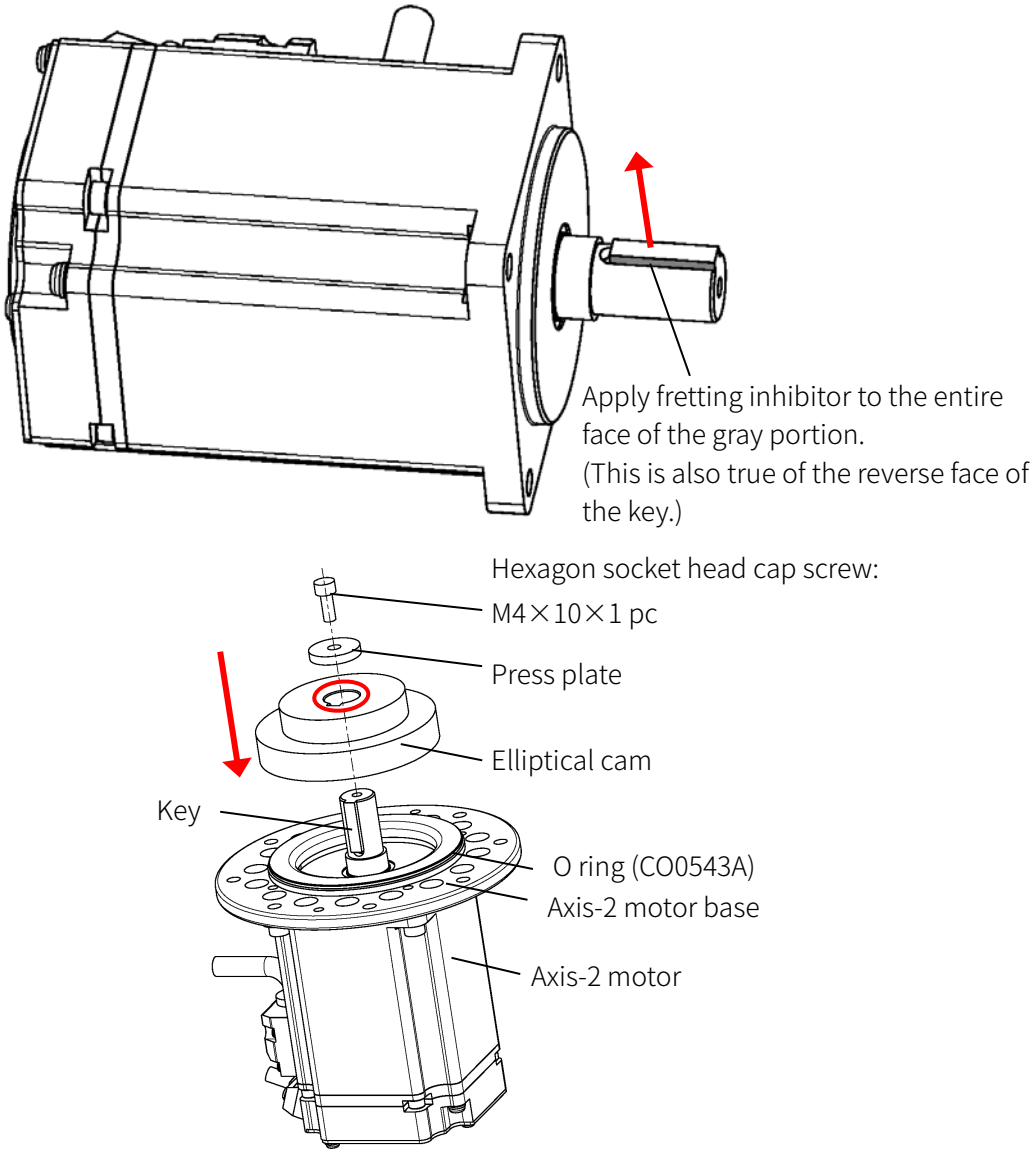


Fig. 8.19 Fitting the elliptical cam

MEMO

Insert the elliptical cam so that the larger chamfered boss (portion shown by red circle in Fig. 8.19) faces the outside.

5 Apply grease to the axis-2 motor assembly.

For application of grease, see 5 of "8.7.6 Fitting the Axis-2 Reduction Gear."

6 Fit the axis-2 motor assembly.

Be careful about the fitting phase of the motor, the orientation of the reduction gear body and elliptical cam when the motor is inserted (match the reduction gear body and the long portion of ellipse of the elliptical cam) and the misaligned O ring to insert the axis-2 motor assembly into the axis-2 arm.

Then, secure the axis-2 motor assembly with the hexagon socket head cap screws (M4×10×8 pcs)

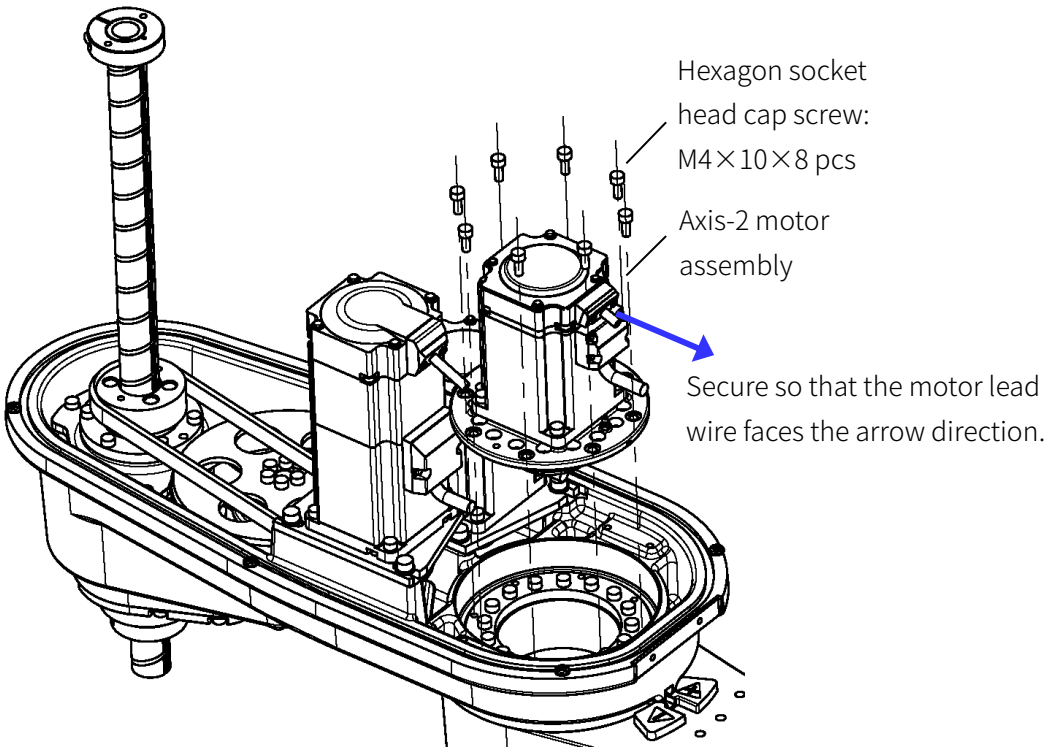


Fig. 8.20 Fitting the axis-2 motor assembly

7 Move the arm 2 by hand.

Make sure that there is no abnormal sound.

8 Fit the harness guide and support plate.

Secure the harness guide and support plate with the hexagon socket head cap screws (M4×8×8 pcs).

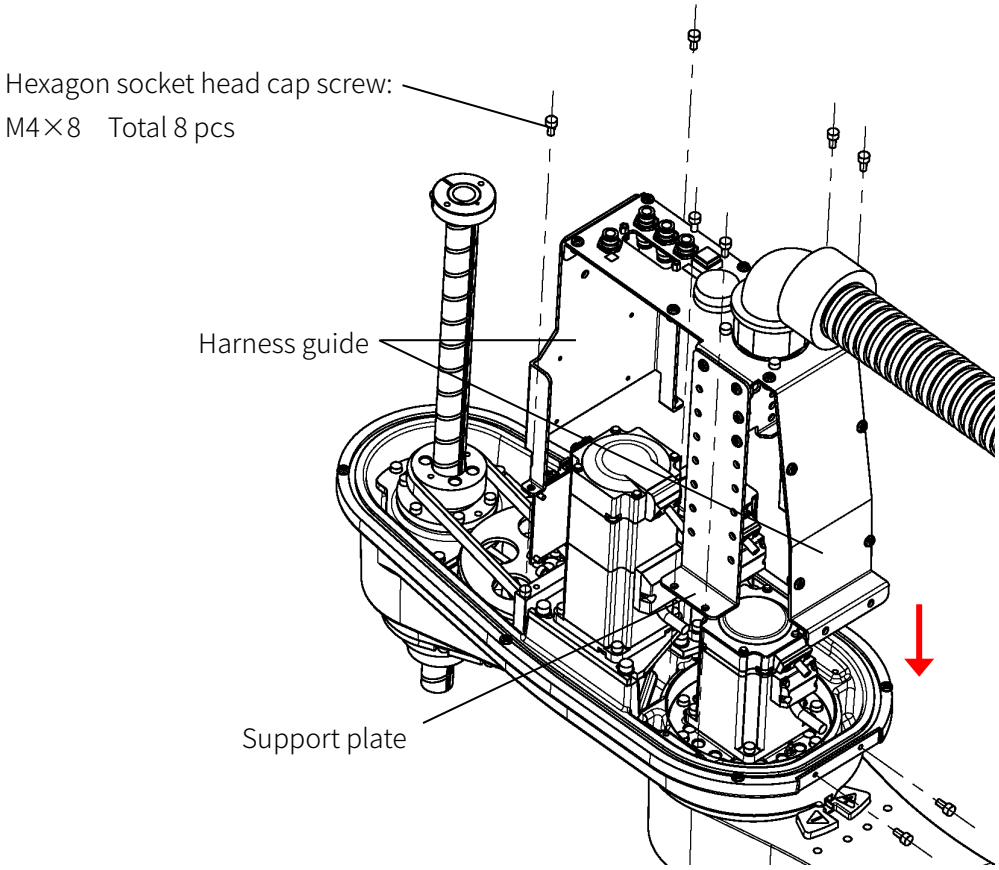


Fig. 8.21 Fitting harness guide and support plate

9 Connect the connectors and return the cable to its original state.

See the pictures taken in 2 of "8.4.5 Removing the Axis-2 Motor."

10 Fit the arm-2 cover.

See "8.3.1 Arm-2 Cover."

11 Set the zero position of the axis 2.

Turn on the power to set the zero position of the axis 2 to complete the replacement of the axis-2 motor. (Check the coordinates of axes 1, 3, and 4 and set the zero position as required. See "9 Robot Zero Point and Position Detector Error.")

12 Perform operation at a low speed.

Perform running-in operation at 20% of override for about one hour to check whether there is any problem with operation of each part.

8.4.7 Removing the Axis-3 Motor

⚠ WARNING

- When the axis-3 motor is replaced, be sure to lower the ball screw spline shaft to the lower limit. When the motor or timing belt is removed, the shaft that drops by its own weight could catch your hand or fingers.
  - When the ball screw spline or axis-4 timing belt or reduction gear is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

1 Remove the arm-2 cover.

See "8.3.1 Arm-2 Cover."

2 Use a nipper or the like to cut off the cable straps of the cables.

It is recommended to photograph where to route the cables and where to use the cable straps beforehand for restoration.

3 Disconnect the cable connectors of the axis-3 motor.

Disconnect the cable connectors of the axis 3, J3AS and J3AP (axis-3 power drive cables), J3BS and J3BP (axis-3 encoder cables), and J3DS and J3DP (axis-3 brake cables).

4 Use a tension meter to measure the tension of the axis-3 timing belt.

Enter the values listed in Table 8.3 as the values input to the tension meter. Be sure to record the results of measuring the tension using the tension meter because it is necessary to match the tension with the current belt tension value when fitting the belt.

Table 8.3 Values Input to Tension Meter

Unit Mass (g/m)	Belt Width (mm)	Span (mm)
2.9	10	192

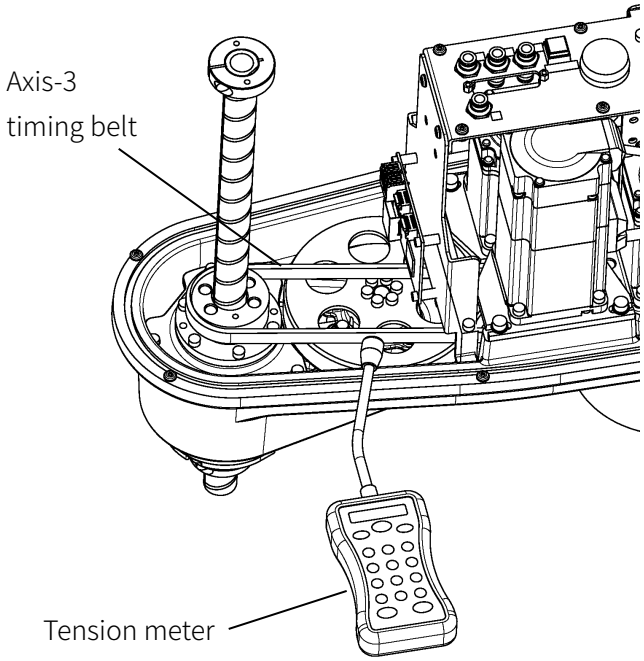


Fig. 8.22 Measuring the Tension of the Axis-3 Timing Belt

5 Remove the harness guide and support guide.

Remove the harness guide secured with the hexagon socket head cap screws (M4×8×4 pcs) and the support guide secured with the hexagon socket head cap screws (M4×8×4 pcs) (see 4 in "8.4.5 Removing the Axis-2 Motor").

6 Remove the axis-3 motor assembly.

Remove the hexagon socket head cap screws (M5×16×3 pcs) and washers securing the axis-3 motor base to release the tension of the axis-3 timing belt. Then, remove the axis-3 motor assembly.

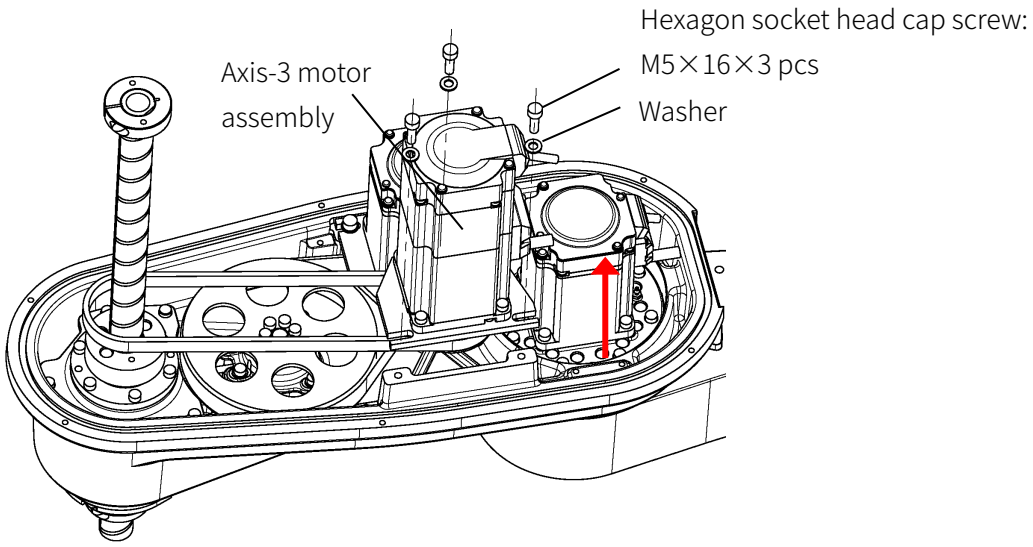


Fig. 8.23 Removing the axis-3 motor assembly

7 Pull out the pulley of the axis-3 motor assembly.

Remove the hexagon socket head cap screw (M4×8×1 pc). Then, pull out the press plate and pulley.

8 Remove the axis-3 motor from the axis-3 motor base.

Remove the hexagon socket head cap screws (M4×12×4 pcs) securing the axis-3 motor to separate the axis-3 motor base and the axis-3 motor.

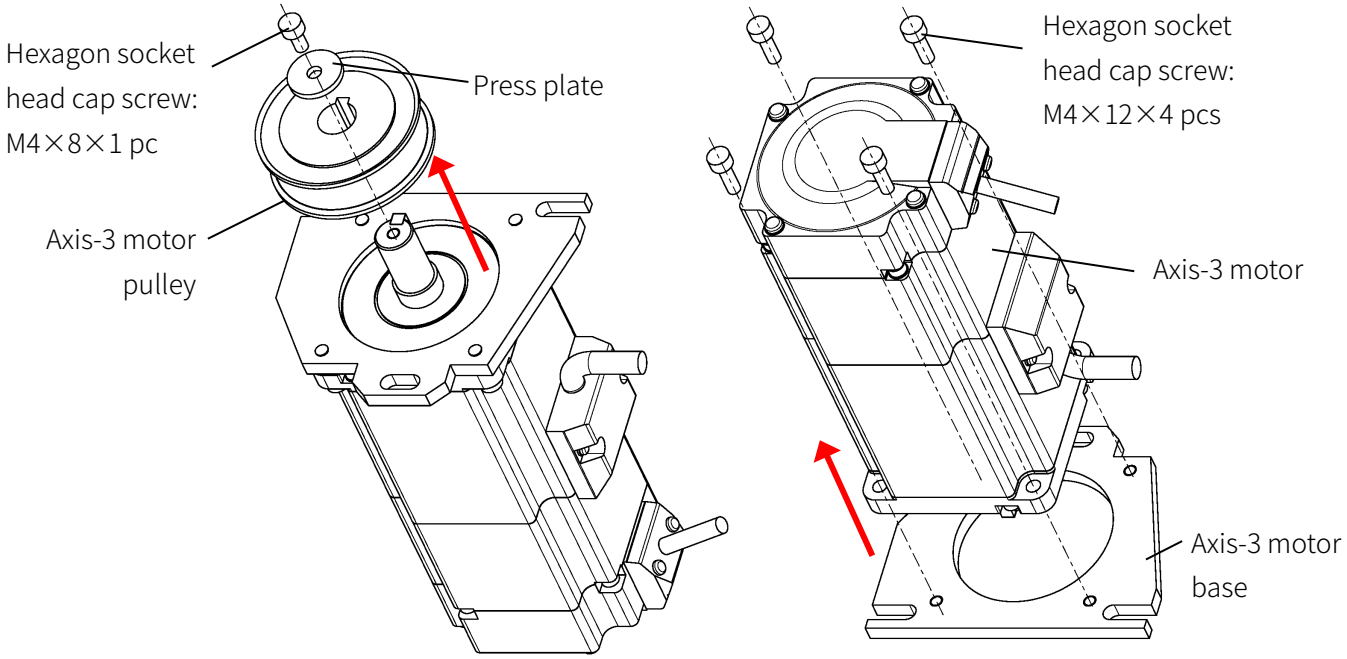


Fig. 8.24 Removing the axis-3 motor pulley and motor base

8.4.8 Fitting the Axis-3 Motor

1 Secure the axis-3 motor to the axis-3 motor base.

Secure the axis-3 motor to the axis-3 motor base with the hexagon socket head cap screws (M4×12×4 pcs). At this time, be careful about the fitting phase of the axis-3 motor and axis-3 motor base.

2 Fit the axis-3 motor pulley.

Remove the key that has been fitted to the axis-3 motor once, apply fretting inhibitor to the side face of the key (Fig. 8.25), and then fit the axis-3 motor pulley and secure it with the press plate and hexagon socket head cap screw (M4×8×1 pc).

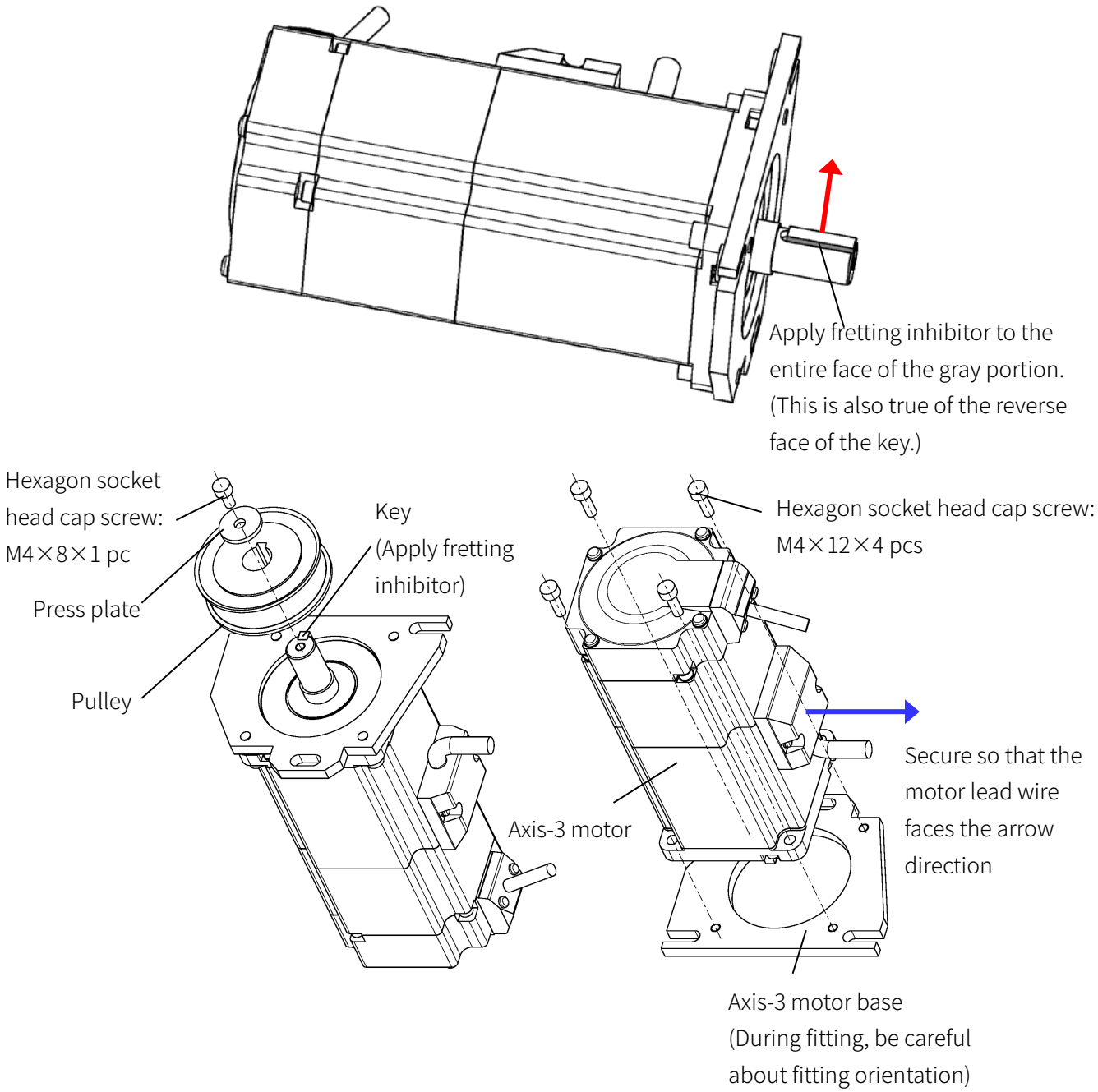


Fig. 8.25 Fitting the axis-3 motor pulley and motor base

3 Fit the axis-3 motor assembly.

Put the axis-3 timing belt on the axis-3 motor pulley, and temporarily fix the axis-3 motor base with the hexagon socket head cap screws (M5×16×3 pcs) and washers. (Application of screw locking adhesive is not required.) At this time, be careful about the fitting phase of the motor.

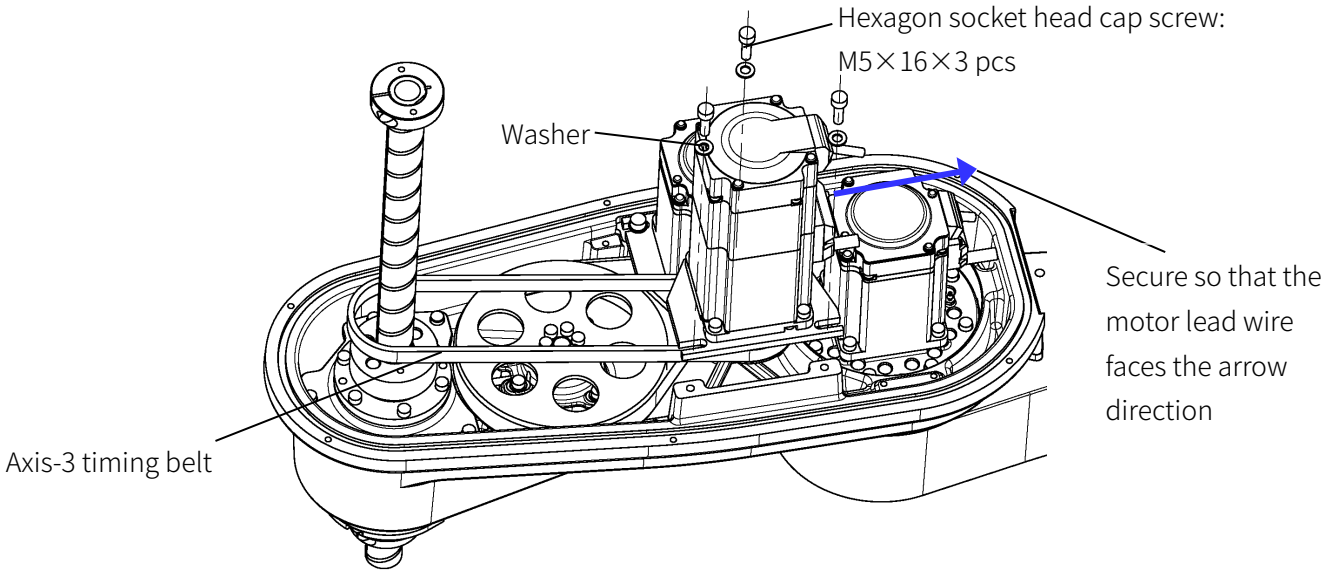


Fig. 8.26 Fitting the axis-3 motor assembly

4 Fit the tension tool 34 assembly.

Move up and down the ball screw spline unit to adjust the axis-3 timing belt. Then, fit the tension tool 34 assembly (option) to the arm 2.

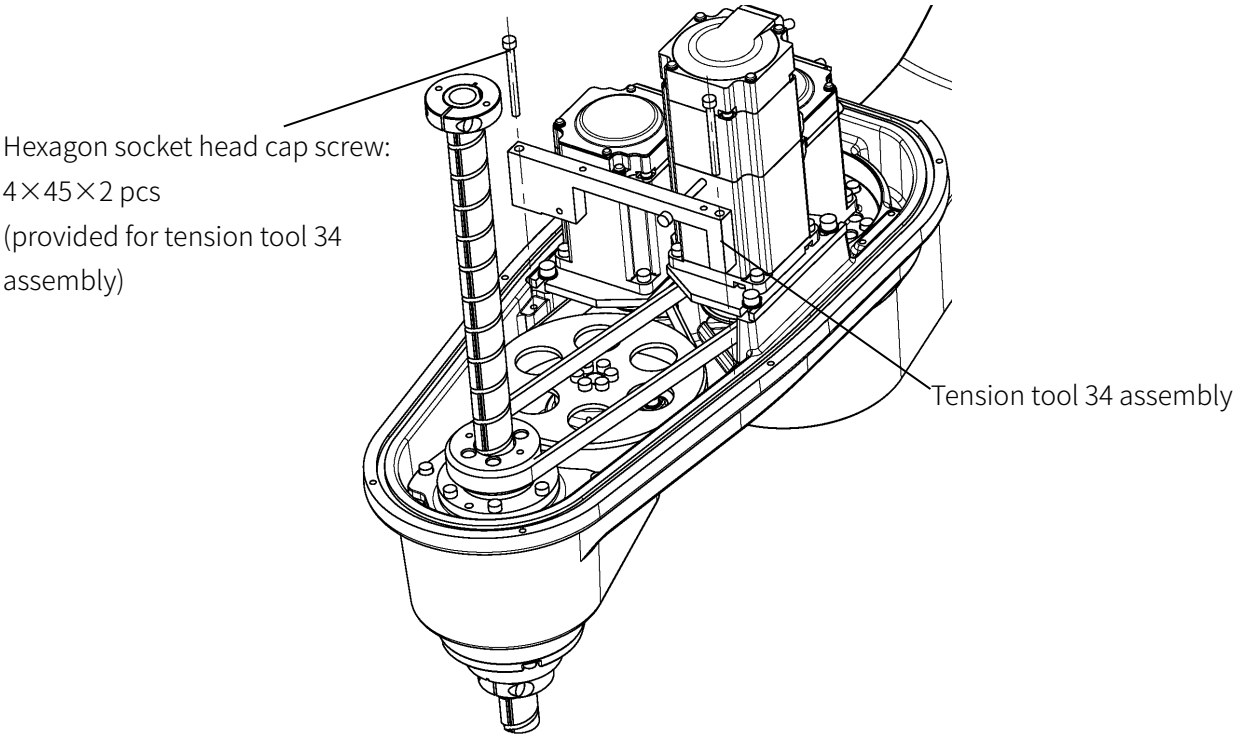


Fig. 8.27 Fitting axis-3 and 4 tension adjustment tool

5 Adjust the tension of the axis-3 timing belt.

Use the tension tool 34 assembly (option) to apply tension to the axis-3 timing belt. While using the screws for axis-3 tension adjustment of tension tool 34 assembly to push the axis-3 motor base, use a tension meter to measure the tension. Tighten the hexagon socket head cap screws to secure the motor base that has been temporarily fixed at a position where the tension has become a little smaller than the value A*(N) measured in step 4 in "8.4.7 Removing the Axis-3 Motor" (because retightening the hexagon socket head cap screws increases the tension). Make adjustment so that the tension becomes A*×0.95 to A*×1.05(N) when the hexagon socket head cap screws are retightened. After the hexagon socket head cap screws are retightened, remove the tension tool 34 assembly. The value to be used in the tension meter is as listed in Table 7.4.

Table 8.4 Tension of the axis-3 timing belt

Tension (N)	Unit Mass (g/m)	Belt Width (mm)	Span (mm)
A*×0.95 to A*×1.05	2.9	10	192

A is the value measured in step 4 of "8.4.7 Removing the Axis-3 Motor."

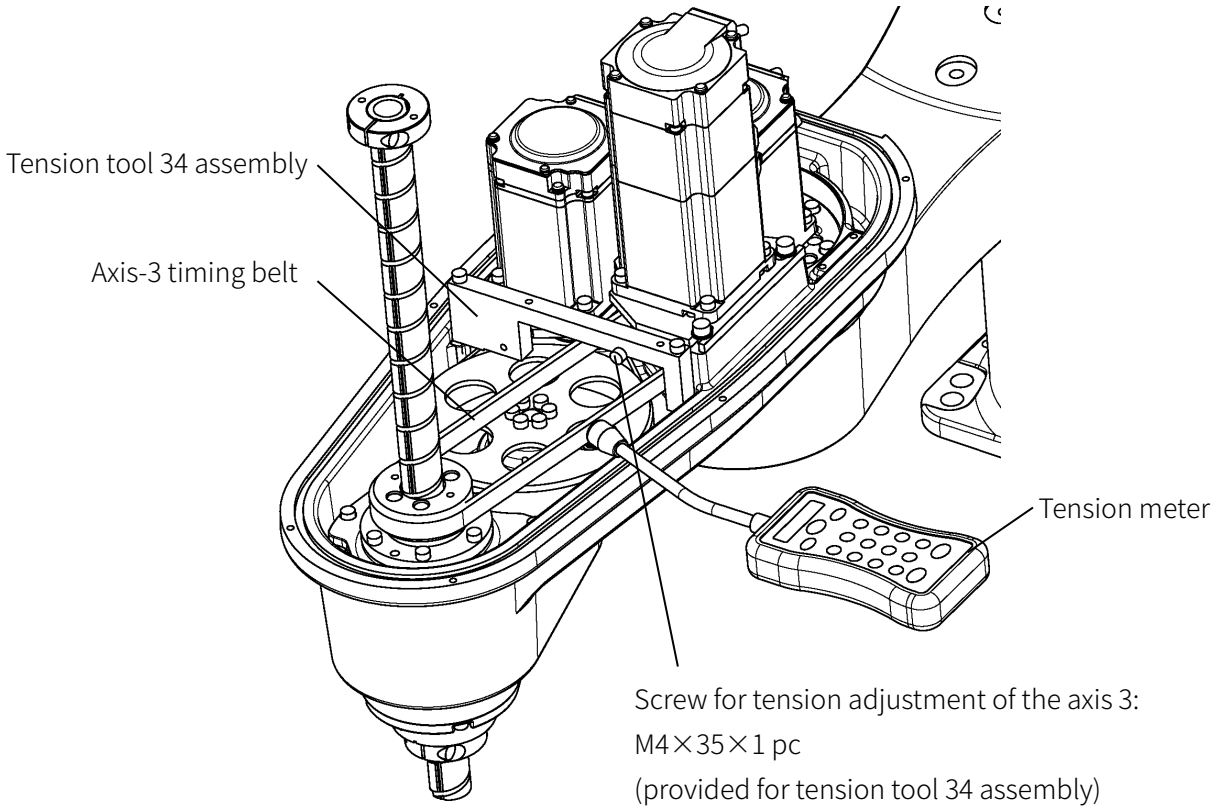


Fig. 8.28 Adjusting the tension of the axis 3

6 Fit the harness guide and support guide.

Use the hexagon socket head cap screws (M4×8×8 pcs) to fit the harness guide and support guide (see 8 in "8.4.6 Fitting the Axis-2 Motor").

7 Fit the removed connectors.

Connect the connectors and return the cable to its original state. (See the pictures taken in 3 of "8.4.7 Removing the Axis-3 Motor."

8 Fit the arm-2 cover.

See "8.3.1 Arm-2 Cover."

9 Set the zero position of the axes 3 and 4.

The replacement of the axis-3 motor is completed. The replacement of the axis-3 motor also always requires setting of the zero position of the axis 4. (See "9 Robot Zero Point and Position Detector Error.")

10 Check the movement of the ball screw.

Move up and down the ball screw while pressing the brake release switch in the servo power OFF condition to make sure that the ball screw moves smoothly.

11 Perform test operation on the axes 3 and 4.

Perform running-in operation at 20% of override for about one hour to check whether there is any problem with operation of each part.

8.4.9 Removing the Axis-4 Motor

1 Remove the arm-2 cover.

See "8.3.1 Arm-2 Cover."

2 Use a plier or the like to cut off the cable straps of the cables.

It is recommended to photograph where to route the cables and where to use the cable straps beforehand for restoration.

3 Disconnect the cable connectors of the axis-4 motor.

Disconnect the cable connectors of the axis-4 motor, J4AS and J4AP (axis-4 power drive cables) and J4BS and J4BP (axis-4 encoder cables).

4 Remove the harness guide and support guide.

Remove the harness guide secured with the hexagon socket head cap screws (M4×8×4 pcs) and the support guide secured with the hexagon socket head cap screws (M4×8×4 pcs) (see 4 in "8.4.5Removing the Axis-2 Motor.").

5 Remove the axis-3 motor assembly.

See 1 to 6 in "8.4.7 Removing the Axis-3 Motor."

6 Measure the tension of the axis-4 timing belt M.

Use a tension meter to measure the tension of the axis-4 timing belt M. Enter the values listed in Table 8.5 as the values input to the tension meter. Be sure to record the results of measuring the tension using the tension meter because it is necessary to match the tension with the current belt tension value when fitting the belt.

Table 8.5 Values Input to Tension Meter

Unit Mass (g/m)	Belt Width (mm)	Span (mm)
2.0	15	101

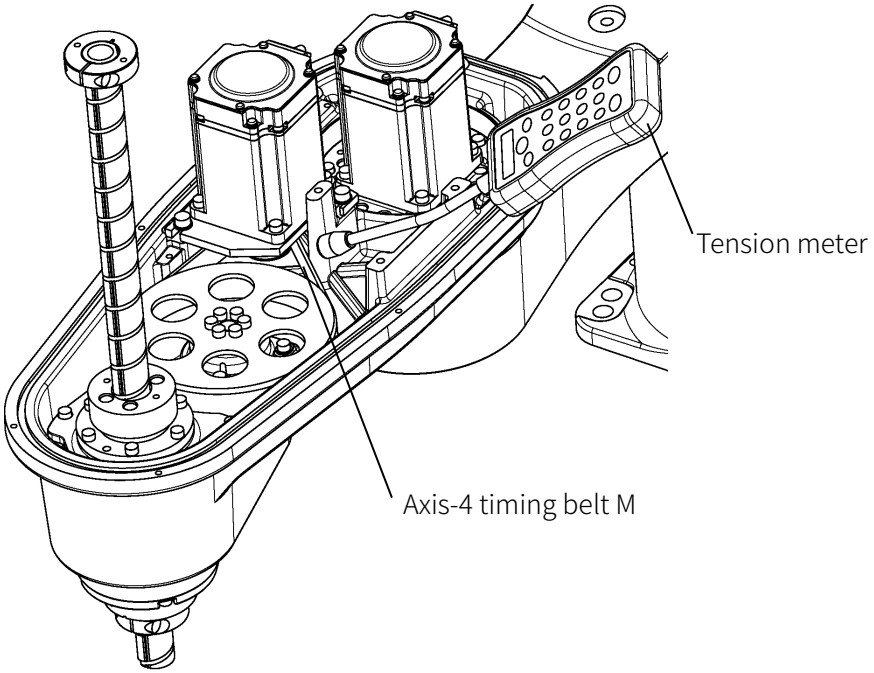


Fig. 8.29 Measuring the Tension of the Axis-4 Timing Belt M

7 Remove the axis-4 motor assembly.

Remove the hexagon socket head cap screws (M5×16×3 pcs) and washers securing the axis-4 motor base to release the tension of the axis-4 timing belt M. Then, remove the axis-4 motor assembly.

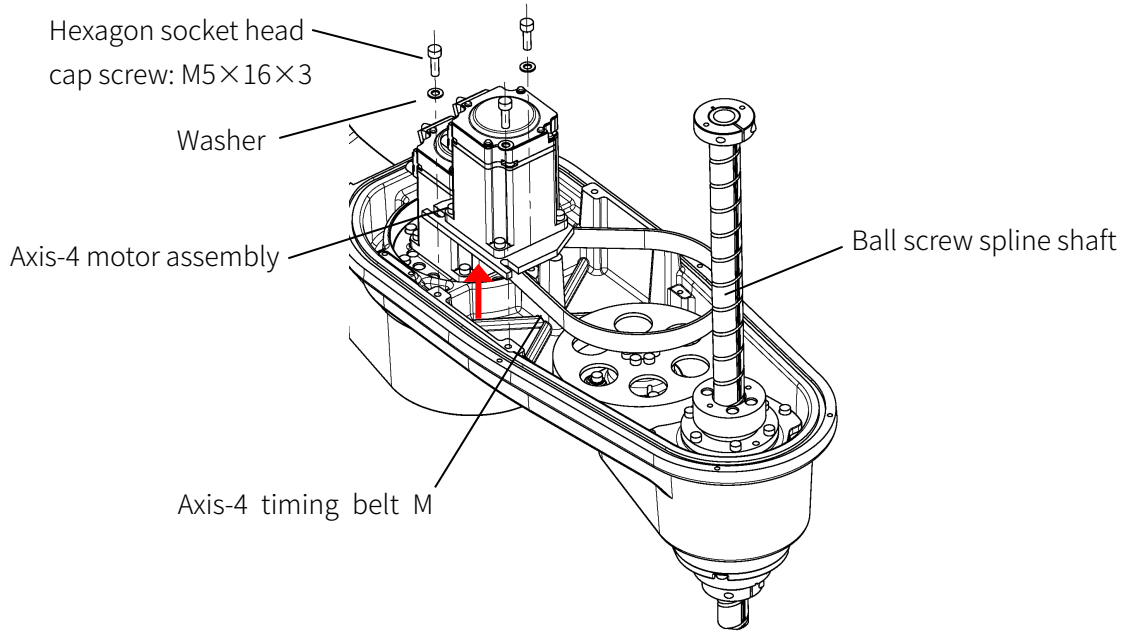


Fig. 8.30 Removing the axis-4 motor assembly

8 Pull out the pulley of the axis-4 motor assembly.

Remove the hexagon socket head cap (M4×8×1 pc). Then, pull out the press plate and pulley.

9 Remove the axis-4 motor from the axis-4 motor base.

Remove the hexagon socket head cap screws (M4×12×4 pcs) securing the axis-4 motor to separate the axis-4 motor base and the axis-4 motor.

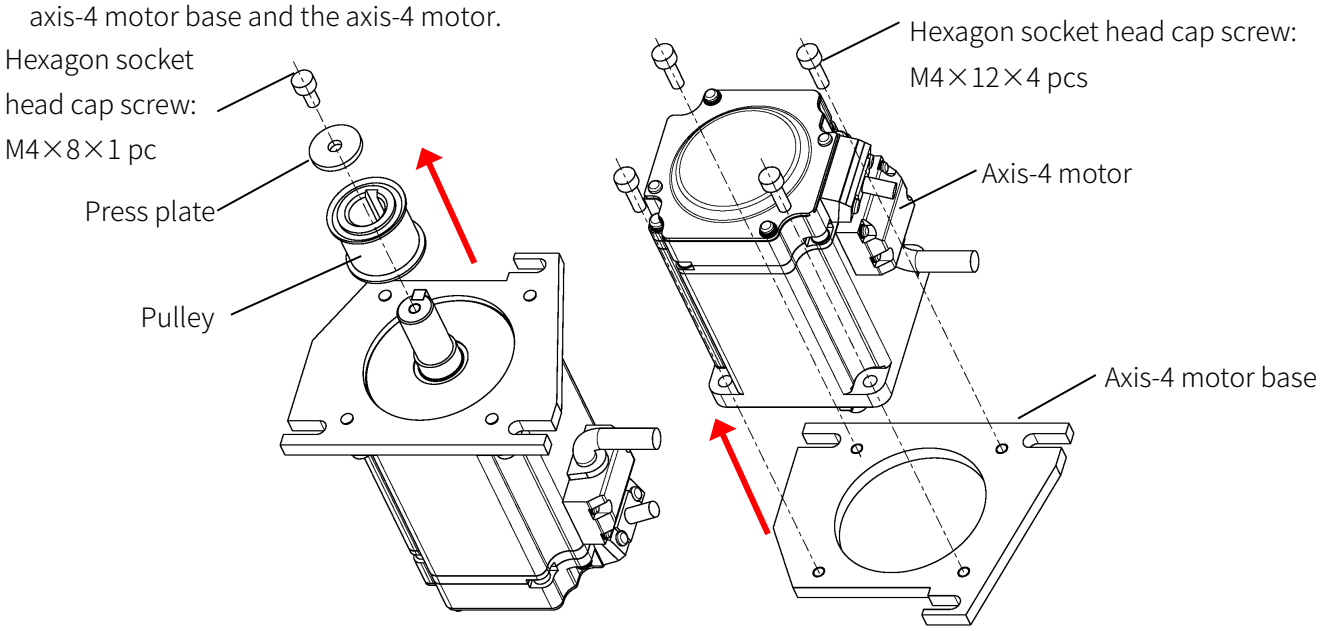


Fig. 8.31 Removing the axis-4 motor

8.4.10 Fitting the Axis-4 Motor

1 Fit the axis-4 motor to the axis-4 motor base.

Secure the axis-4 motor to the axis-4 motor base with the hexagon socket head cap screws (M4×12×4 pcs). At this time, be careful about the fitting phase of the axis-4 motor and axis-4 motor base.

2 Fit the axis-4 motor pulley.

Remove the key that has been fitted to the axis-4 motor once, apply fretting inhibitor to the side face of the key (see Fig. 8.32), and then fit the axis-4 motor pulley and secure it with the press plate and the hexagon socket head cap screw (M4×8×1).

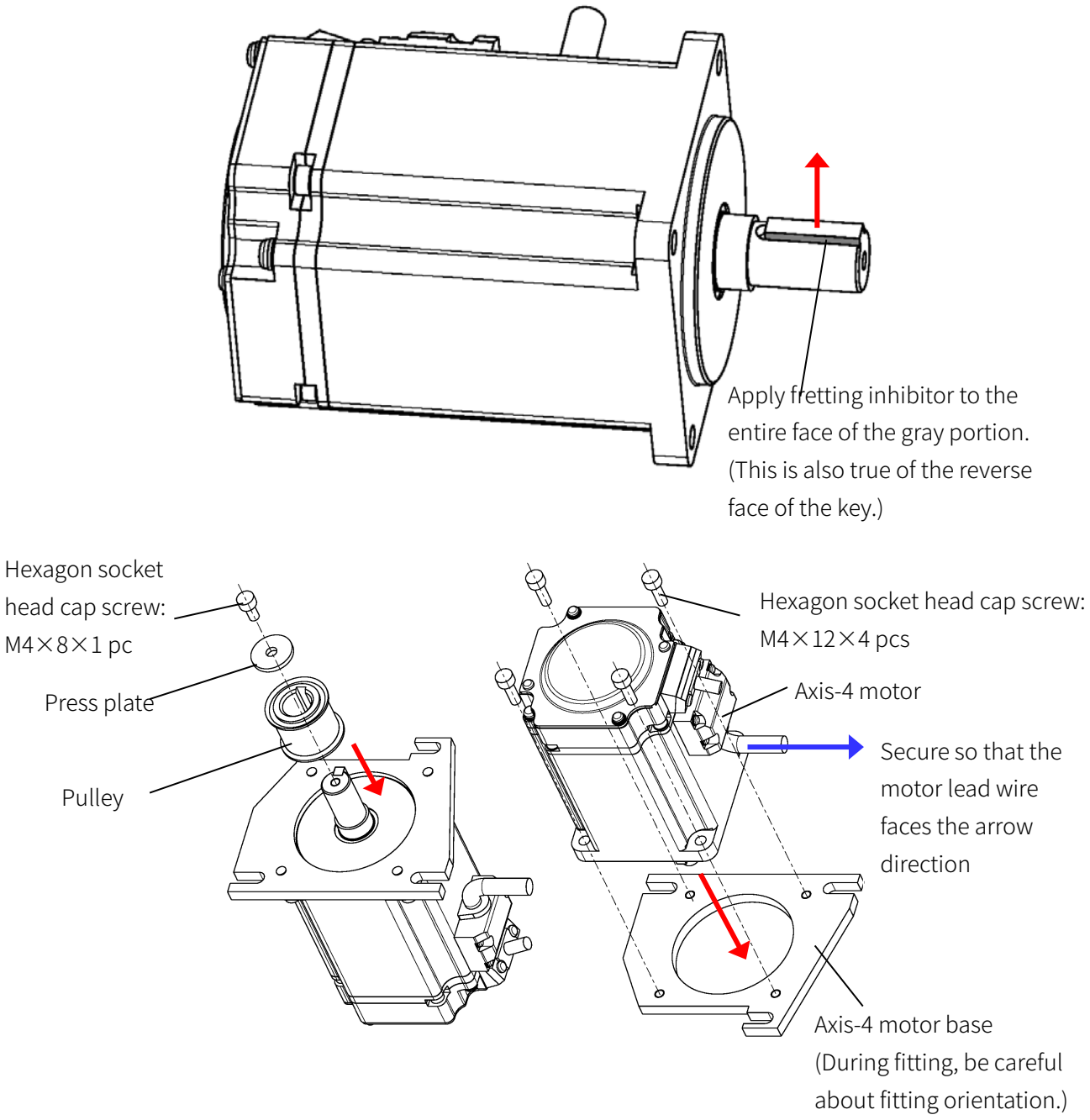


Fig. 8.32 Fitting the axis-4 motor

3 Fit the axis-4 motor assembly.

Put the axis-4 timing belt on the axis-4 motor pulley, and temporary fix the axis-4 motor base with the hexagon socket head cap screws (M5×16×3 pcs) and washers. (Application of screw locking adhesive in not required.) At this time, be careful about the fitting phase of the motor.

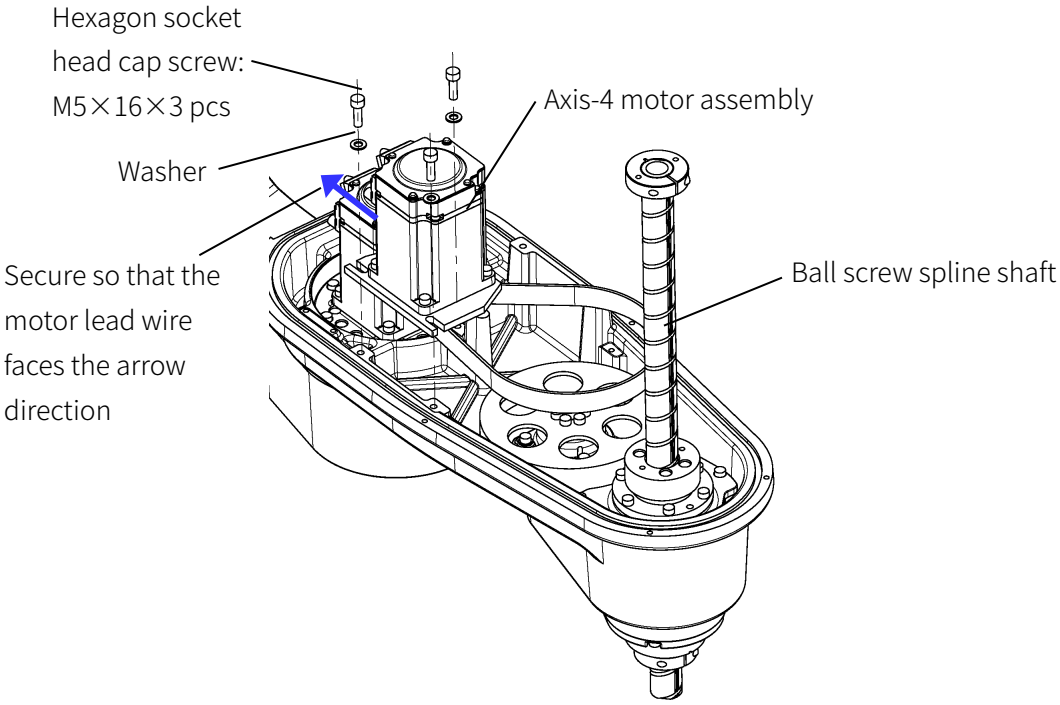


Fig. 8.33 Fitting the axis-4 motor

4 Fit the tension tool 34 assembly.

Move up and down the ball screw spline unit to adjust the axis-4 timing belt. Then, fit the tension tool 34 assembly (option) to the arm 2.

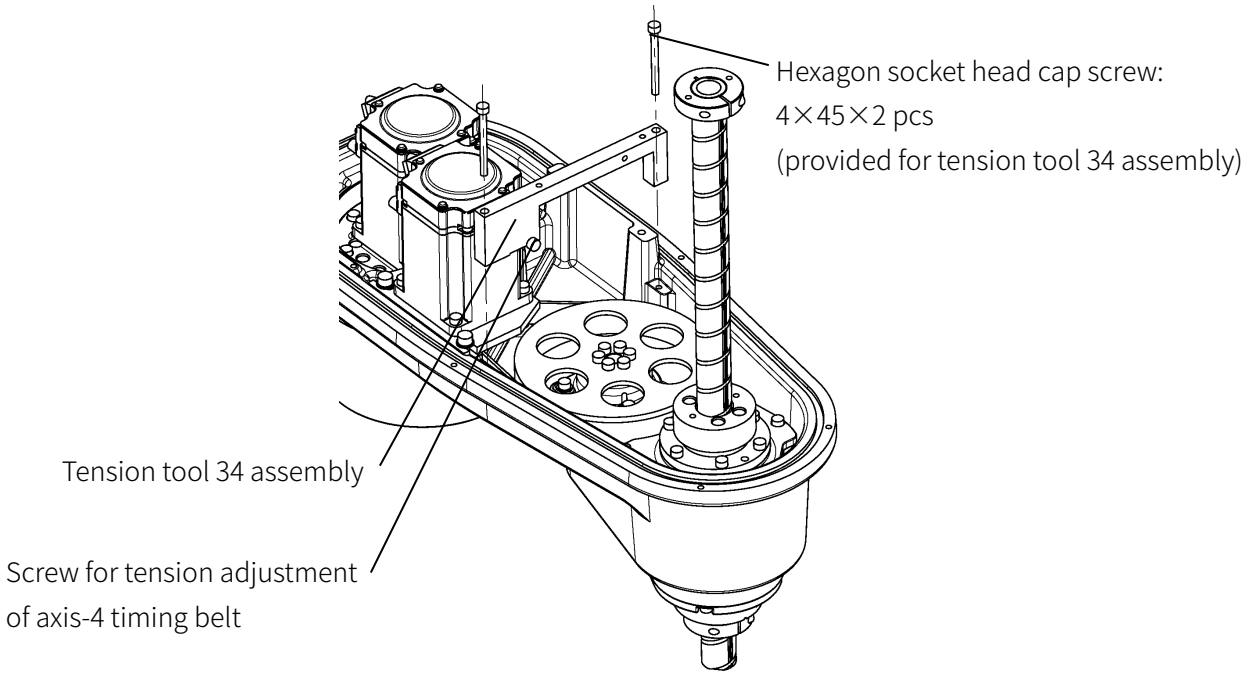


Fig. 8.34 Fitting the tension tool 34 assembly

5 Adjust the tension of the axis-4 timing belt M.

Use the tension tool 34 assembly (option) to apply tension. While using the screws for axis-4 tension adjustment of tension tool 34 assembly to push the axis-4 motor base, use a tension meter to measure the tension. Tighten the hexagon socket head cap screws temporarily fitting the motor base at a position where the tension has become a little smaller than the value B*(N) measured in step 6 in "8.4.9 Removing the Axis-4 Motor" (because retightening the hexagon socket head cap screws increases the tension). Make adjustment so that the tension becomes B*×0.95 to B*1.05(N) when the hexagon socket head cap screws are retightened. After the hexagon socket head cap screws are retightened, remove the tension tool 34 assembly. The value to be used in the tension meter is as listed in Table 8.6.

Table 8.6 Tension of the axis-4 timing belt M

Tension (N)	Unit Mass (g/m)	Belt Width (mm)	Span (mm)
B*×0.95 to B*×1.05	2.0	15	101

B is the value measured in step 6 of "8.4.9 Removing the Axis-4 Motor."

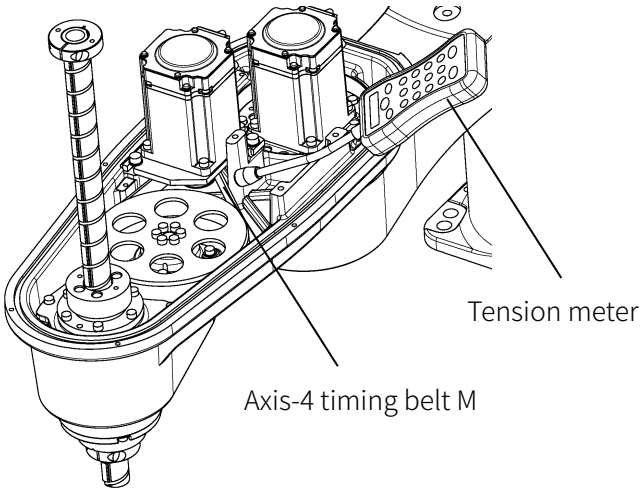


Fig. 8.35 Measuring the tension of axis-4 timing belt M

6 Fit the axis-3 motor assembly.

See 3 to 5 in "8.4.8 Fitting the Axis-3 Motor."

7 Fit the harness guide and support guide.

Use the hexagon socket head cap screws (M4×8×8 pcs) to fit the harness guide and support guide (see 8 in "8.4.6 Fitting the Axis-2 Motor").

8 Fit the removed connectors.

Connect the connectors and return the cable to its original state. (See the pictures taken in 2 of "8.4.9 Removing the Axis-4 Motor.")

9 Fit the arm-2 cover.

See "8.3.1 Arm-2 Cover."

10 Set the zero position of the axes 3 and 4.

The replacement of the axis-4 motor is completed. The replacement of the axis-4 motor also always requires setting of the zero position of the axis 3. (See "9 Robot Zero Point and Position Detector Error.")

11 Check the movement of the ball screw.

Move up and down the ball screw while pressing the brake release switch in the servo power OFF condition to make sure that the ball screw moves smoothly.

12 Perform operation at a low speed.

Perform running-in operation at 20% of override for about one hour to check whether there is any problem with operation of each part.

8.5 Adjustment and Replacement of Timing Belts

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.

⚠DANGER

- When replacing the timing belt, be sure to remove the power plug of the controller.
  The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

- When replacing the timing belt, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.
- After the timing belt is replaced, be sure to re-set the zero position.
  Because a mechanical zero position is shifted with replacement, the robot cannot perform normal operation.
  - For how to re-set the zero position, see "9 Robot Zero Point and Position Detector Error."

8.5.1 Types of Timing Belts

The timing belts used by this robot are as listed in Table 8.7.

When ordering a replacement belt, specify the robot model name, manufacture number, applicable axis, and SHIBAURA drawing number.

For the location of the manufacture number label, refer to "STA1179 Safety Manual" Instruction Manual.

Table 8.7 Types of timing belts

Part name	Applicable Axis	Width	SHIBAURA Dwg. No>
Timing belt	Axis 3	10mm	S968393
	Axis 4 Motor to intermediate shaft	15mm	S968397
	Axis 4 Intermediate shaft to ball screw spline	30mm	S968395

8.5.2 Locations of Timing Belts

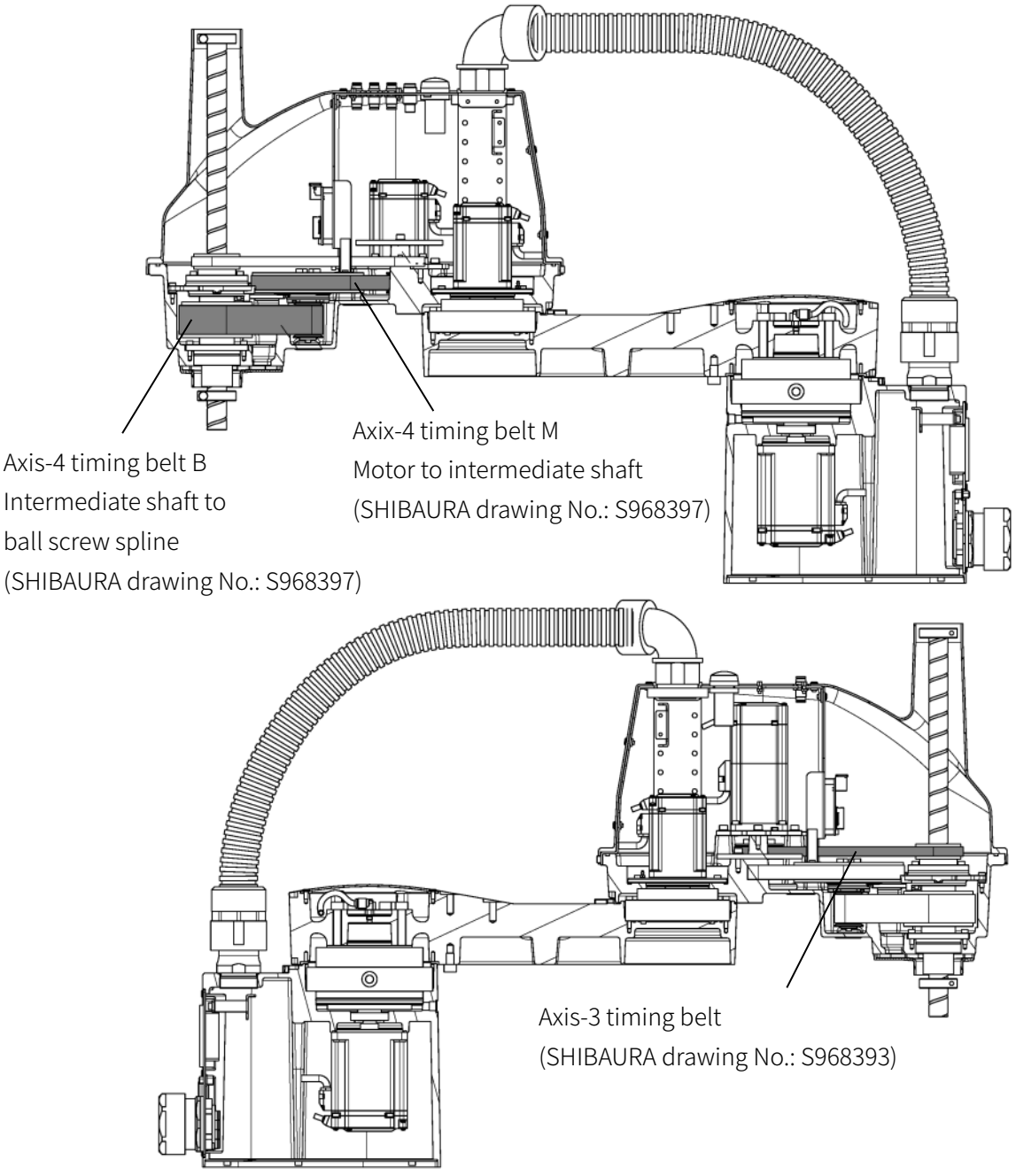


Fig. 8.36 Locations of Timing Belts

8.5.3 Replacing the Axis-3 Timing Belt

⚠ WARNING

- When replacing the axis-3 timing belt, be sure to lower the ball screw spline shaft to the lower limit. When the motor or timing belt is removed, the shaft drops by its own weight.
  - When the ball screw spline or axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

1 Remove the arm-2 cover.

See "8.3.1 Arm-2 Cover."

2 Use a plier or the like to cut off the cable straps of the cables.

It is recommended to photograph where to route the cables and where to use the cable straps for restoration.

3 Remove the axis-3 motor assembly.

See 3 to 6 in "8.4.7 Removing the Axis-3 Motor."

4 Remove the axis-3 timing belt.

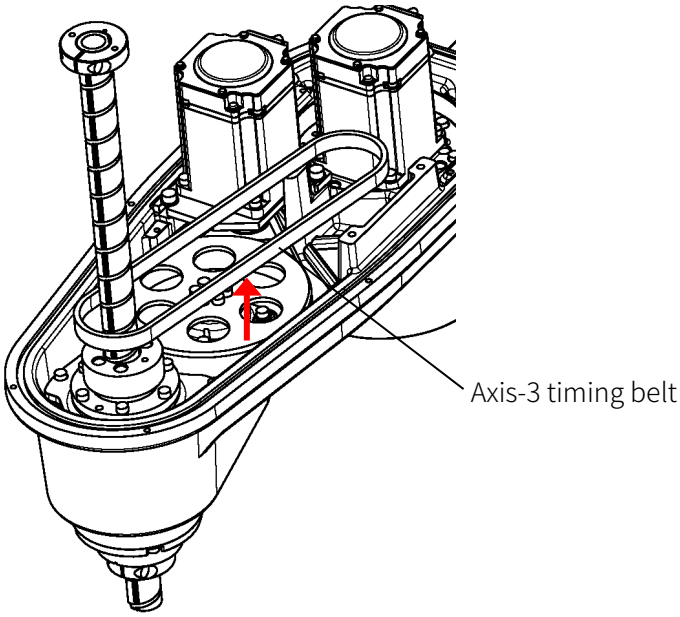


Fig. 8.37 Removing the Axis-3 timing belt

5 Replace the axis-3 timing belt.

Put a newly prepared timing belt on the axis-3 motor pulley of the motor assembly removed in step 3 and ball screw spline side pulley, and temporarily fix the axis-3 motor base to the arm 2 with the hexagon socket head cap screws (M5×16×3 pcs, not requiring application of screw locking adhesive) and washers. At this time, be careful about the fitting phase of the motor.

6 Fit the tension tool 34 assembly.

Move up and down the ball screw spline unit to adjust it. Then, fit the tension tool 34 assembly (option). (See 4 in "8.4.8 Fitting the Axis-3 Motor.")

7 Adjust the tension of the axis-3 timing belt.

Use the tension tool 34 assembly (option) to apply tension. While using the screws for axis-3 tension adjustment of tension tool 34 assembly to push the axis-3 motor base, use a tension meter to measure the tension. Tighten the hexagon socket head cap screws to secure the motor base that has been temporarily fitted at a position where the tension has become a little smaller than 87(N) (because retightening the hexagon socket head cap screws increases the tension). Make adjustment so that the tension becomes 87 to 116(N) when the hexagon socket head cap screws are retightened. After the hexagon socket head cap screws are retightened, remove the tension tool 34 assembly. The value to be used in the tension meter is as listed in Table 8.8.

Table 8.8 Tension of the axis-3 timing belt

Tension (N)	Unit mass (g/m)	Belt width (mm)	Span (mm)
87 to 116	2.9	10	192

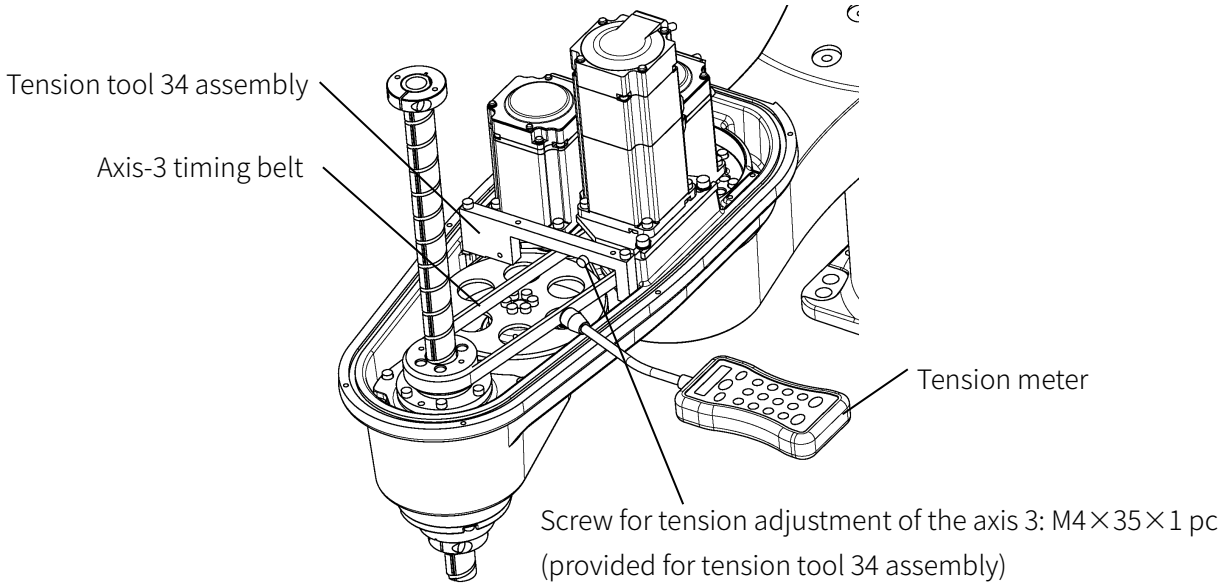


Fig. 8.38 Adjusting the tension of the axis 3

8 Connect the disconnected connectors.

Connect the connectors and return the cable to its original state. (See the pictures taken in 2 in "8.5.3 Replacing the Axis-3 Timing Belt.")

9 Fit the arm-2 cover.

See "8.3.1 Arm-2 Cover."

10 Set the zero position of axes 3 and 4.

The replacement of the axis-3 timing belt is completed. (See "9 Robot Zero Point and Position Detector Error.")

11 Check the movement of the ball screw.

Move up and down the ball screw while pressing the brake release switch in the servo power OFF condition to make sure that the ball screw moves smoothly.

12 Perform operation at a low speed.

Perform running-in operation at 20% of override for about one hour to check whether there is any problem with operation of each part.

8.5.4 Replacing the Axis-4 Timing Belt B

NOTICE

- Never pull out the ball screw nut from the ball screw spline shaft.
  Otherwise, the ball screw spline unit from which balls have come out cannot be used.
- The replacement of the axis-4 timing belt requires the axis-3 motor and timing belt to be removed. Therefore, the precautions for the replacement of the axis-3 motor and timing belt must also be observed.

1 Remove the arm-2 cover.

See "8.3.1 Arm-2 Cover."

2 Use a plier or the like to cut off the cable straps of the cables.

It is recommended to photograph where to route the cables and where to use the cable straps beforehand for restoration.

3 Disconnect the cable connectors of the axis-3 motor and axis-4 motor.

Disconnect the cable connectors of the axis 3, J3AS and J3AP (axis-3 power drive cables), J3BS and J3BP (axis-3 encoder cables), J3DS and J3DP (axis-3 brake cables) and the cable connectors of the axis-4 motor, J4AS and J4AP (axis-4 power drive cables) and J4BS and J4BP (axis-4 encoder cables).

4 Remove the harness guide and support guide.

Remove the harness guide secured with the hexagon socket head cap screws (M4×8×4 pcs) and the support guide secured with the hexagon socket head cap screws (M4×8×4 pcs) (see 4 in "8.4.5 Removing the Axis-2 Motor").

5 Remove the axis-3 motor assembly.

See 3 to 6 in "8.4.7 Removing the Axis-3 Motor."

6 Remove the axis-4 motor assembly.

See 6 and 7 in "8.4.9 Removing the Axis-4 Motor."

7 Release the tension of the axis-4 timing belt B.

Remove the hexagon socket head cap screws (M4×12×4 pcs) securing the axis-4 intermediate shaft to release the tension of the timing belt.

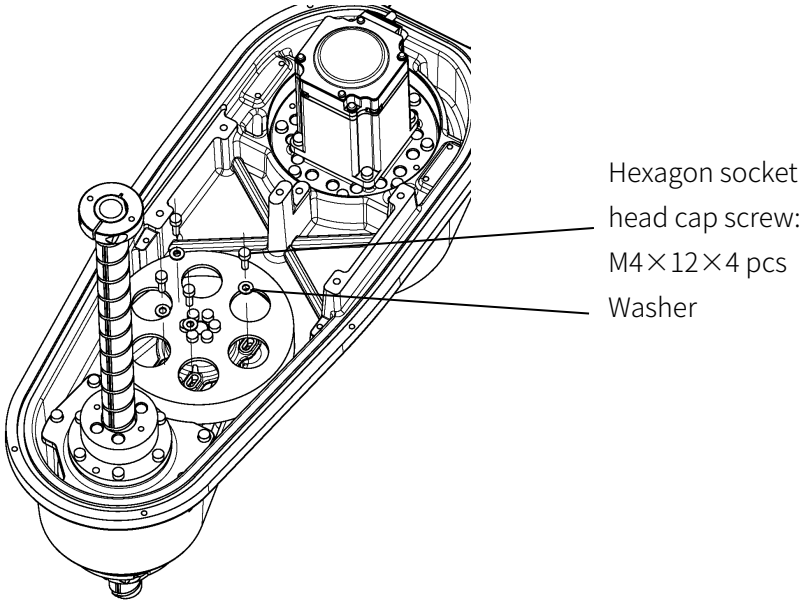


Fig. 8.39 Releasing the tension of axis-4 intermediate shaft

8 Remove the ball screw spline unit.

Remove the upper-limit mechanical stopper, ball screw spline shaft, ball screw nut, and axis-3 bracket. For removal, see "8.6.4 Removing the Ball Screw Spline Unit."

9 Pull out the axis-4 intermediate shaft assembly.

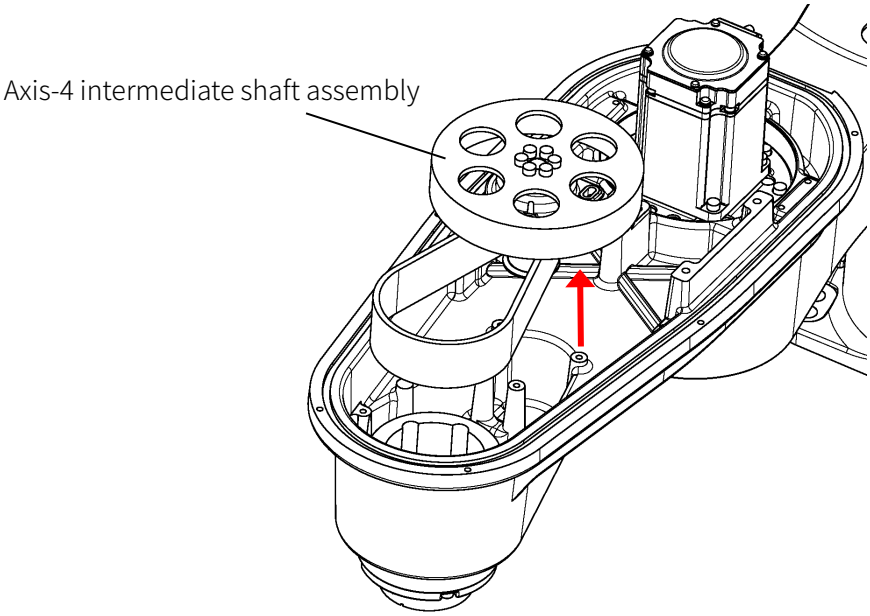


Fig. 8.40 Removing the axis-4 intermediate shaft assembly

10 Temporarily fix the axis-4 intermediate shaft assembly to which a new axis-4 timing belt B has been fitted.

Put the new axis-4 timing belt B on the pulley and temporarily fix the axis-4 intermediate shaft assembly to the arm 2 with the hexagon socket head cap screws (M4×12×4 pcs) and washers.

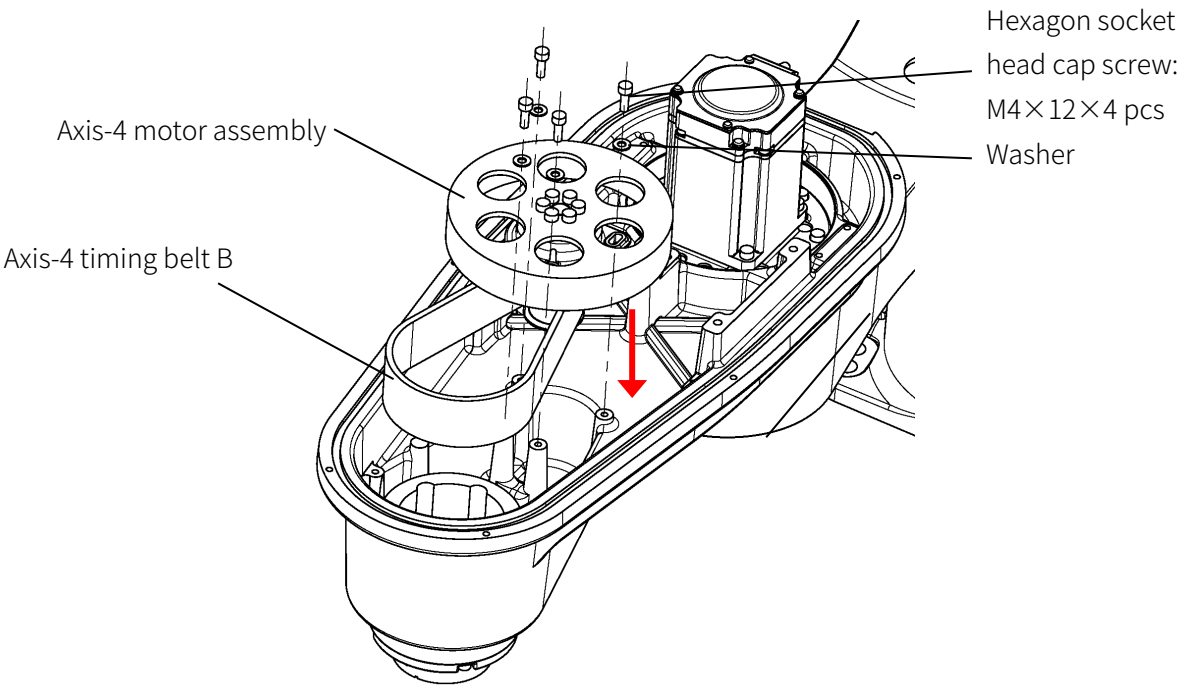


Fig. 8.41 Fitting the axis-4 intermediate shaft assembly

11 Fit the tension tool intermediate assembly (option).

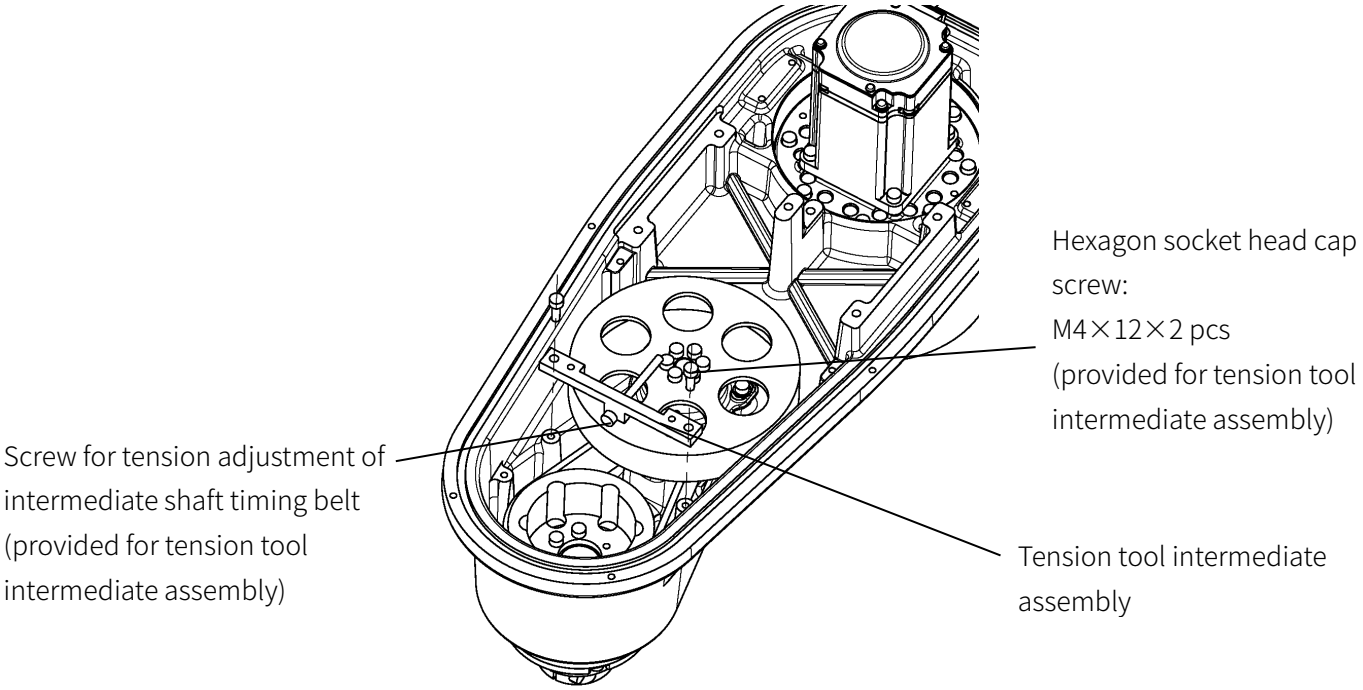


Fig. 8.42 Fitting the tension tool intermediate assembly

12 Remove the spline nut cover.

Remove the spline nut cover secured with the hexagon socket head cap screws (M4×8×6 pcs).

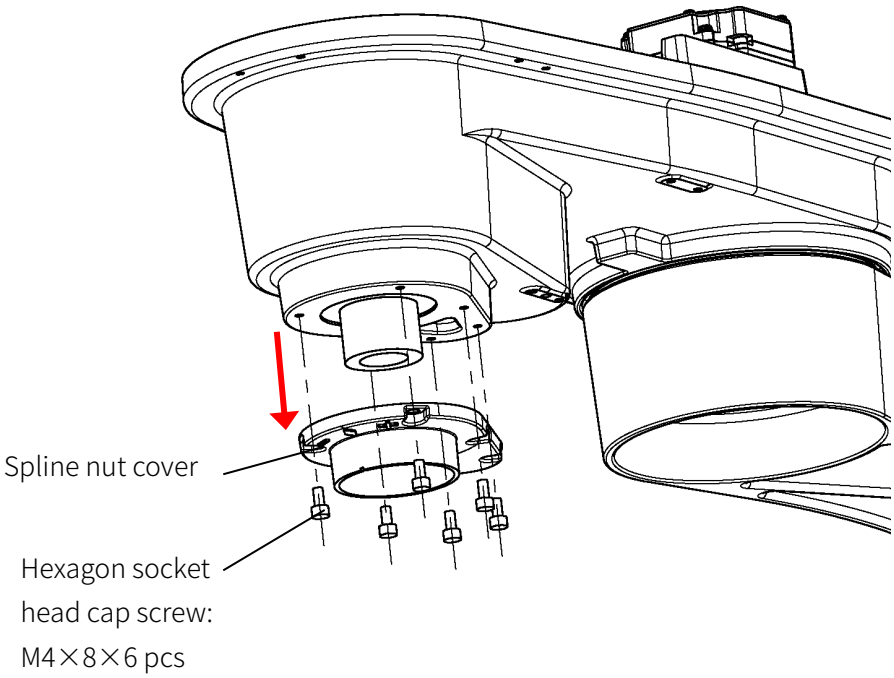


Fig. 8.43 Removing the spline nut cover

13 Adjust the tension of the axis-4 timing belt B.

Use the tension tool intermediate assembly (option) to apply tension. While using the screws for tension adjustment of intermediate shaft timing belt of the tension tool intermediate assembly to push the intermediate shaft, use a tension meter to measure the tension. At this time, the tension meter can be inserted from the hole on the lower side of the arm 2 (shown in Fig. 8.44).

Tighten the hexagon socket head cap screws to secure the intermediate shaft that has been temporarily fixed at a position where the tension has become a little smaller than 304(N) (because retightening the hexagon socket head cap screws increases the tension). Make adjustment so that the tension becomes 304 to 406(N) when the hexagon socket head cap screws are retightened. After the hexagon socket head cap screws are retightened, remove the tension tool intermediate assembly. The value to be used in the tension meter is as listed in Table 8.9.

Table 8.9 Tension of the axis-4 timing belt B

Tension (N)	Unit Mass (g/m)	Belt Width (mm)	Span (mm)
304 to 406	2.9	30	95

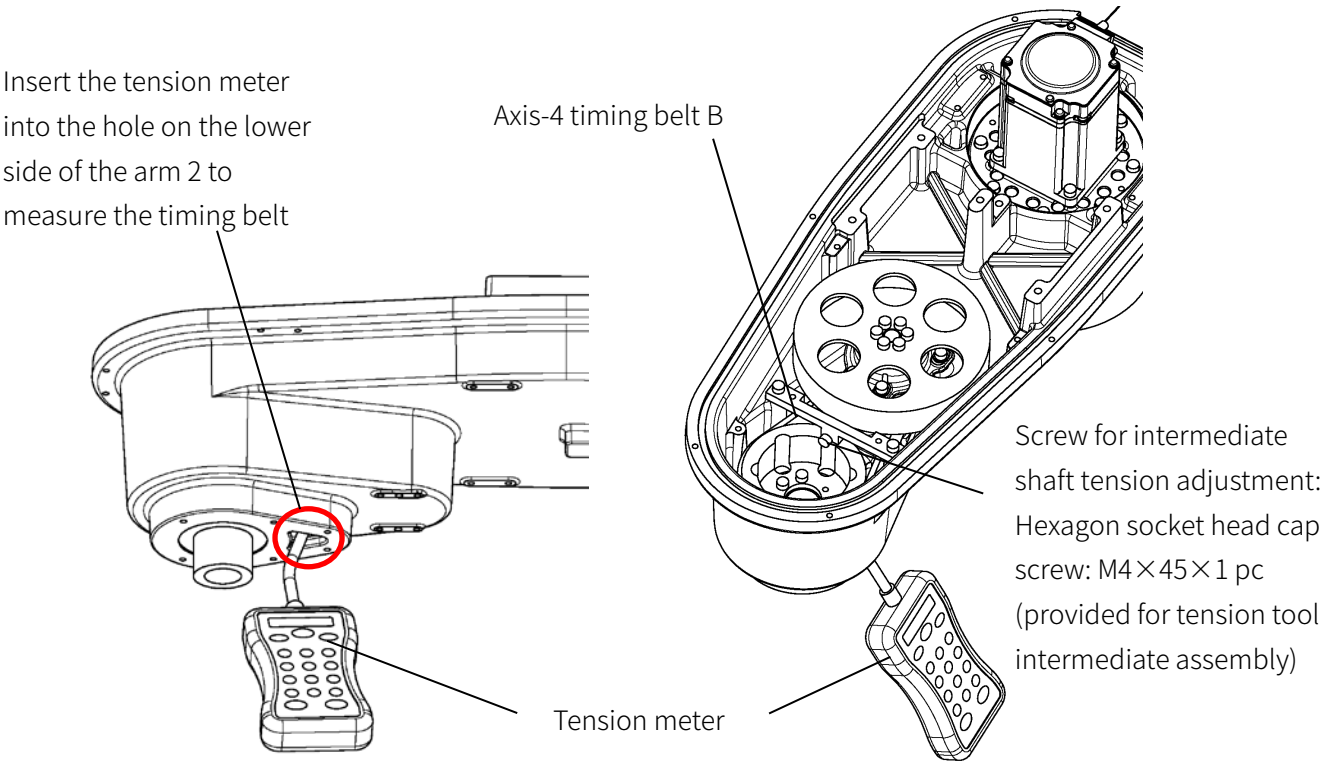


Fig. 8.44 Adjusting the tension of axis 4

14 Fit the spline nut cover.

Fit the spline nut cover with the hexagon socket head cap screws (M4×8×6 pcs). At this time, because too strong clamping torque damages the spline nut cover, fit the spline nut cover with a clamping torque of 2.0[N].

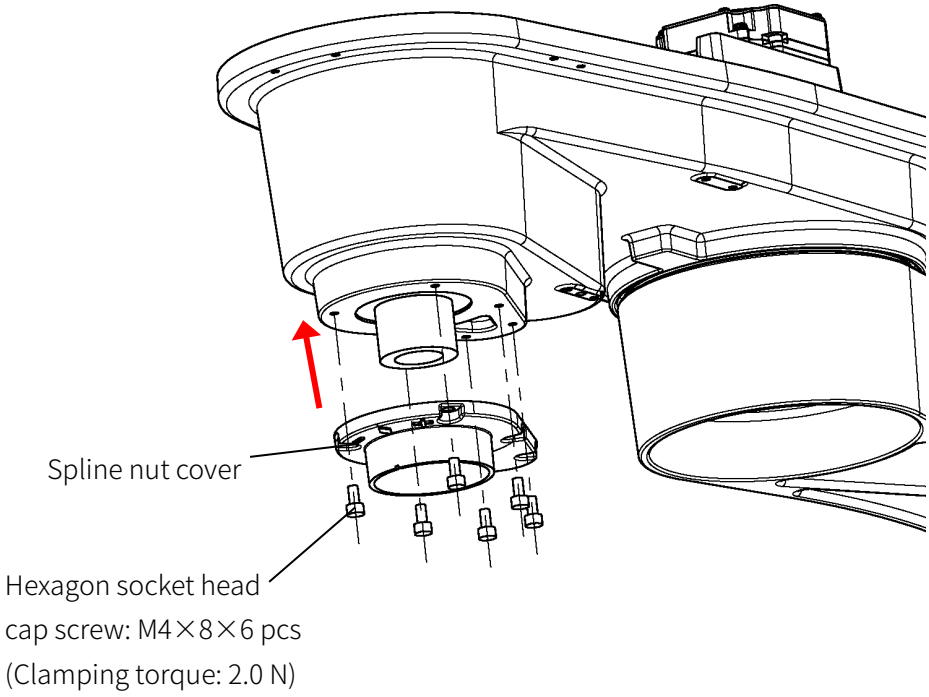


Fig. 8.45 Fitting the spline nut cover

15 Temporarily fix the axis-3 bracket.

Temporarily fix the axis-3 bracket to the arm 2 with the hexagon socket head cap screws (M4×12×5 pcs).

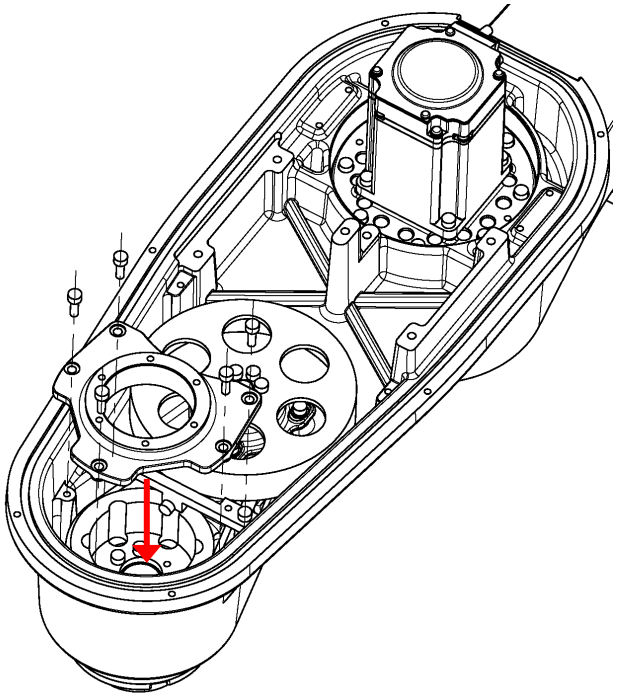


Fig. 8.46 Fitting the axis-3 bracket

16 Fit the ball screw spline unit.

Secure the ball screw nut, ball screw spline shaft, and stopper removed in step 8 to the arm 2. For the fitting of the ball screw spline unit and stopper, see "8.6.5 Fitting the Ball Screw Spline Unit."

17 Fit the axis-4 motor assembly.

Fit the axis-4 motor assembly and timing belt. For the fitting, see 3 to 5 in "8.4.10 Fitting the Axis-4 Motor."

18 Fit the axis-3 motor assembly.

Fit the axis-3 motor assembly and timing belt. For the fitting, see "8.5.3 Replacing the Axis-3 Timing Belt."

19 Fit the harness guide and support guide.

Use the hexagon socket head cap screws (M4×8×8 pcs) to fit the harness guide and support (see 8 in "8.4.6 Fitting the Axis-2 Motor.")

20 Connect the disconnected connectors.

Connect the connectors and return the cable to its original state. (See the pictures taken in 3 in "8.5.4 Replacing the Axis-4 Timing Belt").

21 Fit the arm-2 cover.

See "8.3.1 Arm-2 Cover."

22 Set the zero position of axes 3 and 4.

The replacement of the axis-4 timing belt is completed. (See "9 Robot Zero Point and Position Detector Error.")

23 Check the movement of the ball screw.

Move up and down the ball screw while pressing the brake release switch in the servo power OFF condition to make sure that the ball screw moves smoothly.

24 Perform operation at a low speed.

Perform running-in operation at 20% of override for about one hour to check whether there is any problem with operation of each part.

8.5.5 Replacing the Axis-4 Timing Belt M

NOTICE

- The replacement of the axis-4 timing belt requires the axis-3 motor and timing belt to be removed. Therefore, the precautions for the replacement of the axis-3 motor and timing belt must also be observed.

1 Remove the arm-2 cover.

See "8.3.1 Arm-2 Cover."

2 Use a plier or the like to cut off the cable straps of the cables.

It is recommended to photograph where to route the cables and where to use the cable straps beforehand for restoration.

3 Disconnect the cable connectors of the axis-3 motor and axis-4 motor.

Disconnect the cable connectors of the axis 3, J3AS and J3AP (axis-3 power drive cables), J3BS and J3BP (axis-3 encoder cables), J3DS and J3DP (axis-3 brake cables) and the cable connectors of the axis-4 motor, J4AS and J4AP (axis-4 power drive cables) and J4BS and J4BP (axis-4 encoder cables).

4 Remove the harness guide and support guide.

Remove the harness guide secured with the hexagon socket head cap screws (M4×8×4 pcs) and the support guide secured with the hexagon socket head cap screws (M4×8×4 pcs) (see 4 in "8.4.5 Removing the Axis-2 Motor").

5 Remove the axis-3 motor assembly.

See 3 to 6 in "8.4.7 Removing the Axis-3 Motor."

6 Remove the axis-4 motor assembly.

See 6 and 7 in "8.4.9 Removing the Axis-4 Motor."

7 Fit the new axis-4 timing belt M and the axis-4 motor assembly.

Put the new axis-4 timing belt M on the axis-4 motor pulley, and temporary fix the axis-4 motor base with the hexagon socket head cap screws (M5×16×3 pcs) and washers. (Application of screw locking adhesive in not required.) At this time, be careful about the fitting phase of the motor.

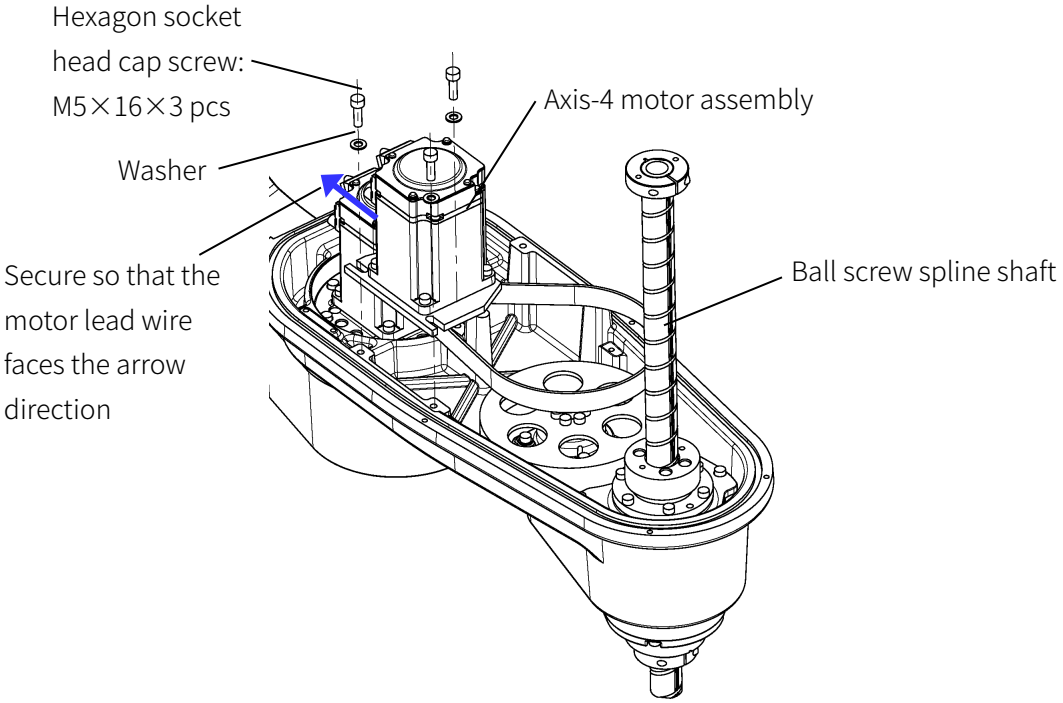


Fig. 8.47 Fitting the axis-4 motor

8 Fit the tension tool 34 assembly.

Move up and down the ball screw spline unit to adjust the axis-4 timing belt. Then, fit the tension tool 34 assembly (option) to the arm 2.

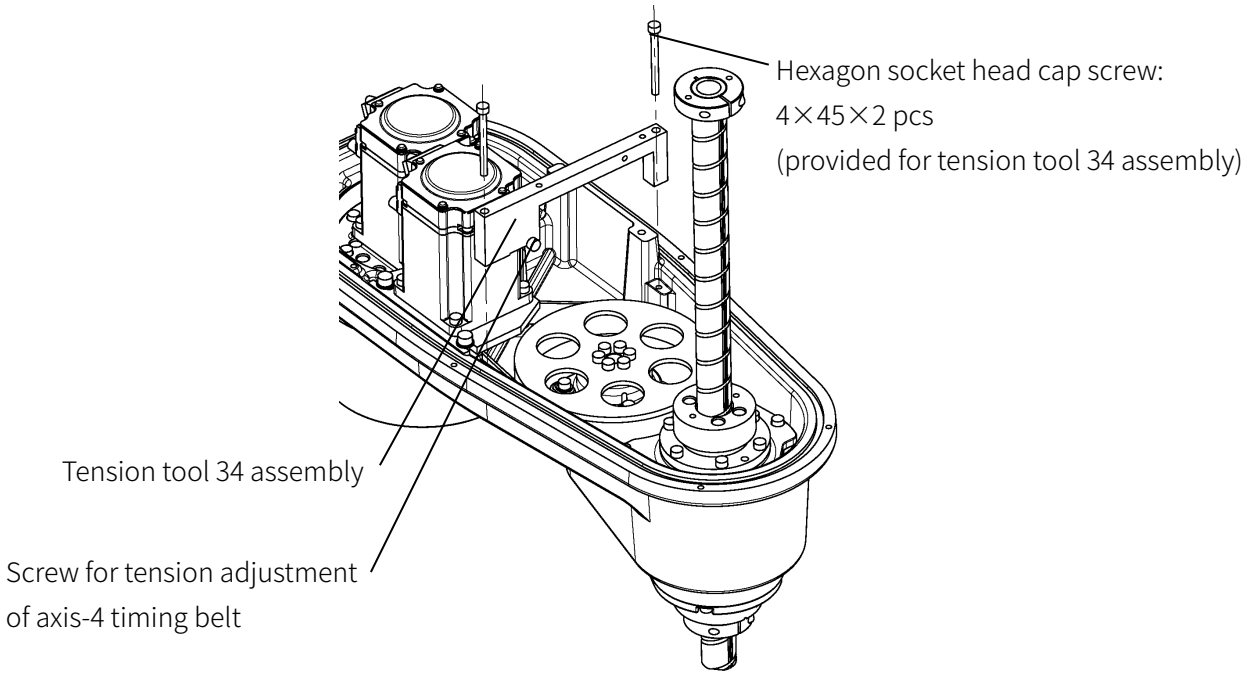


Fig. 8.48 Fitting the tension tool 34 assembly

9 Adjust the tension of the axis-4 timing belt M.

Use the tension tool 34 assembly (option) to apply tension. While using the screws for axis-4 tension adjustment of tension tool 34 assembly to push the axis-4 motor base, use a tension meter to measure the tension. Tighten the hexagon socket head cap screws temporarily fitting the motor base at a position where the tension has become a little smaller than the value 55(N) (because retightening the hexagon socket head cap screws increases the tension). Make adjustment so that the tension becomes 55 to 74 when the hexagon socket head cap screws are retightened. After the hexagon socket head cap screws are retightened, remove the tension tool 34 assembly. The value to be used in the tension meter is as listed in Table 8.10.

Table 8.10 Tension of the axis-4 timing belt M

Tension (N)	Unit Mass (g/m)	Belt Width (mm)	Span (mm)
55 to 74	2.0	15	101

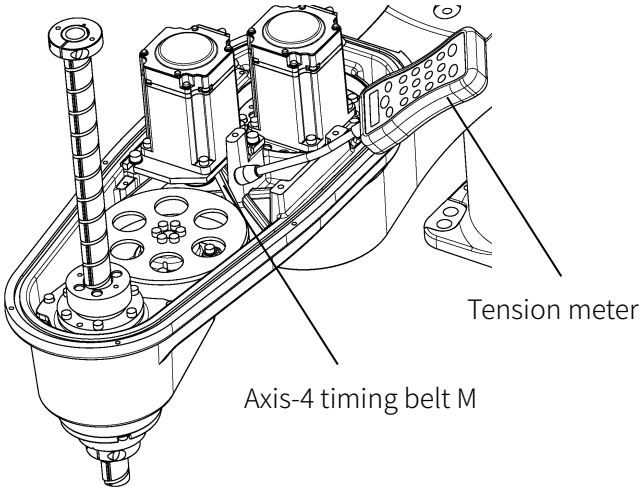


Fig. 8.49 Measuring the tension of axis-4 timing belt M

10 Fit the axis-3 motor assembly.

See 3 to 5 in "8.4.8 Fitting the Axis-3 Motor."

11 Fit the harness guide and support guide.

Use the hexagon socket head cap screws (M4×8×8 pcs) to fit the harness guide and support guide (see 8 in "8.4.6 Fitting the Axis-2 Motor").

12 Fit the removed connectors.

Connect the connectors and return the cable to its original state. (See the pictures taken in 2 of "8.4.9 Removing the Axis-4 Motor.")

13 Fit the arm-2 cover.

See "8.3.1 Arm-2 Cover."

14 Set the zero position of the axes 3 and 4.

The replacement of the axis-4 motor is completed. The replacement of the axis-4 motor also always requires setting of the zero position of the axis 3. (See "9 Robot Zero Point and Position Detector Error.")

15 Perform operation at a low speed.

Perform running-in operation at 20% of override for about one hour to check whether there is any problem with operation of each part.

8.6 Greasing up and Replacement of the Ball Screw Spline Unit

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.

⚠DANGER

- When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
  The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

- When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.
- After the ball screw spline unit is replaced, be sure to re-set the zero position.
  Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
  - For how to re-set the zero position, see "9 Robot Zero Point and Position Detector Error."

8.6.1 Type of the Ball Screw Spline Unit

The ball screw spline units used by this robot are listed in the Table 8.11. When ordering a replacement product, specify the robot model name, manufacture number, SHIBAURA drawing number and unit code. For the location of the manufacture number label, see the INSTRUCTION MANUAL "SAFETY MANUAL".

Table 8.11 Ball screw spline unit

Part name	Stroke	SHIBAURA Dwg. No.	Unit code
Ball screw spline unit	210 mm	H880457	Y610D0CP0

8.6.2 Location of the Ball Screw Spline Unit

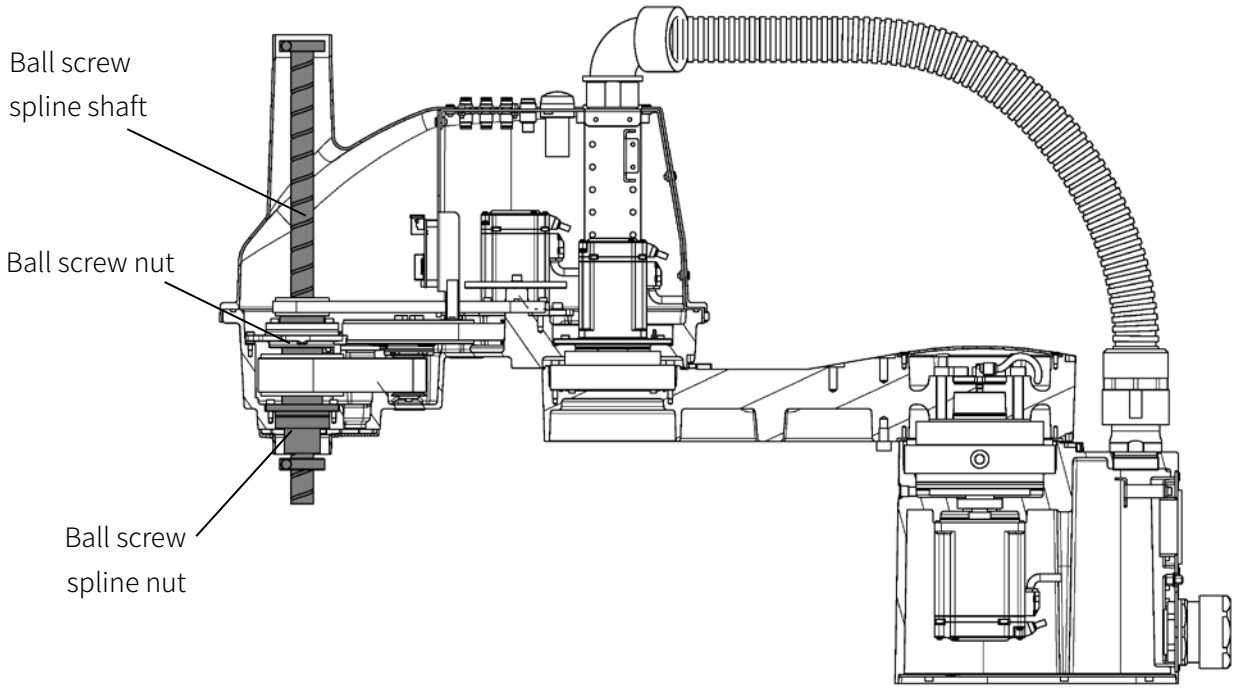


Fig. 8.50 Location of the ball screw spline unit

8.6.3 Greasing Up the Ball Screw Spline Unit and Applying Corrosion Inhibitor

⚠ CAUTION

- When moving the ball screw spline shaft by hand, never insert your hand or finger inside the mechanical stopper or under the robot hand.
  Otherwise, your hand or finger may get caught in the mechanical stopper and hand.

NOTICE

- Never touch the ball screw spline unit with bare hands.
  Touching it with bare hands may cause the ball screw spline unit to rust.
  - Be sure to wear gloves.
- When greasing up the ball screw spline unit and apply corrosion inhibitor, be sure to cover the peripheral equipment, etc.
  Grease and corrosion inhibitor may drip on the peripheral equipment.
  - Wipe out excess grease and corrosion inhibitor with waste cloth.
- Be sure to check the application state of grease daily.
  A shortage of grease and development of rust may cause flaws in the sliding surface, resulting in a reduction in the performance and lifespan of the robot. The robot may malfunction.
- Be sure to use the specified type of grease.
  Otherwise, the performance and lifespan of the robot may be reduced. The robot may malfunction.
- Be sure to check the application state of corrosion inhibitor daily.
  Otherwise, rust may develop on the ball screw spline unit.
  - Corrosion inhibitor must be applied only when it is used up.

MEMO

- *The greasing up of the ball screw spline unit should be basically done every three months, but grease must be applied when it is used up.*
- *During a daily inspection, the grease of the ball screw spline unit must also be checked.*

The state of corrosion inhibitor should be checked during a daily inspection, and if corrosion inhibitor is used up, it must be applied.

Table 8.12 Recommended corrosion inhibitors

Recommended corrosion inhibitor	Maker
Kluber A20	NOK
WD-40	ST Trading

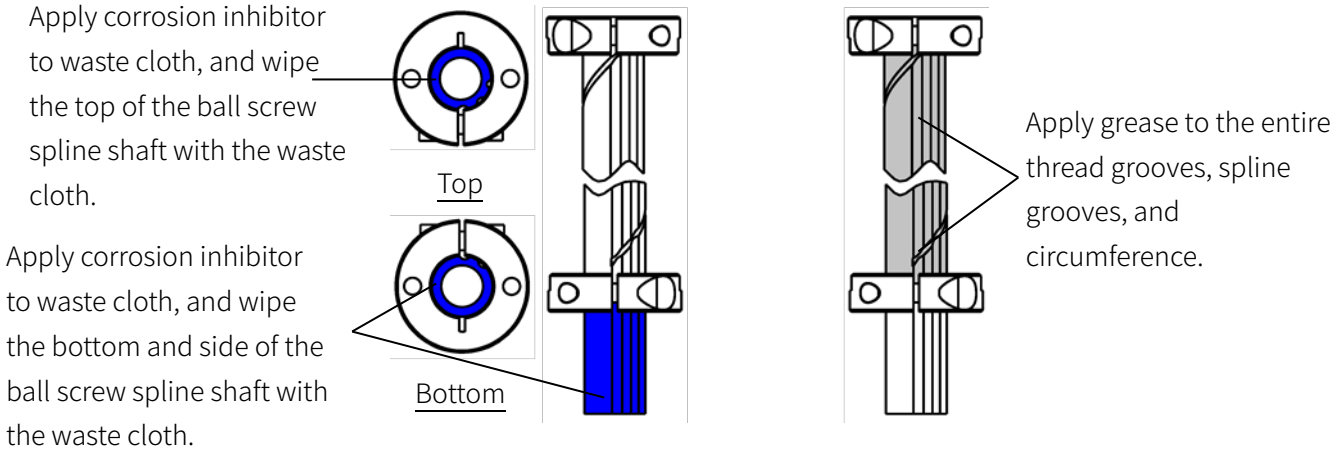


Fig. 8.51 Parts of the ball screw spline unit to which corrosion inhibitor or grease is applied

1 **Remove the arm-2 cover.**
See "8.3.1 Arm-2 Cover"

2 **Connect the power plug of the controller, and turn off the servos.**

3 **Move the arm by hand to a position where it can be moved along its full stroke length on the axis 3.**

4 **Press down the ball screw spline shaft to the lower limit while pressing the axis-3 brake release switch.**

5 Apply grease to the visible part of the shaft.

Apply grease, with a brush, etc., to the visible part of the shaft. Apply it so that shaft grooves are filled.
For the part to apply grease to, see Fig. 8.52.

Table 8.13 Recommended grease

Recommended grease	Maker
AFF grease	THK

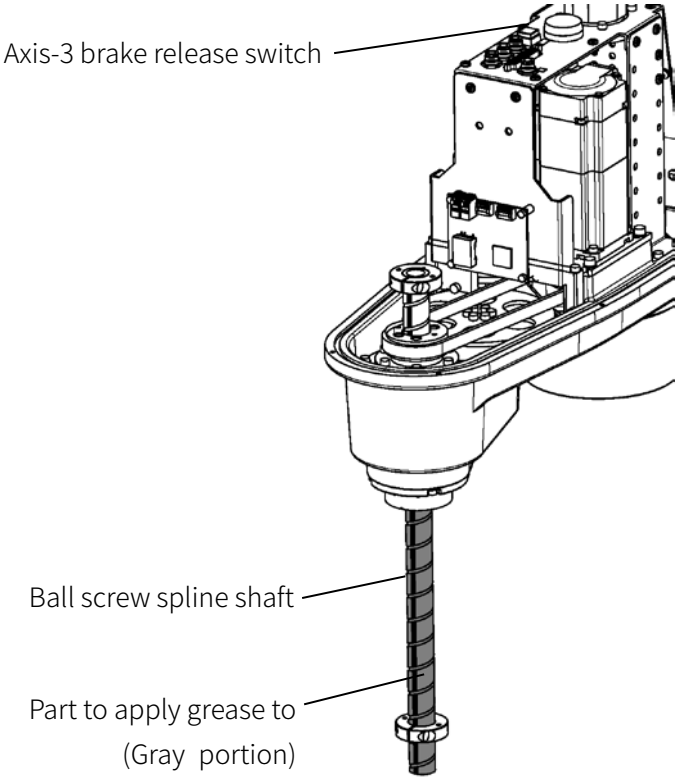


Fig. 8.52 Greasing up the lower side of the ball screw spline unit

6 Press up the shaft to the upper limit while pressing the axis-3 brake release switch.

7 Use a brush or the like to apply grease to the visible part of the shaft on the top of the ball screw nut.

Apply it so that shaft grooves are filled. For the part to apply grease to, see Fig. 8.53.

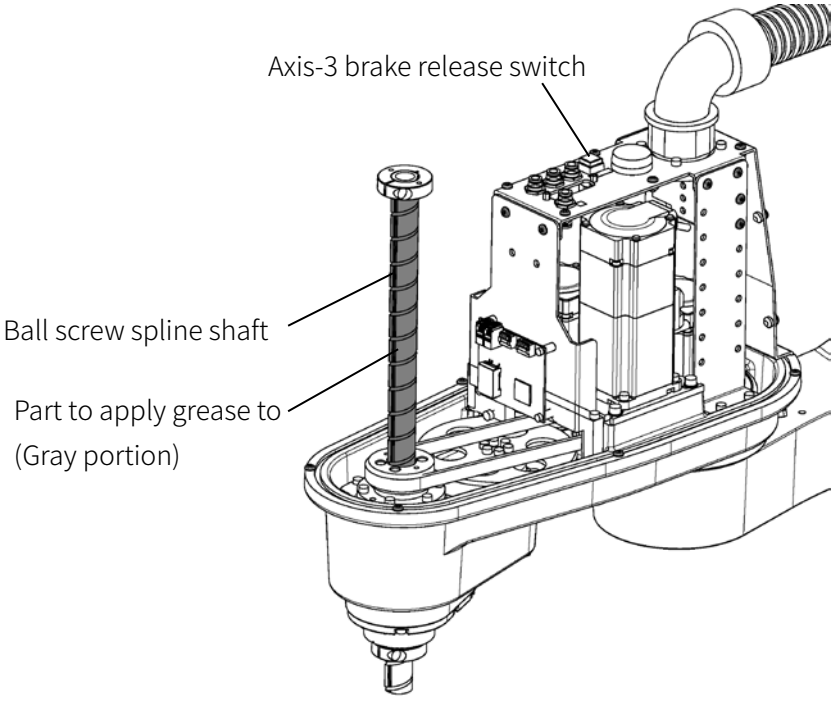


Fig. 8.53 Greasing up the upper side of the ball screw spline unit

8 Smooth out the grease.

Move up and down the shaft several times while pressing the axis-3 brake release switch to smooth out the grease. Then, wipe out excess grease to finish the work.

8.6.4 Removing the Ball Screw Spline Unit

⚠ DANGER

- When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
  The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

- Never touch the ball screw spline unit with bare hands.
  Touching it with bare hands causes the ball screw spline unit to rust.
  - Be sure to wear gloves.
- Be sure to handle the ball screw spline unit carefully.
  Drop impact or application of excessive external force could ruin the unit.
- After the ball screw spline unit is replaced, be sure to re-set the zero position.
  Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
  - For how to re-set the zero position, see "9 Robot Zero Point and Position Detector Error."
- The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
  Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

The description on how to replace the ball screw spline unit does not refer to any replacement procedure for the motor or timing belt of the axis 3 or axis 4. Therefore, see the sections of their respective replacement procedures. (See "8.4.7 Removing the Axis-3 Motor", "8.4.9 Removing the Axis-4 Motor", "8.5.3 Replacing the Axis-3 Timing Belt", and "8.5.4 Replacing the Axis-4 Timing Belt.")

1 Remove the axis-2 arm cover.

See "8.3.1 Arm-2 Cover."

2 Use a plier or the like to cut off the cable straps of the cables.

It is recommended to take pictures beforehand so that where to route the cables and where to use the cable straps are identifiable in restoration work.

3 Disconnect the cable connectors from the axis-3 and axis-4 motors.

Disconnect the cable connectors of the axis 3, J3AS and J3AP (axis-3 power drive cables), J3BS and J3BP (axis-3 encoder cables) and J3DS and J3DP (axis-3 brake cables) and the cable connectors of the axis-4 motor, J4AS and J4AP (axis-4 power drive cables), J4BS and J4BP (axis-4 encoder cables).

4 Use a tension meter to measure the tension of the axis-3 timing belt.

Be sure to record the results of measuring the tension using the tension meter. (For the measurement method, see 4 in "8.4.7 Removing the Axis-3 Motor.")

5 Remove the harness guide and support guide.

Remove the harness guide secured with the hexagon socket head cap screws (M4×8×4 pcs) and the support guide secured with the hexagon socket head cap screws (M4×8×4 pcs) (see 4 in "8.4.5 Removing the Axis-2 Motor").

6 Remove the hand, etc. and remove the mechanical stopper fitted to the lower side of the axis 3.

7 Remove the axis-3 motor assembly.

See 6 in "8.4.7 Removing the Axis-3 Motor."

8 Remove the axis-4 motor assembly.

See 6 and 7 in "8.4.9 Removing the Axis-4 Motor."

9 Use a tension meter to measure the tension of the axis-4 timing belt.

Be sure to record the results of measuring the tension using the tension meter. (For the measurement method, see 11 to 14 in 8.5.4 Replacing the Axis-4 Timing Belt.")

10 Release the tension of axis-4 timing belt B.

Remove the hexagon socket head cap screws (M4×12×4 pcs) securing the axis-4 intermediate shaft to release the tension of the timing belt. (See 7 in 8.5.4 Replacing the Axis-4 Timing Belt.")

11 Pull out the ball screw nut and ball screw spline shaft.

Remove the hexagon socket head cap screws (M4×12×6 pcs), and pull out the ball screw nut and ball screw spline shaft together upward. Be careful when pulling them out.

12 Remove the axis-3 bracket from the arm 2.

Remove the hexagon socket head cap screws (M4×12×5 pcs) securing the axis-3 bracket, and remove the axis-3 bracket from the arm 2.

13 Remove the axis-3 nut pulley and the axis-3 mechanical stopper.

Remove the hexagon socket head cap screws (M4×14×4 pcs), and remove the axis-3 nut pulley secured to the ball screw nut and the mechanical stopper fitted to the upper side of the axis 3.

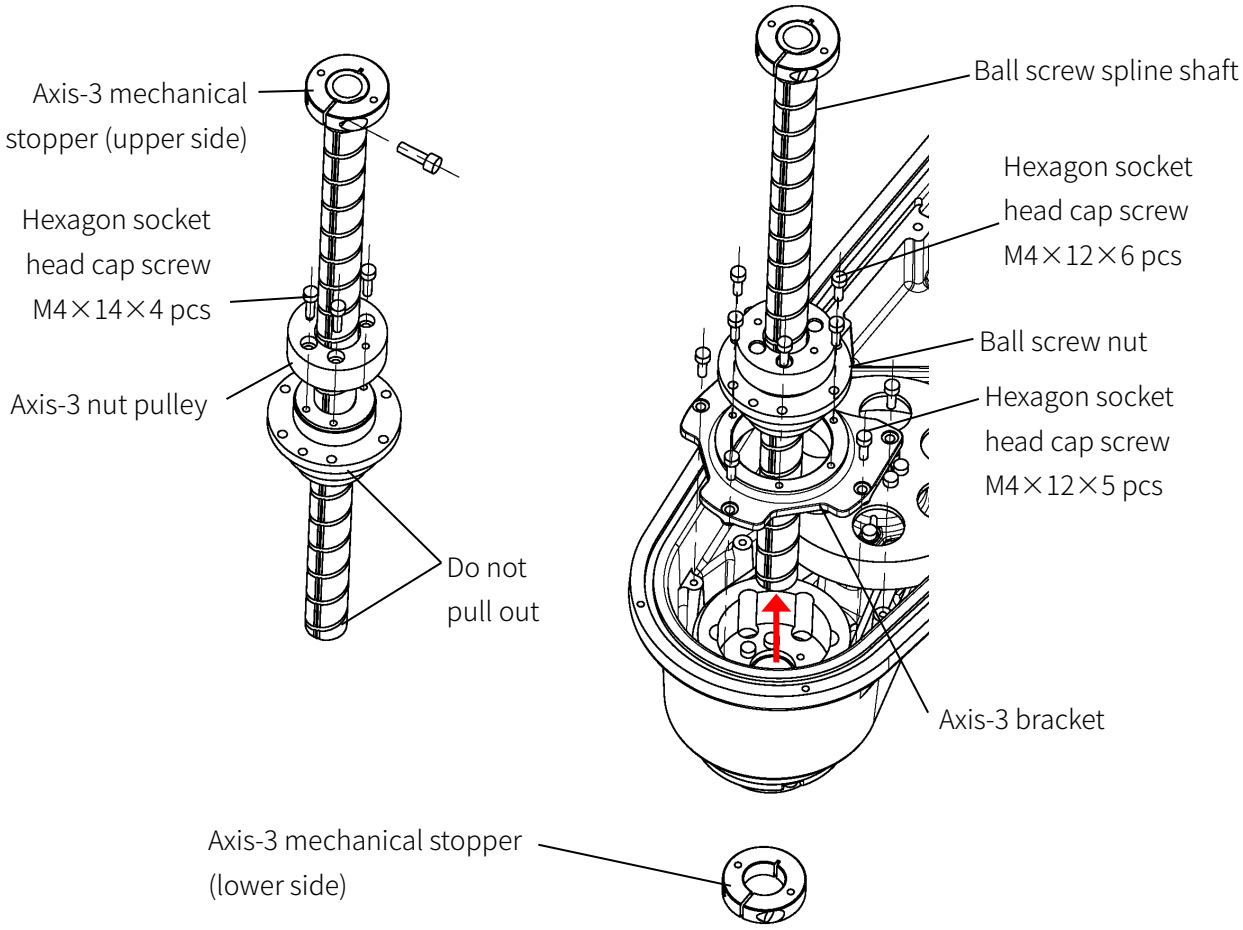


Fig. 8.54 Removing the ball screw spline unit

NOTICE

- Never pull out the ball screw nut from the ball screw spline shaft.
  Otherwise, the ball screw spline unit from which balls have come out cannot be used.

14 Remove the ball spline nut.

Remove the hexagon socket head cap screws (M4×12×6 pcs) securing the ball spline nut, and pull out the ball spline nut. If the ball spline nut is difficult to come off, hit the pulley softly with a plastic hammer or the like that is unlikely to damage the pulley to remove the ball spline nut. If the pulley is hit strongly with the hammer or the ball spline nut is forcibly pulled out, the arm 2 is deformed and a new ball screw cannot be fitted accurately. Leave the axis-4 intermediate shaft assembly inside the arm 2.

15 Remove the axis-4 nut pulley.

Remove the axis-4 nut pulley secured to the ball spline nut with the hexagon socket head cap screws (M4 ×14×4 pcs).

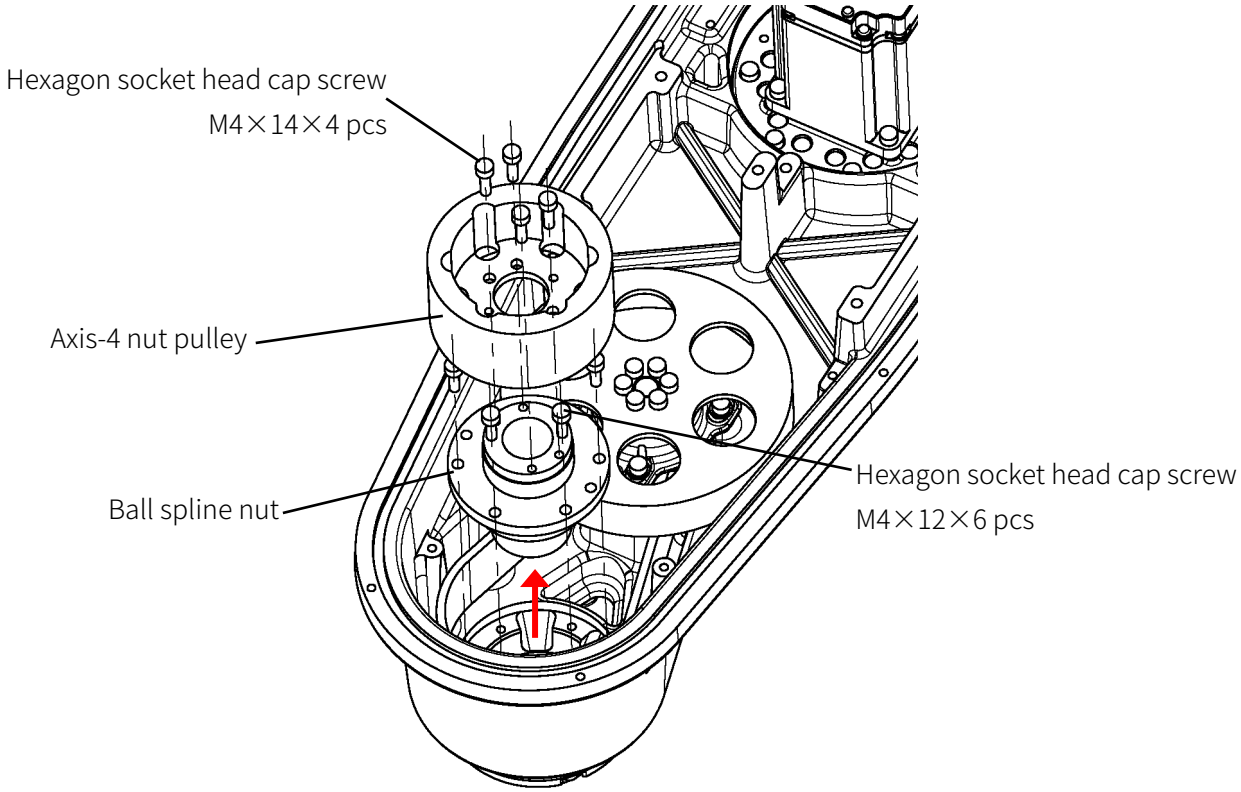


Fig. 8.55 Removing the ball screw spline nut

16 Insert the removed ball spline nut into the ball screw spline shaft.

After removing the ball spline nut, insert it into the ball screw spline shaft. When inserting it, match the phases of the marks engraved on both the ball screw spline shaft and the ball spline nut. Wrapping a cable strap around the both ends of the ball screw spline shaft prevents the nut from falling off.

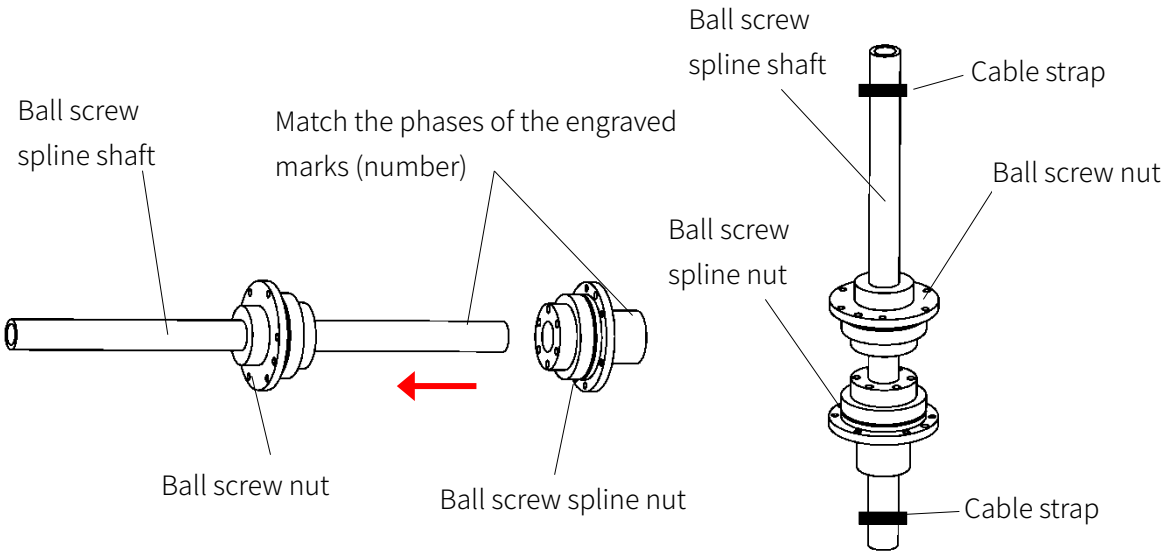


Fig. 8.56 Ball screw spline unit

8.6.5 Fitting the Ball Screw Spline Unit

⚠DANGER

- When replacing the ball screw spline, be sure to remove power plug of the controller.

The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

- Never touch the ball screw spline with bare hands.
  Touching it with bare hands causes the ball screw spline to rust.
  - Be sure to wear gloves.
- Never pull out the ball screw nut from the ball screw spline shaft.
  Otherwise, the ball screw spline unit from which balls have come out cannot be used.
- Be sure to handle the ball screw spline unit carefully.
  Drop impact or application of excessive external force could ruin the unit.
- After the ball screw spline unit is replaced, be sure to re-set the zero position.
  Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
  - For how to re-set the zero position, see “9 Robot Zero Point and Position Detector Error.”
- The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
  Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

1 Remove the ball spline nut from the new ball screw spline unit.

2 Fit the axis-4 nut pulley to the ball spline nut.

Fit the axis-4 nut pulley to the ball spline nut with the hexagon socket head cap screws (M4×14×4 pcs). The pulley rotates making it difficult to tighten the screws; therefore, secure the pulley with a plier, etc., while protecting it with a waste cloth so that it is not damaged when tightening the screws.

3 Fit the ball spline nut.

Fit the ball spline nut to the arm 2 with the hexagon socket head cap screws (M4×12×6 pcs).
Fit it so that the arm-4 timing belt in the arm 2 is engaged with the pulley.

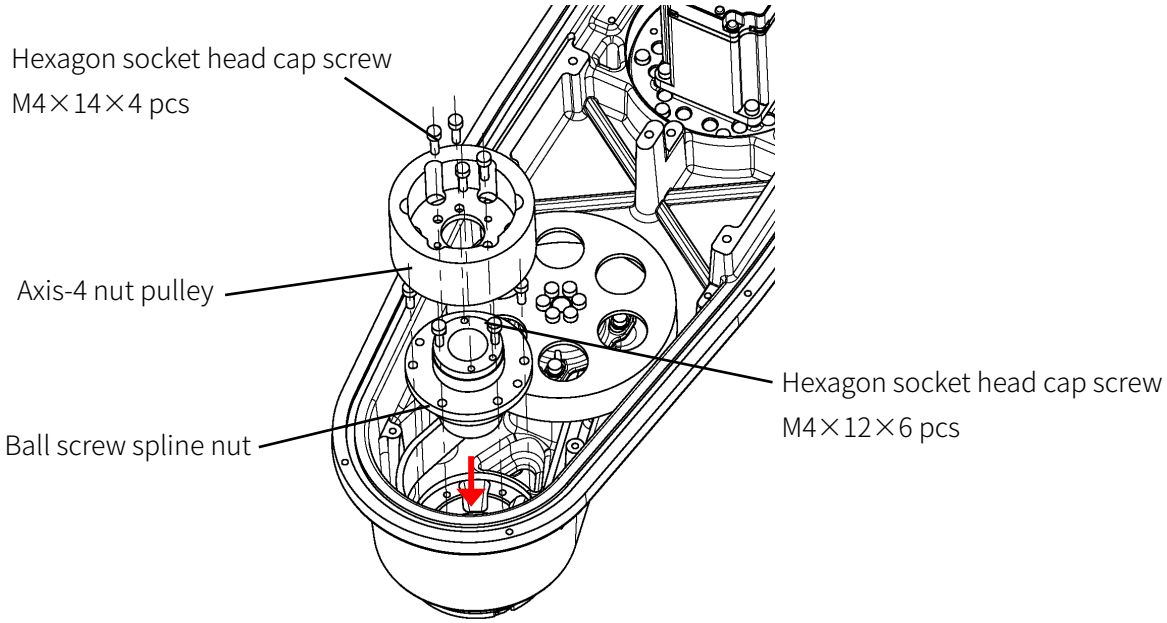


Fig. 8.57 Fitting the ball spline nut

4 Adjust the tension of the axis-4 timing belt.

See 10 to 13 in "8.5.4 Replacing the Axis-4 Timing Belt."

5 Fit the axis-4 motor assembly.

See 3 to 5 in "8.4.10 Fitting the Axis-4 Motor."

6 Temporarily fix the axis-3 bracket.

Temporarily fix the axis-3 bracket to the arm 2 with the hexagon socket head cap screws (M4×12×5 pcs).

7 Secure the axis-3 nut pulley and the axis-3 upper side mechanical stopper to the ball screw nut.

Secure the axis-3 nut pulley and the lower-limit mechanical stopper to the ball screw nut with the hexagon socket head cap screws (M4×14×4 pcs). Be careful about the fitting orientation of the upper side stopper (see Fig. 8.58).

8 Insert the ball screw nut and ball screw spline shaft into the ball spline nut.

9 Fit the ball screw nut to the axis-3 bracket.

Secure the axis-3 bracket with the hexagon socket head cap screws (M4×12×6 pcs). At this time, put the axis-3 timing belt on the axis-3 pulley.

10 Fit the arm-3 lower side mechanical stopper.

Fit the stopper to the portion of 30 mm from the lower end of the ball screw spline shaft. When fitting the stopper, align the dividing portion of the stopper with the left end of the spline groove.

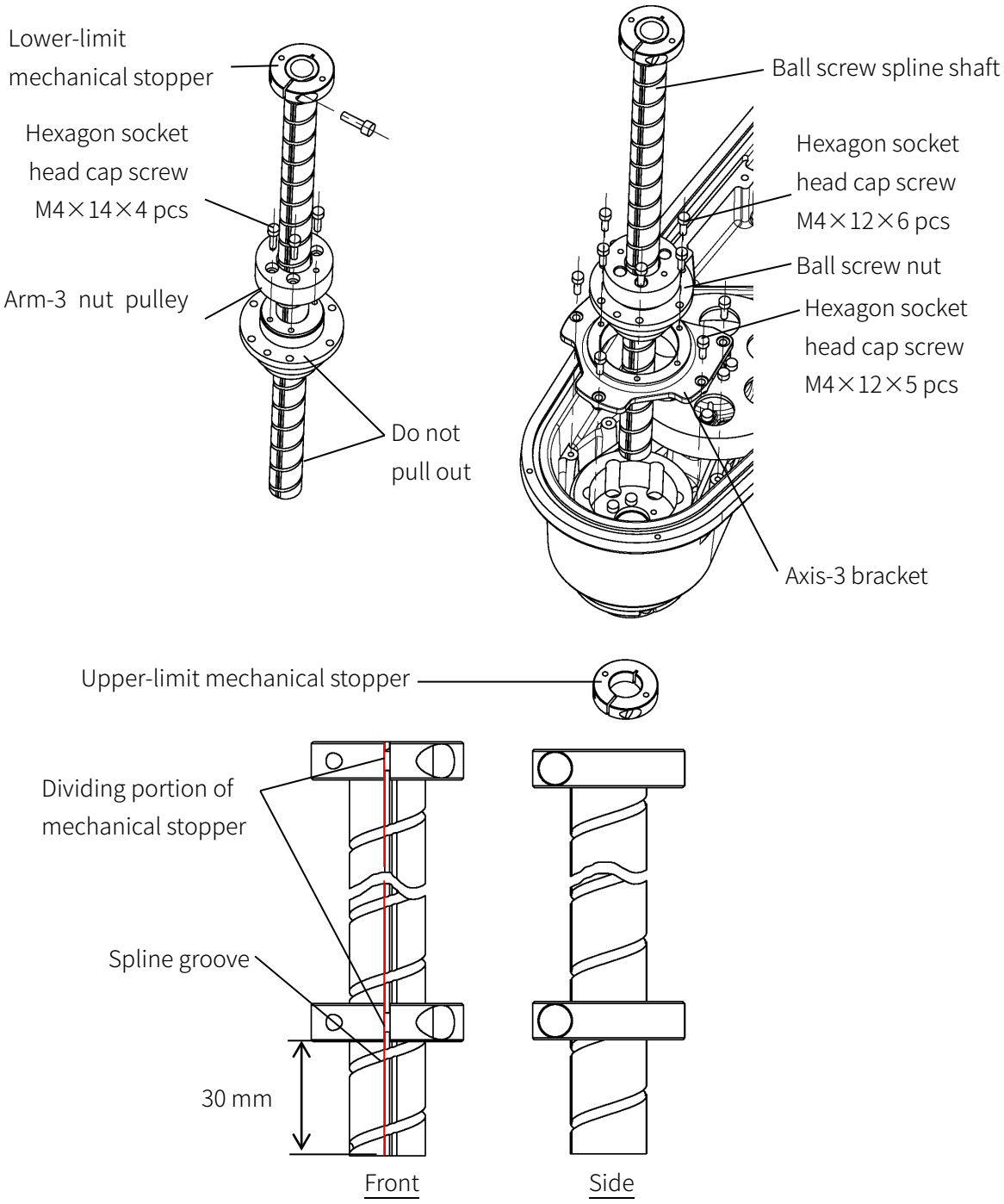


Fig. 8.58 Fitting the ball screw spline unit

11 Move up and down the ball screw spline shaft to adjust it.

After the shaft is adjusted, retighten the hexagon socket head cap screws (M4×12×5 pcs) securing the axis-3 bracket while pressing the axis-3 bracket to prevent it from moving.

12 Fit the axis-3 motor assembly and the axis-3 timing belt.

Fit the axis-3 motor assembly and the axis-3 timing belt and adjust the belt tension. (See 3 to 5 in "8.4.8 Fitting the Axis-3 Motor.")

13 Fit the harness guide and the support plate.

Secure the harness guide and support plate with the hexagon socket head cap screws (M4×8×8 pcs). (See 8 in "8.4.6 Fitting the Axis-2 Motor.")

14 Connect the disconnected connectors.

Connect the connectors and return the cable to its original state. (See the pictures taken in 2 in "8.6.4 Removing the Ball Screw Spline Unit.")

15 Fit the arm-2 cover.

See "8.3.1 Arm-2 Cover."

16 Set the zero positions of the axes 2, 3 and 4.

Turn on the power and set the zero positions of axes 2, 3 and 4. (See "9 Robot Zero Point and Position Detector Error.") When setting the zero position of the axis 4, align the dividing portion of the stopper with the zero point match-mark. After setting the zero position, attach the match-mark from the old ball spline nut to the new ball spline nut. For the location of the zero point match-mark, see "9.5.1 Locations of Zero Point Match-Marks of the Robot."

17 Check the movement of the ball screw.

Move up and down the ball screw while pressing the brake release switch in the servo power OFF condition to make sure that the ball screw moves smoothly.

18 Perform operation at a low speed.

Perform running-in operation at 20% of override for about one hour to check whether there is any problem with operation of each part.

8.7 Replacing Reduction Gears

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.

⚠DANGER

- When replacing the reduction gear, be sure to remove the power plug of the controller.
  The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

- When replacing the reduction gear, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.
- After the reduction gear is replaced, be sure to re-set the zero position.
  Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
  - For how to re-set the zero position, see "9 Robot Zero Point and Position Detector Error."

8.7.1 Types of Reduction Gears

The reduction gears used by this robot are as listed in Table 8.14.
When ordering a replacement reduction gear, specify the robot model name, manufacture number, applicable axis, SHIBAURA drawing number, and unit code.
For the location of the manufacture number label, refer to “STA1179 Safety Manual” Instruction Manual.

Table 8.14 Types of reduction gears

Part name	Applicable axis	SHIBAURA Dwg. No.	Unit code
Reduction gear	Axis 1	S968522	Y610D0CZ0
	Axis 2	S981478	Y610D0D00

8.7.2 Locations of Reduction Gears

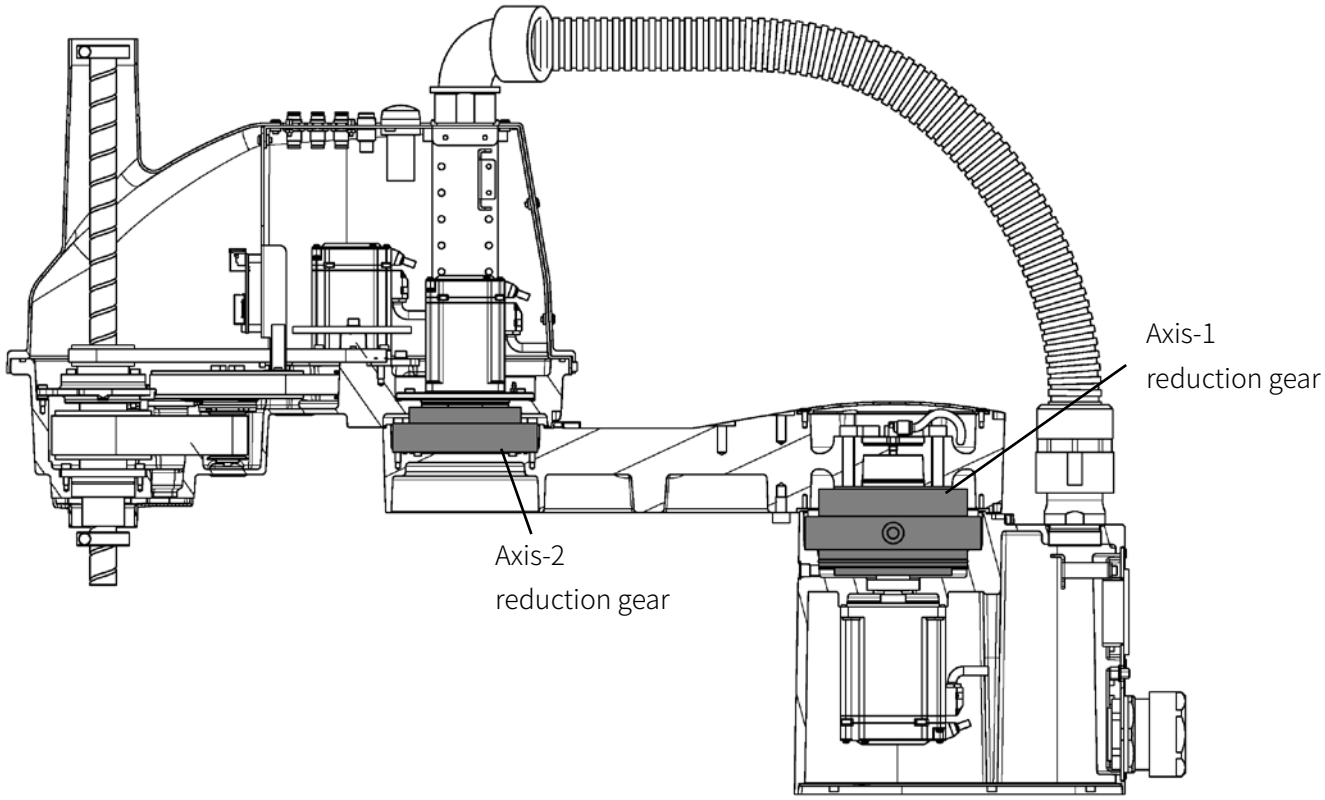


Fig. 8.59 Locations of reduction gears

8.7.3 Removing the Axis-1 Reduciton Gear

1 Remove the base rear cover and base bottom cover.

For removal method, see "8.3.3 Base Cover."

The base rear cover is connected with the connectors inside.

Therefore, do not pull it forcibly.

2 Remove the axis-1 motor.

For the removal method, see "8.4.3 Removing the Axis-1 Motor."

3 Remove the arm-1 cover.

The arm-1 cover is secured with the hexagon socket head cap screws (M4×8×6 pcs).

For the removal method, see "8.3.2 Arm-1 Cover."

4 Remove the arm 1.

The arm 1 is secured with the hexagon socket head cap (M8×50×12 pcs).

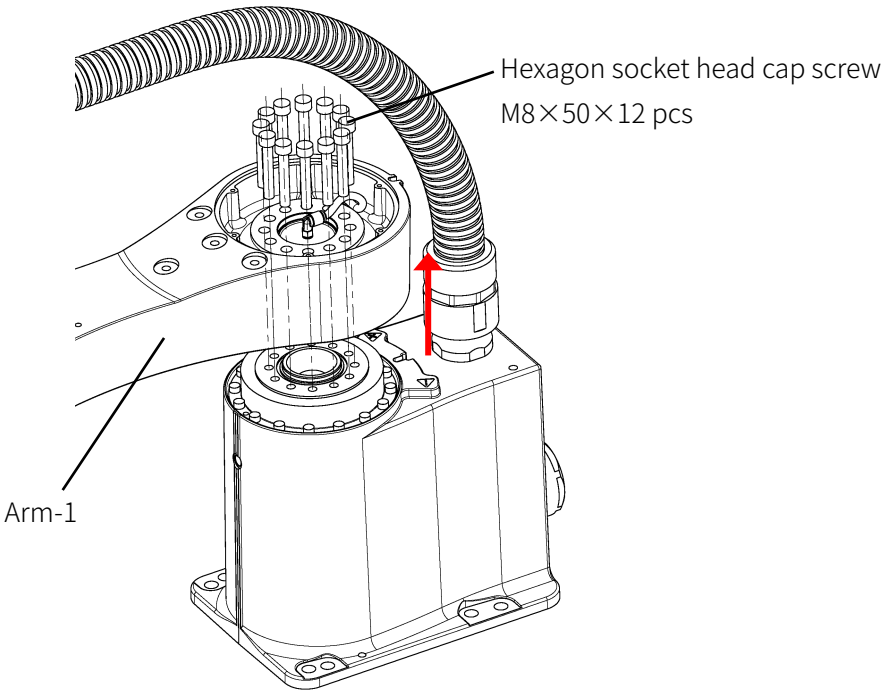


Fig. 8.60 Removing the arm 1

⚠ CAUTION

- When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
  If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

- Be sure to handle the arm carefully.
  Drop impact or application of excessive external force could ruin the arm.

5 Remove the axis-1 reduction gear.

The base and the axis-1 reduction gear are secured with the hexagon socket head cap screws (M5×35×16 pcs).

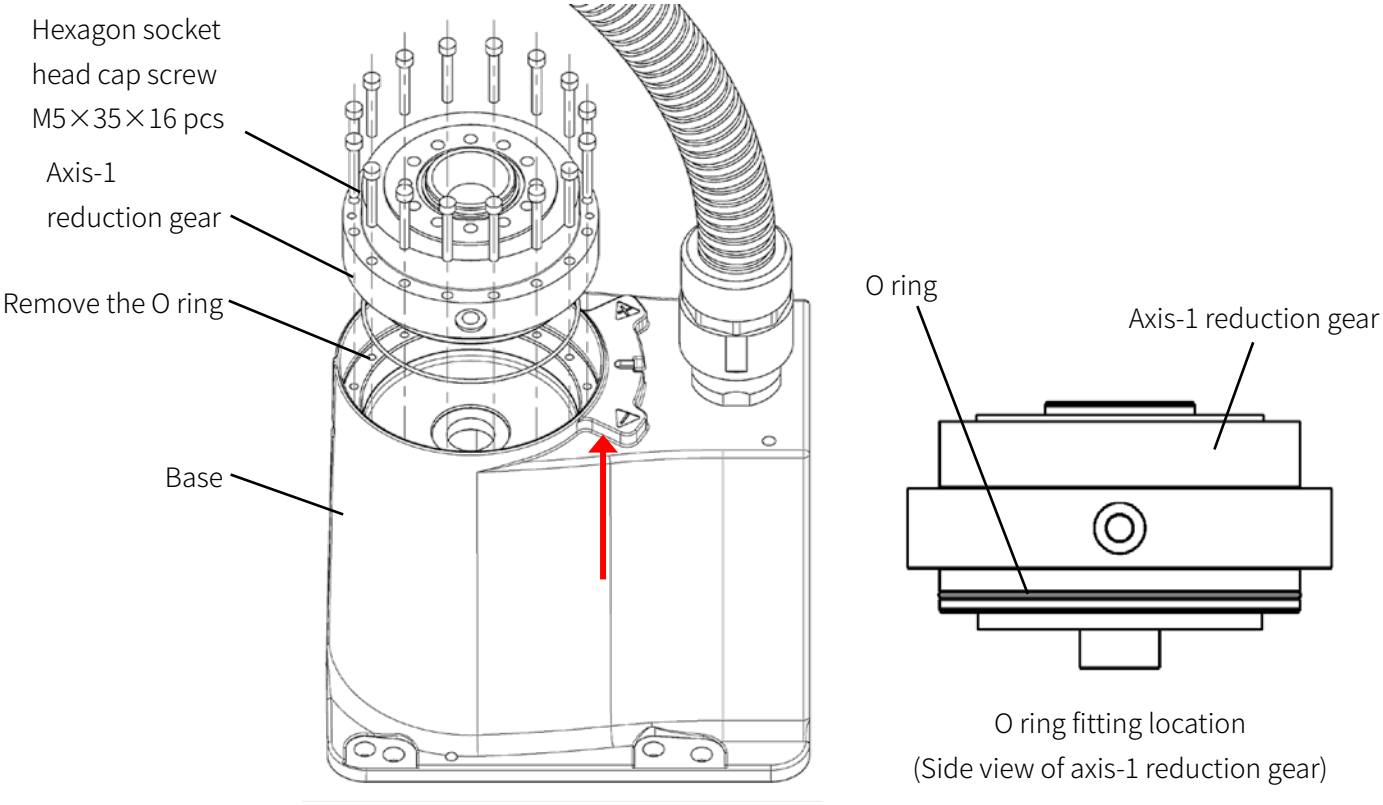


Fig. 8.61 Removing the axis-1 reduction gear

8.7.4 Fitting the Axis-1 Reduction Gear

NOTICE

- Never forget fitting the O ring, and never use the cut or damaged O ring.
  Otherwise, grease will leak from the fitting surface.
  - Use the new O ring provided for the reduction gear.
- Be sure to handle the reduction gear carefully.
  Drop impact or application of excessive external force could ruin the reduction gear.
- Be sure to use elliptical cam provided for new reduction gear to be replaced.
  The combination of the reduction gear and elliptical cam is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced.

1 Wipe off dust, dirt, remaining grease, etc. well.

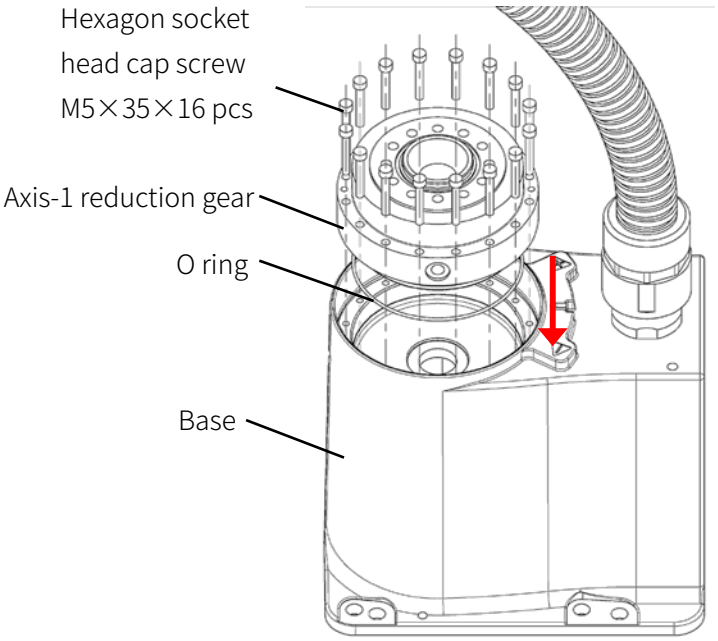
Wipe off the fitting portion of the axis-1 reduction gear in the arm 1 and the O ring groove of the new reduction gear.

2 Fit the O ring to the axis-1 reduction gear.

Apply grease to the O ring provided for the new reduction gear, and fit it into the O ring groove.

3 Fit the arm-1 reduction gear.

Fit the arm-1 reduction gear with the hexagon socket head cap screws (M5×35×16 pcs). When fitting it, create the tool shown in Fig. 8.63, and fit it.



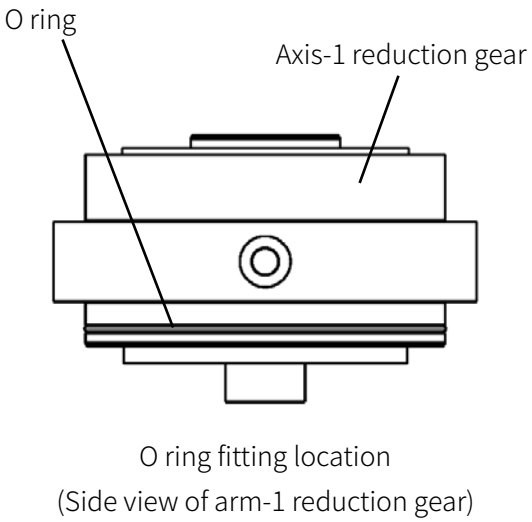


Fig. 8.62 Fitting the axis-1 reduction gear

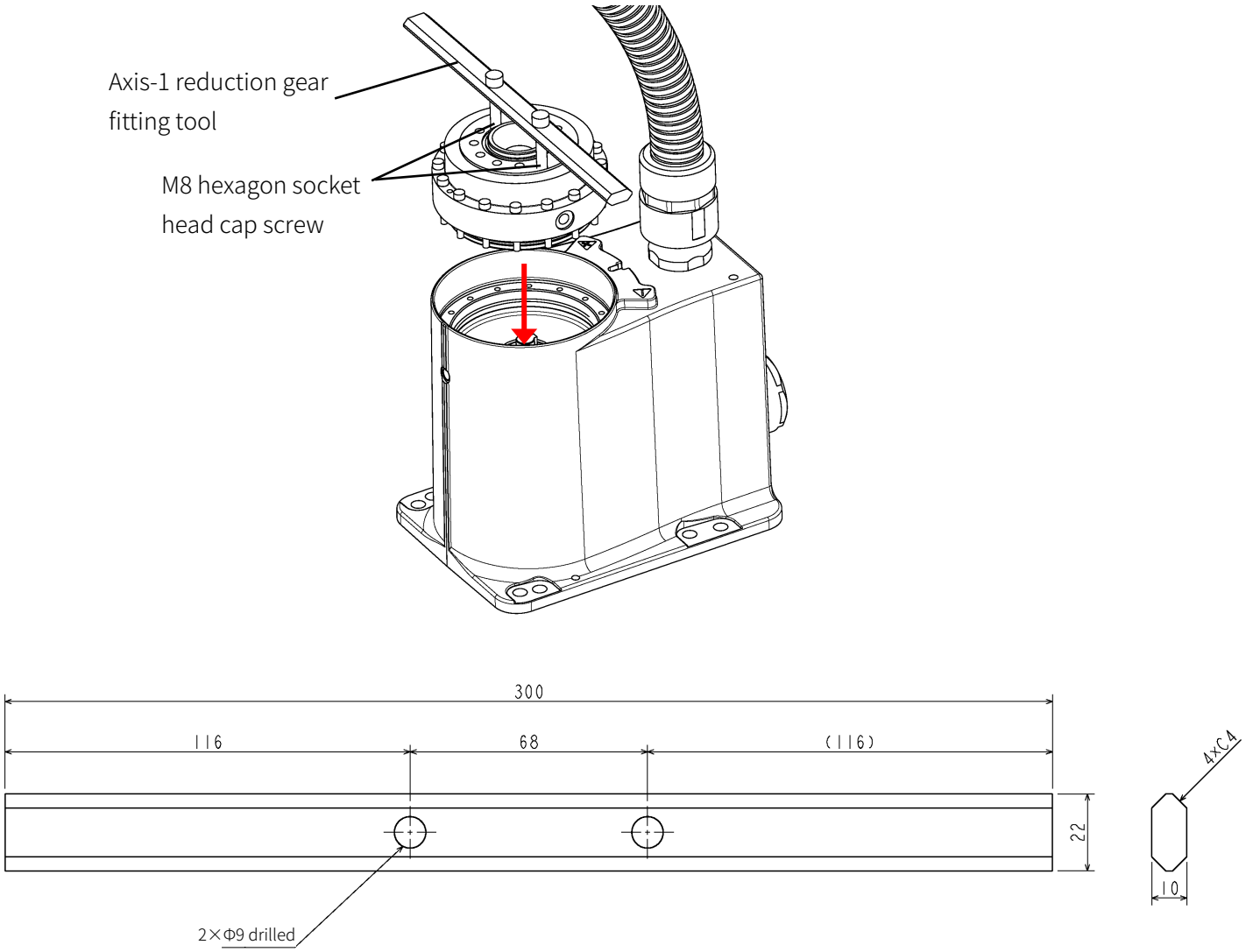


Fig. 8.63 Example of creating axis-1 reduction gear fitting tool

4 Fit the axis-1 motor to the axis-1 reduction gear.

For the fitting method, see "8.4.4 Fitting the Axis-1 Motor."

5 Fit the base rear cover and base bottom cover to the base.

For the fitting method, see "8.3.3 Base Cover."

6 Fit the arm 1 to the arm-1 reduction gear.

Fit the arm 1 with the hexagon socket head cap screws (M8×50×12 pcs).

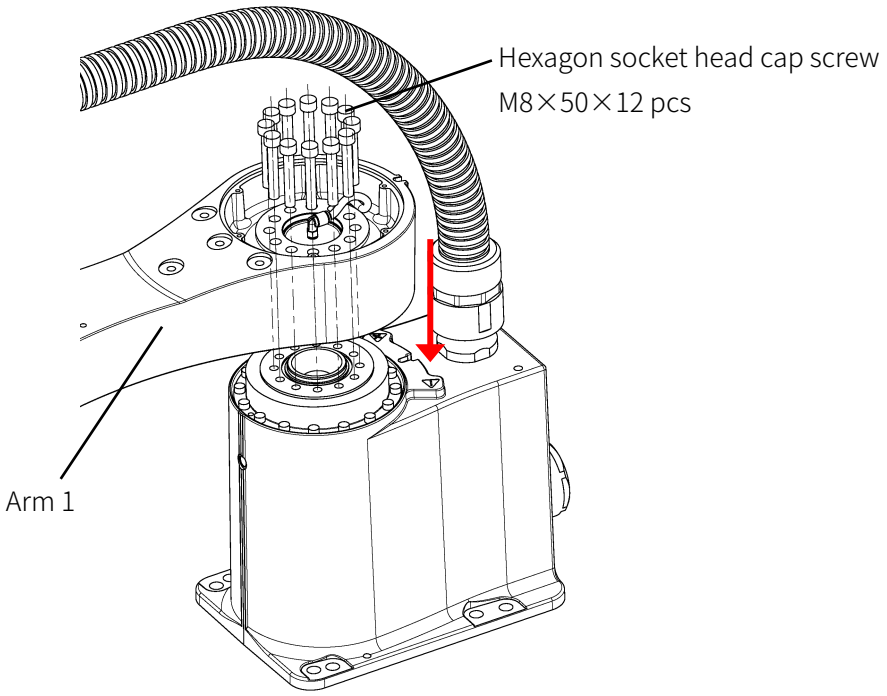


Fig. 8.64 Fitting the arm 1

7 Fit the arm-1 cover to the arm 1.

For the fitting method, see "8.3.2 Arm-1 Cover."

8 Remove the hexagon socket head sunk plug (Rc1/8) from the front of the base.

9 Fit the user-prepared grease nipple.

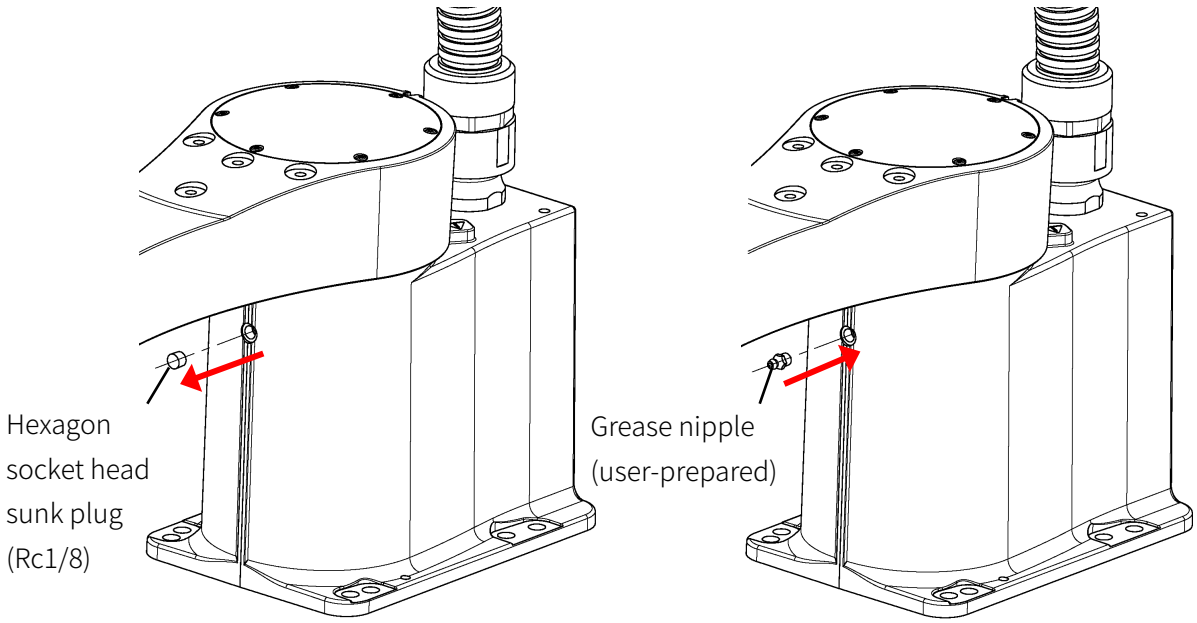


Fig. 8.65 Fitting grease nipple

10 Fill grease in the arm-1 reduction gear.

Use the grease gun to fill grease. For the filling amount, see Table 8.15.

11 Remove the grease nipple.

12 Fit the hexagon socket head sunk plug to the base.

Wind the seal tape around the hexagon socket head sunk plug to prevent grease from leaking.

Table 8.15 Filling amount of grease

Recommended grease	Maker	Filling amount
Multemp FZ No.00	Kyodo Yushi Co., Ltd.	60 g

13 Move the arm 1 by hand.

Make sure that it moves smoothly.

14 Set the zero position of the arm 1.

For how to set the zero position, see "9.8 ②Restoring Zero Position Data by ZEROP Function Using the Zero Setting Tool."

The replacement of the axis-1 reduction gear.

15 Perform operation at a low speed.

Perform running-in operation at 20% of overrun for about one hour.

Make sure that there is any problem with operation of each part.

NOTICE

- Be sure to use the specified types of grease.
  Using grease of other than specified types causes malfunction due to poor lubrication.
- Be sure to observe the filling amount of grease.
  If the filling amount is tool large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the filling amount is too small, malfunction is caused by poor lubrication.

8.7.5 Removing the Axis-2 Reduciton Gear

1 Remove the arm-2 cover.

For the removal method, see "8.3.1 Arm-2 Cover."

2 Use a plier or the like to cut off the cable straps of the cables.

It is recommended to take pictures beforehand so that where to route the cables and where to use the cable straps are identifiable in restoration work.

3 Disconnect J2AS and J2AP (axis-2 power drive cables) and J2BS and J2BP (axis-2 encoder cables).

4 Remove the harness guide and support plate.

5 Remove the axis-2 motor assembly.

Remove the O ring (CO0543A) from the O ring groove as well. (See "8.4.5 Removing the Axis-2 Motor.")

6 Remove the arm 2.

Remove the hexagon socket head cap screws (M4×20×16 pcs) securing the arm 2 and reduction gear.

7 Remove the axis-2 reduction gear.

Remove the hexagon socket head cap screws (M4×20×16 pcs) securing the arm 1 and axis-2 reduction gear.

Remove the O ring (provided for the reduction gear) from the O ring groove as well.

8 Remove the elliptical cam from the axis-2 motor assembly.

For the removal method, see "8.4.5 Removing the Axis-2 Motor."

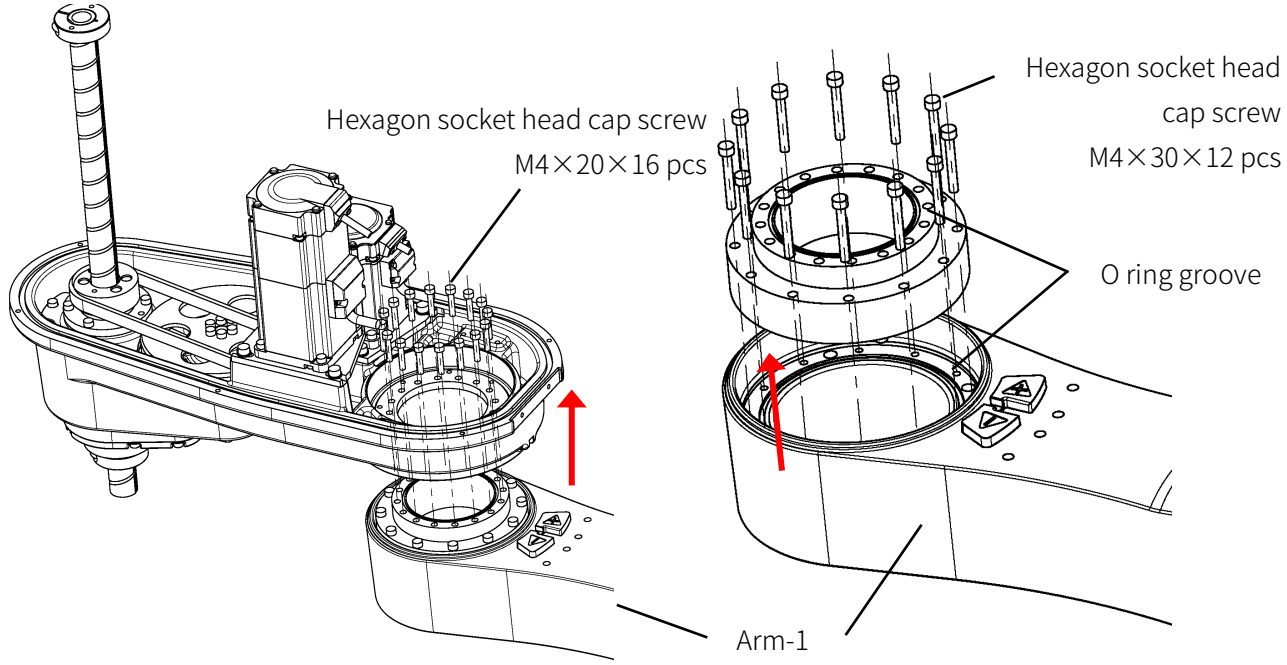


Fig. 8.66 Removing the axis-2 reduction gear

⚠ CAUTION

- When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
  If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

- Be sure to handle the arm carefully.
  Drop impact or application of excessive external force could ruin the arm.

8.7.6 Fitting the Axis-2 Reduction Gear

NOTICE

- Never forget fitting the O ring, and never use the cut or damaged O ring.
  Otherwise, grease will leak from the fitting surface.
  - Use the new O ring provided for the reduction gear.
- Be sure to handle the reduction gear carefully.
  Drop impact or application of excessive external force could ruin the reduction gear.
- Be sure to use the elliptical cam provided for the new reduction gear to be replaced.
  The combination of the reduction gear and elliptical cam is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced.

1 Fit the elliptical cam to the axis-2 motor assembly.

Use the elliptical cam provided for the new reduction gear. (See "8.4.6 Fitting the Axis-2 Motor.")

2 Wipe off dust, dirt, remaining grease, etc.

Wipe off the fitting portion of the reduction gear in the arm 1 and the O ring groove of the axis-2 reduction gear.

3 Fit the O ring to the arm 1 and axis-2 reduction gear.

Use the O ring provided for the new reduction gear to be replaced.
Before fitting, apply grease to the O ring.

4 Fit the axis-2 reduction gear to the arm 1.

Fit it to the arm 1 with the hexagon socket head cap screws (M4×20×16 pcs).

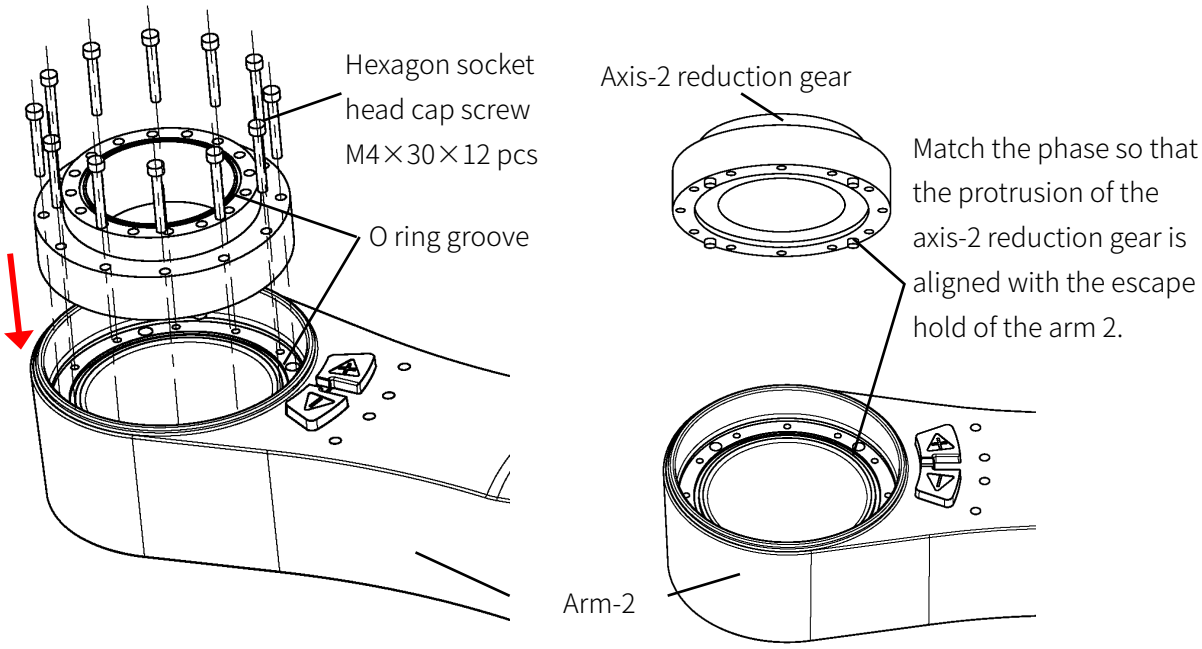


Fig. 8.67 Fitting the Axis-2 Reduction Gear

5 Apply grease to the axis-2 reduction gear and axis-2 motor assembly.

Apply grease as shown in Fig. 8.68.

For the application amount, see Table 8.16.

Table 8.16 Application amount of grease

Recommended grease	Maker	Application amount to axis-2 reduction gear side	Application amount to axis-2 motor side
SFB No. 1	Nidec-Shimpo Corp.	40 g	12.5 g

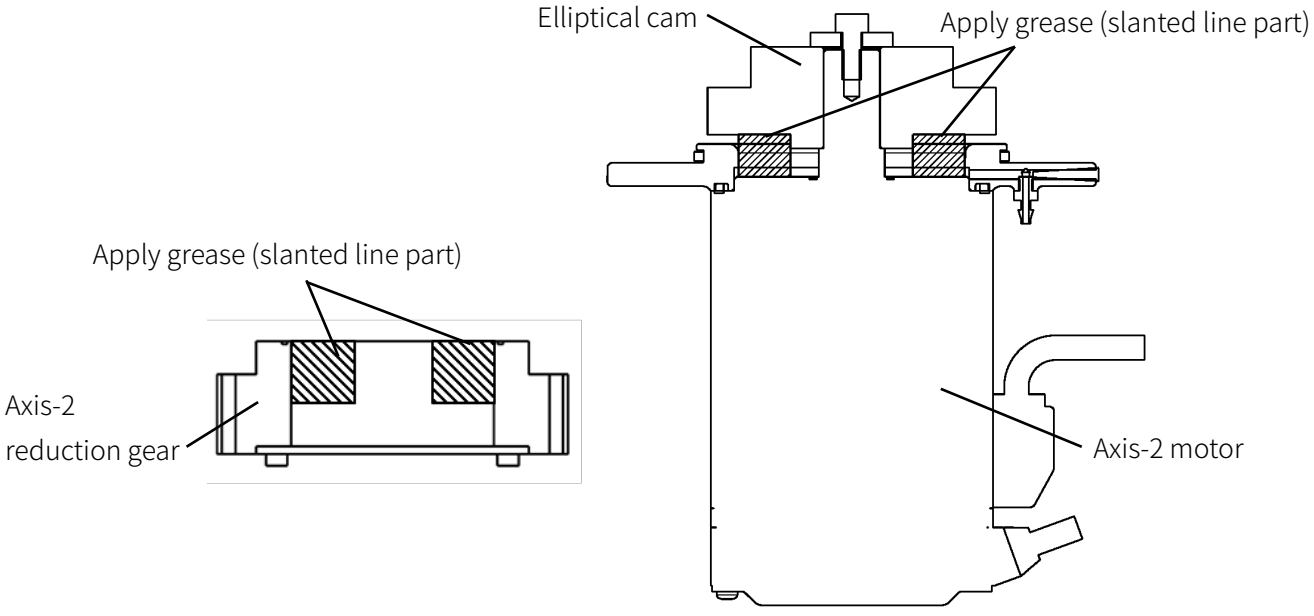


Fig. 8.68 Applying Grease to Axis-2 Reduction Gear

NOTICE

- Be sure to use the specified types of grease.
  Using grease of other than specified types cause malfunction due to poor lubrication.
- Be sure to observe the application amount of grease.
  If the application amount is too large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the application amount is too small, malfunction is caused by poor lubrication.

6 Fit the arm 2 to the axis-2 reduction gear.

Fit the arm 2 with the hexagon socket head caps (M4×20×16 pcs).

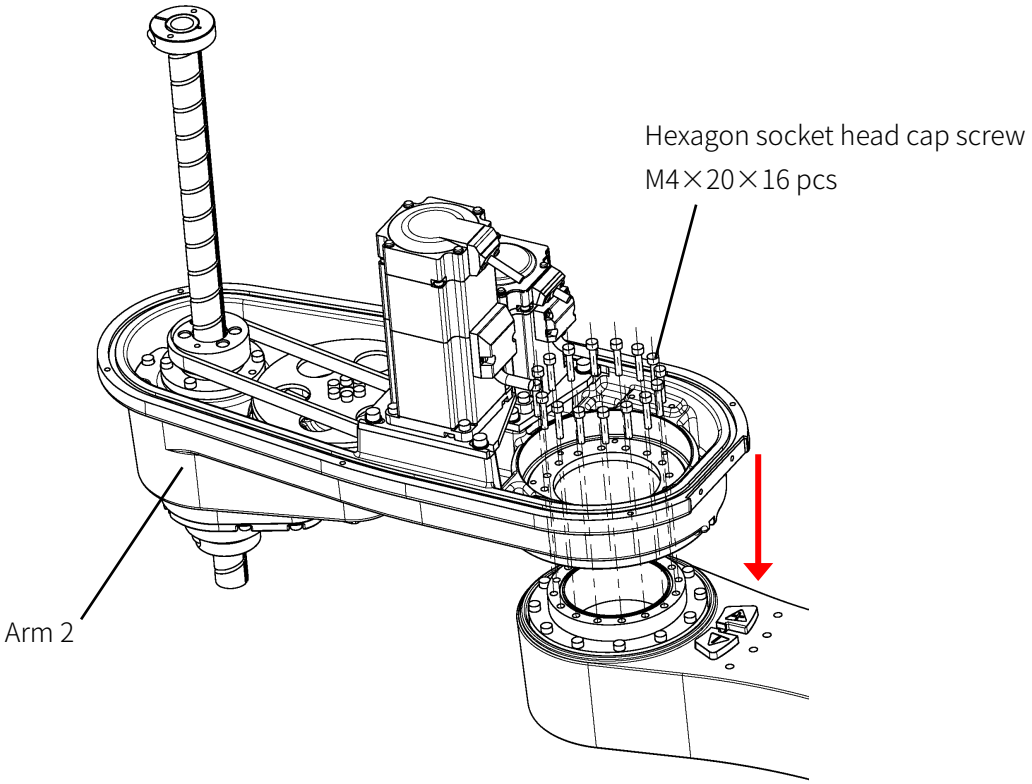


Fig. 8.69 Fitting the Arm 2

7 Fit the axis-2 motor assembly to the arm 2.

For the fitting method, see "8.4.6 Fitting the Axis-2 Motor."

8 Move the arm 2 by hand.

Make sure that it moves smoothly.

9 Fit the harness guide and support plate.

10 Connect the connectors.

See the pictures taken in 2 in "8.7.5 Removing the Axis-2 Reduciton Gear" to return the cable to its original state.

11 Fit the arm-2 cover.

For the fitting method, see "8.3.1 Arm-2 Cover."

12 Turn on the power.

13 Set the zero position of the axis 2.

For how to set the zero position, see "9.8 ②Restoring Zero Position Data by ZEROP Function Using the Zero Setting Tool."

The replacement of the axis-2 reduction gear is completed.

14 Perform test operation on the axis 2

Make sure that there isn't any problem with operation of each part.

8.8 Replacing the LED Lamp

The work should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.

⚠DANGER

- When replacing the LED lamp, be sure to remove the power plug of the controller.
  The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

- When replacing the LED lamp, be sure to remove the power plug of the controller.
  If replacement work is done with the power turned on, the robot may malfunction.

The LED lamp used by this robot is as listed in Table 8.17.

When ordering the LED lamp, specify the robot model name, manufacture number, SHIBAURA drawing number and unit code.

Table 8.17 Type of LED lamp

Part name	Voltage	SHIBAURA Dwg. No.	Unit code
Lamp assembly	24 V	M342499	Y610D0DU0

8.8.1 Location of LED Lamp

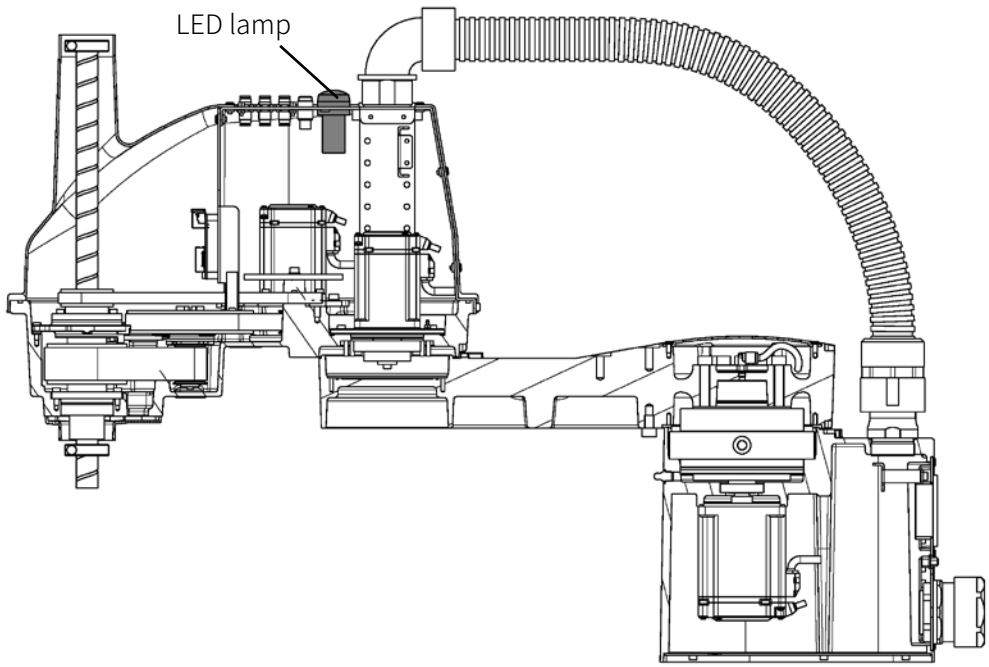


Fig. 8.70 Location of the LED lamp

8.8.2 Removing the LED Lamp

1 Remove the arm-2 cover.

For the removal method, see "8.3.1 Arm-2 Cover."

2 Use a plier or the like to cut off the cable straps of the cables.

It is recommended to take pictures beforehand so that where to route the cables and where to use the cable straps are identifiable in restoration work.

3 Remove the LED lamp.

The "LED" seal is attached to the connector.

4 Remove the harness guide and support guide.

The harness guide is secured with the hexagon socket head cap screws (M4×8×4 pcs).

The support guide is secured with the hexagon socket head cap screws (M4×8×4 pcs).

For the removal method, see 4 in "8.4.5 Removing the Axis-2 Motor."

5 Remove the LED lamp cable.

The LED lamp cable is secured with the screws (M4×2 pcs) on the bottom.

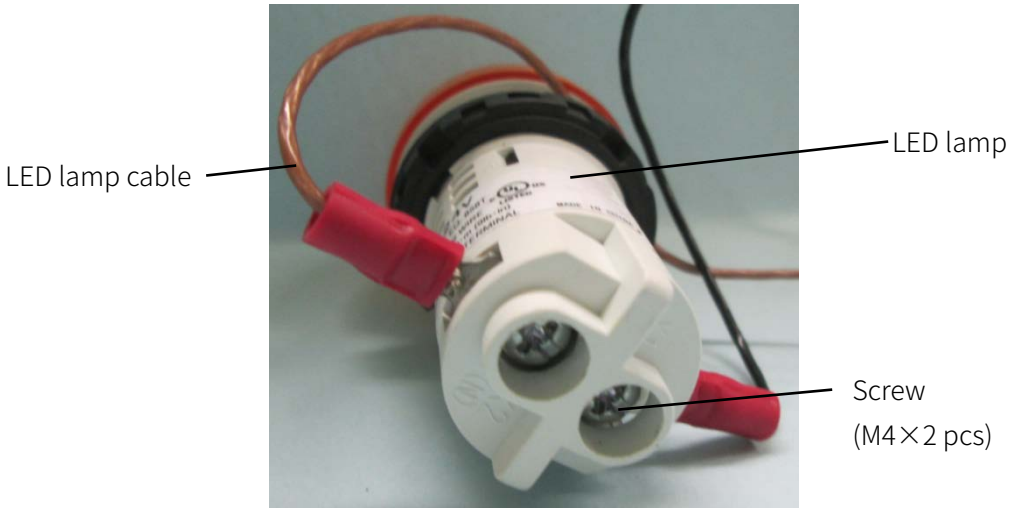


Fig. 8.71 Removing the LED Lamp Cable

6 Remove the LED lamp.

The LED lamp is secured with the clamp.

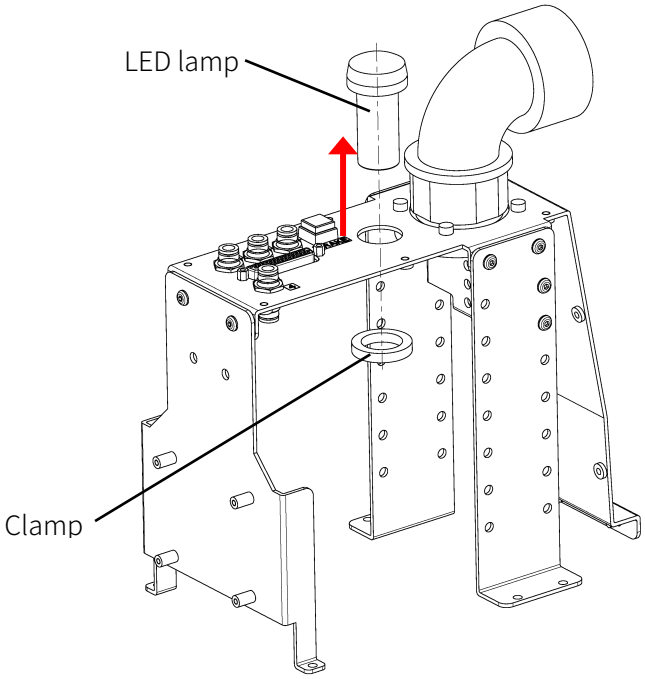


Fig. 8.72 Removing the LED Lamp

8.8.3 Fitting the LED Lamp

1 Fit the LED lamp.

Fit the LED lamp with the clamp.

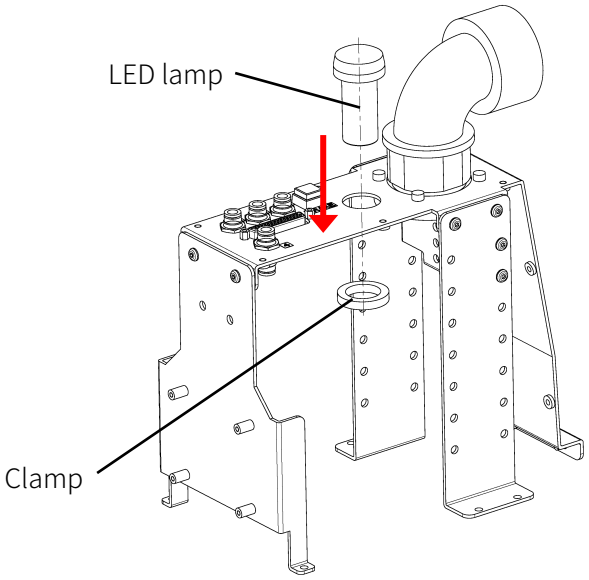


Fig. 8.73 Fitting the LED Lamp

2 Fit the LED lamp cable terminal to the LED lamp.

Loosen the screws (M4×2 pcs) on the bottom of the LED lamp, insert the Y-type terminal, and tighten the screws (M4×2 pcs) to secure the LED lamp cable. (See Fig. 8.71. Any secured combination is acceptable.)

3 Fit the harness guide and support guide.

Fit them with the hexagon socket head cap screws (M4×8×8 pcs).

For the fitting method, see 8 in "8.4.6 Fitting the Axis-2 Motor."

4 Connect the connector.

Connect the LED lamp cable to which the "LED" seal is attached.

5 Return the cable to its original state. (See the pictures taken in 2 in "8.8.2 Removing the LED Lamp.")

6 Fit the arm-2 cover.

For the fitting method, see "8.3.1 Arm-2 Cover"

7 Make sure that the LED lamp lights.

Turn on the power and turn on the servo power.

8 Perform test operation.

Check whether there is any problem with operation of each part.

9. Robot Zero Point and Position Detector Error

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the work done by the customer.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hand and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feed and toes Use: Protect from a falling object	

9.1 Zero Setting of Robot

Before delivery from our plant, zero setting is performed for the robot after its arm is secured with clamp for zero setting.
At the time of zero setting, position data of the motor position detector (i.e., encoder) is backed up by batteries, and coordinates of the robot need not be set each time the power is turned on.
The position data of the motor position detector comes in the two (2) types; servo offset and multi-turn data.

Servo offset: At zero setting operation of the robot (ZEROP operation and REORG operation), the data is written into the motor inside the robot.

Multi-turn data: The data is backed up in the encoder by backup using the robot body side batteries. If the battery voltage drops, correct multi-turn data cannot be guaranteed, and so the encoder itself outputs an alarm. This data is set to "0" during operation for the zero setting of the robot (ZEROP operation and REORG operation). It is also set to "0" during operation for multi-turn data reset.

9.2 Position Detector Error

If a position detector error (encoder error) occurs, errors listed in the table below are shown on the error screen of the robot controller.

Error code	Description	Remarks
8-601	Joint shaft 1 multi count error	
8-602	Joint shaft 2 multi count error	
8-603	Joint shaft 3 multi count error	
8-604	Joint shaft 4 multi count error	

The position detector error is an error detected by the encoder itself.
If the machine coordinates have changed without causing a position detector error, the mechanical connecting position may have shifted (e.g.: Tooth skip of timing belt).
Errors detected by the encoder itself include the battery voltage drop, error caused by temperature rise in the encoder, counter overflow, and internal counter data inconsistency. Among these errors, the error which occurs most frequently is the battery voltage drop which is caused by the absence of maintenance at specified change intervals due to a long-term shutdown or by cable breakage.
To prevent the position detector error, therefore, replace the batteries on a regular basis. For the battery replacement, see "8.3 Replacing Position Detector (or Encoder) Batteries".
Also, a heavy vibration should not be exerted on the robot or the robot should not be moved suddenly by hand during power OFF. Especially, when transporting and storing the robot, secure the robot in a posture for shipment, using the attached clamps.

Restoration from position detector error

There are the following threw methods to recover from a position detector error. Confirm the occurrence of a position detector error on the error screen, and check the type of the position detector error and position data on the encoder status screen. Then, perform restoring operations. For the encoder status screen, see "9.6 Encoder Status Screen".

① Multi-turn data reset operation: See Para. 9.7.
② ZEROP operation using the zero setting tool: See Para. 9.8.
③ Other zero setting operation: See Para. 9.9.

[1] When zero setting is not required

[1.1] The battery has been replaced at the time of annual inspection or when the battery error code is level-1 alarm (see 9.3.3 Battery Error Code)
After the battery is replaced, the alarm must be reset. The alarm is reset on the currently occurring alarm screen displayed when the alarm button is tapped. The alarm can be reset by tapping the alarm reset button on the currently occurring alarm screen. Then, perform "9.10 Check Work."
If the alarm is not reset, contact the SHIBAURA MACHINE Service Department.

[2] When ① multi-turn data reset operation is required

[2.1] If the battery is replaced when the battery error code is level-8 alarm (see 9.3.3 Battery Error Code)
[2.2] If the body harness and capacitor are replaced
[2.3] If the cable connectors of encoder cable and battery cable shown in Fig. 9.1 are removed
After each work is finished, perform ①multi-turn data reset operation (see 9.7 Restoring Zero Position Data by ① Multi-turn Data Reset).

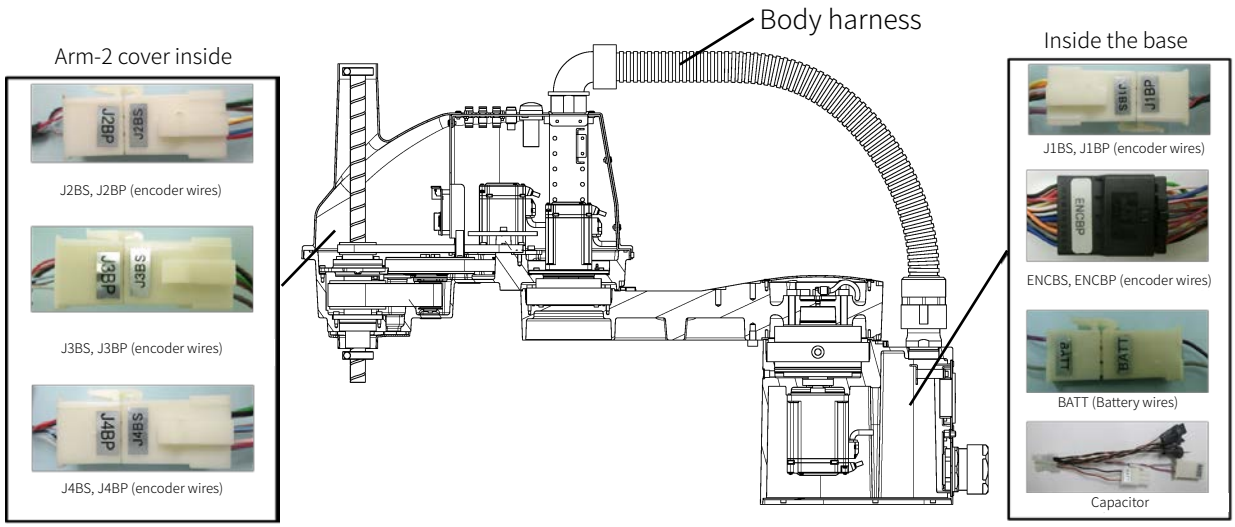


Fig. 9.1 Location where ① Multi-turn Data Reset Operation is Required when the Connectors are Removed

[3] When ② ZEROP operation using the zero setting tool or ③ other zero setting operation is required

[3.1] If the parts shown in Fig. 9.2 are replaced

After the parts are replaced, perform ② ZEROP operation using the zero setting tool (see 9.8 ② Restoring Zero Position Data by ZEROP Function Using the Zero Setting Tool) or ③ other zero setting operation (see 9.9 ③ Other Zero Setting).

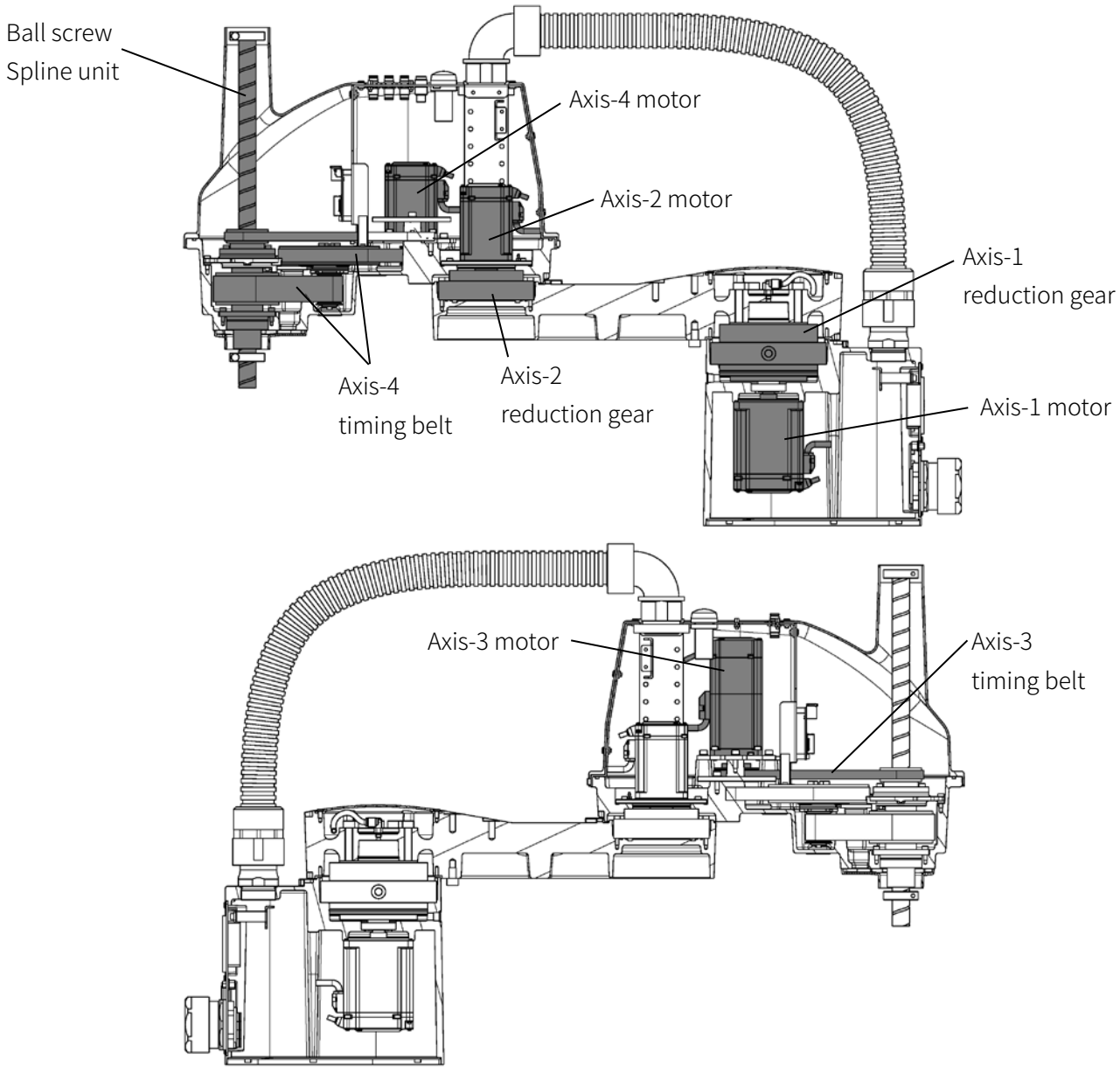


Fig. 9.2 Replacement Parts that Require ② ZEROP Operation Using the Zero Setting Tool or ③ Other Zero Setting Operation

⚠DANGER

- When performing maintenance with the power turned on in the dangerous range of the robot, be sure to switch the master mode key inserted to the teach pendant to “MANUAL” (for TP5000) to pull out the key and carry it with the teach pendant.
  Otherwise, the robot that has operated unpredictably may collide with a worker.
  - The worker should receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.
  - Display the fact that work is in progress near the controller.
- When approaching the robot being stopped, be sure to check the robot operation status.
  If the robot is under operation, the robot that has restarted operation may collide with the worker.
  - Check the operation status indicated on the indicator lamp of the controller and on the teach pendant.
  - The robot operation status is output by signals (servo operation preparation completion, external mode, operation preparation completion, automatic operation, failure and emergency stop). Connect to the display equipment, etc.
- When performing maintenance with the power turned on in the dangerous range of the robot, be sure to arrange a work observer outside the dangerous range of the robot.
  Otherwise, when a worker in the dangerous range of the robot suffers damage, the worker cannot be rescued.
  - The observer should secure safety of the worker and the third party who approaches the dangerous range of the robot.
  - The observer should also receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.

⚠ WARNING

- When rotating the axis 4 by hand, never insert your hand or finger inside the upper and lower mechanical stoppers or under the robot hand.
  Otherwise, your hand or finger may get caught in the mechanical stopper or robot hand.
  - When the axis 4 rotates, the ball screw spline shaft will move up and down.
  - The axis-4 motor is not equipped with the brake. If the robot hand, etc. is touched by hand when the servo power is off, the axis 4 will rotate.
- When releasing the motor brake, be sure to support the ball screw spline shaft.
  Otherwise, the shaft will drop by its own weight concurrently with release of the brake.
  - A worker should support the shaft to prevent hand or finger from getting caught.
  - When something strange is felt during work, release the brake release switch immediately. The brake will be also applied by turning off the power to the controller.
  - The brake release switch is located on the arm-2 top face of the robot. Do not press the switch by mistake.
  - The worker should receive safety training regarding industrial robot teaching- or inspection-related work defined by each country's laws and regulations.

⚠ CAUTION

- Be sure to start maintenance when the surface temperature becomes less than 50°C after power off.
  Otherwise, touching hot motor or periphery of the motor results in burn injury.

NOTICE

- If zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of the teaching point.
  The zero position shifts slightly due to error in setting. If the zero position is used as is, the robot will collide with the peripheral equipment.

9.3 Replacing Position Detector (or Encoder) Batteries

⚠DANGER

- Never touch liquid that leaked from the battery.
  If liquid adheres to your skin or eye, inflammation results.
  - When liquid adheres to your skin, wash it away with flowing water immediately.
  - When liquid adheres to your eye, wash it away with flowing water immediately and then see a doctor.
  - Replace the parts to which liquid has adhered.

NOTICE

- Be sure to replace the position detector battery during annual inspection.
  If position detector data is lost, the robot cannot be operated.
  - When the battery voltage has dropped, a “battery alarm” will occur. At this stage, data is not lost. If the batteries are replaced, the alarm is automatically released.
  - When the battery voltage has dropped further, a “battery error” will occur. At this stage, data is lost. A position detector error alarm will also occur.
  - If data is lost, re-set the zero position. For how to set the zero position, see “9 Robot Zero Point and Position Detector Error.”
- When the robot is not used with the power turned off, be sure to replace the position detector battery every two (2) months.
  Because the battery consumption becomes large in the power off condition, position detector data will be lost before periodic inspection.

9.3.1 Location of Battery Box

The battery box is secured to the base rear cover on the base rear.

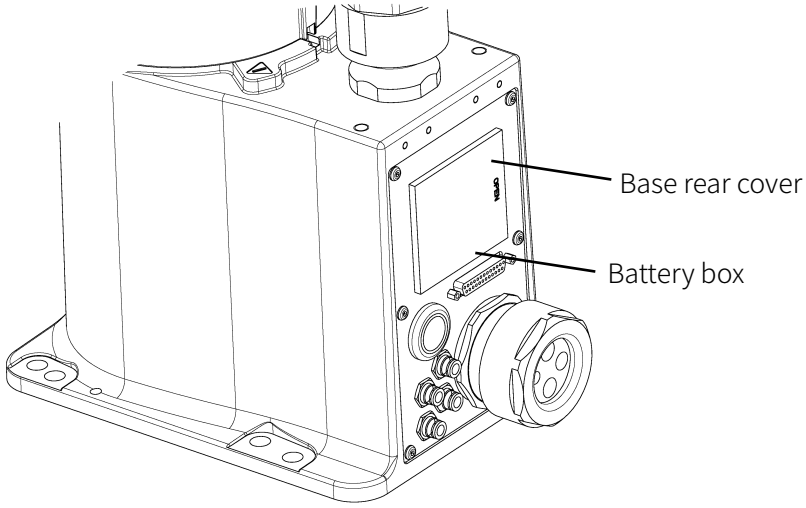


Fig. 9.3 Location of the Battery Box

9.3.2 Replacing Batteries

⚠DANGER

- When replacing the battery with the power kept on, be sure to emergency-stop the robot.
  Otherwise, the robot that has operated unpredictably will collide with a worker.
  - The worker should receive special training, etc. regarding industrial robot inspection-related work defined by each country's laws and regulations.

1 If safety can be ensured, keep the power on and press the emergency stop button (move the emergency stop switch to the ON position) to place the robot in stop state. If safety cannot be ensured, turn off the power.

2 Open the cover of the battery box, and take out the battery.

3 Fit three new batteries, being careful about their polarity.

MEMO

If you replace the batteries with the power turned off, finish the replacement within 5 minutes.

4 Reset the alarm.

If the alarm "1-40* Joint Axis* Encoder Battery low," which is listed in Table 9.1 does not disappear, take step 5. If the alarm disappears, close the lid to finish the replacement.

5 Remove the base rear cover (see Fig. 9.3).

As shown in the following picture, pull out the connector and put the terminals of a tester to measure voltage.

6 Make sure that the voltage is about 4.5 V.

If the voltage is low, it may be caused by a poor fit.

7 If the voltage is low, remove the batteries, wipe the metal battery contact of the battery box with waste cloth, and then fit the batteries again.

Measure the voltage again using a tester, and if the voltage is about 4.5 V, take step 4 again.

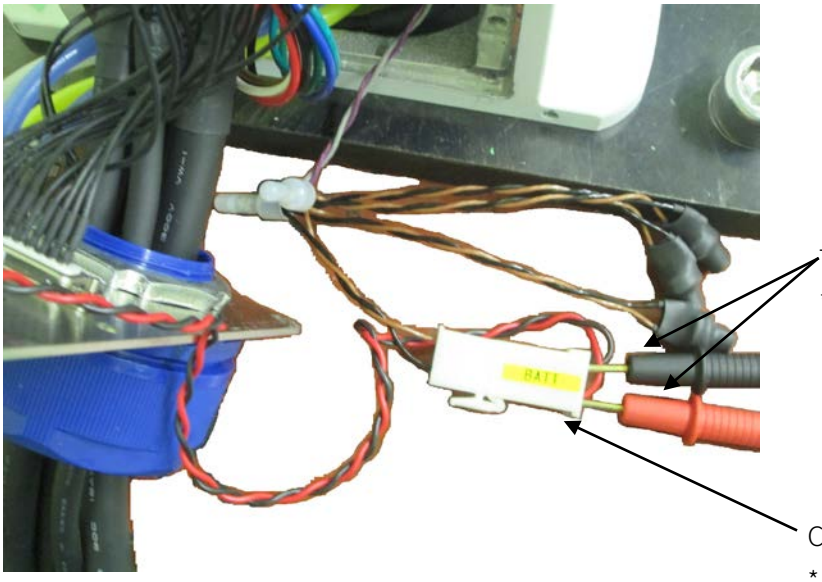


Fig. 9.4 Checking the Battery for Fitting Failure

9.3.3 Battery Error Code

When a position detector error, including a “battery error”, has occurred, the following alarm code is shown on the currently occurring alarm screen.

When the battery voltage has been reduced, the following alarm code is shown on the currently occurring alarm screen. If one of the following alarm codes is displayed in the alarm history when a position detector error has occurred, it indicates that a “battery error” has occurred.

Table 9.1 1-40* battery error codes

Alarm code	Alarm message	Remarks
1-401	Joint Axis 1 Encoder Battery low	
1-402	Joint Axis 2 Encoder Battery low	
1-403	Joint Axis 3 Encoder Battery low	
1-404	Joint Axis 4 Encoder Battery low	

If an alarm of level 1 alarm level listed in Table 9.1 has occurred, the early battery replacement is recommended. If the state in which this alarm appears is left as is, an alarm of level 8 alarm listed in Table 9.2 to Table 9.4 will occur. Be aware that the information on the zero position of the robot may disappear in some cases.

Table 9.2 8-60* battery error codes

Alarm code	Alarm message	Remarks
8-601	Joint Axis 1 multi count error	
8-602	Joint Axis 2 multi count error	
8-603	Joint Axis 3 multi count error	
8-604	Joint Axis 4 multi count error	

If the alarm listed in Table 9.2 appears, counting skips may have occurred due to battery power shortage. In the first place, replace the battery. If the alarm does not disappear even after the battery replacement, the encoder may be faulty. In such case, the motor needs to be replaced. Contact SHIBAURA MACHINE Service Department.

Table 9.3 8-61* battery error codes

Alarm code	Alarm message	Remarks
8-611	Joint Axis 1 Encoder Battery empty	
8-612	Joint Axis 2 Encoder Battery empty	
8-613	Joint Axis 3 Encoder Battery empty	
8-614	Joint Axis 4 Encoder Battery empty	

If this alarm has appeared, the battery voltage may have been reduced below the specified value. Replace the battery. If the battery voltage is reduced below the specified value, the information on the zero position of the robot may have disappeared. If the alarm is not reset after battery replacement, see “9 Robot Zero Point and Position Detector Error” to perform zero setting.

Table 9.4 8-40* battery error codes

Alarm code	Alarm message	Remarks
8-401	Joint Axis 1 Encoder error	
8-402	Joint Axis 2 Encoder error	
8-403	Joint Axis 3 Encoder error	
8-404	Joint Axis 4 Encoder error	

Among level 8 battery errors, the two types of alarms listed in Table 9.2 and Table 9.3 appear in principle. However, the alarms listed in Table 9.4 may appear. If this alarm appears, replace the battery. If the battery voltage is reduced below the specified value, the information on the zero position of the robot may have disappeared. If the alarm is not reset even after battery replacement, see “9 Robot Zero Point and Position Detector Error” to perform zero setting.

The battery error is one of the position detector errors (i.e., encoder errors), and there is other cause leading to the position detector error. The restoring method from the position detector error varies with the error contents generated. To properly execute the restoring method from the position detector error, therefore, read through and completely understand the descriptions carried in "Section 9 Robot Zero Point and Position Detector Error ".

9.4 Backing up Data

Save all parameters in the controller. The factory-set parameters are saved on the instruction manual DVD. Use the data as well.
For how to back up data, refer to “7.3 Backing up Data” in “STA1161 Controller Manual.”

9.5 Zero Position

This section describes the zero position required for zero setting. When moving the axes 1, 2, and 4 of the robot to the zero position, see "9.5.1 Locations of Zero Point Match-Marks of the Robot", and align the zero point match-mark of each axis. The axis 3 does not have a zero point match-mark. See "9.5.2 Axis-3 Zero Position" to move it to the zero position. The installation of the zero setting tool (optional) is explained in "9.5.3 Installing the Zero Setting Tool (Optional)."

9.5.1 Locations of Zero Point Match-Marks of the Robot

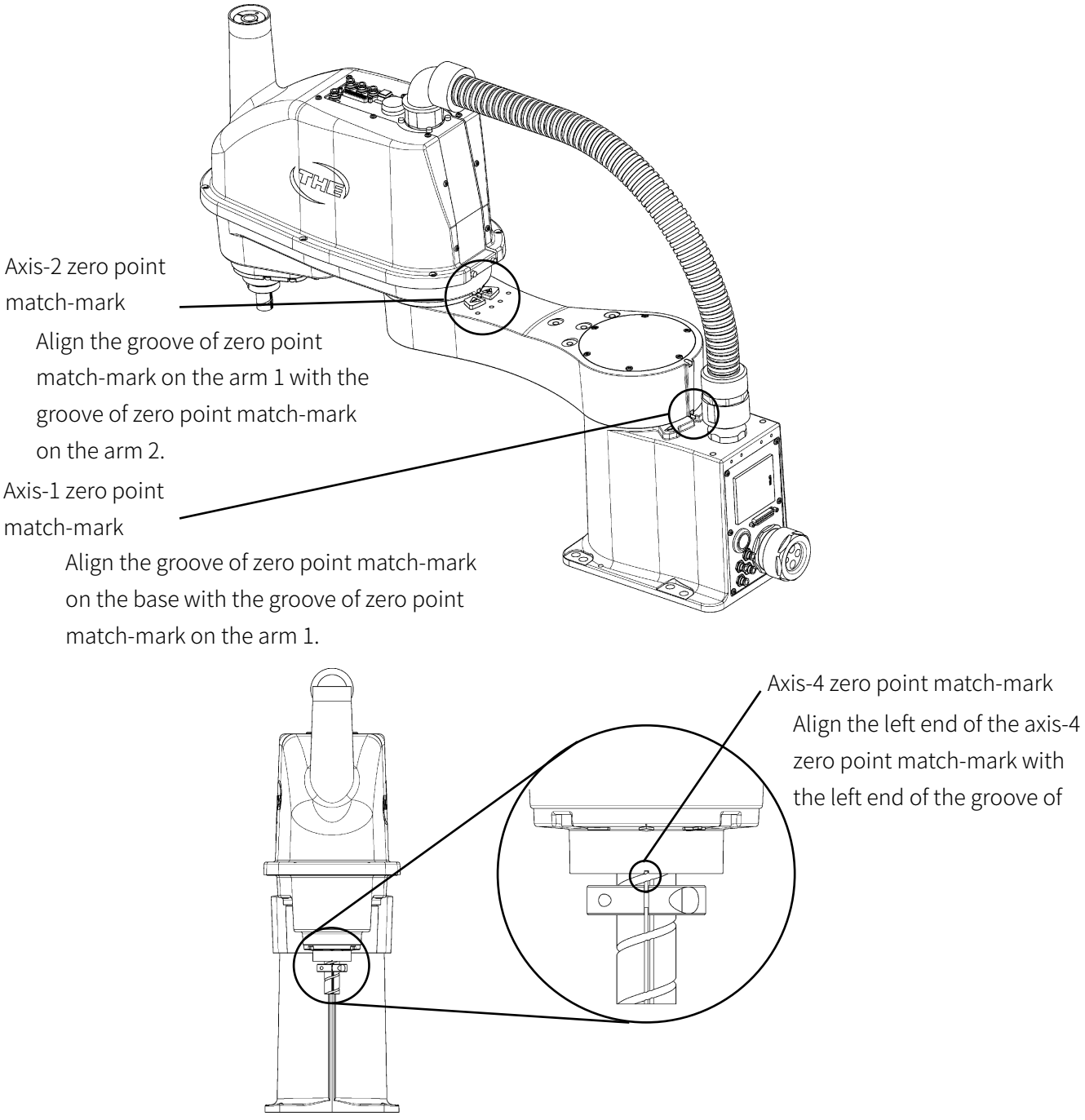


Fig. 9.5 Locations of zero point match-marks

9.5.2 Axis-3 Zero Position

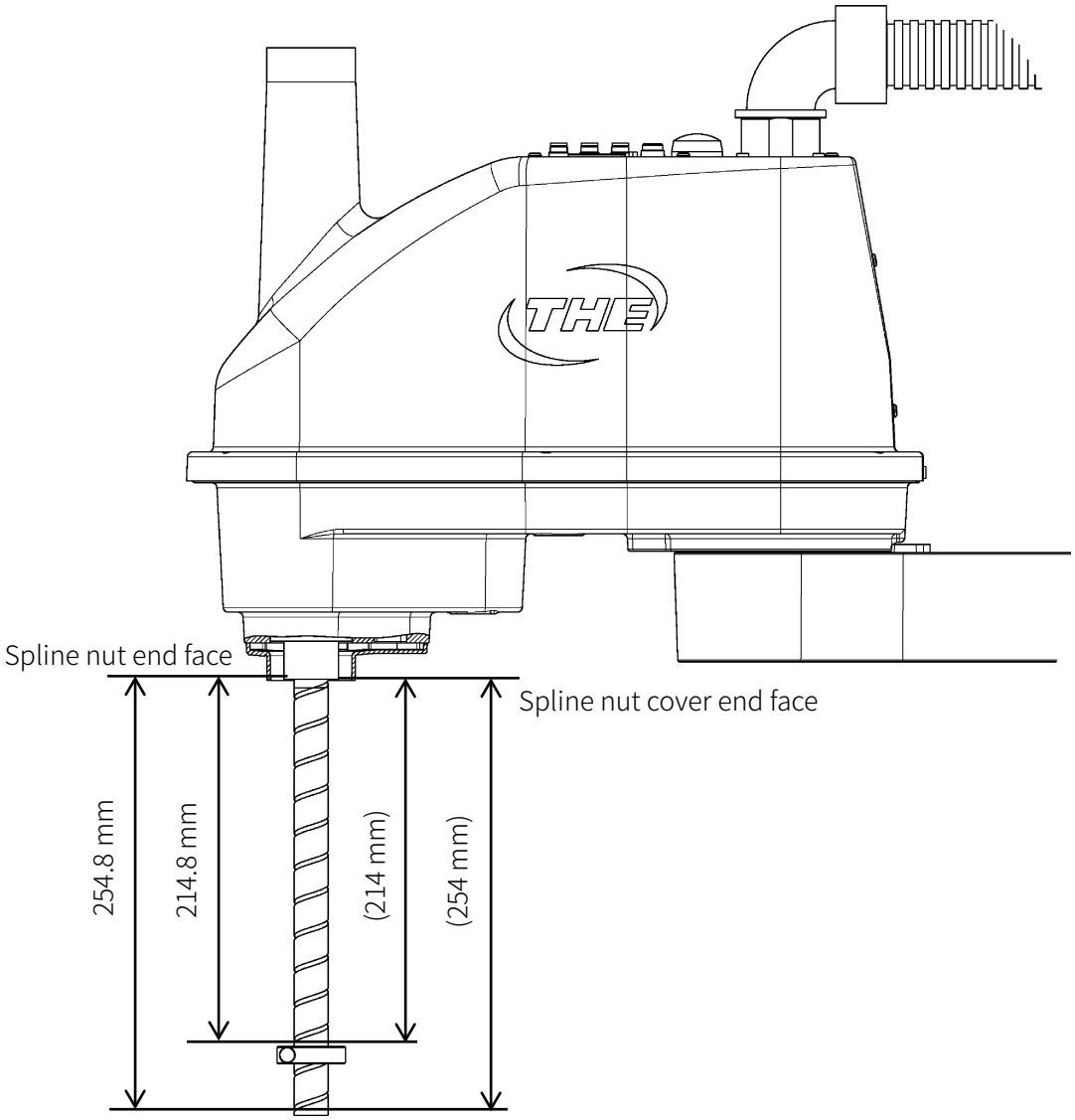


Fig. 9.6 Dimensions of the Axis-3 Zero Point

9.5.3 Installing the Zero Setting Tool (Optional)

This paragraph shows the installation location of the zero setting tool (optional) meter described in "9.8 ②Restoring Zero Position Data by ZEROP Function Using the Zero Setting Tool." For the procedure for attaching/detaching the axes-3 and-4 zero setting tool, see "9.5.4 Attaching/Detaching the Axes-3 and -4 Zero Setting Tool." The zero setting tools, etc. used in this paragraph are as listed in Table 9.5. When ordering a zero setting tool, specify the robot model name, SHIBAURA drawing number, and unit code.

Table 9.5 Types of Zero Setting Tools, etc.

Part name	Model	SHIBAURA Dwg. No.	Unit code
Axes-1 and -2 zero setting tool		M342399	Y610D0DQ0
Axes-3 and -4 zero setting tool		L23752G01	Y610D0DR0
Tool flange*		L12515G01	Y610D0300

*The hexagon socket head cap screws (M4×16×4 pcs) for fitting the tool flange are also included.

The parallel pin and parallel key used by the axes-3 and -4 zero setting tool are as listed in Table 9.6.

Table 9.6 Types of Parallel Pin and Parallel Key

Part name	Model	Size	Material
Parallel pin	Type B	4×8	SUS
Parallel key	Round	4×4×16 (p6)	S45C

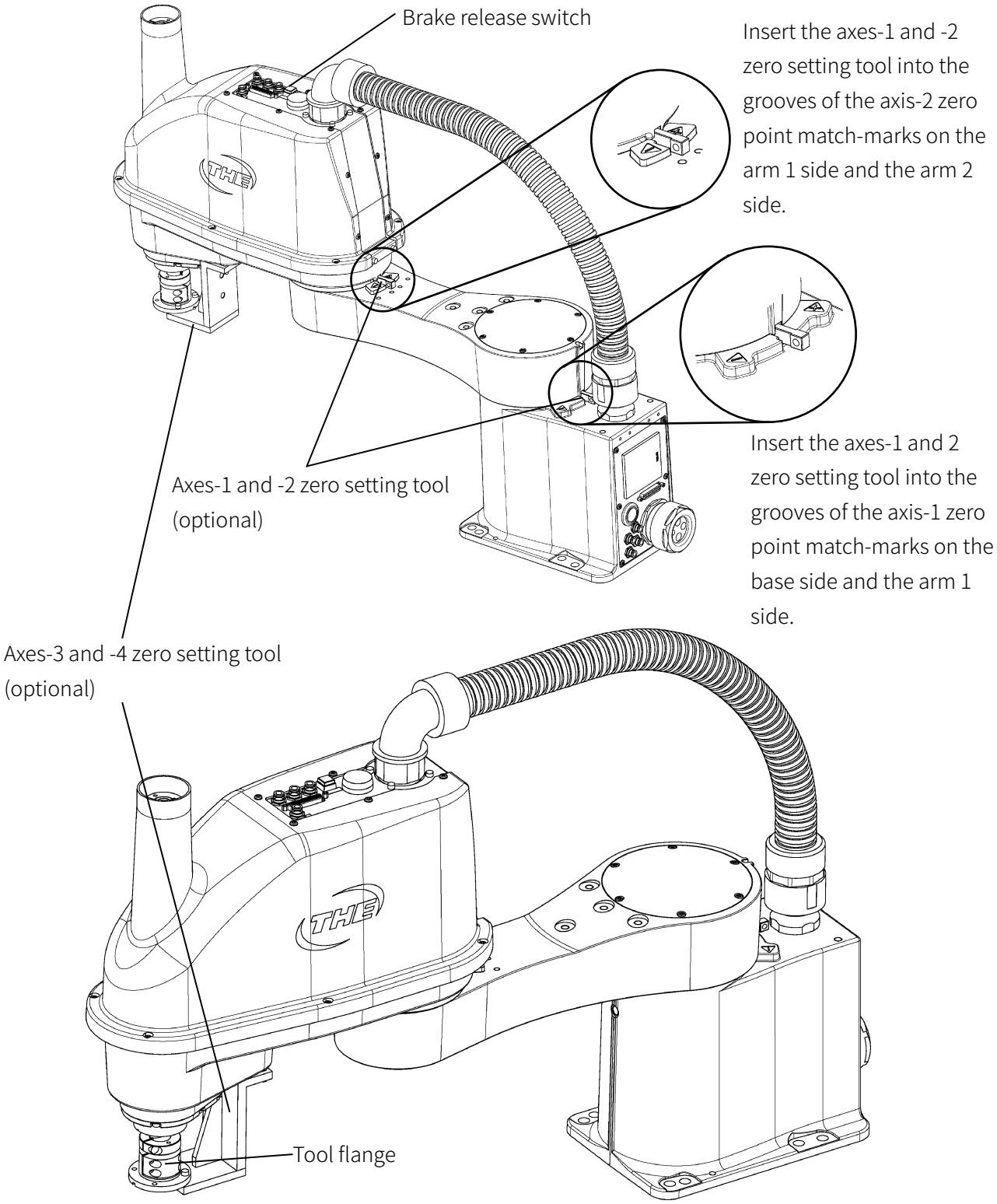


Fig. 9.7 Installing the Zero Setting Tool (Optional)

9.5.4 Attaching/Detaching the Axes-3 and -4 Zero Setting Tool

1 Fit the tool flange (optional).

Fit the tool flange so that the left side of the dividing portion of the stopper is aligned with the left side of the dividing portion of the tool flange as shown in Fig. 9.8

.

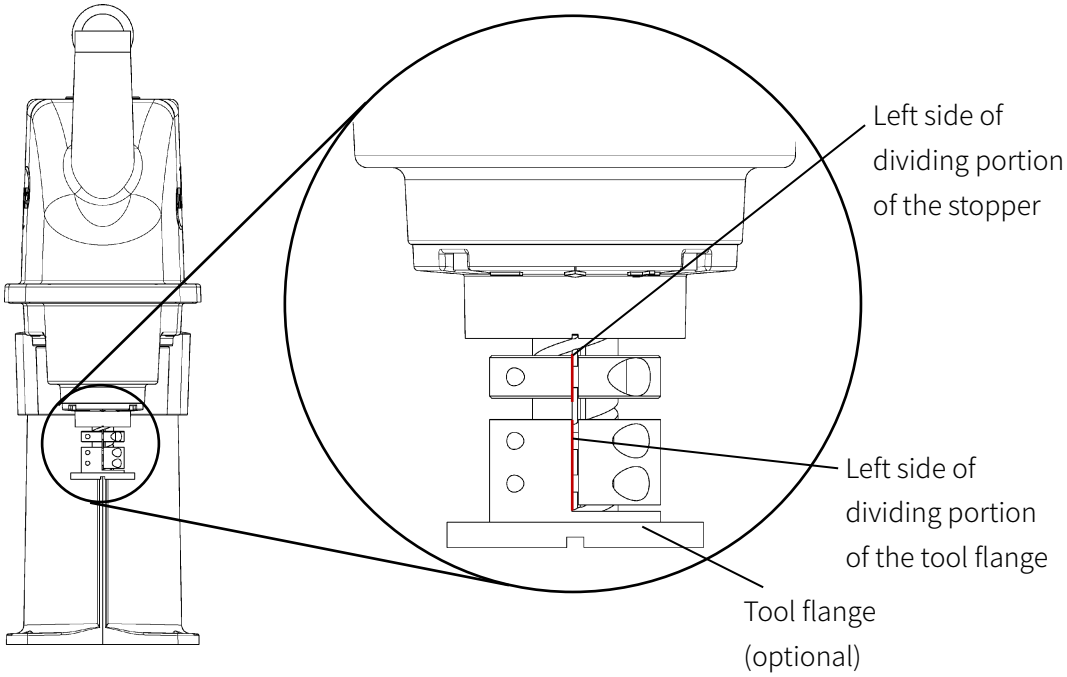


Fig. 9.8 Fitting the Tool Flange

2 Fit the axis-3 and -4 zero setting tool into the lower side of the arm 2.

Insert the parallel pins of the axis-3 and -4 zero setting tool (optional) into the pinholes, and secure it with the hexagon socket head cap screws (M4×12×2 pcs).

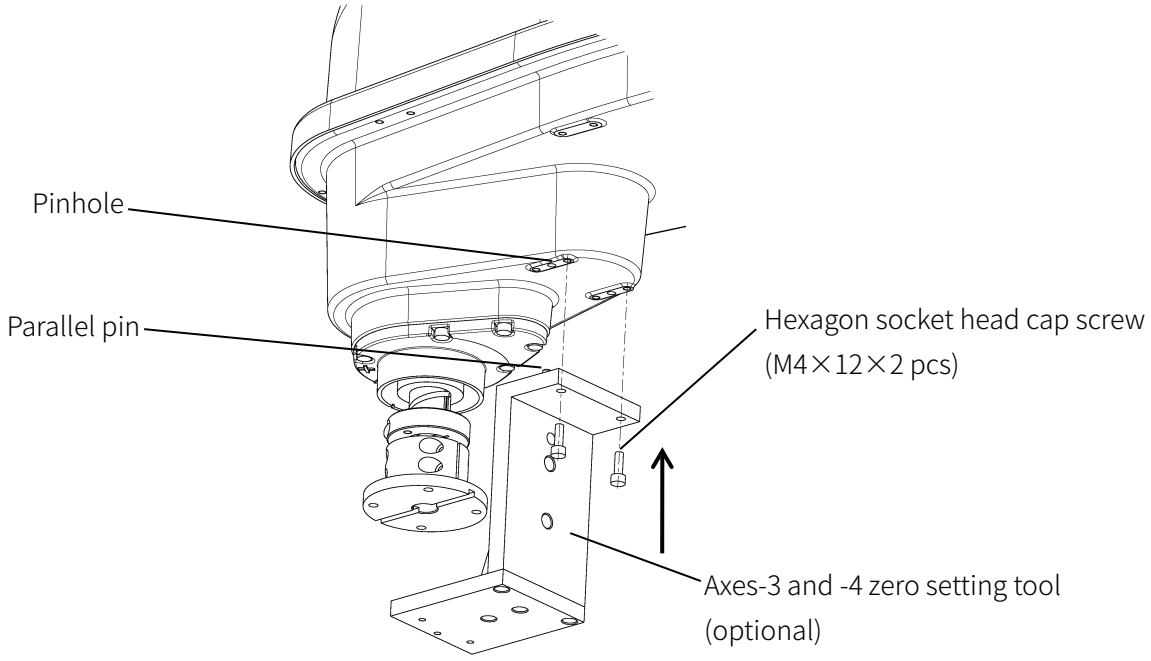


Fig. 9.9 Secure the Axes-3 and -4 Zero Setting Tool

3 Lower the tool flange so that the parallel key of the axis-3 and -4 zero setting tool fits into the lower key groove of the tool flange.

Press the brake release switch to rotate the ball screw spline and match the orientation.

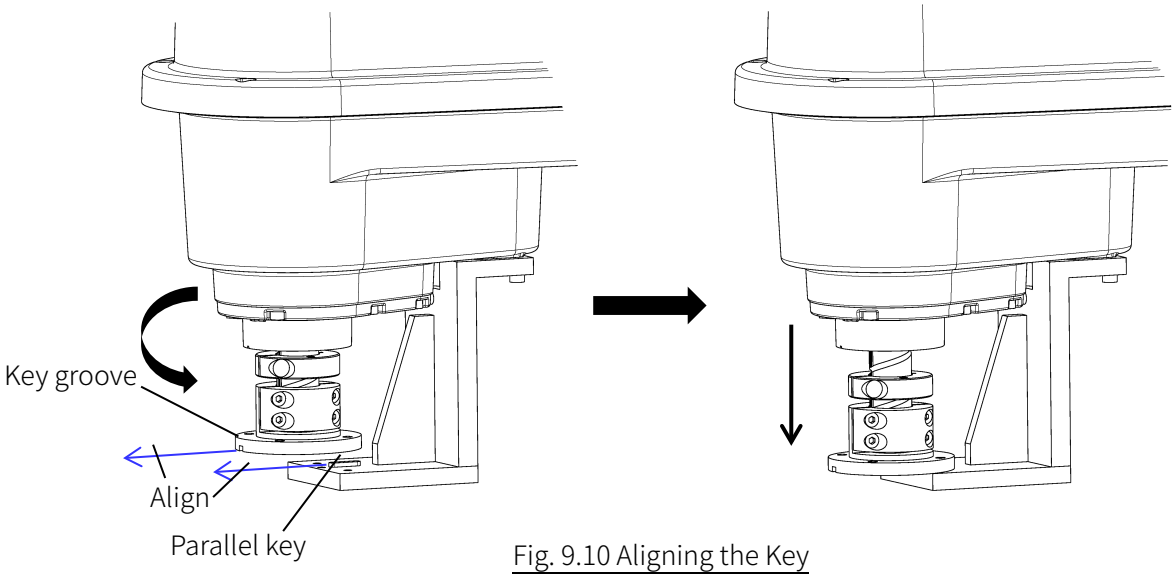


Fig. 9.10 Aligning the Key

4 Secure the tool flange to the axis-3 and -4 zero setting tool.

Use the hexagon socket head cap screws (M4×12×2 pcs).

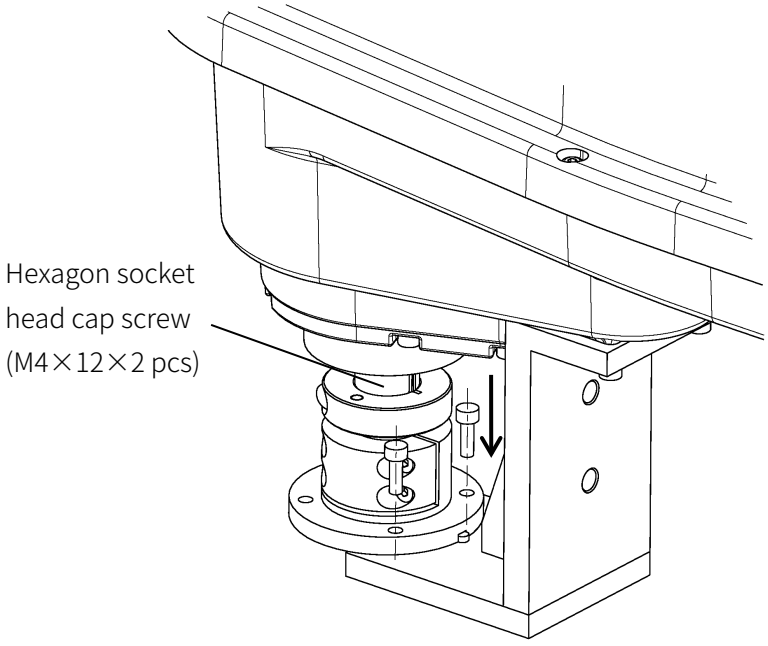


Fig. 9.11 Securing the Tool Flange

5 Set the zero position according to the procedure explained in "9.8 ②Restoring Zero Position Data by ZEROP Function Using the Zero Setting Tool."

6 Reversing the fitting procedure 1 to 4, remove the axis-3 and -4 zero setting tool.

9.6 Encoder Status Screen

Position data and the type of position detector error can be confirmed by changing the teach pendant screen to encoder information screen. Types of position detector errors are expressed as status shown below along with a bit graph. The motor multi-turn data, position data within one (1) turn of the motor and status are displayed under the column of multi-turn data, position data and status, respectively. The procedure for displaying the encoder information screen is shown below.

Table 9.7 Status table

Status	Detail
0000	Signifies the normal condition.
0100	This error occurs when a heavy vibration has been exerted on the robot or when the robot has been moved fast by hand while the power drive cable was disconnected during power OFF. The multi-turn data has possibly shifted.
8000	The battery voltage has dropped. Replace the batteries.
4000	The battery voltage has dropped further than the above. An error has possibly occurred in the multi-turn data.
C000	Error statuses 8000 and 4000 have occurred at the same time.

<Displaying encoder information screen>

1 Tap the maintenance button on the top screen.

2 Tap the encoder button.

The encoder information screen is displayed.

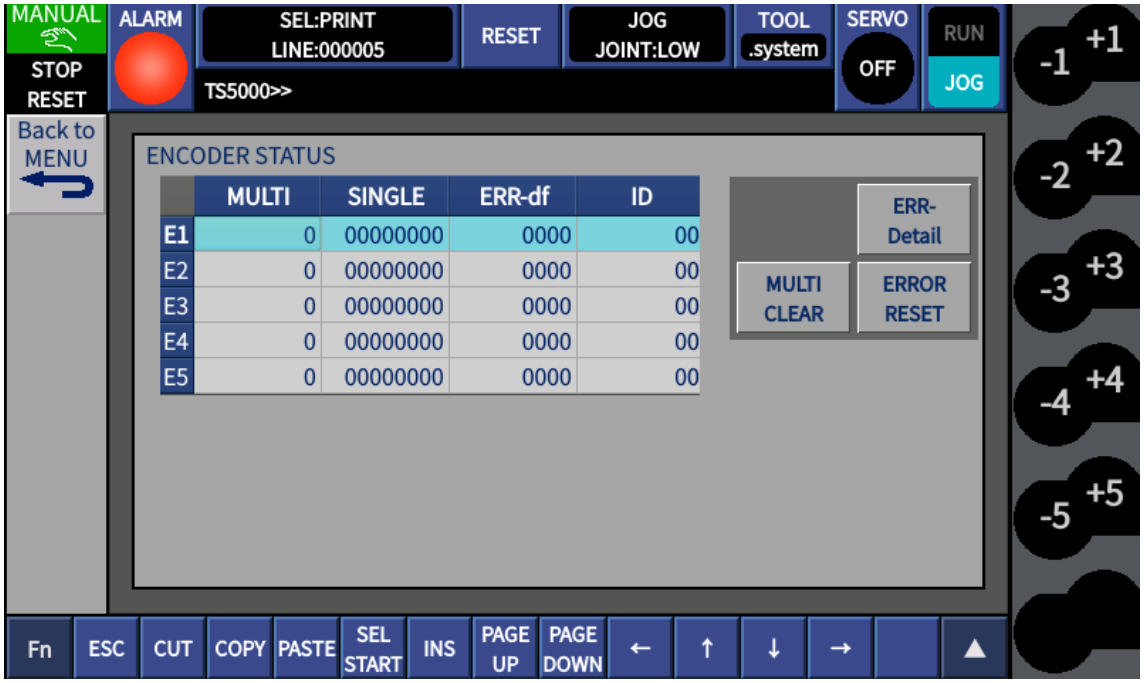


Fig. 9.12 The encoder information screen

If a status listed in the above status table displayed, the system can be restored from the position detector error by the user.

9.7 Restoring Zero Position Data by ① Multi-turn Data Reset

When the mechanical connection position of the servo motor and mechanical unit remains unchanged, execute this operation after battery replacement due to battery voltage drop or cable replacement due to cable breakage, for example.

<Resetting multi-turn data>

1 See "9.11TP5000 Operation Restriction," and change the TP5000 user type to "Maintenance."

2 See "9.5 Zero Position", and move the robot to the zero position by hand.

3 Change the master mode to MANUAL.

4 See "9.6 Encoder Status Screen" to display the encoder information screen.

5 Move the cursor on the axis for which multi-turn data is to be reset to (zero), and tap the multi-turn reset button.

The confirmation message appears.

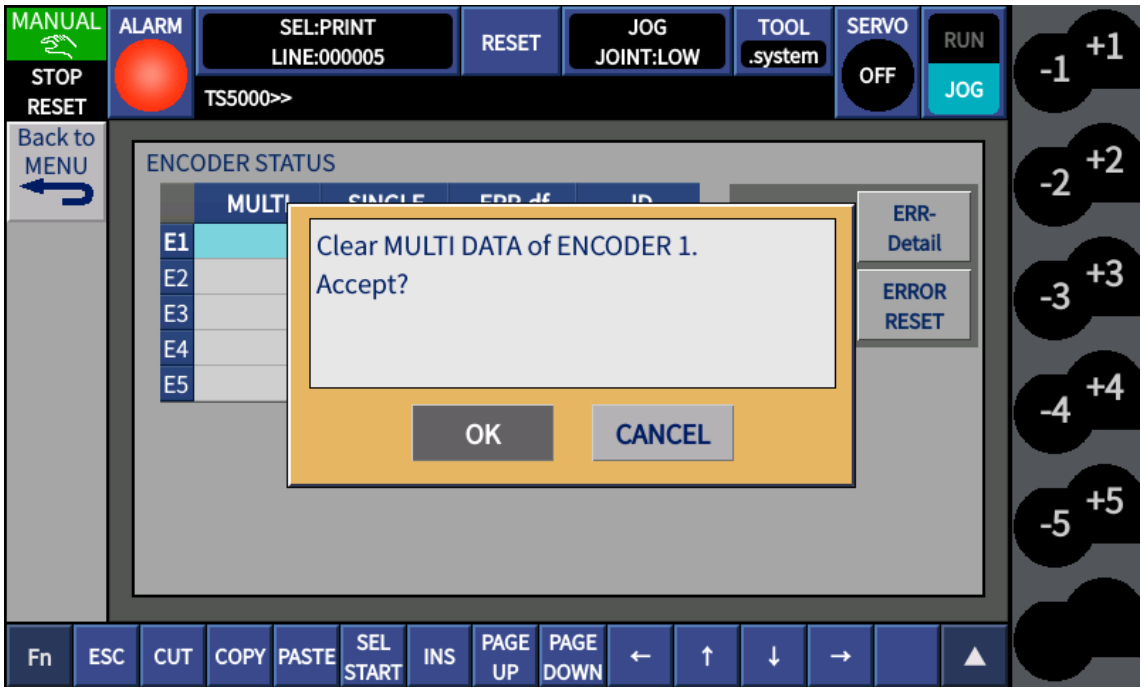


Fig. 9.13 Encoder Status Screen

6 Confirm the description and tap the OK button.

Multi-turn data is reset (set to zero).

7 Turn off the power to the controller, and then turn it on.

8 Perform "9.10 Check Work."

9 Move the robot to a teaching point, etc. to make sure that zero setting has been performed correctly. (In this operation, the zero position is restored to its original state completely.)

9.8 ②Restoring Zero Position Data by ZEROP Function Using the Zero Setting Tool

If there is a change in mechanical connection due to motor replacement, etc., and the re-teaching of teaching points is assumed, then perform this operation.

NOTICE

- If the zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of teaching points.
  The zero position shifts slightly due to the error in setting. If the zero position is used as is, the robot will collide with the peripheral equipment, etc.

9.8.1 Restoring Zero Position Data for All Axes

<Restoring zero position data>

1 See "9.11TP5000 Operation Restriction," and change the TP5000 user type to "Maintenance."

2 Change the master mode to MANUAL mode.

3 Fit the zero setting tool (optional) as described in "9.5.3 Installing the Zero Setting Tool (Optional)."

4 Tap the maintenance button on the top screen.

5 Tap the zero position button.

The zero position screen appears.

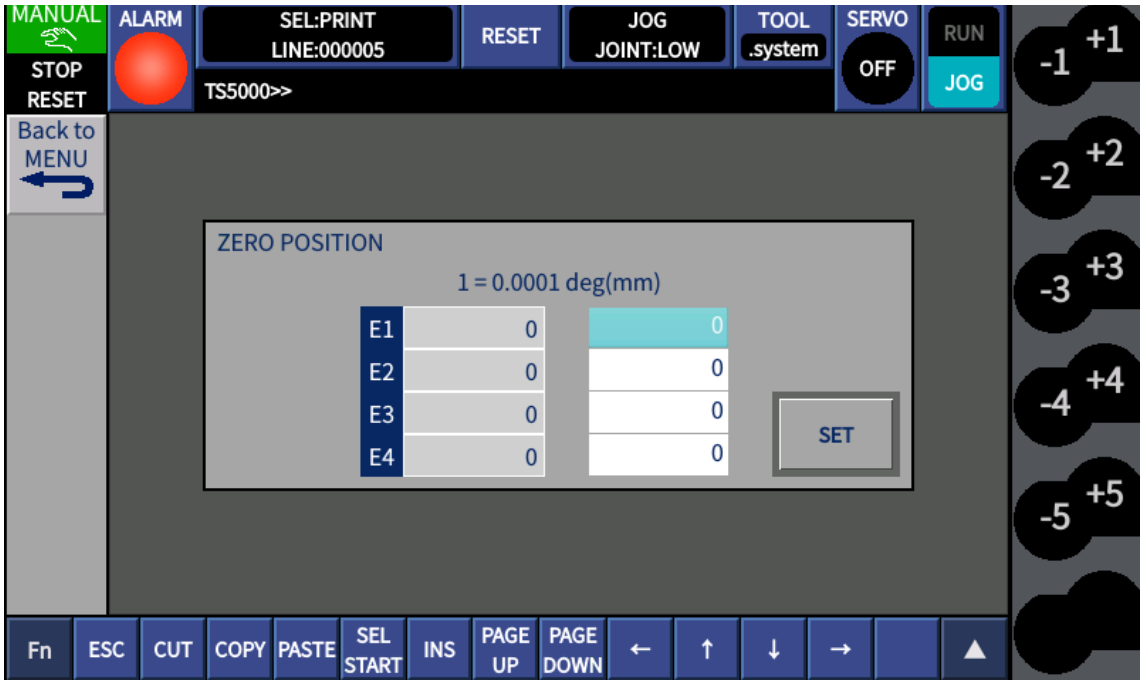


Fig. 9.14 The zero position screen(1)

6 Tap the [↓] key on the keyboard three times or tap the E4 cell.

The selection cursor moves to the E4 (axis 4) cell.

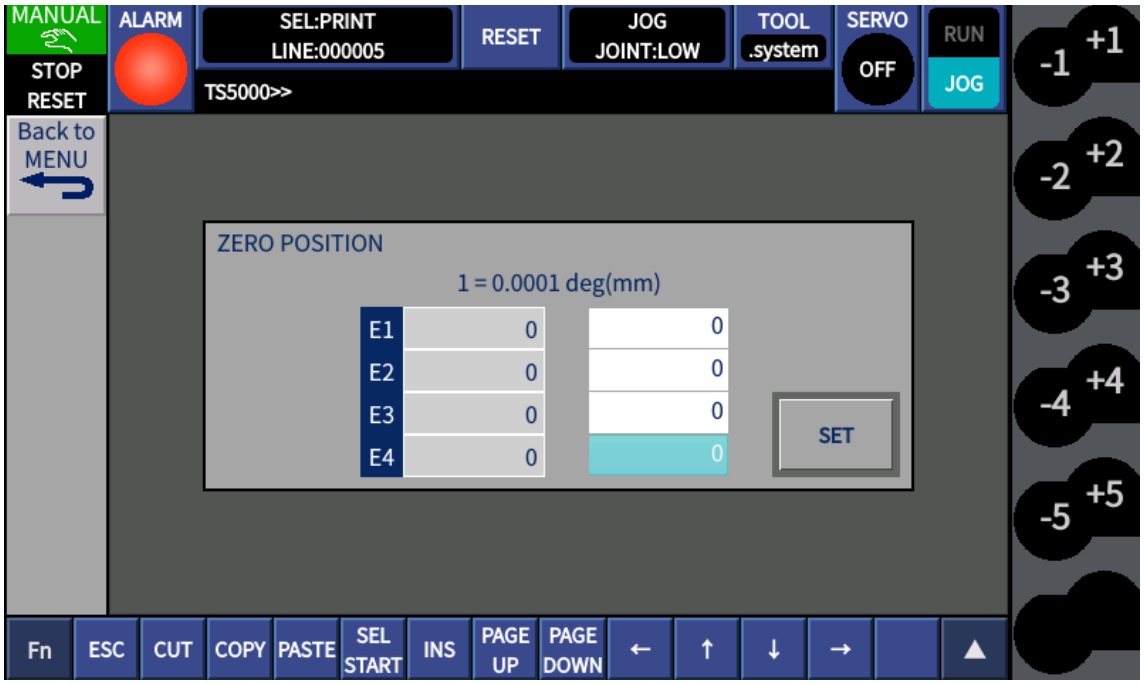


Fig. 9.15 The zero position screen(2)

7 Tap the E4 cell.

The cursor is displayed in the cell, which can be edited.

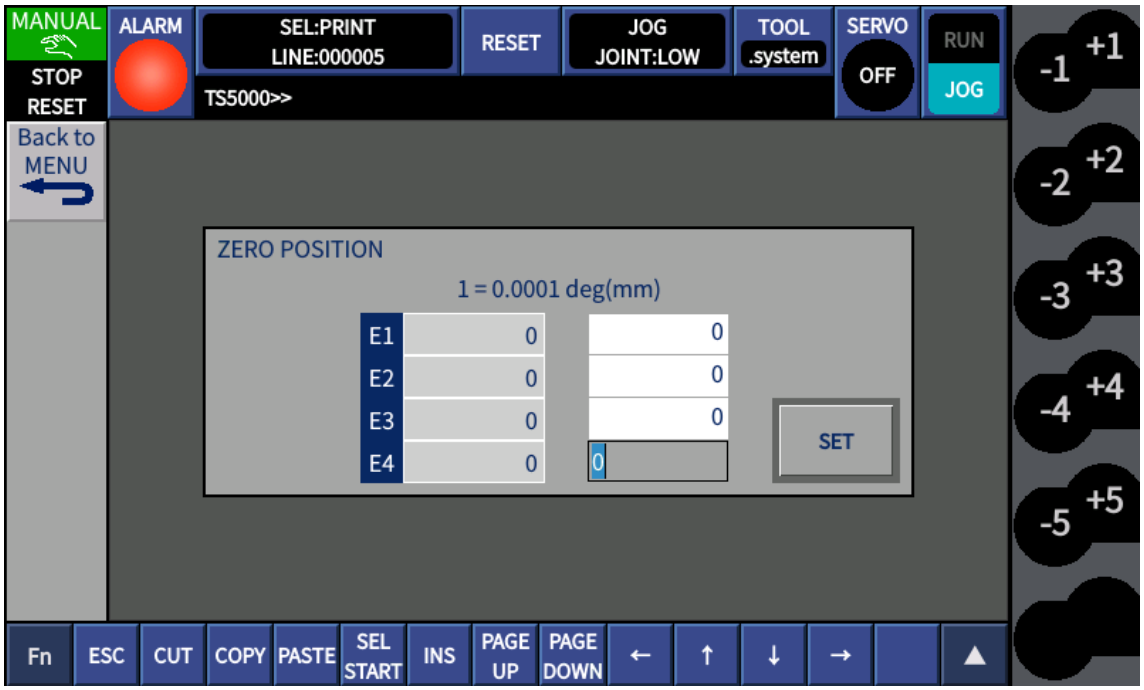


Fig. 9.16 The zero position screen(3)

8 Enter “0” and tap the [Enter] key or tap the place other than the cell being edited.

9 Tap the setting button.

The confirmation screen appears.

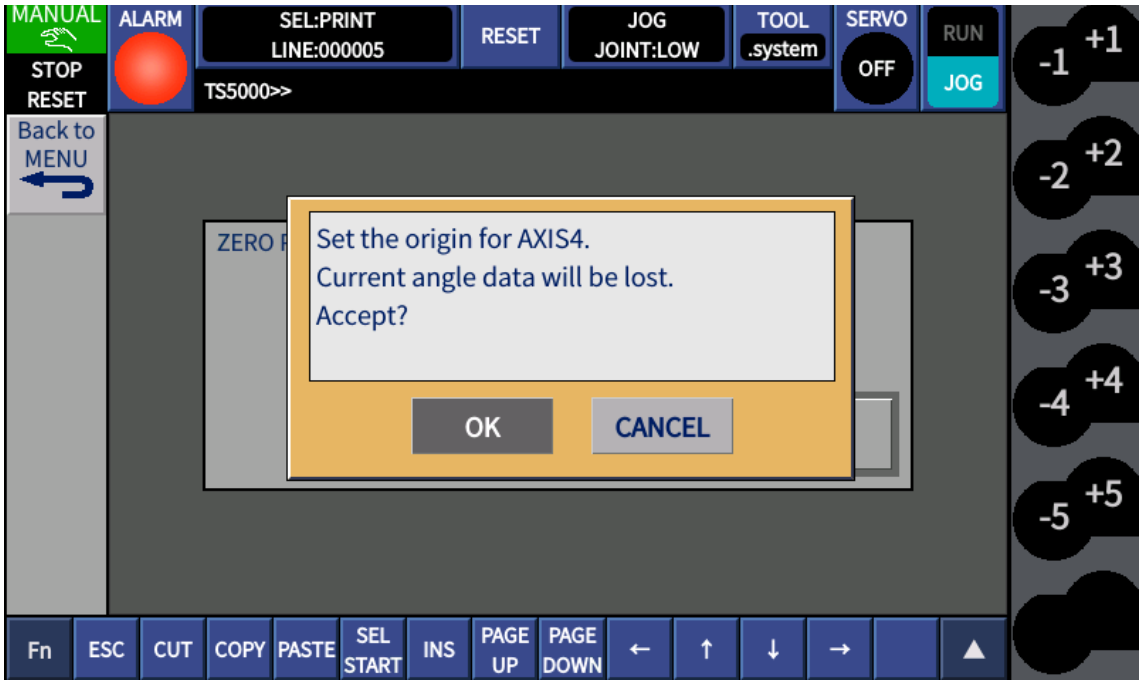


Fig. 9.17 The zero position screen(4)

10 Tap the OK button.

The value of E4 becomes 0.

11 Similarly, take steps 1 to 9 for E3 as well to restore zero position data.

MEMO

For the axis 3, set the value in step 7 to 2000000.

12 Similarly, take steps 1 to 9 for E1 and E2 as well to restore zero position data.

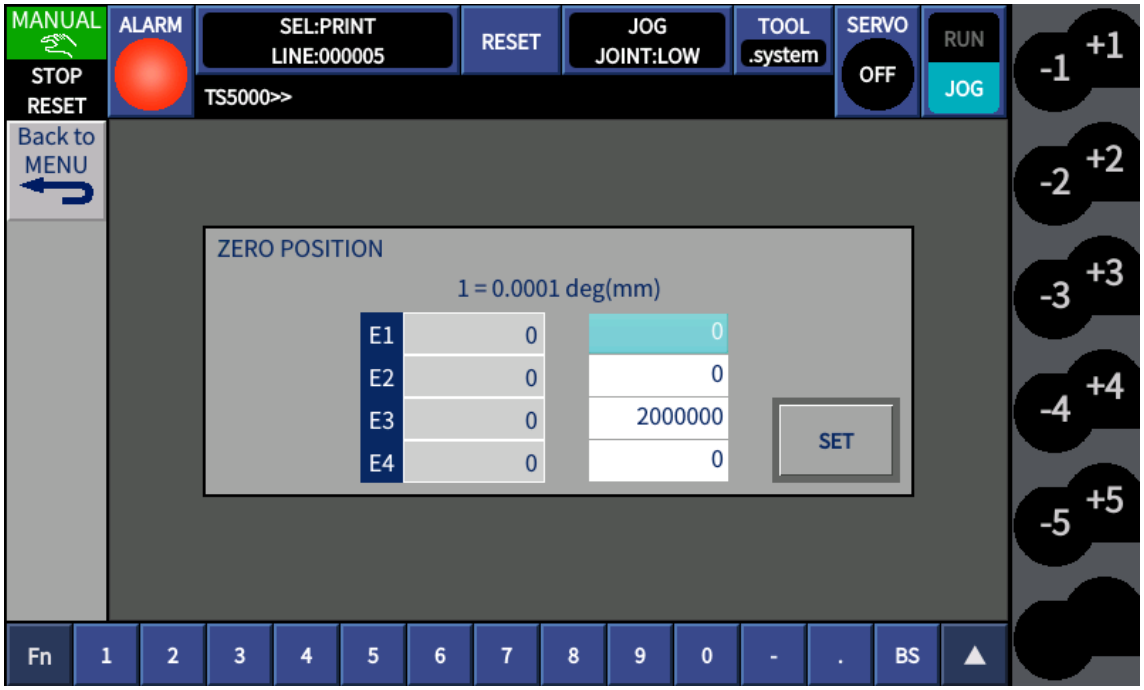


Fig. 9.18 The zero position screen(5)

NOTICE

- If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.
  The axis 3 operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
  3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
  - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.

13 Turn off the power to the controller and then turn it on.

14 Remove the zero setting tool (optional) that has been fitted in step 3.

15 Perform "9.10 Check Work."

16 Move the robot to a teaching point, etc. to make sure that zero setting has been performed correctly.

9.9 ③ Other Zero Setting

9.9.1 Restoring Zero Position Data by HOME Function

9.9.1.1 Outline of HOME Function

This function memorizes a specific coordinate value data in each axis and restores machine zero position data to the original zero position data by using the HOME position when the machine zero position data has been destroyed.

Four set points are provided for HOME1, HOME2, SYSTEM1 and SYSTEM2, and the plus (+) side mechanical stopper positions of the axis 1 to axis 3 are factory-set in SYSTEM1, and the minus (-) side mechanical stopper positions of the axis 1 to axis 3 are factory-set in SYSTEM2, respectively. The axis 4 is set at the 0° position according to the zero point match-mark. For HOME1 and HOME2, the customer can specify any values. It is recommended to specify any position for securing the robot end when building up a system. Two patterns can be set: pattern 1 in HOME1 and pattern 2 in HOME2.

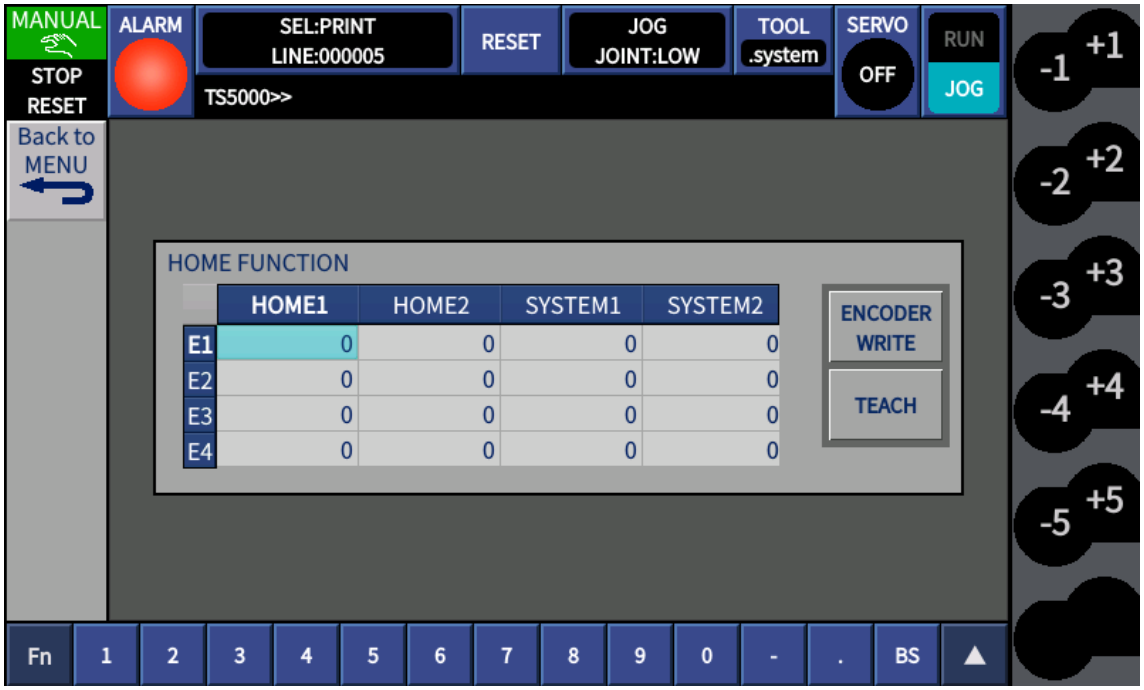


Fig. 9.19 HOME Function screen

MEMO

This screen is used to restore the machine zero point. Never operate the screen other than when restoring the zero point.

9.9.1.2 Setting HOME1 and HOME2

NOTICE

- Be sure to enter data to HOME1 and HOME2 for the REORG function when installing the robot.
  Even if data is entered after the zero point has shifted, it cannot be used as data for restoring the zero point.

1 See "9.11TP5000 Operation Restriction," and change the TP5000 user type to "Maintenance."

2 Guide the robot to secure the hand at any position.

Fig. 9.20 shows an example of securing the hand using the tool flange (optional).
Fig. 9.21 shows an example of the shape that should be prepared by the user when the tool flange (optional) is secured.

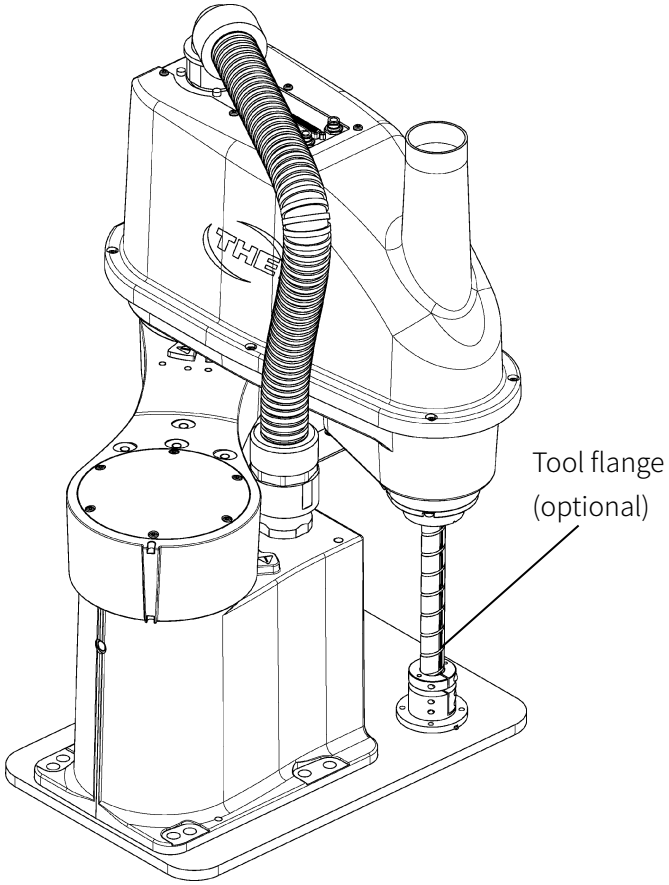


Fig. 9.20 Example of securing the hand

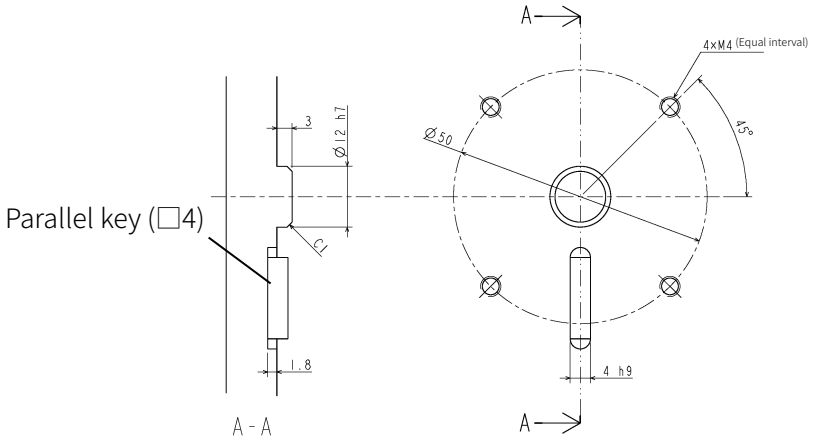


Fig. 9.21 Example of Shape of Securing the Tool Flange (Optional)

3 Turn off the servo power.

4 Tap the maintenance button on the top screen.

5 Tap the zero position button.

The zero position screen appears.

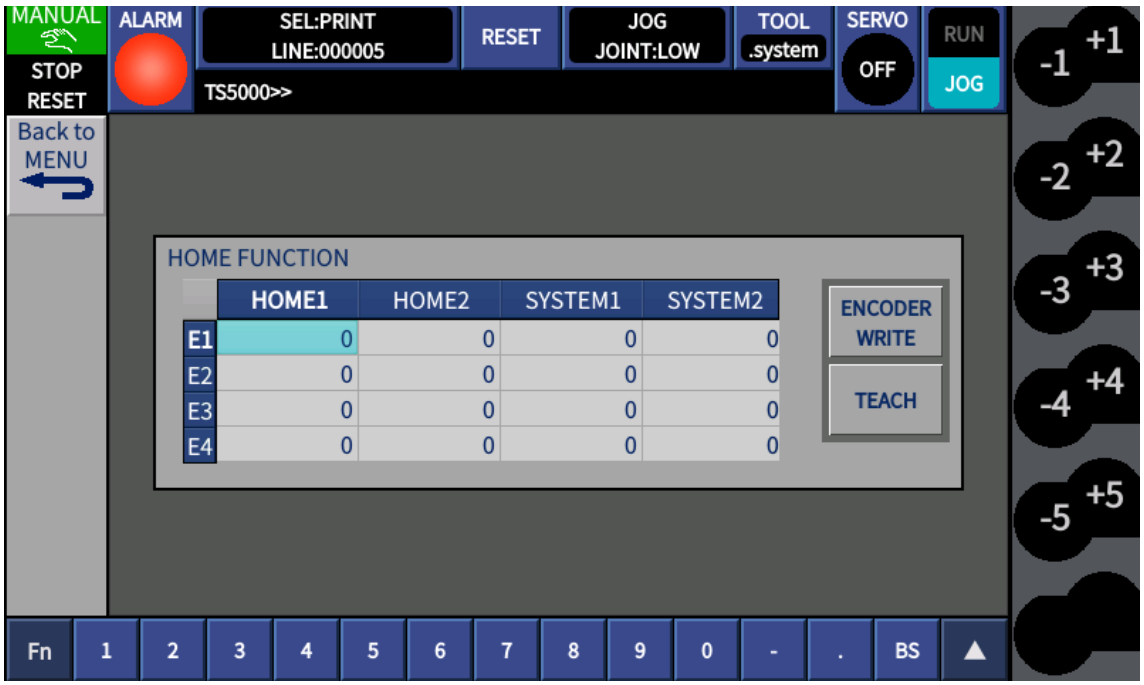


Fig. 9.22 HOME Function screen(1)

6 Tap the cell corresponding to the joint axis for which zero position is to be restored.

The selection cursor moves the tapped cell.

7 Tap the setting button.

The zero position setting confirmation screen appears.

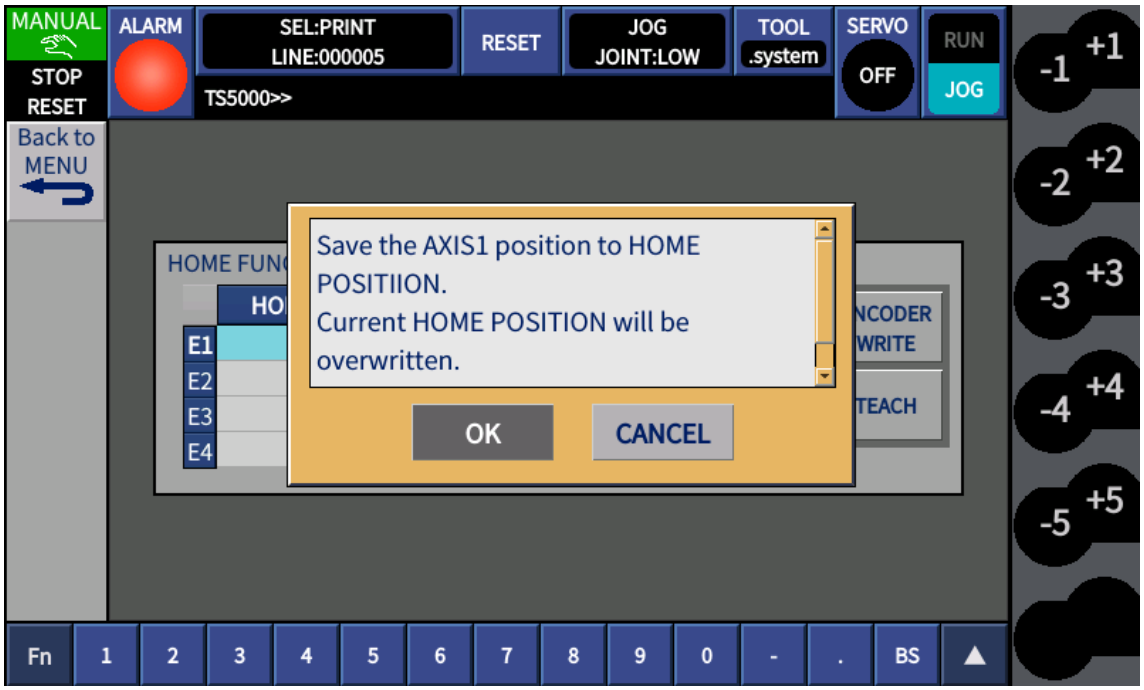


Fig. 9.23 HOME Function screen(2)

8 Tap the OK button.

The value of the cell at the selection cursor is rewritten to the value of the current joint axis.

9.9.1.3 How to Restore Data by HOME1 or HOME2

If there is a change in mechanical connection due to motor replacement, etc., and the re-teaching of teaching points is assumed, then perform this operation. This paragraph shows the method of restoring data using HOME1. If HOME2 is used, data is also restored in the same way.

NOTICE

- If the zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of teaching points.
  The zero position shifts slightly due to the error in setting. If the zero position is used as is, the robot will collide with the peripheral equipment, etc.

1 See "9.11TP5000 Operation Restriction," and change the TP5000 user type to "Maintenance."

2 Move the robot to the position set by HOME1, and fix it at the position. (See Fig. 9.20)

3 Turn off the servo power.

4 Tap the maintenance button on the top screen.

5 Tap the zero position button.
The zero position screen appears.

6 Tap the cell corresponding to the joint axis for which zero point is to be restored at the HOME1 column. The selection cursor moves to the tapped cell.

7 Tap the zero position button.

The zero position confirmation screen appears.

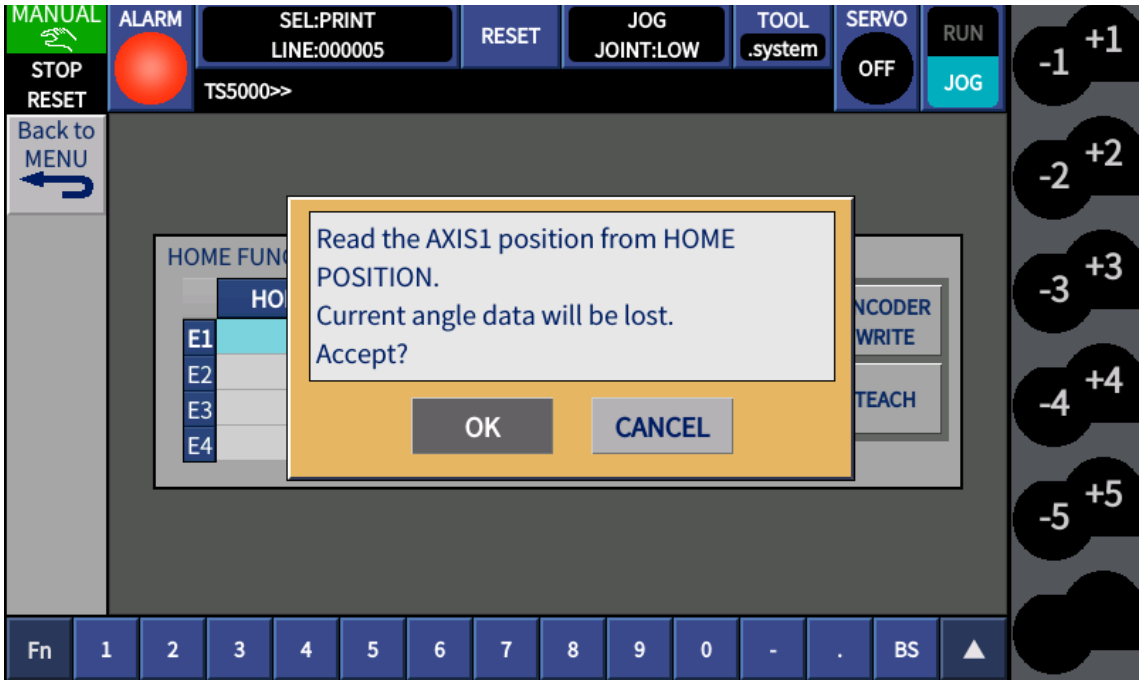


Fig. 9.24 zero position confirmation screen

8 Tap the OK button.

The current position data of the joint axis in the cell at the selection cursor is rewritten to the value of the cell.

MEMO

Even after HOM1 is performed, the robot can be used. However, the data format is different from the factory default setting. If it is used as is, then normal zero setting cannot be performed even by resetting multi-turn data when the batteries run out. See "9.9.1.5 Rewriting Zero Position Data" to rewrite the zero position data.

9 Perform "9.10 Check Work."

10 Move the robot to a teaching point, etc. to make sure that zero setting has been performed correctly.

NOTICE

- If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.
  The axis 3 operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
  3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
  - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.

9.9.1.4 How to Restore Data by SYSTEM1 or SYSTEM2

If there is a change in mechanical connection due to motor replacement, etc., and the re-teaching of teaching points is assumed, then perform this operation. The plus (+) side mechanical stopper positions of the axis 1 to axis 3 are factory-set in SYSTEM1, and the minus (-) side mechanical stopper positions of the axis 1 to axis 3 are factory-set in SYSTEM2. The axis 4 is set at the position of the zero point match-mark. (Because the axis 4 is set according to the zero point match-mark, it may be unable to be restored to original state completely.)
This section describes how to restore data using SYSTEM1.

NOTICE

- If the zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of teaching points.
  The zero position shifts slightly due to the error in setting. If the zero position is used as is, the robot will interfere with the peripheral equipment, etc.

1 Turn off the servo power.

2 Tap the maintenance button on the top screen.

3 Tap the zero position button.

The zero position screen appears.

4 Manually rotate the axis-1 arm in the plus (+) direction to bring the mechanical stopper into contact with the axis 1.

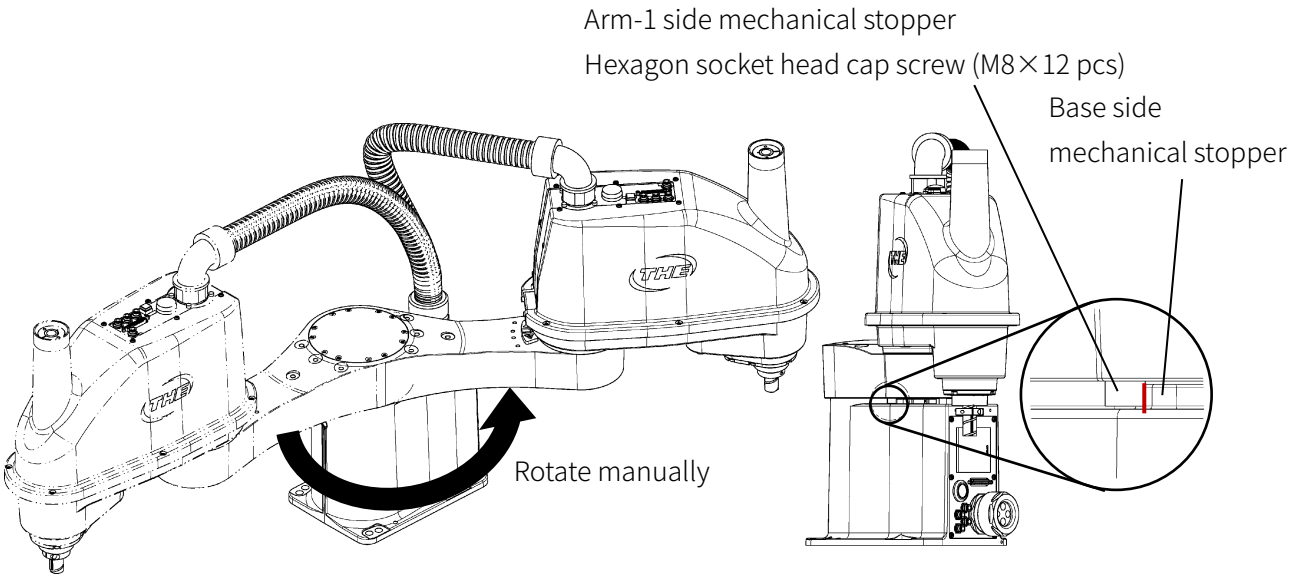


Fig. 9.25 Work to Bring the Mechanical Stopper into Contact with the Axis 1 (SYSTEM1)

5 Tap the (E1) cell of SYSTEM1.

6 Tap the zero position button.

The machine zero point of the axis 1 is restored.

7 Similarly, manually rotate the arm 2 in the plus (+) direction to bring the mechanical stopper into contact with the axis 2.

8 Tap the (E2) cell of SYSTEM1.

9 Tap the zero position button.

The machine zero point of the axis 2 is restored.

10 Align the axis 4 with the axis-4 zero point match-mark while pressing the brake release switch.

(See "9.5.1 Locations of Zero Point Match-Marks of the Robot.")

11 Tap the (E4) cell of SYSTEM1.

12 Tap the zero position button.

The axis 4 is restored.

13 Bring the mechanical stopper into contact with the spline nut while pressing the brake release switch.

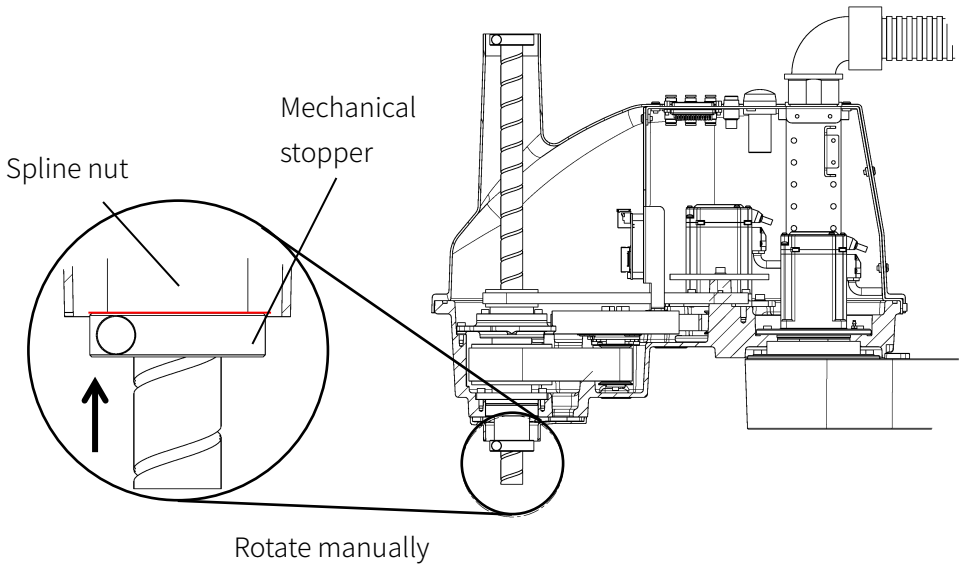


Fig. 9.26 Work to Bring the Mechanical Stopper into Contact with the Axis 3 (SYSTEM1)

14 Tap the (E3) cell of SYESTEM1.

15 Tap the zero position button.

The axis 3 is restored.

16 Return to the maintenance screen.

17 Perform "9.9.1.5 Rewriting Zero Position Data"

Even after SYSTEM 1 and 2 are performed, the robot can be used, but the data format is different from the factory default setting. If you continue to use it along with the data format, then no normal zero setting can be performed even by clearing multi-turn data when the batteries run out.

18 Perform "9.10 Check Work."

19 Move the robot to a teaching point, etc. to make sure that zero setting has been performed correctly.

9.9.1.5 Rewriting Zero Position Data

After the operation to restore a zero point is performed, servo offset and multi-turn data formats are different from the factory default setting. Even if the data formats are used without performing this operation, the robot operates normally. However, normal zero setting cannot be performed by resetting multi-turn data when the batteries run out. After HOME operation, perform this operation to return zero point data to the factory default setting.

1 Change the master mode to MANUAL mode.

2 Execute the MOVEA command directly to move each joint axis to the "0" position.

At this time, perform OVRD at a low speed while checking safety.

For how to execute the command directly, refer to "STA1161 TP5000 Operation Manual."

3 Tap the maintenance button on the top screen.

4 Turn off the servo power.

5 Tap the zero position button.

The zero position screen appears.

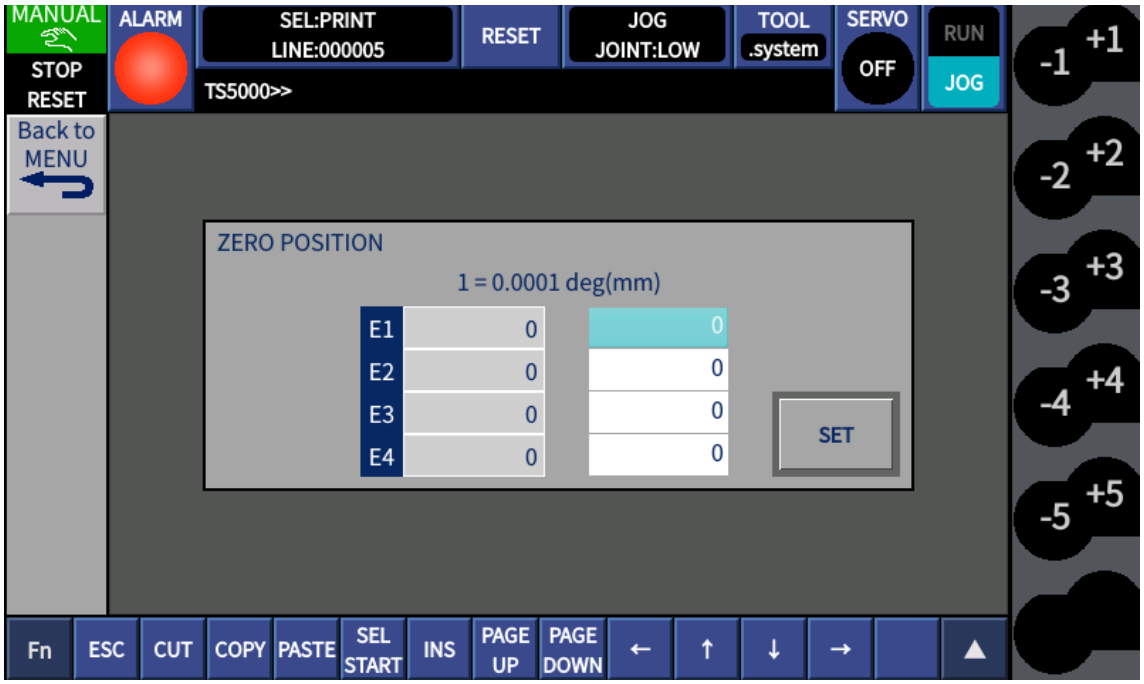


Fig. 9.27 zero position screen(1)

6 Tap the [↓] key on the keyboard three times or tap the E4 cell.

The selection cursor moves to the E4 (axis 4) cell.

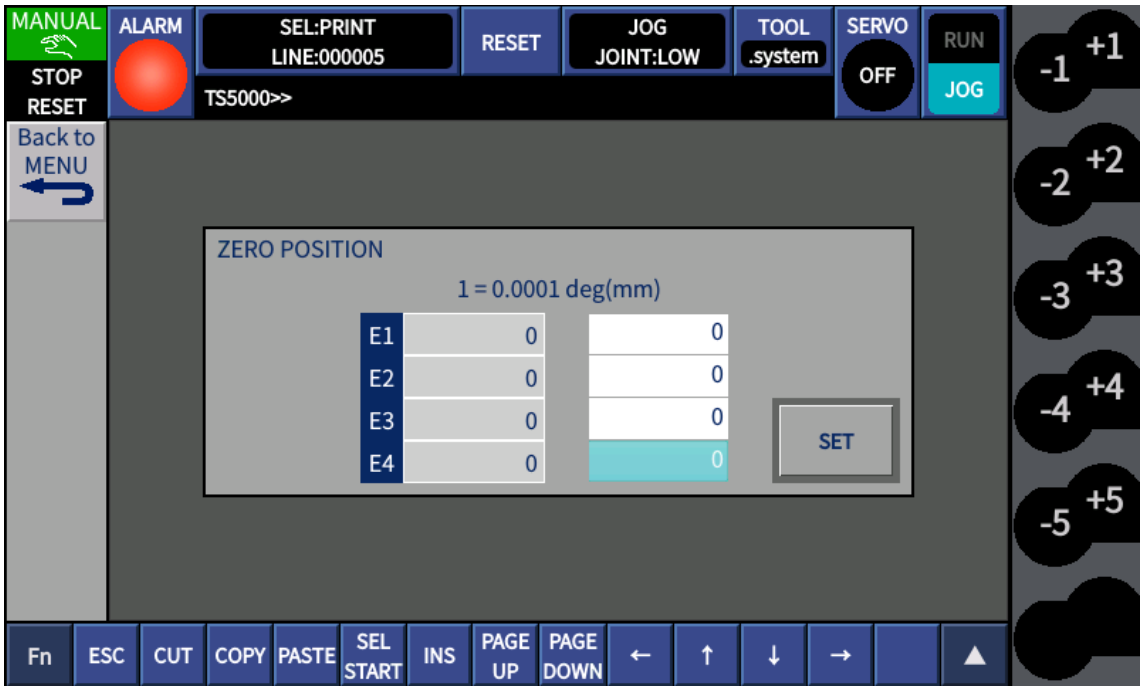


Fig. 9.28 zero position screen(2)

7 Tap the E4 cell.

The cursor is displayed in the cell, which can be edited.

8 Enter “0” and tap the [Enter] key or tap the place other than the cell being edited.

9 Tap the setting button.

The confirmation screen appears.

10 Tap the OK button.

The value of E4 becomes 0.

11 Similarly, take steps 1 to 8 for E3 as well to rewrite zero position data.

12 Similarly, take steps 1 to 8 for E1 and E2 as well to rewrite zero position data.

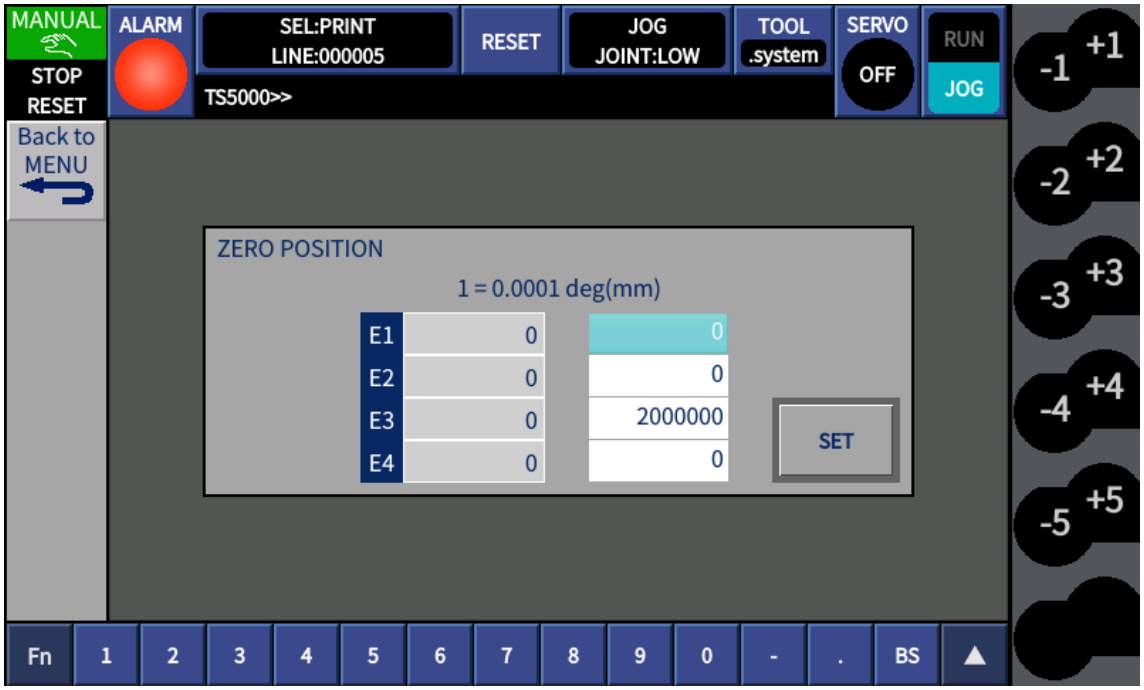


Fig. 9.29 zero position screen(3)

NOTICE

- If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.
  The axis operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
  3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
  - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.

13 Turn off the power to the controller and then turn it on.

14 Perform "9.10 Check Work."

15 Move the robot to a teaching point, etc. to make sure that zero setting has been performed correctly.

9.9.2 Restoring Zero Position Data by Zero Point Match-Mark

NOTICE

- If the zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of teaching points.
  The zero position shifts slightly due to the error in setting. If the zero position is used as is, the robot will collide with the peripheral equipment, etc.

1 Change the master mode to MANUAL mode.

2 See "9.5.1 Locations of Zero Point Match-Marks of the Robot" and "9.5.2 Axis-3 Zero Position," and move the robot to the zero position.

3 Tap the maintenance button on the top screen.

4 Turn off the servo power.

5 Tap zero position button.

The zero position screen appears.

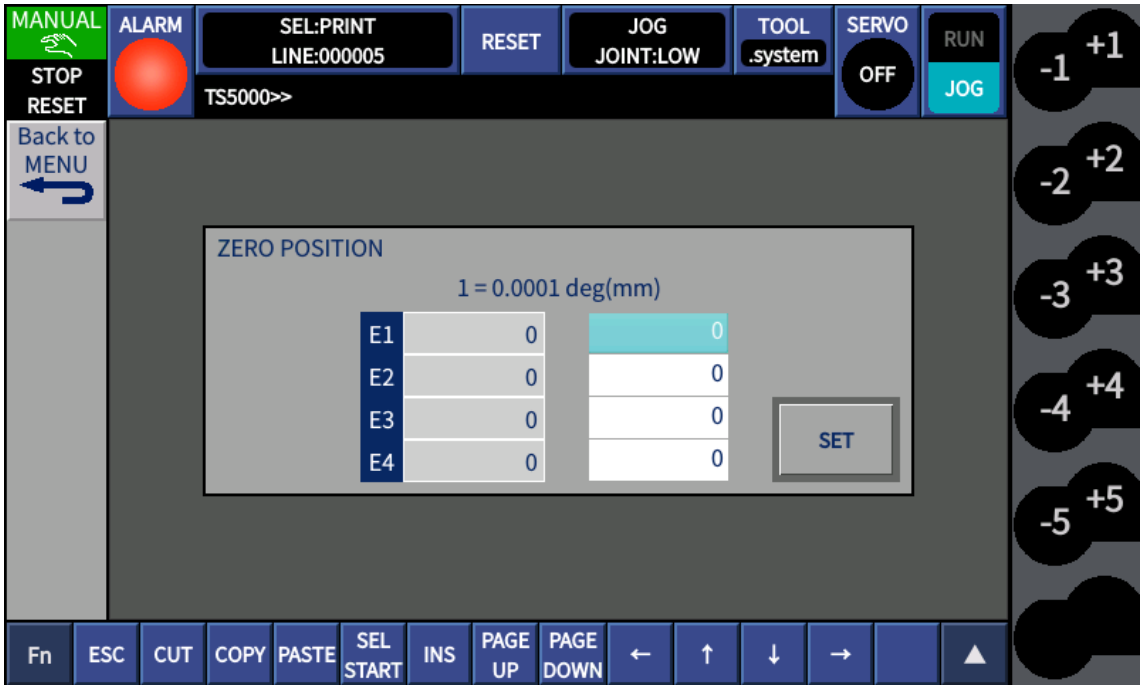


Fig. 9.30 zero position screen(1)

6 Tap the [↓] key on the keyboard three times or tap the E4 cell.

Move the selection cursor to the E4 (axis 4) cell.

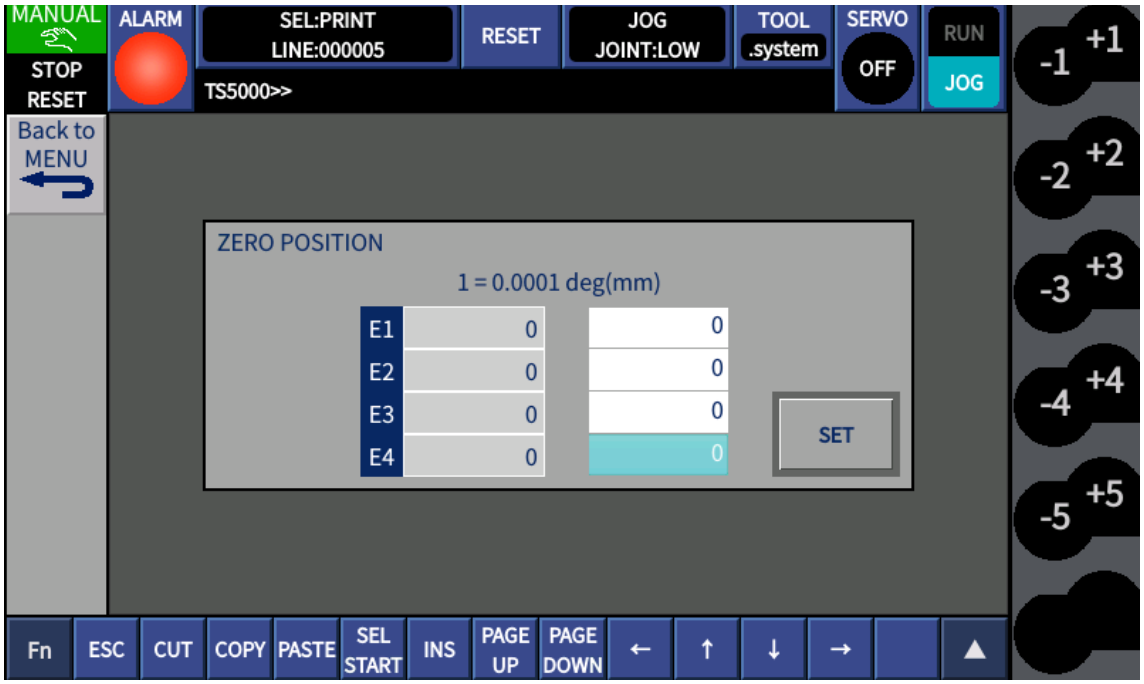


Fig. 9.31 zero position screen(2)

7 Tap the E4 cell.

The cursor is displayed in the cell, which can be edited.

8 Enter “0” and tap the [Enter] key or tap the place other than the call being edited.

9 Tap the setting button.

The confirmation screen appears.

10 Tap the OK button.

The value of E4 becomes 0.

11 Similarly, take steps 1 to 8 for E3 as well to restore zero position data.

12 Similarly, take steps 1 to 8 for E1 and E2 as well to restore zero position data.

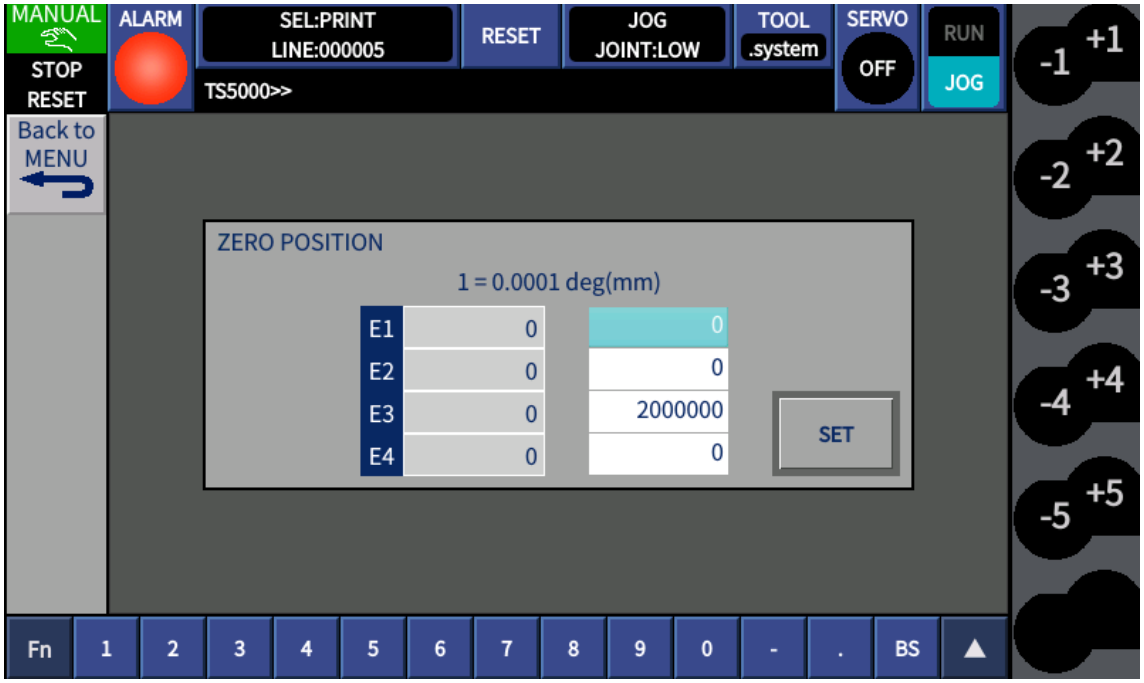


Fig. 9.32 zero position screen(3)

NOTICE

- If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.
  The axis 3 operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
  3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
  - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.

13 Turn off the power to the controller, and then turn it on.

14 Perform "9.10 Check Work."

15 Move the robot to a teaching point, etc. to make sure that zero setting has been performed correctly.

9.10 Check Work

Carry out check work in order of the axis 1, axis 2, axis 3 and axis 4. After all axes are checked completely, check whether there is a deviation in teaching points as well.

NOTICE

- After the zero point has been re-set, be sure to perform check work.
  Unless the zero point is restored correctly, the robot will collide with the mechanical stopper or peripheral equipment when it operates.

9.10.1 Checking the Axis 1 and Axis 2

Manually move the axes 1 and 2 to the travel limit in an emergency stop state, and check that a soft limit error occurs. Also, when manually bringing an arm into contact with the mechanical stopper, check on the POS screen of the utility that the position is not largely deviated from the values of axis 1 and axis 2 in SYSTEM1 or SYSTEM2 used in 9.9.1.4 How to Restore Data by SYSTEM1 or SYSTEM2." Fig. 9.33 shows check work that rotates the arm 1 in the plus (+) direction to bring the mechanical stopper into contact with the arm 1. Rotate the arm 1 in the minus (-) direction and check the axis 2 similarly.

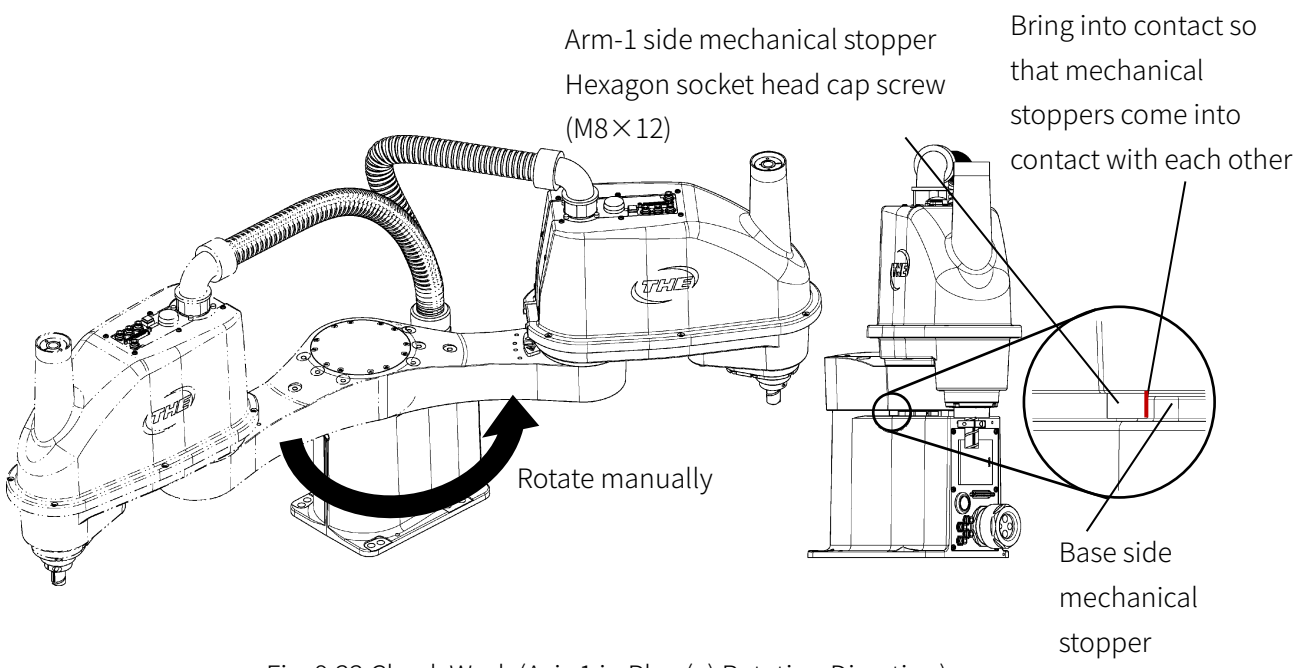


Fig. 9.33 Check Work (Axis 1 in Plus (+) Rotation Direction)

9.10.2 Checking the Axis 3

While pressing the brake release switch in an emergency stop state, manually move the axis 3 to the travel limit to check that a soft limit error occurs. Also, when manually bringing an arm into contact with the mechanical stopper, check on the POS screen of the utility that the value of the axis 3 is not largely deviated from the value of axis 3 in SYSTEM1 or SYSTEM2 used in 9.9.1.4 How to Restore Data by SYSTEM1 or SYSTEM2 in "9.9.1 Restoring Zero Position Data by HOME Function." Fig. 9.34 shows check work for the axis 3.

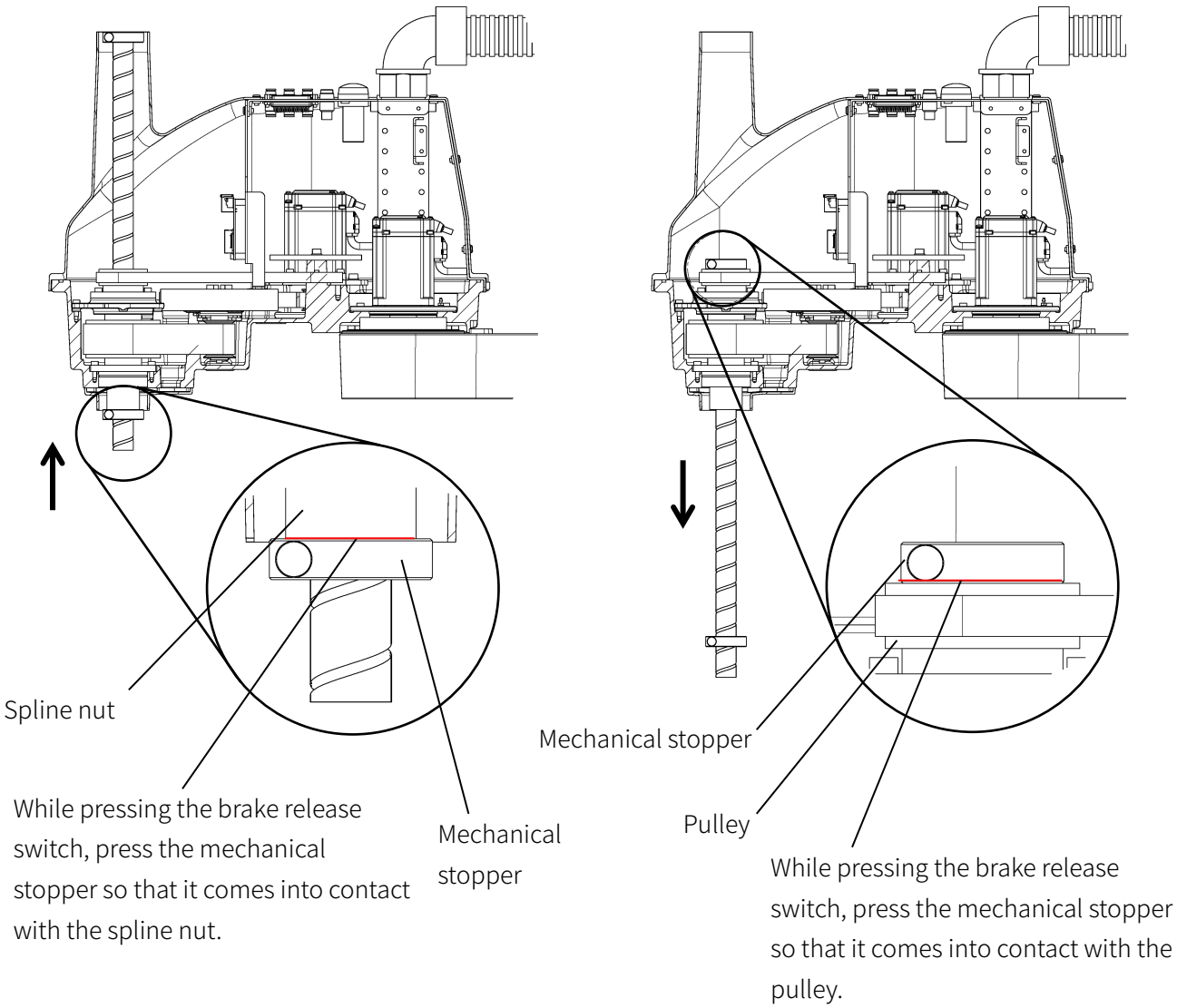


Fig. 9.34 Check Work (Axis 3)

9.10.3 Checking the Axis 4

While pressing the brake release switch in an emergency stop state, manually align the axis 4 with the zero position as shown in Fig. 9.35. At this time, check on the POS screen of the utility that the value of the axis 4 is not deviated from 0.

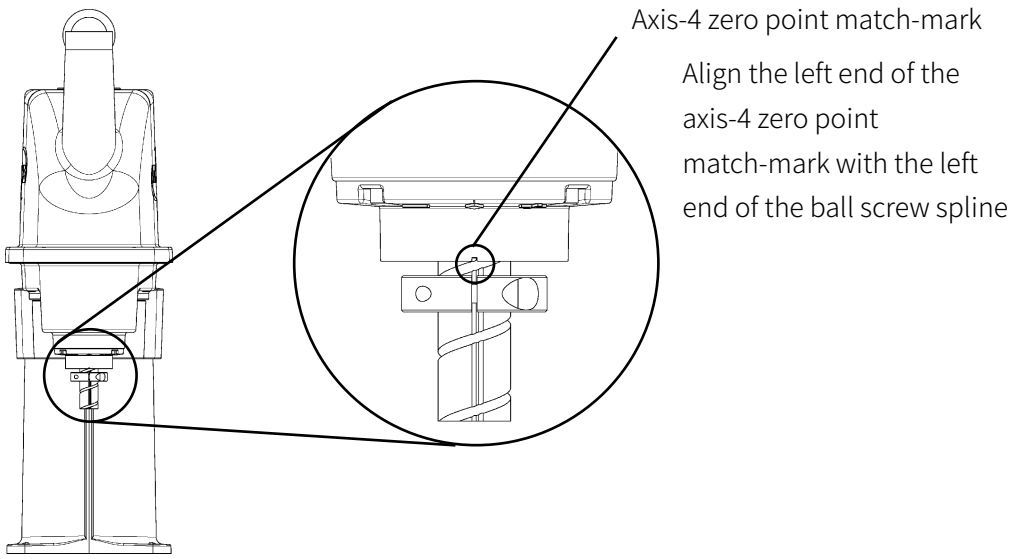


Fig. 9.35 Check Work (Axis 4)

9.11 TP5000 Operation Restriction

A screen operation restriction based on the user type is provided on the teach pendant TP5000 to prevent data (zero point data, etc.) from being lost due to misoperation by the third party.

Performing operation for maintenance on the maintenance screen requires the user type to be changed to “maintenance.”

<Changing user type>

1 Tap the maintenance button on the top screen.

The maintenance menu screen appears.

2 Tap the authority setting button.

The password input screen appears.

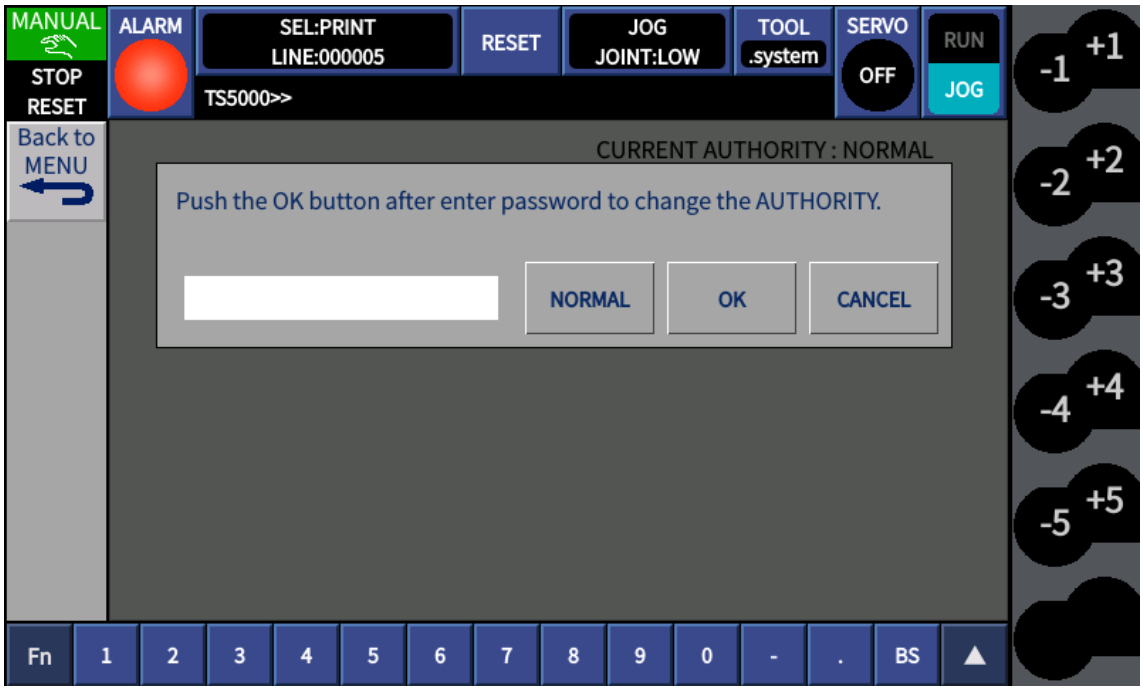


Fig. 9.36 Password input screen

3 Tap the text box.

The cursor is displayed in the text box, which can be edited.

4 Use the keyboard to enter “0,” and tap the OK button.

The user type is changed to “maintenance.”

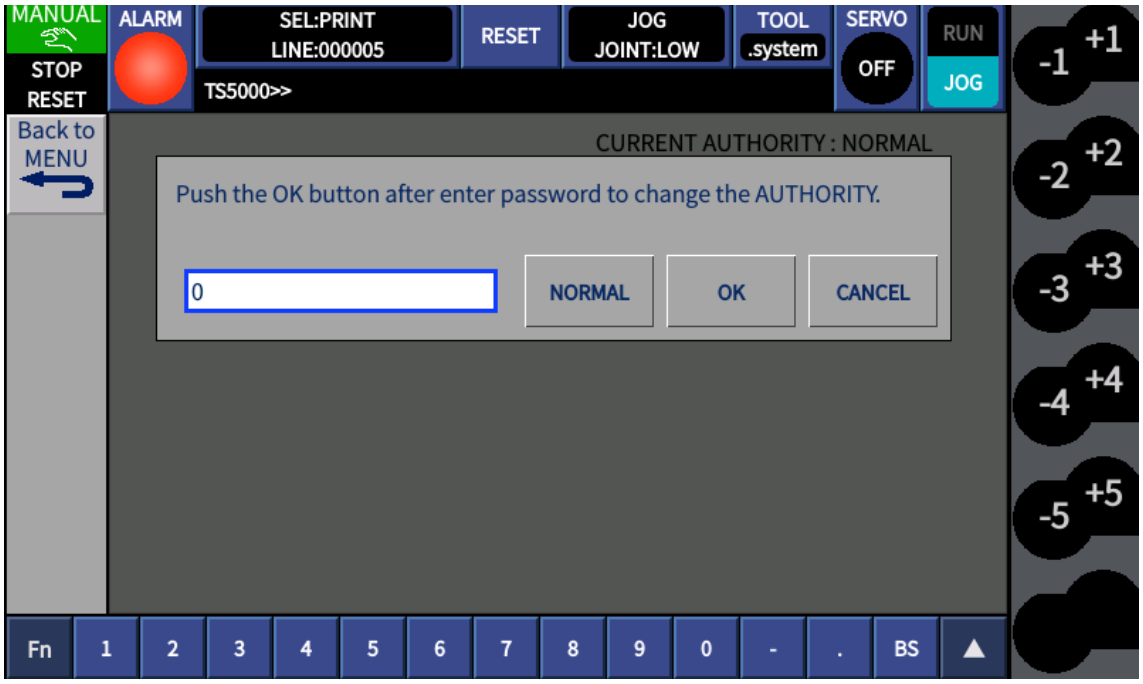


Fig. 9.37 Password input screen 2

MEMO

- The changed user type is valid until it is changed to other user type or the power is turned off.

- When the power to TP5000 is turned on, the user type is “general.”

10. Replacement Parts for Maintenance

10.1 Notes on Replacement Parts for Maintenance

⚠DANGER

- Be sure to replace the parts with SHIBAURA MACHINE-specified replacement parts for maintenance. If parts other than the specified parts are used, unpredictable disaster will occur.
  - Items other than the position detector battery are all special-specification products. When purchasing or ordering them, be sure to contact SHIBAURA MACHINE.
- Be sure to contact SHIBAURA MACHINE service company about work other than replacement of the position detector battery. We accepts the work at an extra cost.
  SHIBAURA MACHINE's warranty does not cover accidents and damage resulting from the replacement performed by the customer and its consequence.

NOTICE

- Be sure to replace the parts with SHIBAURA MACHINE-specified replacement parts for maintenance. If parts other than the specified parts are used, the performance and lifespan of the robot will be reduced. The robot will malfunction.

10.2 List of Replacement Parts – Robot

No	Part name	Model	Drawing No.	Unit code	Manufacturer	Q'ty	Remarks
1	AC servo motor		S968416	Y610D0CW0	SHIBAURA MACHINE	1	Axis 1
2			S968417	Y610D0CX0		1	Axis 2 and Axis 4
3			S968418	Y610D0CY0		1	Axis 3
5	Reduction gear		S968522	Y610D0CZ0	SHIBAURA MACHINE	1	Axis 1
6			S981478	Y610D0D00		1	Axis 2
8	Timing pulley		S968379	Y610D0DD0	SHIBAURA MACHINE	1	Axis 3 (ball screw side)
9			S968380			1	Axis 3 (motor side)
10			S968381			1	Axis 4 (lower side of intermediate shaft)
11			M342404			1	Axis 4 (ball screw side)
12			M342405			1	Axis 4 (motor side)
13			M342406			1	Axis 4 (upper side of intermediate shaft)
14	Timing belt		S968393			1	Axis 3
15			S968395			1	Axis 4 (ball screw side)
16			S968397			1	Axis 4 (motor side)
17	Ball screw spline unit		H880457	Y610D0CP0	SHIBAURA MACHINE	1	
18	Body harness		F127810	Y610D0DF0	SHIBAURA MACHINE	1	
19	LED Lamp		M342499	Y610D0DU0	SHIBAURA MACHINE	1	
20	O ring	CO0540A			NOK Corp.	1	Used for Axis-2 motor assembly
21		CO0543A				1	

22	Grease	Multemp FZNo.00			Kyodo Yushi Co., Ltd.		Axis 1
23		SFB No.1 (For reduction gear)			Nidec-Shimpo Corp.		Axis 2
24		AFF grease (For ball screw)			THK		Axis 3
25		LGAF 3E (For key)			Japan SKF Co., Ltd.		All axes
26	Screw locking adhesive	LOCTITE 241			Henkel Japan Ltd.		
27		LOCTITE 221					
28	Encoder backup battery	Alkaline AA-cell battery				3	All axes
29	Zero setting tool		M342399	Y610D0DQ0	SHIBAURA MACHINE		Axis 1 and Axis 2
30			L23752G01	Y610D0DR0			Axis 3 and Axis 4
31	Tension adjustment tool		L27611G01	Y610D0DW0	SHIBAURA MACHINE		Axis 4
32			L27612G01	Y610D0DX0			Axis 3 and Axis 4
33	Tool flange		L12515G01	Y610D0300	SHIBAURA MACHINE		

11. Disposal

11.1 Notes on Disposal

When disposing of the robot body, pallets and packing materials that have become unnecessary after the product is unpacked, parts that have become unnecessary after replacement, bundling tie cut during maintenance, waste cloth to which grease has adhered, etc., handle them as industrial waste.
When disposing of them, observe the customer-defined rules. SHIBAURA MACHINE does not collect and dispose of them.

⚠DANGER

- Never burn, short-circuit, charge, disassemble, and heat the battery.
  If liquid that leaks after the battery blows out adheres to your skin or eye, inflammation results.
  - When liquid adheres to your skin, wash it away with flowing water immediately.
  - When liquid adheres to your eye, wash it away with flowing water immediately and then see a doctor.
  - Dispose of the battery according to the customer-defined rules.

NOTICE

- Be sure to dispose of the packing materials that have become unnecessary according to the customer-defined rules.
  When storing the packing materials, be sure to manage them to prevent them from being used for other than packing the robot.